5

VECTOR ALGEBRA, WITH APPLICATIONS TO ANALYTIC GEOMETRY

5.1 Historical introduction

In the foregoing chapters we have presented many of the basic concepts of calculus and have illustrated their use in solving a few relatively simple geometrical and physical problems. Further applications of the calculus require a deeper knowledge of analytic geometry than has been presented thus far and therefore we turn our attention now to a more detailed investigation of some fundamental geometric ideas.

Analytic geometry is the name given to the method introduced by Descartes in 1637 for the application of algebra to geometry. It employs the idea of expressing the position of any point in space by three coordinate numbers (x, y, z) which describe how far the point is, in three perpendicular directions, from some convenient origin. A geometric object, such as a curve or a surface, is a locus of points satisfying one or more special conditions; by translating these conditions into formulas involving the coordinates x, y, and z we obtain one or more equations which characterize the object in question.

A *surface* can be thought of as the locus of a point moving in space with two degrees of freedom and this suggests that we describe a surface analytically by imposing one restriction on a variable point (x, y, z) by writing an equation of the form

$$F(x, y, z) = 0 . \tag{5.1}$$

For example, we shall prove later that the equation $x^2 + y^2 + z^2 = r^2$ represents the surface of a sphere of radius r and center at the origin. An equation like (5.1) is called an *implicit representation* of the surface. If we are fortunate enough to be able to solve this equation for one of the coordinates in terms of the other two, say for z in terms of x and y, we then obtain one or more equations of the form $z = f(x, y)$, each of which is said to provide an *explicit representation* of the surface (or of a portion of the surface). For example, if we solve for z in the equation for the sphere, we find two solutions, $z = \sqrt{r^2 - x^2 - y^2}$ and $z = -\sqrt{r^2 - x^2 - y^2}$. The first gives an explicit representation of the upper hemisphere and the second of the lower hemisphere.

A *space curve* may be regarded as the locus of a point moving in space with one degree of freedom and therefore to describe it analytically we impose two conditions $F(x, y, z) = 0$

and $G(x, y, z) = 0$. Each of these equations represents a surface, and the curve may be thought of as the intersection of the two surfaces. If the curve happens to lie in the xy-plane, then one of the conditions is $z = 0$ and the other condition takes the form $f(x, y) = 0$. For example, the pair of equations $x^2 + y^2 = r^2$, $z = 0$, represents a circle, the intersection of the sphere $x^2 + y^2 + z^2 = r^2$ with the xy-plane, $z = 0$. Other methods for describing curves and surfaces will be described later. In all these methods we deal with equations involving functions of one or more variables. The algebraic and analytic properties of these functions enable us to conduct investigations regarding surfaces and curves and to discover many geometric properties that might otherwise escape notice.

As we have pointed out earlier in this book, calculus and analytic geometry were intimately related throughout their historical development. Every new discovery in one subject led to an improvement in the other. The problem of drawing tangents to curves resulted in the discovery of the derivative; that of area led to the integral; and partial derivatives were introduced to investigate curved surfaces in space. Along with these accomplishments came other parallel developments in mechanics and mathematical physics. In 1788 Lagrange published his masterpiece *Mécanique analytique* (Analytical Mechanics) which showed the great flexibility and tremendous power attained by using analytical methods in the study of mechanics. Later on, in the 19th century, the Irish mathematician William Rowan Hamilton (1805–1865) introduced his *Theory of Quaternions*, a new method and a new point of view that contributed much to the understanding of both algebra and physics. The best features of quaternion analysis and Cartesian geometry were later united, largely through the efforts of J. W. Gibbs (1839–1903) and O. Heaviside (1850–1925), and a new subject called *vector analysis* sprang into being. It was soon realized that vectors are the ideal tools for the exposition and simplification of many important ideas in geometry and physics. In this chapter we propose to discuss the elements of vector analysis along with applications to the analytic geometry of planes and straight lines. In Chapter 6, vector methods will be combined with the methods of calculus to study more general surfaces and curves.

5.2 Geometric description of vectors

In physics certain entities such as *force*, *displacement*, *velocity*, and *acceleration* possess both *magnitude* and *direction*, and they may be represented geometrically by drawing an arrow having the magnitude and direction of the quantity in question. Physicists refer to the arrow as a *vector*, and those quantities that can be represented by arrows are called *vector quantities*.

In mathematics vectors can be described in two ways, *geometrically* and *analytically*. The geometric procedure is this: A line segment joining two distinct points P and Q is said to be *directed* if one of the points, say P, is called the *initial point* or *base* and the other, Q, the *terminal point* or *tip*. We then denote the segment by $\overrightarrow{PQ}$ and refer to it as the *vector* from P to Q. This is substantially the same as the physics definition since all it amounts to is a technical description of the word "arrow." (See Figure 5.1.) Sometimes vectors are denoted by a single letter with a bar or small arrow over the symbol, thus: $\overline{A}$ or $\vec{A}$. Boldface type is often used to denote vectors in print. However, in this book we shall use lightface type with an arrow over the symbol.

For our purposes it will be convenient to regard two vectors as being *equal* if they have the same length and the same direction. In other words, we shall consider a vector

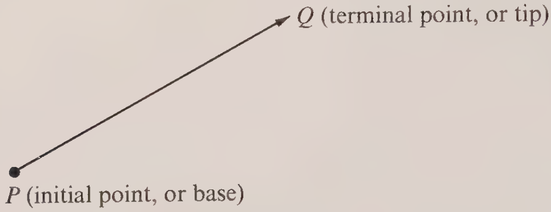


FIGURE 5.1 *The vector $\overrightarrow{PQ}$ extends from the point P to the point Q.*

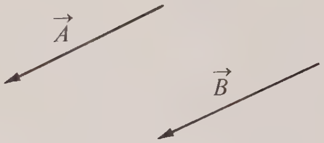


FIGURE 5.2 *$\vec{A} = \vec{B}$ because $\vec{A}$ and $\vec{B}$ have same length and direction.*

as remaining unchanged if it is moved parallel to itself.† (See Figure 5.2.) With this agreement we can assume, if we wish, that every vector emanates from some fixed reference point O. If we introduce a rectangular coordinate system with this point as origin, we are led to an analytical description of a vector, that is, a description that can be given entirely in terms of *numbers*.

5.3 Analytic description of vectors. Components

Consider a vector $\vec{A}$ with its initial point at the origin of our coordinate system and let the coordinates of the terminal point be (a_1, a_2, a_3). (See Figure 5.3.) The three real numbers a_1, a_2, a_3 are called the *components* of $\vec{A}$: a_1 is the *first* or x-component, a_2 the *second* or y-component, and a_3 the *third* or z-component. Now let $P = (p_1, p_2, p_3)$ and $Q = (q_1, q_2, q_3)$ be two points in space such that the vector $\overrightarrow{PQ}$ is equal to $\vec{A}$ (according to the above definition of equality). By examining corresponding sides of congruent triangles in Figure 5.3, we see that we must have

$$q_1 - p_1 = a_1\,, \qquad q_2 - p_2 = a_2\,, \qquad q_3 - p_3 = a_3\,.$$

The three numbers $q_1 - p_1$, $q_2 - p_2$, and $q_3 - p_3$ are called the components of $\overrightarrow{PQ}$. Thus, equal vectors have the same components, and conversely, two vectors that agree in their respective components must necessarily be equal in magnitude and direction. This means that a vector is completely determined by specifying the three numbers which constitute its components. If a vector $\vec{A}$ has components a_1, a_2, a_3, we indicate this by writing $\vec{A} = [a_1, a_2, a_3]$. Similarly, if $\vec{B}$ has components b_1, b_2, b_3, we write $\vec{B} = [b_1, b_2, b_3]$.

† Vectors that conform to this definition of equality are often called *free vectors* because they are not fixed to any position in space. For some applications of vectors, this definition of equality is not advisable. For example, suppose a rigid bar is pivoted at its center. A force applied at one end of the bar will cause it to rotate clockwise whereas a force of the same magnitude and direction applied at the other end will cause it to rotate counterclockwise. In situations like this, where a translation of the vector quantity alters its effect, it is preferable to define vectors to be equal whenever they have the same length, direction, *and* position in space. Such vectors are often referred to as *bound* vectors. In this book all vectors are assumed to be free vectors unless something is said to the contrary.

A vector equation such as $\vec{A} = \vec{B}$ is simply a brief way of writing the following three equations for real numbers:

$$a_1 = b_1 , \qquad a_2 = b_2 , \qquad a_3 = b_3 .$$

It is easy to calculate the length of a vector from a knowledge of its components. If $\vec{A} = [a_1, a_2, a_3]$, we may refer to Figure 5.3 again and apply the theorem of Pythagoras twice to find that

$$\text{(5.2)} \qquad \text{length of } \vec{A} = \sqrt{a_1^2 + a_2^2 + a_3^2} .$$

The symbol $|\vec{A}|$ will be used to denote the length of A.

It is convenient to admit the possibility of a *zero vector*, that is, one whose components are all zero. This vector is denoted by the symbol $\vec{0}$ and it is assigned the length 0, which is consistent with (5.2), but we do not assign the zero vector any particular direction. Geometrically, the zero vector can be thought of as a limiting case of a directed line segment in which the two endpoints are brought into coincidence. Note that all *nonzero* vectors have a *positive* length, that is, $|\vec{A}| > 0$ if $\vec{A} \neq \vec{0}$.

5.4 Direction angles and direction cosines

Next we want to assign numbers which can be used to describe the direction of a vector. This is usually done by considering the three angles α, β, and γ determined by a vector $\vec{A}$ and the positive x-, y-, and z-axes as shown in Figure 5.4. Sometimes it is more convenient to work with the *cosines* of these angles rather than with the angles themselves because the cosines are proportional to the components. In fact, we have

$$\text{(5.3)} \qquad \cos \alpha = \frac{a_1}{|\vec{A}|}, \qquad \cos \beta = \frac{a_2}{|\vec{A}|}, \qquad \cos \gamma = \frac{a_3}{|\vec{A}|} .$$

The three numbers $\cos \alpha$, $\cos \beta$, and $\cos \gamma$ are called the *direction cosines* of $\vec{A}$, the angles themselves being referred to as the *direction angles*. If the direction cosines and the length of $\vec{A}$ are known, we can compute the components from the equations in (5.3).

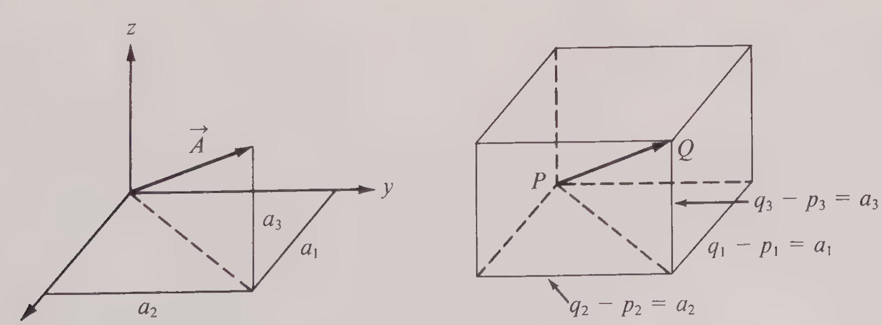


FIGURE 5.3 *Equal vectors have the same components.*

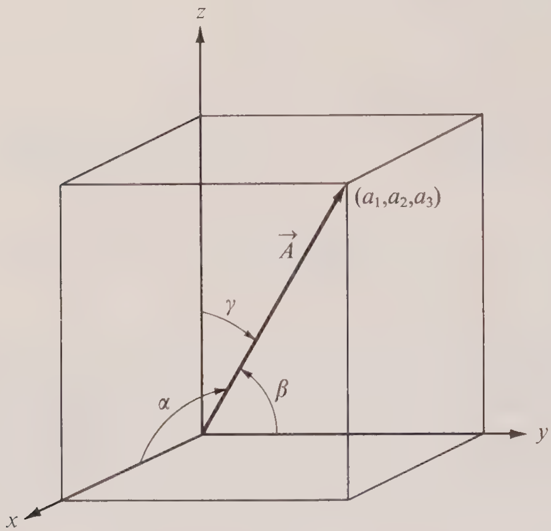


FIGURE 5.4 *Direction angles of a vector.*

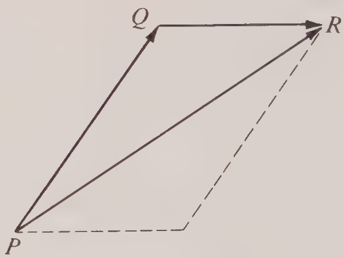


FIGURE 5.5 *The parallelogram law.*

The direction cosines of a vector cannot be arbitrary. They are related, in fact, by the equation

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 ,$$

which is nothing but a restatement of Equation (5.2).

Notice that the above discussion implicitly assumes that $\vec{A} \neq \vec{0}$. We do not assign any direction angles or direction cosines to the zero vector.

5.5 The parallelogram law

If an object is displaced along a line from one point to another, say from P to Q, we may denote this displacement by the vector $\overrightarrow{PQ}$. Suppose this is followed by another displacement from Q to R say, which we denote by the vector $\overrightarrow{QR}$. (See Figure 5.5.) The net result of the two displacements is the same as the single displacement from P to R given by the vector $\overrightarrow{PR}$. It is customary to refer to the displacement $\overrightarrow{PR}$ as the *resultant* of the two displacements $\overrightarrow{PQ}$ and $\overrightarrow{QR}$. Notice that the resultant $\overrightarrow{PR}$ is a diagonal of the parallelogram determined by $\overrightarrow{PQ}$ and $\overrightarrow{QR}$. For this reason, displacements are said to combine according to the *parallelogram law*. The importance of vectors in physics stems from the remarkable fact that a large number of physical quantities (such as displacements, forces, velocities, and accelerations) combine by the parallelogram law.

When the parallelogram law is translated into the language of components we find a very simple relationship holding between the components of the various vectors involved. Figure 5.6 shows two vectors $\vec{A} = [a_1, a_2, 0]$ and $\vec{B} = [b_1, b_2, 0]$ lying in the xy-plane.

It is evident from the figure that the resultant $\vec{C}$ has the components $[a_1 + b_1, a_2 + b_2, 0]$. In other words, the components of the resultant are obtained by adding the corresponding components of $\vec{A}$ and $\vec{B}$. The same holds true even when $\vec{A}$ and $\vec{B}$ are arbitrary vectors in space. For this reason the resultant is usually called the *sum* of $\vec{A}$ and $\vec{B}$ and the process described by the parallelogram law is known as *vector addition.*

From the mathematical point of view, the parallelogram law is only the starting point for introducing an entirely new algebraic system called *vector algebra* in which we study not only *addition* of vectors but also *subtraction* and three different kinds of *multiplication.* As we investigate these new operations in the pages that follow we shall find that they satisfy many of the familiar laws of ordinary algebra and, as a result, many formulas involving vectors may be manipulated as though they were ordinary algebraic formulas for real numbers. On the other hand, the analogy between vectors and numbers is not entirely complete—some of the laws for numbers do not carry over to vectors and careful attention must be paid to these exceptions.

5.6 The basic ideas of vector algebra

The study of vector algebra can proceed in any one of several ways. We may define the various algebraic operations in a geometric fashion and then derive (as theorems)

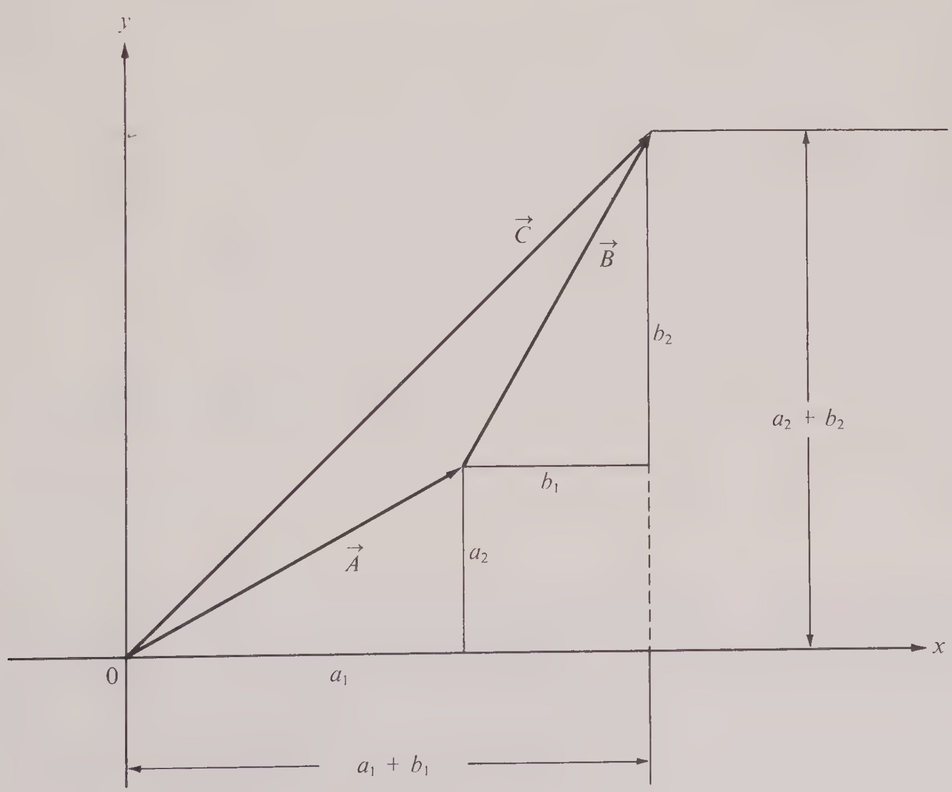


FIGURE 5.6 *The parallelogram law expressed in terms of components.*

the corresponding formulas which express these operations in the language of components. Or we may begin by defining all the operations entirely in terms of components and then interpret the resulting formulas geometrically. These procedures are equivalent, but we shall adopt the latter, first of all, because it makes it very easy to prove the fundamental laws governing vector algebra, and secondly, because it shows us the way to generalize most of these operations to higher-dimensional spaces.†

In line with this point of view it should be mentioned that the study of vectors can be divorced entirely from geometry simply by defining a vector from the outset to be a collection of numbers written down in a certain order. Thus, for example, a two-dimensional vector can be defined as an *ordered pair* $[a_1, a_2]$ of real numbers, a three-dimensional vector as an *ordered triple* $[a_1, a_2, a_3]$ and, in general, an n-dimensional vector as an *ordered n-tuple* $[a_1, a_2, \ldots, a_n]$, the individual numbers $a_1, a_2, \ldots, a_n$ being referred to as the *components* of the vector. The totality of all n-dimensional vectors is called *n-dimensional space* or, simply,‡ *n-space*. This is consistent with the geometric definition in 3-space given above if we agree to represent $[a_1, a_2, a_3]$ geometrically by a directed line segment from some initial point $P = (p_1, p_2, p_3)$ to a terminal point Q with coordinates $(p_1 + a_1, p_2 + a_2, p_3 + a_3)$.

Although our primary interest in this book is the study of spaces of dimension three or less, it is just as easy to formulate most of the introductory ideas for vectors in n-space as well. Unfortunately, the geometric pictures which are a great help in motivating and illustrating the vector operations in 3-space are not available when $n > 3$; therefore the study of vectors in higher-dimensional spaces must proceed entirely by analytic means.

The reader may well ask at this stage why we are interested in spaces of dimension greater than three. One answer is that many problems which involve a large number of simultaneous equations are more easily analyzed by introducing vectors in a suitable n-space and replacing all these equations by a single vector equation. Another advantage, especially useful to us here, is that if we speak about a general n-dimensional space we are able to deal in one stroke with many properties common to both 2-space and 3-space. This is in keeping with the spirit of modern mathematics which favors the development of comprehensive methods for attacking problems on a wide front.

The first notion to be defined is that of *equality* of two vectors. If $\vec{A} = [a_1, a_2, \ldots, a_n]$ and $\vec{B} = [b_1, b_2, \ldots, b_n]$ are any two vectors in n-space, we say that $\vec{A}$ and $\vec{B}$ are *equal* whenever they agree in their respective components. That is, the vector equation

$$\vec{A} = \vec{B}$$

† There is also another approach to vector algebra called the *abstract* or *axiomatic* approach, analogous to the axiomatic development of the real-number system given in Chapter 1. Here no attempt is made to describe the nature of a vector or of the algebraic operations on vectors. Instead, vectors and vector operations are thought of as *undefined concepts* of which we know nothing except that they satisfy a certain set of axioms. Such an algebraic system, with appropriate axioms, is called a *linear space* or a *linear vector space*. Examples of linear spaces occur in all branches of mathematics; the vector algebra developed here is only one such example. The study of linear spaces from the abstract point of view is ordinarily carried out in courses in abstract algebra.

‡ The real-number system itself may be thought of as *one-dimensional space*, and the real numbers as one-dimensional vectors.

means exactly the same thing as the n numerical (or scalar†) equations

$$a_1 = b_1\,, \quad a_2 = b_2\,, \quad \ldots, \quad a_n = b_n\,.$$

Next we define *addition* of vectors in n-space. If $\vec{A}$ and $\vec{B}$ are two given vectors, say $\vec{A} = [a_1, \ldots, a_n]$ and $\vec{B} = [b_1, \ldots, b_n]$, the *sum* $\vec{A} + \vec{B}$ is defined to be the vector obtained by adding corresponding components. Thus we have

$$\vec{A} + \vec{B} = [a_1 + b_1\,, a_2 + b_2\,, \ldots, a_n + b_n]\,.$$

From this definition it is easy to verify the following properties of vector addition.

5–1 THEOREM.

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \qquad \text{(commutative law)},$$
$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \qquad \text{(associative law)}.$$

The proof follows quickly from the definition and is left as an exercise for the reader.

Any algebraic system consists of a collection of objects along with certain operations for combining these objects. These operations are usually of two types: *binary* operations, which are used to combine two objects at a time, and *unary* operations, where the operation applies to just one object at a time.‡ Whenever we study a binary operation (for example, vector addition) we often ask whether or not there is an *identity element* for the operation, that is, an object which leaves all other objects unchanged by the operation. In the case of vector addition an identity element exists and is provided by the so-called *zero vector*. This vector, denoted by $\vec{0}$, has all its components zero. Thus, $\vec{0} = [0, \ldots, 0]$, and it has the property that $\vec{A} + \vec{0} = \vec{A}$ for every vector $\vec{A}$. We leave it as an exercise for the reader to show that there is only one identity element for vector addition.

An example of a unary operation is that of taking the *negative* of a vector. If a vector $\vec{A} = [a_1, a_2, \ldots, a_n]$, then by $-\vec{A}$ we mean the vector

$$-\vec{A} = [-a_1\,, -a_2\,, \ldots, -a_n]\,.$$

Note that $\vec{A} + (-\vec{A}) = \vec{0}$ and that $-\vec{A}$ is the only vector which gives zero when added to $\vec{A}$. We define the *difference* $\vec{A} - \vec{B}$ to be $\vec{A} + (-\vec{B})$.

In spaces of dimension three or less the vector $-\vec{A}$ may be represented geometrically by an arrow which points exactly opposite to that representing $\vec{A}$, as shown in Figure 5.7. Both vectors $\vec{A}$ and $-\vec{A}$ have the same length. The operation of vector subtraction is illustrated in Figure 5.8.

Incidentally, the notion of "length" can also be introduced in n-space. One way of doing this is by analogy with formula (5.2) above. If $\vec{A} = [a_1, a_2, \ldots, a_n]$, we define the length of $\vec{A}$ to be the nonnegative number $|\vec{A}|$ given by the following formula:

† The word "scalar" is often used as a synonym for "real number."

‡ A binary operation is really a *function* whose domain consists of ordered pairs of objects a, b taken from some set S and whose range consists of members of S. Such an operation is an example of a function of two variables. By the same token, a unary operation is an example of a function of one variable—it assigns to certain objects in S other objects in S.

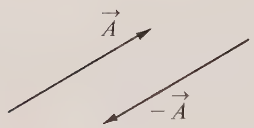


FIGURE 5.7 *A vector $\vec{A}$ and its negative $-\vec{A}$.*

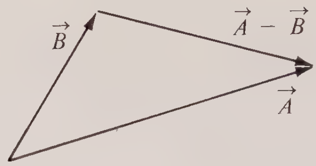


FIGURE 5.8 *Geometric meaning of subtraction of vectors.*

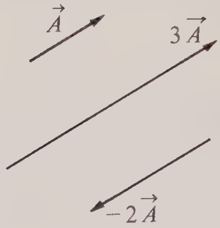


FIGURE 5.9 *Multiplication of vectors by scalars.*

$$|\vec{A}| = \left(\sum_{k=1}^{n} a_k^2\right)^{1/2}. \tag{5.4}$$

In other words, the length of $\vec{A}$ is the nonnegative square root of the sum of the squares of its components. The justification for this definition is, first of all, that it agrees with the geometric notion of length for spaces of dimension ≤ 3, and secondly, it generalizes the important properties of length from 3-space to n-space. In particular, one can prove the following theorem as a consequence of this definition:

5- 2 THEOREM. (a) $|\vec{0}| = 0$, and $|\vec{A}| > 0$ whenever $\vec{A} \neq \vec{0}$,
(b) $|-\vec{A}| = |\vec{A}|$,
(c) $|A + \vec{B}| \leq |\vec{A}| + |\vec{B}|$ (triangle inequality).

Part (c) is the only statement that is not immediately obvious; its proof is given in Section 5.9. The proofs of (a) and (b) are left to the reader.

When $\vec{A} = \vec{B}$, the formula for addition of vectors states that

$$\vec{A} + \vec{A} = [2a_1, 2a_2, \ldots, 2a_n].$$

This suggests that we write $2\vec{A}$ for $\vec{A} + \vec{A}$, as in ordinary algebra. More generally, if $\vec{A} = [a_1, a_2, \ldots, a_n]$ and if c is any real number (positive, negative, or zero), we define the product $c\vec{A}$ to be the vector obtained by multiplying each component of $\vec{A}$ by c. Thus, we have

$$c\vec{A} = [ca_1, ca_2, \ldots, ca_n].$$

This operation is known as *multiplication of vectors by scalars.* (It is customary to write the scalar to the left of the vector.)

As an immediate consequence of this definition we have the following theorem whose proof is left as an exercise.

5–3 THEOREM. $c(\vec{A} + \vec{B}) = c\vec{A} + c\vec{B}$, $(c + d)\vec{A} = c\vec{A} + d\vec{A}$,
$(cd)\vec{A} = c(d\vec{A})$, $|c\vec{A}| = |c|\,|\vec{A}|$,
$0\vec{A} = \vec{0}$, $1\vec{A} = \vec{A}$, $(-1)\vec{A} = -\vec{A}$, $-\vec{0} = \vec{0}$.

It is easy to interpret this operation geometrically in 3-space by examining direction cosines. If $\vec{A}$ is a nonzero vector with direction cosines $\cos\alpha$, $\cos\beta$, $\cos\gamma$, and if c is a nonzero scalar, then [because of (5.3)] the vector $c\vec{A}$ has direction cosines

$$\frac{c}{|c|}\cos\alpha\,, \qquad \frac{c}{|c|}\cos\beta\,, \qquad \frac{c}{|c|}\cos\gamma\,.$$

This means that $\vec{A}$ and $c\vec{A}$ have the same direction cosines if $c > 0$. Otherwise the direction cosines of $c\vec{A}$ are the negatives of those of $\vec{A}$, in which case the direction angles of $c\vec{A}$ can be taken to be

$$\alpha + \pi\,, \qquad \beta + \pi\,, \qquad \gamma + \pi\,.$$

Geometrically this means that $c\vec{A}$ and $\vec{A}$ have the same direction if $c > 0$ and opposite directions if $c < 0$. Examples are shown in Figure 5.9.

In n-space we shall define two vectors $\vec{A}$ and $\vec{B}$ as having the *same direction* if $\vec{B} = c\vec{A}$ for some positive scalar c. The vectors are said to have *opposite directions* if $\vec{B} = c\vec{A}$ for a negative c. They are called *parallel* if $\vec{B} = c\vec{A}$ for some *nonzero* c.

Note that this definition makes every vector have the same direction as itself—a property which we surely want. Note also that this definition ascribes the following properties to the zero vector: The only vector having the same direction as $\vec{0}$ is $\vec{0}$ itself. The zero vector is the only vector having the same direction as its negative and therefore the only vector having the opposite direction to itself. The zero vector is the only vector parallel to $\vec{0}$.

5.7 Exercises

1. (a) By drawing vectors in a plane, illustrate the geometric meanings of the commutative and associative laws for vector addition (Theorem 5–1). (b) Prove Theorem 5–1 for vectors in n-space.

2. The vector sum $\vec{A} + \vec{B}$, when interpreted geometrically in 2-space or in 3-space, is a diagonal of the parallelogram determined by $\vec{A}$ and $\vec{B}$. Show, on the same parallelogram, the meanings of the differences $\vec{A} - \vec{B}$ and $\vec{B} - \vec{A}$.

3. Suppose $\vec{A}$, $\vec{B}$, $\vec{C}$ denote vectors in the plane drawn from the origin O to points A, B, C, respectively.

(a) If the quadrilateral $OABC$ is a parallelogram having A and C as opposite vertices, show that $\vec{A} + \frac{1}{2}(\vec{C} - \vec{A}) = \frac{1}{2}\vec{B}$.

(b) What geometrical theorem about parallelograms can you deduce from the vector equation in part (a)?

4. (a) By drawing vectors in a plane, illustrate the geometric meaning of the properties $(c + d)\vec{A} = c\vec{A} + d\vec{A}$ and $c(\vec{A} + \vec{B}) = c\vec{A} + c\vec{B}$.

(b) Prove, for vectors in n-space, all the properties of multiplication by scalars given in Theorem 5–3.

5. By drawing vectors in a plane, explain the geometric significance of each of the following vector equations:

(a) $\vec{A} + \vec{B} + \vec{C} = \vec{0}$; (b) $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$; (c) $\vec{A}_1 + \vec{A}_2 + \cdots + \vec{A}_n = \vec{0}$.

6. Draw the vectors $\vec{A} = [2, 1]$ and $\vec{B} = [1, 3]$ emanating from the origin in the plane. On the same figure, draw the vector $\vec{C} = \vec{A} + t\vec{B}$ for each of the following values of t: $t = \frac{1}{3}$; $t = \frac{1}{2}$; $t = \frac{3}{4}$; $t = 1$; $t = 2$; $t = -1$; $t = -2$.

7. On a new figure, draw the vectors $\vec{A}$ and $\vec{B}$ as in Exercise 6 and then solve Exercise 6 if $\vec{C} = t\vec{A} + \vec{B}$.

8. On a new figure, draw the vectors $\vec{A}$ and $\vec{B}$ as in Exercise 6. Let $\vec{C} = x\vec{A} + y\vec{B}$, where x and y are real numbers.

(a) Draw the vector $\vec{C}$ for each of the following pairs of values of x and y: $x = y = \frac{1}{2}$; $x = \frac{1}{4}, y = \frac{3}{4}$; $x = \frac{1}{3}, y = \frac{2}{3}$; $x = 2, y = -1$; $x = 3, y = -2$; $x = -\frac{1}{2}, y = \frac{3}{2}$; $x = -1, y = 2$.

(b) Let C denote the tip of $\vec{C}$. What do you think is the locus of all points C obtained as x and y run through all real numbers such that $x + y = 1$? (Just make a guess and show the locus on the figure. No proof is requested.)

(c) Make a guess for the locus of all points C obtained as x and y range independently over the intervals $0 \leq x \leq 1$, $0 \leq y \leq 1$, and draw a sketch of this locus.

(d) What do you think is the locus of C if x ranges through the interval $0 \leq x \leq 1$ and y ranges through all real numbers?

(e) What do you think is the locus if x and y both range over all real numbers?

9. Let $\vec{A} = [2, 1]$ and $\vec{B} = [1, 3]$. Show that for every vector $\vec{C} = [c_1, c_2]$ in the plane there exist real numbers x and y such that $\vec{C} = x\vec{A} + y\vec{B}$. Express x and y in terms of c_1 and c_2.

10. Let $\vec{A} = [a_1, a_2]$ and $\vec{B} = [b_1, b_2]$ be two nonzero and nonparallel vectors in the plane. Show that for every vector $\vec{C} = [c_1, c_2]$ there exist real numbers x and y such that $\vec{C} = x\vec{A} + y\vec{B}$ and express x and y in terms of c_1 and c_2. In your proof, point out where you used the assumption that $\vec{A}$ and $\vec{B}$ are nonparallel.

11. (a) Prove that two nonzero vectors $\vec{A}$ and $\vec{B}$ in n-space are parallel if and only if there exist two real numbers x and y, not both zero, such that $x\vec{A} + y\vec{B} = \vec{0}$.

(b) If $\vec{A}$ and $\vec{B}$ have the same direction, and if $\vec{B}$ and $\vec{C}$ have the same direction, prove that $\vec{A}$ and $\vec{C}$ have the same direction.

(c) If $\vec{A}$ and $\vec{B}$ have opposite directions, and if $\vec{B}$ and $\vec{C}$ have opposite directions, prove that $\vec{A}$ and $\vec{C}$ have the same direction.

12. Suppose that, instead of defining the length of a vector $\vec{A} = [a_1, \ldots, a_n]$ as we did in Equation (5.4), we used the following definition:

$$|\vec{A}| = \sum_{k=1}^{n} |a_k| .$$

(a) Prove that this definition of length satisfies all the properties given in Theorem 5–2.

(b) Use this definition of length in 2-space and describe on a figure the locus of all points (x, y) such that the vector $\vec{A} = [x, y]$ has length 1.

(c) Which of the properties of Theorem 5–2 would hold if we used the following definition:

$$|\vec{A}| = \left| \sum_{k=1}^{n} a_k \right| ?$$

13. Suppose that, instead of the definition of length given in Equation (5.4), we used the following definition:

$$|\vec{A}| = \max_{1 \leq k \leq n} |a_k| ,$$

where the symbol on the right means the maximum (largest) of the n numbers $|a_1|, |a_2|, \ldots, |a_n|$.

(a) Which properties of Theorem 5–2 are valid with this definition?

(b) Use this definition of length in 2-space and describe on a figure the locus of all points (x, y) such that the vector $\vec{A} = [x, y]$ has length 1.

14. Let

$$|\vec{A}|_1 = \sum_{k=1}^{n} |a_k| \qquad \text{and} \qquad |\vec{A}|_2 = \max_{1 \le k \le n} |a_k|,$$

if $\vec{A} = [a_1, \ldots, a_n]$. Prove that $|\vec{A}|_2 \le |\vec{A}| \le |\vec{A}|_1$. Interpret this inequality geometrically in 2-space.

15. If $ABCD$ is a quadrilateral in space, show that the mid-points of its sides are the vertices of a parallelogram whose center is the terminal point of the vector $\frac{1}{4}(\vec{A} + \vec{B} + \vec{C} + \vec{D})$, where $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$ denote vectors from the origin to the points A, B, C, D, respectively.

16. Let $\vec{P}_1, \ldots, \vec{P}_k$ be vectors from the origin to the points $P_1, \ldots, P_k$. If masses $m_1, \ldots, m_k$ (positive, negative, or zero) are located at the points $P_1, \ldots, P_k$, respectively, we define the *centroid* of the system to be the point C determined by the vector

$$\vec{C} = \frac{m_1\vec{P}_1 + \cdots + m_k\vec{P}_k}{m_1 + \cdots + m_k},$$

provided $m_1 + \cdots + m_k \neq 0$. Show that the position of the centroid is independent of the location of the origin but depends only on $P_1, \ldots, P_k$ and the masses.

17. If A, B, C are three points not on a line, show that any point in their plane can be made their centroid by locating suitable masses at A, B, and C.

18. Given k fixed points $A_1, \ldots, A_k$ in a plane and a movable point P. Let C be the centroid of $A_1, \ldots, A_k$ (assuming equal masses). Show that if P varies on a circle of radius $1/k$ about C, then the vector $\vec{PA_1} + \cdots + \vec{PA_k}$ will have unit length. Describe the path for P which will give this sum a length r. [*Hint.* Use Exercise 16.]

5.8 The dot product

We introduce now a new kind of multiplication called the *dot product*.† This is a binary operation that combines two vectors to produce a *real number*. If we have two vectors $\vec{A} = [a_1, a_2, \ldots, a_n]$ and $\vec{B} = [b_1, b_2, \ldots, b_n]$ in n-space, their dot product is denoted by $\vec{A} \cdot \vec{B}$ and is defined by the equation

$$\vec{A} \cdot \vec{B} = \sum_{k=1}^{n} a_k b_k. \tag{5.5}$$

That is to say, we multiply corresponding components of $\vec{A}$ and $\vec{B}$ and then add all the products to get $\vec{A} \cdot \vec{B}$. This multiplication has the following algebraic properties:

5– 4 THEOREM.

$$\begin{aligned} \vec{A} \cdot \vec{B} &= \vec{B} \cdot \vec{A} && \text{(commutative law),} \\ \vec{A} \cdot (\vec{B} + \vec{C}) &= \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C} && \text{(distributive law),} \\ c(\vec{A} \cdot \vec{B}) = (c\vec{A}) \cdot \vec{B} &= \vec{A} \cdot (c\vec{B}) && \text{(c being any scalar).} \end{aligned}$$

The proofs are easy consequences of the definition in (5.5) and are left as exercises for the reader. Note that $\vec{A} \cdot \vec{A} = a_1^2 + a_2^2 + \cdots + a_n^2$. That is, the dot product of a vector with itself is the square of its length:

† The dot product is also called the *inner product* or the *scalar product*.

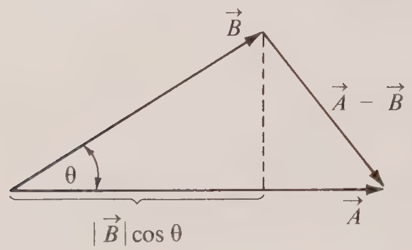


FIGURE 5.10 $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos \theta$.

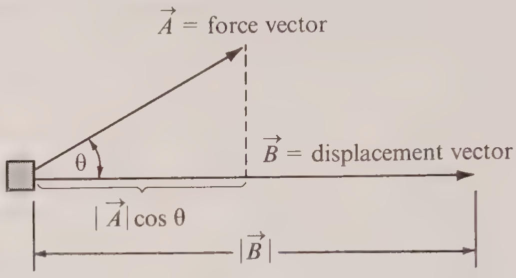


FIGURE 5.11 *Work* $= \vec{A} \cdot \vec{B}$.

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2 . \tag{5.6}$$

In spaces of dimension two or three the dot product has an interesting geometric interpretation which can be seen by referring to the triangle in Figure 5.10. The sides of the triangle have lengths $|\vec{A}|$, $|\vec{B}|$, and $|\vec{A} - \vec{B}|$ and the law of cosines of trigonometry tells us that these lengths are related by the equation

$$|\vec{A} - \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}|\,|\vec{B}| \cos \theta ,$$

where θ is the angle between $\vec{A}$ and $\vec{B}$, $0 \le \theta \le \pi$. Using (5.6) and the algebraic properties of dot products, we also have

$$|\vec{A} - \vec{B}|^2 = (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B}) = \vec{A} \cdot \vec{A} - 2\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{B} = |\vec{A}|^2 + |\vec{B}|^2 - 2\vec{A} \cdot \vec{B}.$$

Comparing this with the foregoing formula for $|\vec{A} - \vec{B}|^2$, we find

$$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos \theta . \tag{5.7}$$

In other words, the dot product of two nonzero vectors $\vec{A}$ and $\vec{B}$ in 3-space is equal to the product of three numbers: the length of $\vec{A}$, the length of $\vec{B}$, and the cosine of the angle between $\vec{A}$ and $\vec{B}$. The lengths $|\vec{A}|$ and $|\vec{B}|$ are nonnegative but $\cos \theta$ may be positive, negative, or zero. Therefore the dot product $\vec{A} \cdot \vec{B}$ is negative only when $\cos \theta$ is negative, that is, only when $\frac{1}{2}\pi < \theta \le \pi$.

Notice that when $\vec{A}$ and $\vec{B}$ are perpendicular we have $\cos \theta = 0$ and hence $\vec{A} \cdot \vec{B} = 0$. Conversely, if $\vec{A}$ and $\vec{B}$ are two nonzero vectors such that $\vec{A} \cdot \vec{B} = 0$, then $\cos \theta$ must necessarily be zero and hence $\vec{A}$ and $\vec{B}$ are perpendicular.

We cannot use (5.7) when either $\vec{A}$ or $\vec{B}$ is the zero vector because the zero vector has no particular direction and it does not make sense to talk about the angle θ. However, since the dot product is zero in this case, mathematicians have adopted the convention of saying that the zero vector is *perpendicular to every vector*. This enables us to say that $\vec{A} \cdot \vec{B} = 0$ if and only if $\vec{A}$ and $\vec{B}$ are perpendicular.

Formula (5.7) permits us to compute $A \cdot \vec{B}$ without using components if we know the two lengths $|\vec{A}|$ and $|\vec{B}|$ and the angle θ. Conversely, it gives us an easy way to compute the angle between two nonzero vectors if we are given their components. In fact, solving (5.7) for $\cos\theta$ we find

$$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{|\vec{A}|\,|\vec{B}|} = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\,\sqrt{b_1^2 + b_2^2 + b_3^2}}. \tag{5.8}$$

In later sections we shall see many applications of the dot product to analytic geometry. It is also useful in physics where it appears in calculations involving *work*. For example, if $\vec{A}$ represents a constant force which acts on a body to move it through a displacement represented by a vector $\vec{B}$ (see Figure 5.11), then the number $|\vec{A}|\cos\theta$ represents the component of the force in the direction of motion and $|\vec{B}|$ represents the distance moved by the body. The product $(|\vec{A}|\cos\theta)(|\vec{B}|)$ is defined to be the *work* done by this force during its action on the body. Because of (5.7) the amount of work done is exactly equal to the dot product $\vec{A}\cdot\vec{B}$.

5.9 A proof of the triangle inequality using properties of the dot product

In this section we show how the algebraic properties of the dot product in n-space can be used to prove the triangle inequality,

$$|\vec{A}+\vec{B}| \le |\vec{A}| + |\vec{B}|, \tag{5.9}$$

mentioned earlier in Theorem 5–2. To avoid square roots we square both sides of (5.9) to obtain

$$|\vec{A}+\vec{B}|^2 \le (|\vec{A}| + |\vec{B}|)^2. \tag{5.10}$$

If we prove (5.10), then (5.9) follows by taking the positive square root of each side. For the left member of (5.10) we have

$$|\vec{A}+\vec{B}|^2 = (\vec{A}+\vec{B})\cdot(\vec{A}+\vec{B}) = \vec{A}\cdot\vec{A} + 2\vec{A}\cdot\vec{B} + \vec{B}\cdot\vec{B} = |\vec{A}|^2 + 2\vec{A}\cdot\vec{B} + |\vec{B}|^2,$$

and for the right member we have

$$(|\vec{A}| + |\vec{B}|)^2 = |\vec{A}|^2 + 2|\vec{A}|\,|\vec{B}| + |\vec{B}|^2.$$

Therefore the inequality in (5.10) is valid if and only if we have

$$\vec{A}\cdot\vec{B} \le |\vec{A}|\,|\vec{B}|. \tag{5.11}$$

If we write $\vec{A} = [a_1, \ldots, a_n]$ and $\vec{B} = [b_1, \ldots, b_n]$ and express (5.11) in terms of components, it becomes

$$\sum_{k=1}^{n} a_kb_k \le \left(\sum_{k=1}^{n} a_k^2\right)^{1/2}\left(\sum_{k=1}^{n} b_k^2\right)^{1/2}.$$

But this last inequality follows immediately from the Cauchy-Schwarz inequality (1.22) which we proved earlier in Section 1.23.

If we write the Cauchy-Schwarz inequality in vector notation, it becomes

$$(\vec{A}\cdot\vec{B})^2 \le |\vec{A}|^2|\vec{B}|^2, \tag{5.12}$$

and by taking square roots we obtain not only (5.11) but also the inequality

(5.13) $$-\vec{A} \cdot \vec{B} \leq |\vec{A}|\,|\vec{B}|\,.$$

This proves that the triangle inequality is a consequence of the Cauchy-Schwarz inequality. Conversely, it is easy to show that (5.9) implies (5.12) so the triangle inequality and the Cauchy-Schwarz inequality are logically equivalent. Incidentally, both (5.11) and (5.13) may be incorporated into one inequality which may be written as follows:

(5.14) $$|\vec{A} \cdot \vec{B}| \leq |\vec{A}|\,|\vec{B}|\,.$$

Although a proof of the Cauchy-Schwarz inequality was given in Section 1.23, we shall present another proof here in vector form that makes no use of components. Such a proof is of interest because it shows that the Cauchy-Schwarz inequality is a consequence of the algebraic properties of the dot product and does not depend on the particular definition of dot product that was used in (5.5). To carry out this proof we notice first that (5.12) holds trivially when either $\vec{A}$ or $\vec{B}$ is $\vec{0}$ since both sides reduce to zero. Therefore to prove (5.12) in general it suffices to assume that both $\vec{A}$ and $\vec{B}$ are nonzero vectors. Assuming this, let

$$a = |\vec{A}|\,, \qquad b = |\vec{B}|\,,$$

and introduce the vector $\vec{C} = b\vec{A} - a\vec{B}$. The dot product of $\vec{C}$ with itself is a nonnegative number and it is zero only if $\vec{C} = \vec{0}$. When we translate this statement in terms of a and b, it will yield (5.11). To express $\vec{C} \cdot \vec{C}$ in terms of a and b we write

$$\begin{aligned}\vec{C} \cdot \vec{C} &= (b\vec{A} - a\vec{B}) \cdot (b\vec{A} - a\vec{B}) = b^2\vec{A} \cdot \vec{A} - 2ab\vec{A} \cdot \vec{B} + a^2\vec{B} \cdot \vec{B} \\ &= b^2a^2 - 2ab\vec{A} \cdot \vec{B} + a^2b^2 = 2ab(ab - \vec{A} \cdot \vec{B})\,.\end{aligned}$$

The inequality $\vec{C} \cdot \vec{C} \geq 0$ becomes $2ab(ab - \vec{A} \cdot \vec{B}) \geq 0$ and, since both a and b are *positive*, we may divide by $2ab$ to get $ab - \vec{A} \cdot \vec{B} \geq 0$, or $\vec{A} \cdot \vec{B} \leq ab$. In other words, we have shown that (5.11) holds. If we replace $\vec{A}$ by $-\vec{A}$ in (5.11), we obtain (5.13). In one of the two inequalities (5.11) or (5.13) the dot product $\vec{A} \cdot \vec{B}$ is nonnegative, and we may square both sides of this particular inequality to obtain (5.12).

The foregoing proof shows that the equality sign in (5.11) holds only if $\vec{C} = \vec{0}$, which means $b\vec{A} = a\vec{B}$. In other words, $\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}|$ only if the two vectors $\vec{A}$ and $\vec{B}$ have the same direction or one is zero. Therefore the triangle inequality becomes an equality if and only if $\vec{A}$ and $\vec{B}$ have the same direction or one is zero. Equality holds in (5.13) only if the two vectors have opposite directions or one is zero. Therefore the equality sign holds in (5.12) if and only if $\vec{A}$ and $\vec{B}$ are parallel or at least one of the two is the zero vector.

At this point we shall abandon the further study of vectors in n-space for $n > 3$ and confine our attention to spaces of dimension two or three. Actually we shall state and prove most of the results only for 3-space since the corresponding results for 2-space may be obtained by setting the z-component equal to zero throughout. A vector in 3-space of the form $[a_1, a_2, 0]$ acts just like the two-dimensional vector $[a_1, a_2]$ in all algebraic formulas involving the operations discussed thus far. In fact, we have

$$[a_1, a_2, 0] + [b_1, b_2, 0] = [a_1 + b_1, a_2 + b_2, 0]\,,$$

$$c[a_1, a_2, 0] = [ca_1, ca_2, 0]\,,$$

and

$$[a_1, a_2, 0] \cdot [b_1, b_2, 0] = a_1b_1 + a_2b_2\,.$$

For this reason we ordinarily make no distinction between the three-dimensional vector $[a_1, a_2, 0]$ and the two-dimensional vector $[a_1, a_2]$.

5.10 The unit coordinate vectors

There are three special vectors, called the *unit coordinate vectors* $\vec{i}$, $\vec{j}$, and $\vec{k}$, that are of particular importance in 3-space. They are defined as follows:

$$\vec{i} = [1, 0, 0]\,, \qquad \vec{j} = [0, 1, 0]\,, \qquad \vec{k} = [0, 0, 1]\,.$$

These vectors lie along the positive x-, y-, and z-axes, respectively, and they each have length 1. Their importance stems from the fact that *every* vector in 3-space can be expressed in one and only one way as a linear combination of these three. In fact, we have the following theorem:

5–5 THEOREM. If $\vec{A} = [a_1, a_2, a_3]$ is an arbitrary vector in 3-space, then we have

$$\vec{A} = a_1 i + a_2 j + a_3\vec{k}\,. \tag{5.15}$$

The geometric meaning of the theorem is illustrated in Figure 5.12. Its proof follows at once by adding the three vectors

$$a_1\, i = [a_1, 0, 0]\,, \qquad a_2\vec{j} = [0, a_2, 0]\,, \qquad a_3\vec{k} = [0, 0, a_3]\,.$$

At this point we shall dispense with the notation $\vec{A} = [a_1, a_2, a_3]$. Hereafter, to indicate that a vector $\vec{A}$ has components a_1, a_2, a_3 we shall write a formula like (5.15). The main advantage of expressing vectors in terms of $\vec{i}$, $\vec{j}$, and $\vec{k}$ is that manipulations involving addition and multiplication by scalars can be performed by the ordinary rules of algebra. The various components can be recognized at any stage of the calculation by collecting the coefficients of $\vec{i}, \vec{j}$, and $\vec{k}$. For example, to add two vectors, say $\vec{A} = a_1\,\vec{i} + a_2\vec{j} + a_3\,\vec{k}$ and $B = b_1\, i + b_2\vec{j} + b_3\,\vec{k}$, we may apply associative and commutative laws repeatedly to get

$$\begin{aligned}\vec{A} + \vec{B} &= (a_1\,\vec{i} + a_2\vec{j} + a_3\,\vec{k}) + (b_1\,\vec{i} + b_2\vec{j} + b_3\,\vec{k})\\ &= (a_1 + b_1)\,\vec{i} + (a_2 + b_2)\vec{j} + (a_3 + b_3)\,\vec{k}\,.\end{aligned}$$

The end result agrees, of course, with the definition of addition given in Section 5.6.

Similarly, to compute the dot product of $\vec{A}$ and $\vec{B}$ we dot multiply $a_1\,\vec{i} + a_2\vec{j} + a_3\,\vec{k}$ with $b_1\,\vec{i} + b_2\vec{j} + b_3\,\vec{k}$, and use the distributive law repeatedly. The terms involving $\vec{i}\cdot\vec{j}$, $\vec{i}\cdot\vec{k}$, and $\vec{j}\cdot\vec{k}$ all drop out since $\vec{i}$, $\vec{j}$, and $\vec{k}$ are mutually perpendicular. The terms remaining are

$$a_1b_1\,\vec{i}\cdot\vec{i} + a_2b_2\vec{j}\cdot\vec{j} + a_3b_3\,\vec{k}\cdot\vec{k}\,.$$

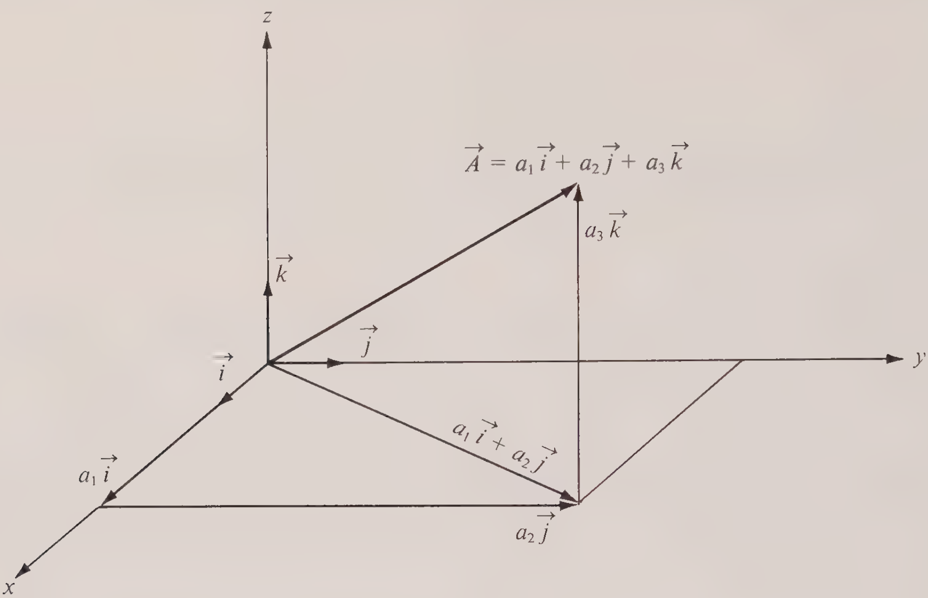


FIGURE 5.12 *A vector $\vec{A}$ expressed as a linear combination of $\vec{i}, \vec{j}, \vec{k}$.*

Since $\vec{i}\cdot\vec{i} = \vec{j}\cdot\vec{j} = \vec{k}\cdot\vec{k} = 1$, this yields $\vec{A}\cdot\vec{B} = a_1b_1 + a_2b_2 + a_3b_3$, as it should.

The triple of vectors $\vec{i}, \vec{j}, \vec{k}$ is said to form a *basis* for 3-space because every vector in 3-space can be expressed *in one and only one way* as a linear combination of these three. Other triples of vectors may serve as a basis, for example the triples $\vec{i}, \vec{j}, -\vec{k}$ and $\vec{i}, \vec{i} + \vec{j}, i + 2\vec{k}$ and many others. However, not *every* triple may be used as a basis. Thus the triple $i, \vec{j}, \vec{i} + \vec{j}$ is obviously *not* a basis for 3-space because any linear combination of these three vectors is a vector with z-component 0. Note that in this case there is more than one linear combination that yields the zero vector, for example,

$$0\vec{i} + 0\vec{j} + 0(\vec{i}+\vec{j}) = \vec{0}, \qquad \vec{i} + \vec{j} - (\vec{i}+\vec{j}) = \vec{0}, \qquad 2\vec{i} + 2\vec{j} - 2(\vec{i}+\vec{j}) = \vec{0}.$$

This is due to the fact that one of the vectors, $\vec{i} + \vec{j}$, is a linear combination of the other two. When this happens we say the three vectors are *linearly dependent.* If, on the other hand, we have three vectors $\vec{A}, \vec{B}, \vec{C}$ in 3-space such that no one of them is a linear combination of the other two, then we say that the three vectors are *linearly independent.* In this case there is only one linear combination of $\vec{A}, \vec{B}, \vec{C}$ that yields $\vec{0}$, namely $0\vec{A} + 0\vec{B} + 0\vec{C}$. In fact, if there were another we could write

$$x\vec{A} + y\vec{B} + z\vec{C} = \vec{0} \tag{5.16}$$

with x, y, z not all zero, say with $x \neq 0$; then we could multiply Equation (5.16) by $1/x$ and solve for $\vec{A}$ in terms of $\vec{B}$ and $\vec{C}$, contradicting the fact that $\vec{A}$, $\vec{B}$, and $\vec{C}$ are linearly independent. In Exercise 22 of Section 5.11 the reader is asked to prove that three vectors

$\vec{A}, \vec{B}, \vec{C}$ form a basis for 3-space if and only if they are linearly independent. When we study the applications of vectors to analytic geometry we shall find that linear dependence of $\vec{A}, \vec{B}, \vec{C}$ is equivalent to saying that $\vec{A}$, $\vec{B}$, and $\vec{C}$ are coplanar—that is, when they represent vectors from the origin O to points A, B, C, respectively, then the four points O, A, B, C lie in the same plane.

5.11 Exercises

1. Prove Theorem 5–4 which describes properties of the dot product of two vectors in n-space.

2. (a) Show that the two vectors $\vec{A} = 4\vec{i} + \vec{j} - 3\vec{k}$ and $\vec{B} = \vec{i} + 2\vec{j} + 2\vec{k}$ are perpendicular.
(b) Let $\vec{A} = \vec{i} + 2\vec{j} - 2\vec{k}$, $\vec{B} = 2\vec{i} + \vec{j} + 2\vec{k}$, $\vec{C} = 2\vec{i} - 2\vec{j} - \vec{k}$. Show that $\vec{A}$, $\vec{B}$, $\vec{C}$ are mutually perpendicular.

3. Find all vectors in 2-space that are perpendicular to $\vec{A}$ and which have the same length as $\vec{A}$ if:
(a) $\vec{A} = \vec{i} + \vec{j}$; (b) $\vec{A} = \vec{i} - \vec{j}$; (c) $\vec{A} = 2\vec{i} - 3\vec{j}$; (d) $\vec{A} = a\vec{i} + b\vec{j}$.

4. (a) Find the direction cosines of the vector $\vec{A} = 6\vec{i} + 3\vec{j} - 2\vec{k}$.
(b) Find all vectors of unit length parallel to $\vec{A}$.

5. Show that the angle between the two vectors $\vec{A} = \vec{i} + 2\vec{j} + \vec{k}$ and $\vec{B} = 2\vec{i} + \vec{j} - \vec{k}$ is twice that between the two vectors $\vec{C} = \vec{i} + 4\vec{j} + \vec{k}$ and $\vec{D} = 2\vec{i} + 5\vec{j} + 5\vec{k}$.

6. If $\vec{A} = 2\vec{i} - \vec{j} + \vec{k}$ and $\vec{B} = 3\vec{i} - 4\vec{j} - 4\vec{k}$, find a vector $\vec{C}$ in 3-space such that the terminal points of $\vec{A}, \vec{B}, \vec{C}$ are the vertices of a right triangle.

7. Given $\vec{A} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{B} = \vec{i} - \vec{j} + 2\vec{k}$, find a nonzero vector $\vec{C}$ perpendicular to both $\vec{A}$ and $\vec{B}$.

8. (a) If $\vec{A} = \vec{i} + 2\vec{j}$, $\vec{B} = 2\vec{i} - 4\vec{j}$, and $\vec{C} = 2\vec{i} - 3\vec{j}$, find real numbers x and y such that $\vec{C} = x\vec{A} + y\vec{B}$. How many such pairs x, y are there?
(b) Express $\vec{C}$ as a linear combination of $\vec{A}$ and $\vec{B}$ if $\vec{A} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{B} = \vec{i} + 2\vec{j} - \vec{k}$, $\vec{C} = 2\vec{i} - 11\vec{j} + 7\vec{k}$.
(c) Can $2\vec{i} + 11\vec{j} + 7\vec{k}$ be expressed as a linear combination of the vectors $\vec{A}$ and $\vec{B}$ in part (b)?

9. (a) Find two perpendicular nonzero vectors $\vec{B}$ and $\vec{C}$, each of which is perpendicular to the vector $\vec{A} = \vec{i} + 2\vec{j} - \vec{k}$.
(b) If $\vec{V} = a\vec{i} + b\vec{j} + c\vec{k}$ is a vector perpendicular to $\vec{A}$, show that $\vec{V}$ is a linear combination of the vectors $\vec{B}$ and $\vec{C}$ you found in part (a), say $\vec{V} = x\vec{B} + y\vec{C}$, and explain how to find x and y when a, b, c are given.

10. If $\vec{A} = \vec{i} - 2\vec{j} + 3\vec{k}$ and $\vec{B} = 3\vec{i} + \vec{j} + 2\vec{k}$, find a nonzero vector $\vec{C}$ which is a linear combination of $\vec{A}$ and $\vec{B}$ and is also perpendicular to $\vec{B}$.

11. Given vectors $\vec{A} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{B} = \vec{i} + 2\vec{j} - \vec{k}$, $\vec{C} = \vec{i} + \vec{j} - 2\vec{k}$, find every vector of unit length which is a linear combination of $\vec{B}$ and $\vec{C}$ and is perpendicular to $\vec{A}$.

12. (a) Given the vectors $\vec{A} = 2\vec{i} - \vec{j} + 2\vec{k}$ and $\vec{B} = \vec{i} + 2\vec{j} - 2\vec{k}$, find two vectors $\vec{C}$ and $\vec{D}$ satisfying all the following conditions:

$$\vec{C} \text{ is parallel to } \vec{B}, \quad \vec{D} \text{ is perpendicular to } \vec{B}, \quad \vec{A} = \vec{C} + \vec{D}.$$

(b) Given two nonzero, nonparallel vectors $\vec{A}$ and $\vec{B}$ in n-space, show that there exist vectors $\vec{C}$ and $\vec{D}$ satisfying the three conditions in part (a) and express $\vec{C}$ and D in terms of $\vec{A}$ and $\vec{B}$.

13. Use vector methods to determine the cosines of the angles of the triangle whose vertices are $(2, -1, 1)$, $(1, -3, -5)$, $(3, -4, -4)$.

14. Prove that for two vectors in n-space we have the identity

$$|\vec{A} + \vec{B}|^2 - |\vec{A} - \vec{B}|^2 = 4\vec{A} \cdot \vec{B},$$

and hence $\vec{A} \cdot \vec{B} = 0$ if and only if $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$. When this is interpreted geometrically in 3-space it states that the diagonals of the parallelogram determined by $\vec{A}$ and $\vec{B}$ are of equal length if and only if the parallelogram is a rectangle.

15. Let $\vec{A}$ and $\vec{B}$ be two vectors in n-space. Show that $\vec{A} \cdot \vec{B} = 0$ if and only if $|A + \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2$. What theorem of geometry does this imply?

16. Prove that for two vectors $\vec{A}$ and $\vec{B}$ in n-space we have

$$|\vec{A} + \vec{B}|^2 + |\vec{A} - \vec{B}|^2 = 2|\vec{A}|^2 + 2|\vec{B}|^2 .$$

What geometric theorem about the sides and diagonals of a parallelogram can you deduce from this identity?

17. The following theorem in geometry suggests a vector identity involving three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$. Guess the identity suggested and prove it algebraically for vectors in n-space. This provides a proof of the theorem by vector methods. (Exercise 16 is a special case.)

"The sum of the squares of the sides of any quadrilateral exceeds the sum of the squares of the diagonals by four times the square of the length of the line segment which connects the mid-points of the diagonals."

18. Prove by vector methods that the diagonals of a rhombus are perpendicular.

19. By forming the dot product of the two vectors

$$\vec{A} = \cos\alpha\, \vec{i} + \sin\alpha\, \vec{j} \qquad \text{and} \qquad \vec{B} = \cos\beta\, \vec{i} + \sin\beta\, \vec{j}$$

derive the trigonometric identity

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta .$$

20. If $\vec{A} = a_1\vec{i} + a_2\vec{j}$ and $\vec{B} = b_1\vec{i} + b_2\vec{j}$ are two vectors in 2-space, define a new vector $\vec{A} \otimes \vec{B}$, called their "complex product," as follows:

$$\vec{A} \otimes \vec{B} = (a_1b_1 - a_2b_2)\vec{i} + (a_1b_2 + a_2b_1)\vec{j} .$$

Show that this product satisfies the following algebraic laws:

(a) $\vec{A} \otimes \vec{B} = \vec{B} \otimes \vec{A}$, (c) $\vec{A} \otimes (\vec{B} + \vec{C}) = \vec{A} \otimes \vec{B} + \vec{A} \otimes \vec{C}$,
(b) $\vec{A} \otimes (\vec{B} \otimes \vec{C}) = (\vec{A} \otimes \vec{B}) \otimes \vec{C}$, (d) $|\vec{A} \otimes \vec{B}| = |\vec{A}|\,|\vec{B}|$.

21. Let $\vec{A}$ and $\vec{B}$ be two nonzero vectors in 2-space with the following property:

If x and y are scalars such that $x\vec{A} + y\vec{B} = \vec{0}$, then $x = y = 0$.

We call such a pair $\vec{A}$, $\vec{B}$ *linearly independent*,

(a) Show that two nonzero vectors in 2-space are linearly independent if and only if they are not parallel.

(b) If $\vec{A}$ and $\vec{B}$ are linearly independent, show that for every vector $\vec{C}$ in 2-space there exist real numbers x and y such that $\vec{C} = x\vec{A} + y\vec{B}$ and that for a given $\vec{C}$ there is only one such pair x, y. We describe this by saying that two linearly independent vectors in 2-space form a *basis* for 2-space. A special case is $\vec{A} = \vec{i}$ and $B = \vec{j}$.

(c) Among the following four vectors, select all possible pairs that can be used as a basis for 2-space:

$$\vec{A} = \vec{i} + 2\vec{j}, \quad \vec{B} = 2\vec{i} - 4\vec{j}, \quad \vec{C} = 2\vec{i} - 3\vec{j}, \quad \vec{D} = \vec{i} - 2\vec{j} .$$

22. Let $\vec{A}$, $\vec{B}$, $\vec{C}$ be three vectors in 3-space which have the following property:

If x, y, z are scalars such that $x\vec{A} + y\vec{B} + z\vec{C} = \vec{0}$, then $x = y = z = 0$.

(a) Prove that none of $\vec{A}$, $\vec{B}$, $\vec{C}$ can be expressed as a linear combination of the other two.
(b) If $\vec{V}$ is any vector in 3-space, prove that there exist three real numbers x, y, z such that

$\vec{V} = x\vec{A} + y\vec{B} + z\vec{C}$. Also, for a given $\vec{V}$ show that there is only one such triple x, y, z. (In other words, a set of three linearly independent vectors in 3-space forms a basis for 3-space.)

(c) Determine x, y, z in part (b) when $\vec{A} = \vec{i}$, $\vec{B} = \vec{i} + \vec{j}$, $\vec{C} = \vec{i} + \vec{j} + 3\vec{k}$, and $\vec{V} = 2\vec{i} - 3\vec{j} + 5\vec{k}$.

23. Given three nonzero mutually perpendicular vectors in 3-space, prove that they are linearly independent. This result, along with Exercise 22(b), shows that three such vectors form a basis for 3-space. Such a basis is often called an *orthogonal basis*. Can you think of any reasons why orthogonal bases are more convenient than others?

24. Consider the following n vectors in n-space:

$$\vec{A}_1 = [1, 0, \ldots, 0], \quad \vec{A}_2 = [0, 1, 0, \ldots, 0], \quad \ldots, \quad \vec{A}_n = [0, 0, \ldots, 0, 1].$$

(These are the generalizations of $\vec{i}, \vec{j}$, and $\vec{k}$.) Prove the following statements:

(a) $|\vec{A}_j| = 1$ and $\vec{A}_i \cdot \vec{A}_j = 0$ if $i \neq j$, for all i and j.

(b) If $c_1, \ldots, c_n$ are n real numbers such that

$$c_1\vec{A}_1 + c_2\vec{A}_2 + \cdots + c_n\vec{A}_n = \vec{0},$$

then $c_1 = c_2 = \cdots = c_n = 0$. Any collection of vectors in n-space with this property is said to be *linearly independent*.

(c) For every vector $\vec{X}$ in n-space there exists exactly one n-tuple of real numbers $x_1, \ldots, x_n$ such that

$$\vec{X} = x_1\vec{A}_1 + \cdots + x_n\vec{A}_n.$$

A set of n vectors in n-space with this property is said to be a *basis* for n-space.

(d) Referring to part (c), show that $x_i = \vec{X} \cdot \vec{A}_i$ for each i.

(e) If $\vec{X}$ is perpendicular to each $\vec{A}_i$, show that $\vec{X} = \vec{0}$.

(f) Find a basis for n-space in which each $\vec{A}_i$ has more than one nonzero component.

25. The identity $\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta$ provides a geometric interpretation of the dot product in 3-space. This identity suggests a way to define angles between vectors in n-space. Let $\vec{A}$ and $\vec{B}$ be two nonzero vectors in n-space.

(a) Prove that

$$-1 \le \frac{\vec{A} \cdot \vec{B}}{|\vec{A}|\,|\vec{B}|} \le 1.$$

(b) Prove that there exists exactly one θ satisfying $0 \le \theta \le \pi$ such that $\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta$. This θ is called the angle between $\vec{A}$ and $\vec{B}$.

(c) Is the law of cosines valid in n-space?

26. Referring to Exercise 25, let $\vec{A} = [1, 1, \ldots, 1]$ and $\vec{B} = [1, 2, \ldots, n]$.

(a) Show that

$$\cos\theta = \frac{\sqrt{3}}{2}\sqrt{\frac{n+1}{n+\frac{1}{2}}}.$$

(b) Find the limiting value of θ as n increases indefinitely (that is, as $1/n \to 0$).

27. Referring to Exercise 25, let $\vec{A} = [2, 4, 6, \ldots, 2n]$ and $\vec{B} = [1, 3, 5, \ldots, 2n - 1]$. Find the limiting value of θ as n increases indefinitely.

5.12 Planes

The study of geometry as a deductive system, as conceived by Euclid around 300 B.C., begins with a set of axioms or postulates which describe properties of points and lines. The concepts "point" and "line" are taken as primitive notions and remain undefined. Other

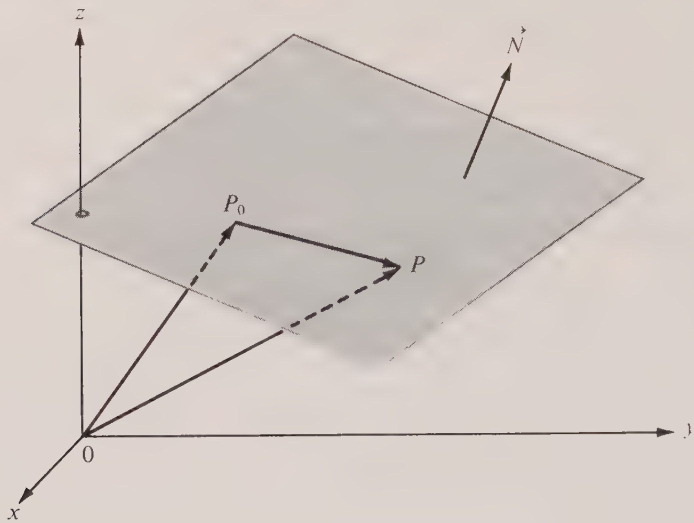


FIGURE 5.13 *A plane through P_0 and P with normal vector $\vec{N}$.*

concepts are defined in terms of points and lines, and theorems are systematically deduced from the postulates. In this book the development of geometry is based instead on the axioms for the real-number system. Fundamental concepts such as points, lines, vectors, parallelism, etc. are described in terms of real numbers, and theorems are deduced with the aid of vector algebra. The whole theory is developed to fit our intuitive ideas about Euclidean geometry. As a consequence of what we assume, all of Euclid's postulates can be derived as theorems. For this reason, 3-space is also called *Euclidean space.*

Our first application of vector algebra will be to the study of planes in 3-space. Of the many equivalent ways of defining a plane we shall adopt the following:

DEFINITION. Let $\vec{N}$ be a given nonzero vector and let P_0 be a given point. The collection of all points P such that the vector $\overrightarrow{P_0P}$ is perpendicular to $\vec{N}$ is called a plane through P_0. The vector $\vec{N}$ is said to be a normal to this plane. (See Figure 5.13.)

Now we shall translate this definition into various types of algebraic equations. For this purpose it is convenient to denote by $\vec{P}$, $\vec{P}_0$, etc. vectors from the origin to points P, P_0, etc. in 3-space. That is, we write $\vec{P}$ for $\overrightarrow{OP}$, and so on. Then the vector $\overrightarrow{P_0P}$ joining a point P_0 to P may be expressed as the difference $\vec{P} - \vec{P}_0$. The condition that $\overrightarrow{P_0P}$ be perpendicular to $\vec{N}$ may be written as an algebraic equation involving the dot product, as follows:

$$(\vec{P} - \vec{P}_0) \cdot \vec{N} = 0 \qquad \text{or} \qquad \vec{P} \cdot \vec{N} = \vec{P}_0 \cdot \vec{N}\,. \tag{5.17}$$

Each of these is called an *equation of the plane* and each is satisfied by *those and only those* points P which lie on the plane through P_0 having $\vec{N}$ as a normal vector.

Sometimes it is convenient to write the equations in (5.17) in terms of the components of the various vectors involved. If the normal vector $\vec{N}$ is given by

$$\vec{N} = A\,\vec{i} + B\vec{j} + C\,\vec{k}$$

and if the points P and P_0 have coordinates (x, y, z) and (x_0, y_0, z_0), respectively, then we have

$$\vec{P} = x\,\vec{i} + y\vec{j} + z\,\vec{k}\,, \qquad \vec{P}_0 = x_0\,\vec{i} + y_0\vec{j} + z_0\,\vec{k}\,,$$

and

$$\vec{P} - \vec{P}_0 = (x - x_0)\,\vec{i} + (y - y_0)\vec{j} + (z - z_0)\,\vec{k}\,.$$

If we form the dot product of $\vec{P} - \vec{P}_0$ and $\vec{N}$, the first equation in (5.17) becomes

$$(5.18) \qquad A(x - x_0) + B(y - y_0) + C(z - z_0) = 0\,.$$

This is called a *Cartesian equation* of the plane and it is satisfied by those and only those points (x, y, z) which lie on the plane.

If $\vec{N}$ is normal to a plane then so is every vector parallel to $\vec{N}$, because $(t\vec{N}) \cdot \vec{V} = 0$ whenever $\vec{N} \cdot \vec{V} = 0$. Therefore the locus represented by (5.18) does not change if we multiply both sides of the equation by a nonzero scalar t. That is why we refer to *an* equation instead of *the* equation of a plane.

We may transpose the terms not involving x, y, and z, and write Equation (5.18) in the form

$$(5.19) \qquad Ax + By + Cz = D\,,$$

where $D = Ax_0 + By_0 + Cz_0$. An equation of this type is said to be *linear* in x, y, and z. We have just shown that every point (x, y, z) on a plane satisfies a linear equation of the form (5.19) in which not all three of A, B, C are zero. Conversely, it is easy to prove that every linear equation with this property represents a plane. In fact, if we are given such an equation, say (5.19), and if $P_0 = (x_0, y_0, z_0)$ is any point satisfying the equation, then $Ax_0 + By_0 + Cz_0 = D$. If we substitute this for D in (5.19), we obtain $\vec{P} \cdot \vec{N} = \vec{P}_0 \cdot \vec{N}$ for every point $P = (x, y, z)$ satisfying (5.19), where $\vec{N} = A\,\vec{i} + B\,\vec{j} + C\,\vec{k}$. If not all three of A, B, C are zero, the collection of all such points P is a plane and the coefficients A, B, and C are the components of a normal $\vec{N}$ to this plane.

If the plane does not pass through the origin, we can always find a point P_1 on the plane such that the vector $\vec{P}_1$ is a normal. (See Figure 5.14.) In fact, if $\vec{N}$ is any normal and if P_0 is any point on the plane, we may choose $\vec{P}_1 = t\vec{N}$, where $t = \vec{P}_0 \cdot \vec{N}/|\vec{N}|^2$. The corresponding point P_1 is on the plane because we have

$$\vec{P}_1 \cdot \vec{N} = t\vec{N} \cdot \vec{N} = t|\vec{N}|^2 = \vec{P}_0 \cdot \vec{N}\,.$$

The length of $\vec{P}_1$ is called the *normal* (or *perpendicular*) *distance* of the plane from the origin. This distance (call it d) is given by

$$d = |\vec{P}_1| = |t\vec{N}| = \frac{|\vec{P}_0 \cdot \vec{N}|}{|\vec{N}|}\,.$$

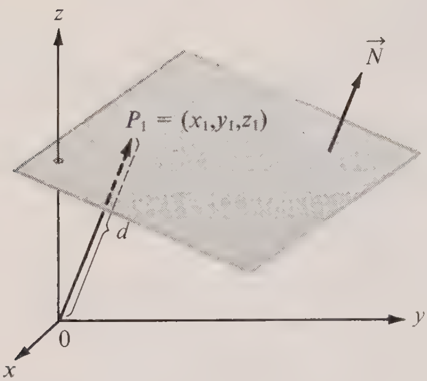


FIGURE 5.14 *The perpendicular distance from the origin to a plane.*

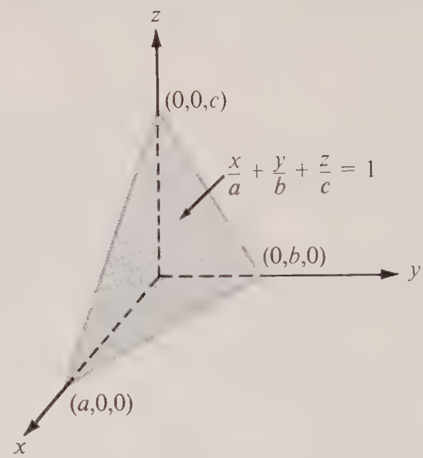


FIGURE 5.15 *A plane with intercepts a, b, c.*

If the plane is described by the linear equation in (5.19), then $D = \vec{P}_0 \cdot \vec{N}$ and the distance d is given by the formula

$$d = \frac{|D|}{|\vec{N}|}\,. \tag{5.20}$$

Two planes are called *parallel* if they have a common normal vector. If the vector $\vec{N} = A\,\vec{i} + B\,\vec{j} + C\,\vec{k}$ is a common normal of two parallel planes, linear equations for these planes can be written as follows:

$$Ax + By + Cz = D_1 \qquad \text{and} \qquad Ax + By + Cz = D_2\,,$$

the only difference being in the right-hand members. If the planes are distinct, a normal vector from a point on one plane to a point on the other has length $|D_1 - D_2|/|\vec{N}|$ [because of (5.20)]. This number represents the perpendicular distance between the two planes. A nonzero vector $\vec{V}$ is said to be parallel to a plane with normal vector $\vec{N}$ if $\vec{N} \cdot \vec{V} = 0$.

Two planes are called *perpendicular* if a normal of one is perpendicular to a normal of the other. If two planes have the linear equations $(\vec{P} - \vec{P}_0) \cdot \vec{N}_1 = 0$ and $(\vec{P} - \vec{Q}_0) \cdot \vec{N}_2 = 0$, then the condition for perpendicularity is $\vec{N}_1 \cdot \vec{N}_2 = 0$.

A plane described by the linear equation in (5.19) passes through the origin if and only if $D = 0$. This follows at once from (5.19) or (5.20). If A and B are zero, the equation in (5.19) can be put in the form $z =$ constant. In this case the vector $\vec{k}$ is a normal and hence the plane is parallel to the xy-plane. Similarly, the plane is parallel to the xz-plane if both A and C are zero and it is parallel to the yz-plane if B and C are zero.

If none of A, B, C is zero and if $D \neq 0$ we can rewrite Equation (5.19) in the form

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1\,,$$

where $a = D/A$, $b = D/B$, $c = D/C$. The three numbers a, b, c are called the *x-, y-, and z-intercepts*, respectively. They tell us where the plane cuts the coordinate axes. An example in which all the intercepts are positive is shown in Figure 5.15.

5.13 Exercises

1. A plane has the Cartesian equation $x + 2y - 2z + 7 = 0$. Find:
   (a) a normal vector of unit length;
   (b) the intercepts of the plane;
   (c) the distance of the plane from the origin;
   (d) the coordinates of that point Q on the plane nearest the origin.

2. Find a Cartesian equation of the plane which passes through $(1, 2, -3)$ and is parallel to the plane given by $3x - y + 2z = 4$. What is the distance between the two planes?

3. Four planes have Cartesian equations $x + 2y - 2z = 5$, $3x - 6y + 3z = 2$, $2x + y + 2z = -1$, and $x - 2y + z = 7$.
   (a) Show that two of them are parallel and the other two are perpendicular.
   (b) Find the distance between the two parallel planes.

4. Two intersecting planes determine two angles, these being the same as those between appropriate normals. Determine these angles if the planes have the Cartesian equations $x + y = 1$ and $y + z = 2$.

5. The three points $A = (1, 1, -1)$, $B = (3, 3, 2)$, and $C = (3, -1, -2)$ determine a plane.
   (a) Find a vector $\vec{N}$ normal to this plane.
   (b) Find a Cartesian equation for this plane.
   (c) Compute the distance of this plane from the origin.

6. Find an equation for the plane determined by the three points $(1, 2, 3)$, $(2, 3, 4)$, and $(-1, 7, -2)$.

7. Find an equation of that plane parallel to the plane given by $2x - y + 2z + 4 = 0$ if the point $(3, 2, -1)$ is equidistant from both planes.

8. Find a Cartesian equation of the plane passing through the point $(2, 3, -7)$, given that the vector joining $(1, 2, 3)$ and $(2, 4, 12)$ is perpendicular to the plane.

9. Find an equation of the plane passing through the point $(1, 1, 1)$ if a normal vector has direction angles $\alpha = \frac{1}{3}\pi$, $\beta = \frac{1}{4}\pi$, $\gamma = \frac{1}{3}\pi$.

10. Use integration to find the volume of the pyramid bounded by the three coordinate planes and the plane given by $x + 2y + 3z = 6$.

11. A plane has the Cartesian equation $2x + 3y + z + 4 = 0$. Find nonzero scalars s and t such that the two vectors $\vec{A} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{B} = s\vec{j} + t\vec{k}$ will lie in a plane perpendicular to the given plane.

12. Given vectors $\vec{A} = 2\vec{i} + 3\vec{j} - 4\vec{k}$ and $\vec{B} = \vec{j} + \vec{k}$.
    (a) Find a nonzero vector $\vec{N}$ perpendicular to each of $\vec{A}$ and $\vec{B}$.
    (b) Find a Cartesian equation of the plane determined by $\vec{A}$ and $\vec{B}$; that is, the plane containing the origin and the terminal points of $\vec{A}$ and $\vec{B}$.
    (c) Let $\vec{C} = s\vec{A} + t\vec{B}$, where s and t are scalars. Show that for each choice of s and t the tip of $\vec{C}$ is in the plane determined by $\vec{A}$ and $\vec{B}$. [*Hint.* Examine $\vec{N} \cdot \vec{C}$.]
    (d) Show, conversely, that each point in this plane is the tip of a vector $\vec{C}$ of the form $s\vec{A} + t\vec{B}$.
    (e) Find the scalars s and t corresponding to the point $(-4, -3, 11)$.

13. Recall that three vectors $\vec{A}, \vec{B}, \vec{C}$ in 3-space are linearly dependent if there exist scalars x, y, z, not all zero, such that $x\vec{A} + y\vec{B} + z\vec{C} = \vec{0}$. Let $\vec{A}, \vec{B}, \vec{C}$ denote vectors from the origin O to points A, B, C, respectively.

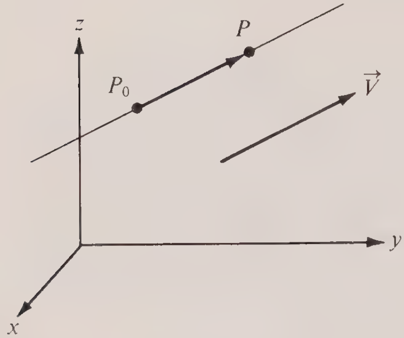


FIGURE 5.16 *A straight line through P_0 parallel to V.*

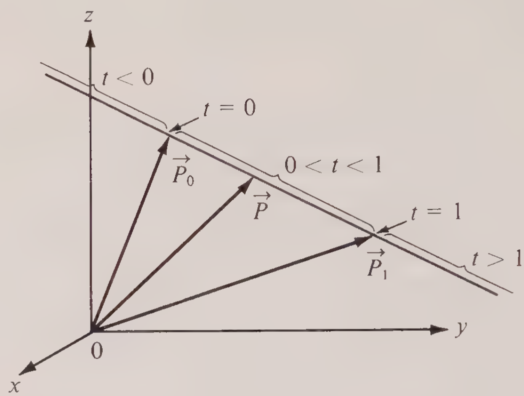


FIGURE 5.17 *A straight line through two points P_0 and P_1.*

(a) If $\vec{A}$, $\vec{B}$, $\vec{C}$ are linearly dependent, prove that O, A, B, C all lie in the same plane.
(b) Prove also the converse of (a).

This result is described by saying that three vectors are *coplanar* if and only if they are linearly dependent.

5.14 Straight lines

As before, we shall denote by $\vec{P}$ the vector $\overrightarrow{OP}$ from the origin to an arbitrary point P in space. Let P_0 be a given point anywhere in space and let $\vec{V}$ be a given nonzero vector. If the vector $\vec{P} - \vec{P}_0$ joining P_0 to P is parallel to $\vec{V}$, then there is a scalar t such that $\vec{P} - \vec{P}_0 = t\vec{V}$ or, what amounts to the same thing, such that

$$\vec{P} = \vec{P}_0 + t\vec{V}\,. \tag{5.21}$$

If we keep P_0 and $\vec{V}$ fixed and let t run through all real numbers, the set of points P satisfying (5.21) is called the straight line through P_0 parallel to $\vec{V}$. The scalar t is called a *parameter* and Equation (5.21) is called a *vector parametric equation* or simply a *vector equation* of this line. (See Figure 5.16.) A point $P \neq P_0$ is on this line if and only if $\vec{P} - \vec{P}_0$ is parallel to $\vec{V}$.

To obtain a vector equation of the line containing two points P_0 and P_1 we use the vector $\vec{P}_1 - \vec{P}_0$ for $\vec{V}$ in (5.21) and we find

$$\vec{P} = \vec{P}_0 + t(\vec{P}_1 - \vec{P}_0) = t\vec{P}_1 + (1 - t)\vec{P}_0\,. \tag{5.22}$$

The point P_0 corresponds to the number $t = 0$ and the point P_1 corresponds to $t = 1$. As t runs through the interval $0 \leq t \leq 1$ the point P traces out a locus called the line segment joining P_0 to P_1. The remaining points on the line come from negative t or from values of $t > 1$, as suggested by Figure 5.17.

By passing to components, the vector equation in (5.21) gives rise to three scalar equations. Thus, if we write $P = (x, y, z)$, $P_0 = (x_0, y_0, z_0)$, and $\vec{V} = a\,\vec{i} + b\,\vec{j} + c\,\vec{k}$, then (5.21) implies

(5.23) $$x - x_0 = ta\,, \qquad y - y_0 = tb\,, \qquad z - z_0 = tc\,.$$

These are sometimes called *scalar parametric equations* or simply *parametric equations* of the line. The line in question consists of the set of all points (x, y, z) determined by the parametric equations as we let t run through all real numbers.

If we multiply the first equation in (5.23) by b and the second by a and subtract, we obtain an equation not involving t, namely

(5.24) $$b(x - x_0) - a(y - y_0) = 0\,.$$

This linear equation represents a plane with normal vector $\vec{N} = b\,\vec{i} - a\,\vec{j}$. Since $\vec{N} \cdot \vec{k} = 0$, this plane is parallel to the z-axis. The line described by (5.23) lies on this plane because any point P which satisfies (5.23) also satisfies (5.24). Similarly, we may eliminate t from the second and third and from the first and third equations in (5.23) to obtain two further equations,

(5.25) $$c(y - y_0) - b(z - z_0) = 0 \qquad \text{and} \qquad c(x - x_0) - a(z - z_0) = 0\,,$$

representing planes parallel to the x- and y-axes, respectively. Points satisfying (5.23) also satisfy these last two equations and hence the line in (5.23) also lies on both these planes. Figure 5.18 shows three such planes intersecting in a line. In Exercise 11 of Section 5.18 we indicate how to prove that the intersection of any two nonparallel planes is a straight line.

There are, of course, many other planes containing a given line. For example, if a line lies on two planes having equations $(\vec{P} - \vec{P}_0) \cdot \vec{N}_1 = 0$ and $(\vec{P} - \vec{P}_0) \cdot \vec{N}_2 = 0$ then it also lies on the plane having the equation $(\vec{P} - \vec{P}_0) \cdot (t_1\vec{N}_1 + t_2\vec{N}_2) = 0$ for all choices of scalars t_1 and t_2 such that $t_1\vec{N}_1 + t_2\vec{N}_2 \neq \vec{0}$. If $\vec{N}_1$ and $\vec{N}_2$ are not parallel, then the linear combination $t_1\vec{N}_1 + t_2\vec{N}_2$ is a nonzero vector unless t_1 and t_2 are both zero. By varying t_1 and t_2, we may obtain a whole family of planes passing through a given line. The planes shown in Figure 5.18 are only three special members of this family.

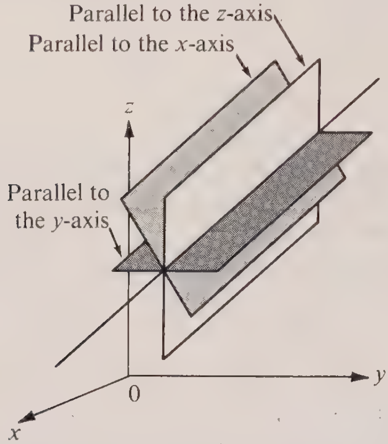
If none of the three numbers a, b, c in (5.24) and (5.25) is zero, we may rewrite these three equations in the form

(5.26) $$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}\,,$$

which is said to provide a *symmetric representation* of the line. These equations merely express the fact that the components of $P - \vec{P}_0$ are proportional to those of $\vec{V}$.

If a line happens to lie in the xy-plane, only the first two parametric equations of (5.23) are needed. We may eliminate t from these two equations to obtain the linear equation

(5.27) $$b(x - x_0) - a(y - y_0) = 0$$



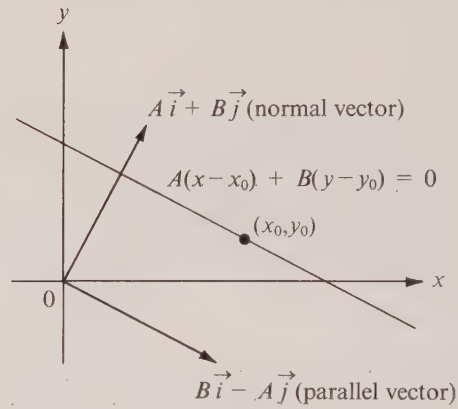


FIGURE 5.18 *Three planes intersecting in a line.* FIGURE 5.19 *A line in the xy-plane through (x_0, y_0) with normal vector $A\vec{i} + B\vec{j}$.*

which represents a line in the xy-plane passing through (x_0, y_0) and parallel to the vector $\vec{V} = a\vec{i} + b\vec{j}$. If we replace b by A and $-a$ by B, Equation (5.27) takes on a form resembling a Cartesian equation of a plane, namely,

$$A(x - x_0) + B(y - y_0) = 0 \qquad \text{or} \qquad Ax + By = C,$$

where $C = Ax_0 + By_0$. The coefficients A and B are the components of a vector $A\vec{i} + B\vec{j}$ perpendicular to $a\vec{i} + b\vec{j}$ (since the dot product of the two vectors is 0). The vector $A\vec{i} + B\vec{j}$ is said to be *normal* to the line. (See Figure 5.19.) If $B \neq 0$, the equation may be written in the form

$$y - y_0 = -\frac{A}{B}(x - x_0).$$

Since $dy/dx = -A/B$, the number $-A/B$ is the slope of the line. This is called the *point-slope form* of the equation of a line.

Note. One of Euclid's famous postulates in his axiomatic treatment of plane geometry is the so-called *parallel postulate* which is logically equivalent to the statement that "through a given point there exists one and only one line parallel to a given line." For a long time mathematicians suspected that this postulate could be deduced from the other postulates of Euclidean geometry, but all attempts to prove this resulted in failure. Then in the early 19th century the mathematicians Karl F. Gauss (1777–1855), J. Bolyai (1802–1860), and N. I. Lobatchevski (1793–1856) became convinced that the parallel postulate could not be derived from the other postulates of Euclid and proceeded to develop non-Euclidean geometries, that is to say, geometries in which the parallel postulate does not hold. The work of these men inspired other mathematicians and scientists to enlarge their points of view about "accepted truths" and to challenge other axioms that had been taken for granted for centuries.

As we mentioned earlier, our treatment of geometry enables us to deduce all of Euclid's postulates as theorems. As an illustration we shall show now that the parallel postulate is a consequence of our theory.

Let P_0 be a given point and let L be a given line. We are to prove that there is one and only one line through P_0 parallel to L. (We say that two lines are parallel if they are parallel to a common vector.) Choose two points on L and let $\vec{V}$ be the vector joining them. Then the line with vector equation $\vec{P} = \vec{P}_0 + t\vec{V}$ passes through P_0 and is parallel to L. To complete the proof we must show that this is the *only* line with this property. For this purpose, suppose that L' is another line passing through P_0 and parallel to L. We shall prove that every point P' on L' satisfies the equation $\vec{P}' = \vec{P}_0 + t\vec{V}$ and this will prove that $L' = L$. If P' is on L' and if $P' \neq P_0$, the vector $\vec{P}' - \vec{P}_0$ is parallel to L and hence there is a scalar t such that $\vec{P}' - \vec{P}_0 = t\vec{V}$. This shows that P' satisfies the vector equation of L and hence $L' = L$, as asserted.

5.15 Exercises

In each of Exercises 1 through 7, find a Cartesian equation for the line L in the xy-plane which satisfies the conditions given:

1. L contains the point (1, 2) and is parallel to the vector $3\vec{i} + 4\vec{j}$.
2. L contains the point (1, 2) and is perpendicular to $3\vec{i} + 4\vec{j}$.
3. L contains the point (3, 4) and has slope $-2/5$.
4. L intersects the x-axis at (4, 0) and the y-axis at (0, -1).
5. L intersects the x-axis at $(a, 0)$ and the y-axis at $(0, b)$.
6. L contains the points (2, 1) and (3, 4).
7. L contains the point (1, -3) and is parallel to the line through (-6, 9) and (3, 0).

8. Write a vector equation for the family of all lines in 3-space parallel to the line through (1, 2, 3) and (4, 1, 0).

9. Write a vector equation for the family of all lines perpendicular to the plane $3x + 2y - 5z = 7$.

10. Two lines in the xy-plane have nonzero slopes m_1 and m_2, respectively. Use vectors to prove that the lines are perpendicular if and only if $m_1m_2 = -1$.

11. A point P moves in space in such a way that at time t we have

$$\overrightarrow{OP} = (1 - t)\vec{i} + (2 - 3t)\vec{j} + (2t - 1)\vec{k}.$$

(a) Show that P moves along a straight line. (Call it L.)

(b) Find a vector $\vec{V}$ parallel to L.

(c) At what time does P strike the plane given by $2x + 3y + 2z + 1 = 0$?

(d) Find an equation for that plane parallel to the one in part (c) which P strikes at time $t = 3$.

(e) Find an equation for that plane perpendicular to L which P strikes at time $t = 2$.

(f) Find a symmetric representation for L using for (x_0, y_0, z_0) the point corresponding to $t = 0$.

12. (a) Show that a line in the xy-plane passing through a point P_0 may be described by an equation of the form $\vec{P} \cdot \vec{N} = \vec{P}_0 \cdot \vec{N}$, where $\vec{N}$ is a normal vector.

(b) Show that the perpendicular distance of the line from the origin is $|\vec{P}_0 \cdot \vec{N}|/|\vec{N}|$.

13. Two parallel lines in the xy-plane have linear equations $Ax + By = C_1$ and $Ax + By = C_2$. Show that the perpendicular distance between them is $|C_1 - C_2|/|A\vec{i} + B\vec{j}|$.

14. Given two points $A = (1, 2, 3)$, $B = (3, 3, 1)$ and a vector $\vec{V} = \vec{i} - 2\vec{j} + 2\vec{k}$. Let L denote that line through A which is parallel to $\vec{V}$.

(a) Show that B does not lie on L.

(b) Find the coordinates of that point Q on L whose distance from B is least. Compute this minimum distance.

15. Given the four points $A = (-2, 0, -3)$, $B = (1, -2, 1)$, $C = (-2, -13/5, 26/5)$, and $D = (16/5, -13/5, 0)$.

(a) Find an equation of the plane through AB which is parallel to CD.

(b) Compute the shortest distance between the lines AB and CD.

In each of Exercises 16 through 19, (a) find a Cartesian equation for the family of all straight lines in the xy-plane described by the given property. This equation should contain an arbitrary constant. (b) Differentiate the equation of the family and eliminate the arbitrary constant to obtain a differential equation for the family.

16. All lines parallel to the vector $2\vec{i} + 3\vec{j}$.

17. All lines perpendicular to the vector $2\vec{i} + 3\vec{j}$.

18. All lines passing through the point $(2, 1)$.

19. All lines tangent to the parabola $y = x^2$. Verify that the parabola is also an integral curve of the differential equation.

20. (a) Show that the equation $y = (1 - Cx)/(1 - C^2)^{1/2}$ represents a family of straight lines tangent to the circle $x^2 + y^2 = 1$. Here C represents a constant such that $C^2 < 1$.

(b) Find a differential equation satisfied by this family and verify that the circle itself is an integral curve of the differential equation. (The differential equation should not contain C.)

21. Find a vector parametric equation for the line which contains the point $(2, 1, -3)$ and is perpendicular to the plane described by the equation $4x - 3y + z = 5$.

22. Find an equation for the plane which is parallel to the y-axis and which passes through the intersection of the planes described by the equations $x + 2y + 3z = 4$ and $2x + y + z = 2$.

23. Find an equation for the plane parallel to the vector $3\vec{i} - \vec{j} + 2\vec{k}$ if it passes through the line of intersection of the planes with equations $x + y = 3$ and $2y + 3z = 4$.

5.16 The cross product

Besides dot multiplication of two vectors, the result of which is a scalar, there is another way of multiplying vectors that is particularly useful in the applications of vector analysis. This is called the *cross product* (or *vector product*) and it combines two vectors $\vec{A}$ and $\vec{B}$ in 3-space to produce a new vector denoted by $\vec{A} \times \vec{B}$. The cross product can be introduced by any of three different methods—the geometric approach, the analytic approach, and the axiomatic approach. We shall adopt the analytic approach but first we wish to make some remarks about the other two methods.

In the geometric approach, $\vec{A} \times \vec{B}$ is defined to be a vector perpendicular to both $\vec{A}$ and $\vec{B}$ with length equal to $|\vec{A}|\,|\vec{B}| \sin\theta$, where θ is the angle between $\vec{A}$ and $\vec{B}$, $0 \le \theta \le \pi$. These properties alone do not fix the direction of $\vec{A} \times \vec{B}$ because there are two such vectors, one being the negative of the other, and some further condition must be added to determine which direction is to be chosen. This is usually done by imposing the "right-hand rule," as illustrated in Figure 5.20. That is to say, when $\vec{A}$ is rotated into $\vec{B}$ in such a way that the fingers of the right hand point in the direction of rotation, then the thumb indicates the direction of $\vec{A} \times \vec{B}$ (assuming, for the sake of the argument, that the thumb is perpendicular to the other fingers). Of course, one could just as well impose a "left-hand rule"—this would simply reverse the direction of $\vec{A} \times \vec{B}$. From this starting point, one can derive as theorems the following algebraic properties of the cross product:

$$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A}), \tag{5.28}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = (\vec{A} \times \vec{B}) + (\vec{A} \times \vec{C}), \tag{5.29}$$

$$c(\vec{A} \times \vec{B}) = (c\vec{A}) \times \vec{B} \qquad (c \text{ being any scalar}). \tag{5.30}$$

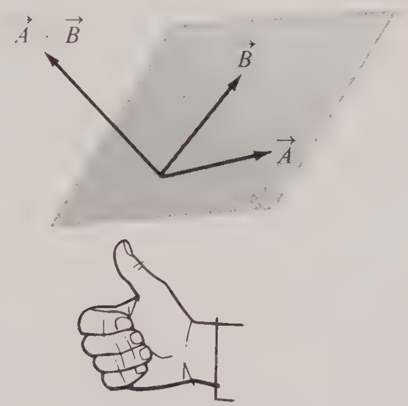


FIGURE 5.20 *Illustrating the right-hand rule for determining the direction of* $\vec{A} \times \vec{B}$.

The property in formula (5.28) takes the place of the usual commutative law for multiplication and is called *skew-symmetry*. Property (5.29) is, of course, a *distributive law.*†

In the axiomatic approach, we make no attempt to define $\vec{A} \times \vec{B}$ explicitly. Instead we say that a vector product is any function which assigns to every ordered pair of vectors $\vec{A}$, $\vec{B}$ in 3-space a new vector $\vec{A} \times \vec{B}$ which has the algebraic properties just mentioned as well as the following two properties:

$$A \cdot (A \times \vec{B}) = 0, \tag{5.31}$$

$$|\vec{A} \times \vec{B}| = \sqrt{|\vec{A}|^2|\vec{B}|^2 - (\vec{A} \cdot \vec{B})^2}. \tag{5.32}$$

These are to hold for all choices of $\vec{A}$ and $\vec{B}$. Property (5.31) makes $\vec{A} \times \vec{B}$ perpendicular to $\vec{A}$. It is not necessary to postulate that $\vec{B} \cdot (\vec{A} \times \vec{B}) = 0$ since this can be derived from (5.31) and (5.28). Property (5.32) gives $\vec{A} \times \vec{B}$ the desired length, since

$$|\vec{A}|^2|\vec{B}|^2 - (\vec{A} \cdot \vec{B})^2 = |\vec{A}|^2|\vec{B}|^2 (1 - \cos^2 \theta) = |\vec{A}|^2|\vec{B}|^2 \sin^2 \theta,$$

† More precisely, (5.29) is a "left" distributive law. A "right" distributive law $(\vec{B} + \vec{C}) \times \vec{A} = (\vec{B} \times \vec{A}) + (\vec{C} \times \vec{A})$ follows from (5.28) and (5.29).

if θ is the angle between $\vec{A}$ and $\vec{B}$. We use (5.32) instead of a formula involving θ because (5.32) makes sense when either $\vec{A}$ or $\vec{B}$ is $\vec{0}$. From this list of axioms about vector products one can prove that there are *at most two* such products—if one exists then a second one also exists and the two are negatives of each other.

In the analytic approach we begin by defining the cross product by a formula which tells us how to calculate the components of $\vec{A} \times \vec{B}$ in terms of the components of $\vec{A}$ and $\vec{B}$ and then we prove as theorems the algebraic properties listed above. The axiomatic approach tells us that there are at most two such formulas and that they differ only in sign. To find what these formulas ought to be, we write

$$\vec{A} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k} \qquad \text{and} \qquad \vec{B} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$$

and then multiply these vectors, using the algebraic properties of the cross product listed above. There are nine terms altogether. The three terms involving $\vec{i} \times \vec{i}$, $\vec{j} \times \vec{j}$, and $\vec{k} \times \vec{k}$ drop out because (5.28) implies that the cross product of a vector with itself is $\vec{0}$. The six remaining terms may be grouped as follows:

$$(a_2b_3 - a_3b_2)(\vec{j} \times \vec{k}) + (a_3b_1 - a_1b_3)(\vec{k} \times \vec{i}) + (a_1b_2 - a_2b_1)(\vec{i} \times \vec{j}) .$$

Therefore the resulting formula for $\vec{A} \times \vec{B}$ depends on the products $\vec{j} \times \vec{k}$, $\vec{k} \times \vec{i}$, and $\vec{i} \times \vec{j}$. From the properties mentioned above we know that $\vec{j} \times \vec{k}$ is to have length 1 and is to be perpendicular to both j and $\vec{k}$. Since $\vec{i}$ and $-\vec{i}$ have these properties we must have $\vec{j} \times \vec{k} = \pm\vec{i}$. Similarly, $\vec{k} \times \vec{i} = \pm\vec{j}$ and $\vec{i} \times \vec{j} = \pm\vec{k}$. It is easy to show that all three signs are $+$ or all three are $-$. To do this we write $\vec{i} \times \vec{j} = a\,\vec{k}$, $\vec{j} \times \vec{k} = b\,\vec{i}$, $\vec{k} \times \vec{i} = c\,\vec{j}$ and apply (5.31) to $\vec{A} = \vec{i} + \vec{j}$ and $\vec{B} = \vec{k}$. Then $\vec{A} \times \vec{B} = \vec{i} \times \vec{k} + \vec{j} \times \vec{k} = -c\,\vec{j} + b\,\vec{i}$ and (5.31) gives us $(\vec{i} + \vec{j}) \cdot (-c\,\vec{j} + b\,\vec{i}) = 0$, or $b = c$. Similarly we find $a = b$ so all three signs are the same. If we want vector multiplication to follow the right-hand rule we choose the $+$ sign in all three cases; otherwise we choose the $-$ sign. This discussion serves to motivate the following analytic definition of the cross product:

DEFINITION. Let $\vec{A} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$ and $\vec{B} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$ be two vectors in 3-space. Their cross product $\vec{A} \times \vec{B}$ (in that order) is defined to be the vector

$$(5.33) \qquad \vec{A} \times \vec{B} = (a_2b_3 - a_3b_2)\,\vec{i} + (a_3b_1 - a_1b_3)\,\vec{j} + (a_1b_2 - a_2b_1)\,\vec{k} .$$

Properties (5.28) through (5.31) follow quickly from this definition and are left as exercises for the reader. To prove (5.32) we write

$$|\vec{A} \times \vec{B}|^2 = (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 ,$$

and

$$|\vec{A}|^2|\vec{B}|^2 - (\vec{A} \cdot \vec{B})^2 = (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 ,$$

and then verify "by brute force" that the two right-hand members are identical. Incidentally, the cross product is *not associative*. For example, $\vec{i} \times (\vec{i} \times \vec{j}) = \vec{i} \times \vec{k} = -\vec{j}$ but $(\vec{i} \times \vec{i}) \times \vec{j} = \vec{0} \times \vec{j} = \vec{0}$.

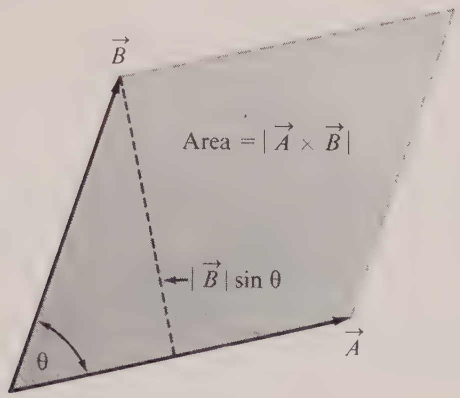


FIGURE 5.21 *The length of $\vec{A} \times \vec{B}$ is the area of the parallelogram determined by $\vec{A}$ and $\vec{B}$.*

If we rewrite (5.32) in the form $|\vec{A} \times \vec{B}| = |\vec{A}|\,|\vec{B}| \sin \theta$, we may interpret the length of $\vec{A} \times \vec{B}$ geometrically as the area of the parallelogram determined by $\vec{A}$ and $\vec{B}$. (See Figure 5.21.)

Because of the way we have defined the cross product, when it is interpreted geometrically its direction depends on the relative positions of the three coordinate axes. In a right-handed coordinate system [illustrated in Figure 5.22(a)] the direction of $\vec{A} \times \vec{B}$ is determined by the right-hand rule; in a left-handed system [Figure 5.22(b)] the direction of $\vec{A} \times \vec{B}$ is reversed and may be determined by the left-hand rule.

5.17 The scalar triple product

The dot and cross products may be combined to form the so-called scalar triple product $\vec{A} \times \vec{B} \cdot \vec{C}$ [which can only mean $(\vec{A} \times \vec{B}) \cdot \vec{C}$]. This product has an interesting geometric interpretation that suggests some of its algebraic properties.

Figure 5.23 shows a parallelepiped determined by three noncoplanar vectors $\vec{A}$, $\vec{B}$, $\vec{C}$.

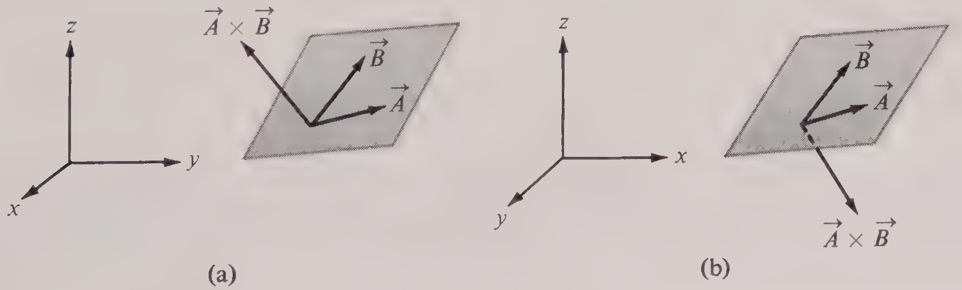


FIGURE 5.22 *The relative positions of $\vec{A}$, $\vec{B}$, and $\vec{A} \times \vec{B}$ in* (a) *a right-handed coordinate system, and* (b) *a left-handed coordinate system.*

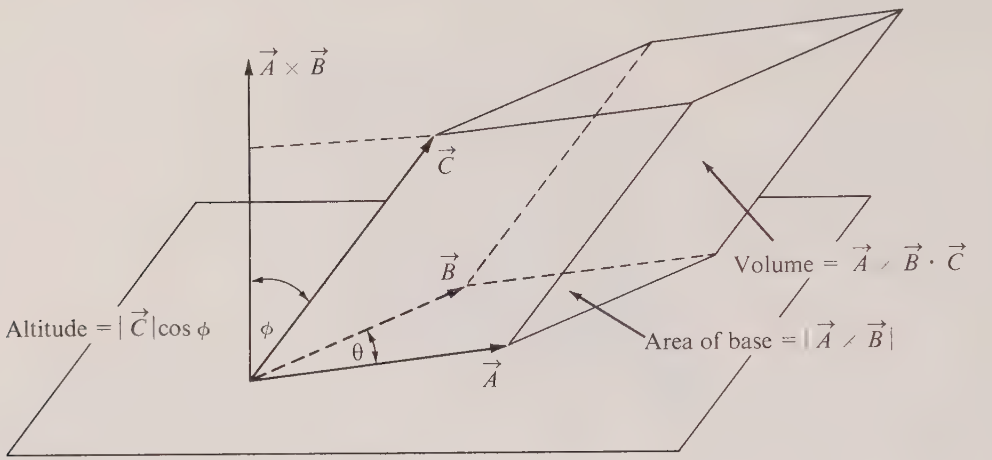


FIGURE 5.23 *Geometric interpretation of the scalar triple product as the volume of a parallelepiped.*

Its altitude is $|\vec{C}| \cos \phi$, where ϕ is the angle between $\vec{A} \times \vec{B}$ and $\vec{C}$. In this figure $\cos \phi$ is positive because $0 \leq \phi < \frac{1}{2}\pi$. The area of the parallelogram which forms the base is $|\vec{A} \times \vec{B}|$ and this is also the area of each cross section parallel to the base. Integrating the cross-sectional area, we find that the volume of the parallelepiped is given by

$$\int_0^{|\vec{C}| \cos \phi} |\vec{A} \times \vec{B}|\, du = |\vec{A} \times \vec{B}|\, |\vec{C}| \cos \phi = (\vec{A} \times \vec{B}) \cdot \vec{C}\,.$$

In other words, the scalar triple product $\vec{A} \times \vec{B} \cdot \vec{C}$ is equal to the volume of the parallelepiped determined by $\vec{A}, \vec{B}, \vec{C}$. (When $\frac{1}{2}\pi < \phi \leq \pi$ the product $\vec{A} \times \vec{B} \cdot \vec{C}$ is the negative of the volume.) This interpretation suggests that $\vec{A} \times \vec{B} \cdot \vec{C}$ is zero if and only if $\vec{A}, \vec{B}, \vec{C}$ are coplanar. (An algebraic proof of this fact is outlined in Exercise 9 of Section 5.18.) The equation $\vec{A} \cdot (\vec{A} \times \vec{B}) = 0$, mentioned earlier in (5.31), is a special case of this property.

Another algebraic property suggested by an examination of Figure 5.23 is that a cyclic permutation of the vectors $\vec{A}, \vec{B}, \vec{C}$ leaves their scalar triple product unchanged. By this we mean that

$$\vec{A} \times \vec{B} \cdot \vec{C} = \vec{B} \times \vec{C} \cdot \vec{A} = \vec{C} \times \vec{A} \cdot \vec{B}\,. \tag{5.34}$$

(For an algebraic proof see Exercise 8 of Section 5.18.) This property implies that the dot and cross are interchangeable in a scalar triple product. In fact, the commutativity of the dot product implies $(\vec{B} \times \vec{C}) \cdot \vec{A} = \vec{A} \cdot (\vec{B} \times \vec{C})$ and when this is combined with the first equation in (5.34) we find

$$\vec{A} \times \vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{B} \times \vec{C}\,. \tag{5.35}$$

We shall illustrate how the scalar triple product may be used to solve a system of three simultaneous linear equations. Suppose the system is written in the form

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1\,, \\ a_2x + b_2y + c_2z &= d_2\,, \\ a_3x + b_3y + c_3z &= d_3\,. \end{aligned} \tag{5.36}$$

Let $\vec{A}$ be the vector with components a_1, a_2, a_3 and define $\vec{B}$, $\vec{C}$, and $\vec{D}$ similarly. Then the three equations in (5.36) are equivalent to the single vector equation

$$x\vec{A} + y\vec{B} + z\vec{C} = \vec{D}\,. \tag{5.37}$$

If we dot multiply both sides of this equation with $\vec{B} \times \vec{C}$, we find

$$x(\vec{A} \cdot \vec{B} \times \vec{C}) + y(\vec{B} \cdot \vec{B} \times \vec{C}) + z(\vec{C} \cdot \vec{B} \times \vec{C}) = \vec{D} \cdot \vec{B} \times \vec{C}\,.$$

The coefficients of y and z drop out and we obtain

$$x = \frac{\vec{D} \cdot \vec{B} \times \vec{C}}{\vec{A} \cdot \vec{B} \times \vec{C}} \quad \text{if} \quad \vec{A} \cdot \vec{B} \times \vec{C} \neq 0\,. \tag{5.38}$$

A similar argument yields analogous formulas for y and z, namely:

$$y = \frac{\vec{A} \cdot \vec{D} \times \vec{C}}{\vec{A} \cdot \vec{B} \times \vec{C}} \quad \text{and} \quad z = \frac{\vec{A} \cdot \vec{B} \times \vec{D}}{\vec{A} \cdot \vec{B} \times \vec{C}} \quad \text{if} \quad \vec{A} \cdot \vec{B} \times \vec{C} \neq 0\,. \tag{5.39}$$

The condition $\vec{A} \cdot \vec{B} \times \vec{C} \neq 0$ means that the three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ are noncoplanar and hence linearly independent. In this case (5.37) shows that every vector $\vec{D}$ in 3-space is a linear combination of $\vec{A}$, $\vec{B}$, $\vec{C}$ and the multipliers x, y, z are uniquely determined by the formulas in (5.38) and (5.39).

If $\vec{A} \cdot \vec{B} \times \vec{C} = 0$ then $\vec{A}$, $\vec{B}$, $\vec{C}$ are coplanar and the system has no solution unless $\vec{D}$ lies in the same plane as $\vec{A}$, $\vec{B}$, $\vec{C}$. In this latter case it is easy to show that there are infinitely many solutions of the system. In fact, the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ are linearly dependent so there exist scalars u, v, w not all zero such that $u\vec{A} + v\vec{B} + w\vec{C} = \vec{0}$. If the triple (x, y, z) satisfies (5.37) then so does the triple $(x + tu, y + tv, z + tw)$ for all real t since we have

$$\begin{aligned} &(x + tu)\vec{A} + (y + tv)\vec{B} + (z + tw)\vec{C} \\ &\qquad = x\vec{A} + y\vec{B} + z\vec{C} + t(u\vec{A} + v\vec{B} + w\vec{C}) = x\vec{A} + y\vec{B} + z\vec{C}\,. \end{aligned}$$

When cross products and scalar triple products are computed in terms of components some of the formulas involved become easier to work with if they are expressed in determinant notation. If a, b, c, d are four real numbers, the difference $ad - bc$ is often denoted by the symbol

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

and is called a *determinant* (of order two). The numbers a, b, c, d are called its *elements* and they are said to be arranged in two *rows*, a, b and c, d and in two *columns*, a, c and b, d. Note that an interchange of two rows or of two columns only changes the sign of the determinant. For example, since $ad - bc = -(bc - ad)$, we have

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = - \begin{vmatrix} b & a \\ d & c \end{vmatrix}.$$

If we express the definition of the cross product in (5.33) using determinants, it assumes the form

$$\vec{A} \times \vec{B} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} + \begin{vmatrix} a_3 & a_1 \\ b_3 & b_1 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k}\,. \tag{5.40}$$

Determinants of order three are written with three rows and three columns and they may be defined in terms of second-order determinants by the formula

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}. \tag{5.41}$$

This is said to be an "expansion" of the determinant along its first row. Note that the determinant on the right that multiplies a_1 may be obtained from that on the left by deleting the row and column in which a_1 appears. The other two determinants on the right are obtained similarly.

Determinants of order n may be defined in terms of determinants of order $n - 1$ by a suitable extension of the expansion in (5.41). However, we do not discuss this extension here because we shall not make any use of determinants of order greater than three. In fact, our only purpose in introducing determinants is to have a useful device for writing certain formulas in a compact form that makes them easier to remember.

Determinants make sense if some of the elements are vectors. For example, if we write the determinant

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

and "expand" this according to the rule prescribed in (5.41) we find that the result is equal to the right member of (5.33). In other words, we may write the definition of the cross product $\vec{A} \times \vec{B}$ in the following compact form:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

For example, to compute the cross product of $\vec{A} = 2\vec{i} - 8\vec{j} + 3\vec{k}$ and $\vec{B} = 4\vec{j} + 3\vec{k}$ we write

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -8 & 3 \\ 0 & 4 & 3 \end{vmatrix} = \begin{vmatrix} -8 & 3 \\ 4 & 3 \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & 3 \\ 0 & 3 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & -8 \\ 0 & 4 \end{vmatrix} \vec{k} = -36\vec{i} - 6\vec{j} + 8\vec{k}.$$

Similarly, the scalar triple product of three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ may be expressed as a determinant involving their respective components as follows:

$$\vec{A} \cdot \vec{B} \times \vec{C} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

The formulas in (5.38) and (5.39), which express the solution of the system of equations in (5.36), involve scalar triple products and when they are rewritten with determinants the resulting formulas for x, y, and z are known as *Cramer's rule.*

5.18 Exercises

1. Let $\vec{A} = -\vec{i} + 2\vec{k}$, $\vec{B} = 2\vec{i} + \vec{j} - \vec{k}$, $\vec{C} = \vec{i} + 2\vec{j} + 2\vec{k}$. Compute the following in terms of $\vec{i}, \vec{j}, \vec{k}$:
   (a) $\vec{A} \times \vec{B}$; (d) $\vec{A} \times (\vec{C} \times \vec{A})$;
   (b) $\vec{B} \times \vec{C}$; (e) $(\vec{A} \times \vec{B}) \times \vec{C}$; (g) $(\vec{A} \times \vec{C}) \times B$.
   (c) $\vec{C} \times \vec{A}$; (f) $\vec{A} \times (\vec{B} \times \vec{C})$;

2. Find a vector of unit length perpendicular to both of the vectors $\vec{A} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{B} = 2\vec{i} + 3\vec{j} - \vec{k}$.

3. (a) Find a vector normal to the plane passing through the points $(0, 2, 2)$, $(2, 0, -1)$, and $(3, 4, 0)$. [*Hint.* Find two vectors $\vec{A}$ and $\vec{B}$ in this plane and compute $\vec{A} \times \vec{B}$.]
   (b) Use cross products to find the area of the triangle determined by the three points in part (a).

4. If $\vec{A} = 2\vec{i} + 5\vec{j} + 3\vec{k}$, $\vec{B} = 2\vec{i} + 7\vec{j} + 4\vec{k}$, and $\vec{C} = 3\vec{i} + 3\vec{j} + 6\vec{k}$, express the cross product $(\vec{A} - \vec{C}) \times (\vec{B} - \vec{A})$ in terms of $\vec{i}, \vec{j}, \vec{k}$.

5. (a) If $\vec{A}$ and $\vec{B}$ are nonzero vectors such that $\vec{A} \times \vec{B} = \vec{0}$, then $\vec{A} = t\vec{B}$ for some nonzero scalar t. Interpret this statement geometrically and then prove it algebraically.
   (b) If $\vec{A} \times \vec{B} \neq \vec{0}$, prove that $\vec{A}$ and $\vec{B}$ are nonzero and nonparallel.

6. If $\vec{A} \times \vec{B} = \vec{0}$ and $\vec{A} \cdot \vec{B} = 0$, then at least one of $\vec{A}$ or $\vec{B}$ is $\vec{0}$. Interpret this statement geometrically and then prove it algebraically.

7. Let $\vec{A} = 2\vec{i} - \vec{j} + 2\vec{k}$ and $\vec{C} = 3\vec{i} + 4\vec{j} - \vec{k}$.
   (a) Find a vector $\vec{B}$ such that $\vec{A} \times \vec{B} = \vec{C}$. Is there more than one solution?
   (b) Find a vector $\vec{B}$ such that $\vec{A} \times \vec{B} = \vec{C}$ and $\vec{A} \cdot \vec{B} = 1$. Is there more than one solution?
   (c) Show that part (b) can be solved when $\vec{A}$ and $\vec{C}$ are any given pair of vectors such that $\vec{A} \neq \vec{0}$ and $\vec{A} \cdot \vec{C} = 0$.

8. Use algebraic properties of the dot and cross products to derive the following properties of the scalar triple product.

(a) $(\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) \times \vec{C} = 0$. [*Hint*. Use (5.31).]

(b) $\vec{A} \cdot \vec{B} \times \vec{C} = -\vec{B} \cdot \vec{A} \times \vec{C}$. (This shows that switching the first two vectors reverses the sign.) [*Hint*. Use part (a) and distributive laws.]

(c) $\vec{A} \cdot \vec{B} \times \vec{C} = -\vec{A} \cdot \vec{C} \times \vec{B}$. (This shows that switching the last two vectors reverses the sign.) [*Hint*. Use (5.28).]

(d) $\vec{A} \cdot \vec{B} \times \vec{C} = -\vec{C} \cdot \vec{B} \times \vec{A}$. (This shows that switching the first and third vectors reverses the sign.) [*Hint*. Use (b) and (c).]

If we equate the right members of (b), (c), and (d), we find

$$\vec{A} \cdot \vec{B} \times \vec{C} = \vec{B} \cdot \vec{C} \times \vec{A} = \vec{C} \cdot \vec{A} \times \vec{B},$$

which shows that a cyclic permutation of $\vec{A}$, $\vec{B}$, $\vec{C}$ leaves their scalar triple product unchanged.

9. (a) If $\vec{A}$, $\vec{B}$, $\vec{C}$ are linearly dependent, prove that $\vec{A} \times \vec{B} \cdot \vec{C} = 0$. [*Hint*. Assume $\vec{C} = a\vec{A} + b\vec{B}$ and compute $(\vec{A} \times \vec{B}) \cdot \vec{C}$.]

(b) If $\vec{A} \cdot \vec{B} \times \vec{C} = 0$, prove that $\vec{A}$, $\vec{B}$, $\vec{C}$ are linearly dependent. [*Hint*. If $\vec{B} \times \vec{C} = \vec{0}$ then $\vec{B} = t\vec{C}$, and $\vec{A}$, $\vec{B}$, $\vec{C}$ are linearly dependent. If $\vec{B} \times \vec{C} \neq \vec{0}$, use part (a) to show that $\vec{B}$, $\vec{C}$, and $\vec{B} \times \vec{C}$ form a basis for 3-space, write $\vec{A} = x\vec{B} + y\vec{C} + z(\vec{B} \times \vec{C})$ and dot multiply with $\vec{B} \times \vec{C}$ to show that $z = 0$.]

10. Let $\vec{N}_1$ and $\vec{N}_2$ be two nonzero nonparallel vectors in 3-space. Interpret each of the following statements geometrically and then prove it algebraically:

(a) There exist no scalars a, b such that $\vec{N}_1 \times \vec{N}_2 = a\vec{N}_1 + b\vec{N}_2$.

(b) $\vec{N}_1 \cdot \vec{N}_2 \times (\vec{N}_1 \times \vec{N}_2) \neq 0$.

(c) The vectors $\vec{N}_1$, $\vec{N}_2$, $\vec{N}_1 \times \vec{N}_2$ form a basis for 3-space.

(d) If $\vec{A} \cdot \vec{N}_1 = 0$ and $\vec{A} \cdot \vec{N}_2 = 0$, then $\vec{A} = t(\vec{N}_1 \times \vec{N}_2)$ for some real t. [*Hint*. Use part (c).]

11. Given two planes with equations $(\vec{P} - \vec{P}_0) \cdot \vec{N}_1 = 0$ and $(\vec{P} - \vec{P}_0) \cdot \vec{N}_2 = 0$, where $\vec{N}_1$ and $\vec{N}_2$ are not parallel. Let $\vec{V} = \vec{N}_1 \times \vec{N}_2$.

(a) Show that the line whose vector equation is $\vec{P} = \vec{P}_0 + t\vec{V}$ lies on both planes.

(b) Show that any point P on both planes satisfies the equation $\vec{P} = \vec{P}_0 + t\vec{V}$, where $\vec{P}$ denotes the vector from the origin to P. (This exercise shows that the intersection of two nonparallel planes is a straight line.)

In Exercises 12 through 15, $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$ denote vectors from the origin to points A, B, C, D, respectively. In each case prove the statement in part (a). The geometric interpretation of the scalar triple product may be helpful in some cases.

12. (a) The volume of the tetrahedron whose vertices are A, B, C, D is

$$\tfrac{1}{6}|(\vec{B} - \vec{A}) \cdot (\vec{C} - \vec{A}) \times (\vec{D} - \vec{A})|.$$

(b) Compute this volume when $A = (1, 1, 1)$, $B = (0, 0, 2)$, $C = (0, 3, 0)$, and $D = (4, 0, 0)$.

13. (a) If $\vec{B} \neq \vec{C}$, the perpendicular distance from A to the line through B and C is

$$|(\vec{A} - \vec{B}) \times (\vec{C} - \vec{B})|/|\vec{B} - \vec{C}|.$$

(b) Compute this distance when $A = (1, -2, -5)$, $B = (-1, 1, 1)$, and $C = (4, 5, 1)$.

14. (a) If B, C, D determine a plane, the perpendicular distance from A to this plane is

$$|(\vec{A} - \vec{B}) \cdot (\vec{C} - \vec{B}) \times (\vec{D} - \vec{B})|/|(\vec{C} - \vec{B}) \times (\vec{D} - \vec{B})|.$$

(b) Compute this distance if $A = (1, 0, 0)$, $B = (0, 1, 1)$, $C = (1, -1, 1)$, and $D = (2, 3, 4)$.

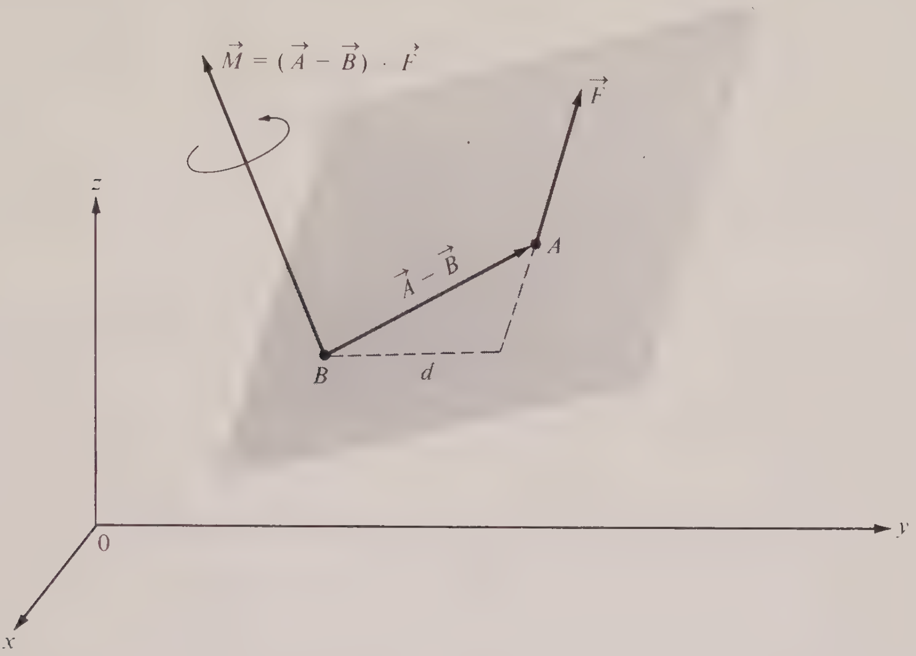


FIGURE 5.24 *If a force $\vec{F}$ is applied at A, its moment about B is $(\vec{A} - \vec{B}) \times \vec{F}$.*

15. (a) If $\vec{A} \neq \vec{B}$ and $\vec{C} \neq \vec{D}$ and if $(\vec{B} - \vec{A}) \times (\vec{C} - \vec{D}) \neq \vec{0}$, the shortest distance between the line through A and B and the line through C and D is

$$|(\vec{C} - \vec{A}) \cdot (\vec{B} - \vec{A}) \times (\vec{C} - \vec{D})| / |(\vec{B} - \vec{A}) \times (\vec{C} - \vec{D})| .$$

Describe a way to find this distance when $(\vec{B} - \vec{A}) \times (\vec{C} - \vec{D}) = \vec{0}$.

(b) Compute this distance when $A = (4, 5, 1)$, $B = (1, -1, 3)$, $C = (1, -2, -5)$, and $D = (-1, 1, 1)$.

16. (a) The vectors $\vec{P}$ and $\vec{P}_0$ join the origin to points P and P_0, respectively. If P is on the plane which is perpendicular to $\vec{V}$ and contains P_0, then $\vec{P} \cdot \vec{V} = \vec{P}_0 \cdot \vec{V}$. Show that if P is on the line which is parallel to $\vec{V}$ and contains P_0 then $\vec{P} \times \vec{V} = \vec{P}_0 \times \vec{V}$.

(b) The plane mentioned in part (a) is at a distance $|\vec{P}_0 \cdot \vec{V}|/|\vec{V}|$ from the origin. Show that the line in part (a) is at a distance $|\vec{P}_0 \times \vec{V}|/|\vec{V}|$ from the origin.

17. Let $\vec{A}$, $\vec{B}$, $\vec{C}$ denote nonzero vectors with lengths a, b, c, respectively, and assume that $\vec{A} + \vec{B} + \vec{C} = \vec{0}$. Let S denote the area of the triangle determined by $\vec{A}$, $\vec{B}$, $\vec{C}$. Show that

(a) $a^2 + \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C} = 0$,
(b) $\vec{A} \cdot \vec{B} = (c^2 - a^2 - b^2)/2$,
(c) $4S^2 = (\vec{A} \times \vec{B}) \cdot (\vec{A} \times \vec{B})$,
(d) $4S^2 = (ab + \vec{A} \cdot \vec{B})(ab - \vec{A} \cdot \vec{B})$,
(e) $S = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = (a + b + c)/2$ (*Heron's formula*).

18. Prove that $(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{C}) \times (\vec{C} \times \vec{A}) = (\vec{A} \cdot \vec{B} \times \vec{C})^2$.

19. Prove that $\vec{A} \times \vec{B} = \vec{A} \cdot (\vec{B} \times \vec{\imath})\,\vec{\imath} + \vec{A} \cdot (\vec{B} \times \vec{\jmath})\,\vec{\jmath} + \vec{A} \cdot (\vec{B} \times \vec{k})\,\vec{k}$.

20. Find the volume of the parallelepiped determined by the vectors $\vec{\imath} + \vec{\jmath}$, $\vec{\jmath} + \vec{k}$, and $\vec{k} + \vec{\imath}$.

21. Prove that $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{C} \cdot \vec{A})\vec{B} - (\vec{B} \cdot \vec{A})\vec{C}$. Use this to deduce:
    (a) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = (\vec{A} \times \vec{B} \cdot \vec{D})\vec{C} - (\vec{A} \times \vec{B} \cdot \vec{C})\vec{D}$.
    (b) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{B} \cdot \vec{D})(\vec{A} \cdot \vec{C}) - (\vec{B} \cdot \vec{C})(\vec{A} \cdot \vec{D})$.
    (c) $\vec{A} \times (\vec{B} \times \vec{C}) + \vec{B} \times (\vec{C} \times \vec{A}) + \vec{C} \times (\vec{A} \times \vec{B}) = \vec{0}$.
    (d) $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \times \vec{C}$ if and only if $\vec{B} \times (\vec{C} \times \vec{A}) = \vec{0}$.

22. Prove or disprove the formula $\vec{A} \times [\vec{A} \times (\vec{A} \times \vec{B})] \cdot \vec{C} = -|\vec{A}|^2 \vec{A} \cdot \vec{B} \times \vec{C}$.

23. If a force, represented by a vector $\vec{F}$, is applied to a body at a point A, the *moment of this force about a point B* is defined to be the vector

$$\vec{M} = (\vec{A} - \vec{B}) \times \vec{F},$$

where $\vec{A}$ and $\vec{B}$ are the vectors from the origin to the points A and B, respectively. This vector $\vec{M}$ measures the tendency of the force to rotate the body about an axis through B perpendicular to the plane determined by $\vec{A} - \vec{B}$ and $\vec{F}$, as suggested in Figure 5.24.

(a) Show that $\vec{M}$ is perpendicular to this plane and that $|\vec{M}| = |\vec{F}|d$, where d can be interpreted as a "lever arm," as indicated in the figure.

(b) If a force $\vec{F} = 2\vec{i} - 3\vec{j} + 4\vec{k}$ is applied at the point (2, 3, 1), find its moment about the point (3, −2, 0).

24. Referring to Exercise 23, the dependence of $\vec{M}$ on A, B, and $\vec{F}$ may be indicated by writing $\vec{M} = \vec{M}(A, B, \vec{F})$. (a) Show that

$$\vec{M}(A, B, \vec{F}_1 + \vec{F}_2) = \vec{M}(A, B, \vec{F}_1) + \vec{M}(A, B, \vec{F}_2)$$

and

$$\vec{M}(A, C, \vec{F}) = \vec{M}(A, B, \vec{F}) + \vec{M}(B, C, \vec{F}),$$

and state a physical principle suggested by each of these equations.

(b) A nonzero force $\vec{F}$ applied at A has equal moments about two points B_1 and B_2. What can you conclude about the positions of B_1 and B_2?

25. Show that

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = \begin{vmatrix} \vec{A} \cdot \vec{C} & \vec{B} \cdot \vec{C} \\ \vec{A} \cdot \vec{D} & \vec{B} \cdot \vec{D} \end{vmatrix}.$$

6

CURVES AND SURFACES

6.1 Methods of representing curves

If a particle moves along a curve in space, its position (x, y, z) at time t may be specified by three equations of the form

$$x = X(t), \qquad y = Y(t), \qquad z = Z(t). \tag{6.1}$$

As t varies over a particular time interval, the points (x, y, z) determined by these equations trace out the path along which the motion takes place. For example, if X, Y, and Z are linear functions, say

$$X(t) = x_0 + at, \qquad Y(t) = y_0 + bt, \qquad Z(t) = z_0 + ct,$$

then the path is along a straight line and the equations in (6.1) may be used as parametric equations for this line, as described in the foregoing chapter. In this chapter we wish to consider parametric representations of more general curves.

Hereafter, when we speak of a curve, we shall mean a set of points (x, y, z) determined by three equations of the form (6.1), called *parametric equations* of the curve. The symbol t, called a *parameter*, is allowed to range through some interval of real numbers, called the *parametric interval.* The capital letters X, Y, and Z in (6.1) denote real-valued functions defined and continuous on this interval. This definition of a curve is an analytic formulation of the intuitive idea that a curve is a locus of a point moving in space with one degree of freedom.

As in the case of straight lines, we may replace the three parametric equations in (6.1) by one vector equation. For this purpose we introduce the vector $\vec{r}(t)$ from the origin to the point (x, y, z) determined by (6.1) and we write

$$\vec{r}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j} + Z(t)\,\vec{k}. \tag{6.2}$$

The vector $\vec{r}(t)$ is called the *radius vector* to the curve. Its terminal point lies on the curve, as shown in Figure 6.1. Equation (6.2) may be referred to as a *vector parametric equation* or simply a *vector equation* of the curve. For curves lying n the xy-plane the z-component, $Z(t)$, is always zero.

Occasionally it will be convenient to think of the curve as being traced out by a moving

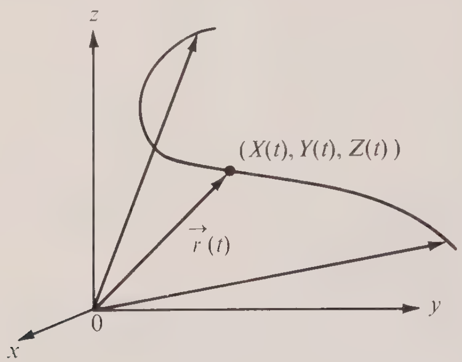


FIGURE 6.1 *Vector representation of a curve.*

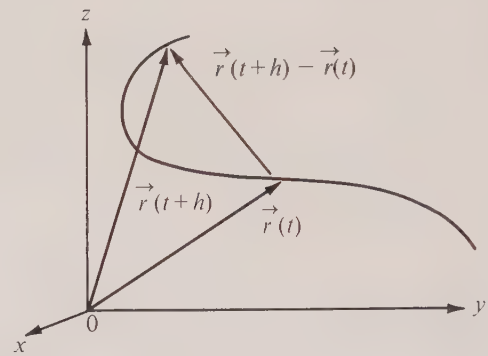


FIGURE 6.2 *The difference quotient $[\vec{r}(t+h) - \vec{r}(t)]/h$ is parallel to $\vec{r}(t+h) - \vec{r}(t)$.*

particle, in which case the parameter t will be referred to as *time*, and the vector $\vec{r}(t)$ will be called the *position vector*. From the mathematical point of view, a "motion" is simply a *vector-valued function* $\vec{r}$ defined on some interval of real numbers. When the vector function is expressed in terms of its components, as in Equation (6.2), a study of its properties is a simultaneous study of the triple of real-valued functions X, Y, and Z.

There are other ways to describe a curve in 3-space. Sometimes we can eliminate t from the parametric equations and obtain two equations relating x, y, and z, say

$$F(x, y, z) = 0 \quad \text{and} \quad G(x, y, z) = 0 . \tag{6.3}$$

A pair of equations like this is called a *Cartesian* or *implicit* representation of the curve. In the examples we shall encounter, each of these equations represents a surface and every point on the curve lies on the intersection of the two surfaces. For example, as we have already seen in the foregoing chapter, the parametric equations of a straight line determine Cartesian equations of two planes which intersect along the given line.

For a curve lying in the xy-plane we have $z = 0$ and so we need consider only the first two parametric equations in (6.1). If t is eliminated from these two, we arrive at an equation of the form $f(x, y) = 0$ which is satisfied by all points on the curve.

For example, if the two parametric equations are

$$x = \sqrt{t}, \qquad y = t, \tag{6.4}$$

elimination of t leads to the Cartesian equation $y = x^2$. Note that the presence of $\sqrt{t}$ in (6.4) requires the parameter t to take only nonnegative values and this results in nonnegative coordinates x and y. However, the Cartesian equation $y = x^2$ makes sense even if x is negative, although points on the curve with $x < 0$ cannot be obtained from (6.4). This example shows that elimination of the parameter may enlarge the locus. In general, if Equations (6.3) are obtained from (6.1) by eliminating t, then every point (x, y, z) which satisfies (6.1) also satisfies (6.3), but the intersection of the surfaces in (6.3) might contain additional points not accounted for by (6.1).

6.2 Derivatives of vector functions. Velocity and acceleration

As t varies through its parametric interval, the radius vector $\vec{r}(t)$ changes, in general, both in length and direction. To study this change we introduce the concept of the *derivative of a vector function.* As in the case of ordinary real-valued functions, we form the "difference quotient"

$$\frac{\vec{r}(t+h)-\vec{r}(t)}{h} \tag{6.5}$$

and investigate its behavior as $h \to 0$. Note that this quotient is a *vector* obtained by multiplying the difference $\vec{r}(t+h)-\vec{r}(t)$ by the scalar $1/h$. The difference $\vec{r}(t+h)-\vec{r}(t)$, illustrated in Figure 6.2, is parallel to the vector representing the difference quotient (6.5).

We may express the difference quotient in terms of components by writing

$$\frac{\vec{r}(t+h)-\vec{r}(t)}{h} = \frac{X(t+h)-X(t)}{h}\vec{i} + \frac{Y(t+h)-Y(t)}{h}\vec{j} + \frac{Z(t+h)-Z(t)}{h}\vec{k}\,.$$

As $h \to 0$, the components tend to $X'(t)$, $Y'(t)$, $Z'(t)$ (assuming these derivatives exist) and hence it seems natural to define the limit of the quotient in (6.5) to be the vector $X'(t)\,\vec{i} + Y'(t)\,\vec{j} + Z'(t)\,\vec{k}$. This is called the derivative of $\vec{r}$ at t and is denoted by $\vec{r}'(t)$. Thus, by definition,

$$\vec{r}'(t) = X'(t)\,\vec{i} + Y'(t)\,\vec{j} + Z'(t)\,\vec{k}\,.$$

This equation defines a *new* vector function $\vec{r}'$ (the first derivative of $\vec{r}$) and we shall have occasion to investigate *its* derivative $\vec{r}''$ (the second derivative of $\vec{r}$), which is given by the formula

$$\vec{r}''(t) = X''(t)\,\vec{i} + Y''(t)\,\vec{j} + Z''(t)\,\vec{k}\,,$$

whenever the derivatives of the individual components exist. In this chapter we shall assume that all functions X, Y, Z which appear in parametric equations may be differentiated as often as is necessary without saying so each time. Sometimes we use the Leibniz notation and write $d\vec{r}/dt$ for $\vec{r}'$ and $d^2\vec{r}/dt^2$ for $\vec{r}''$.

The straight line through the point $(X(t), Y(t), Z(t))$ which is parallel to $\vec{r}'(t)$ is called the *tangent line* to the curve at that point. (It is defined only when $\vec{r}'(t)$ is not the zero vector.) In the terminology of moving particles, $\vec{r}'(t)$ is called the *velocity vector* and is denoted also by $\vec{v}(t)$. If the velocity vector is attached to the curve as shown in Figure 6.3, it lies along the tangent line. The magnitude of the velocity, $|\vec{v}(t)|$, is called the *speed* and is denoted by $v(t)$. This is what the speedometer of an automobile tries to measure. The use of the word "speed" is justified in Section 6.33 where it is shown that $v(t)$ is the time rate of change of arc length along the curve. Thus, the *length* of the velocity vector tells us how fast the particle is moving at every instant, and its *direction* tells us which way it is going. The velocity will change if we alter either the speed or the direction of motion (or both). The derivative of velocity (second derivative of position) is called *acceleration* and is denoted by $\vec{a}(t)$. Acceleration causes the effect one feels when an automobile changes its speed or its direction. Unlike the velocity vector, the acceleration vector does not necessarily lie along the tangent line.

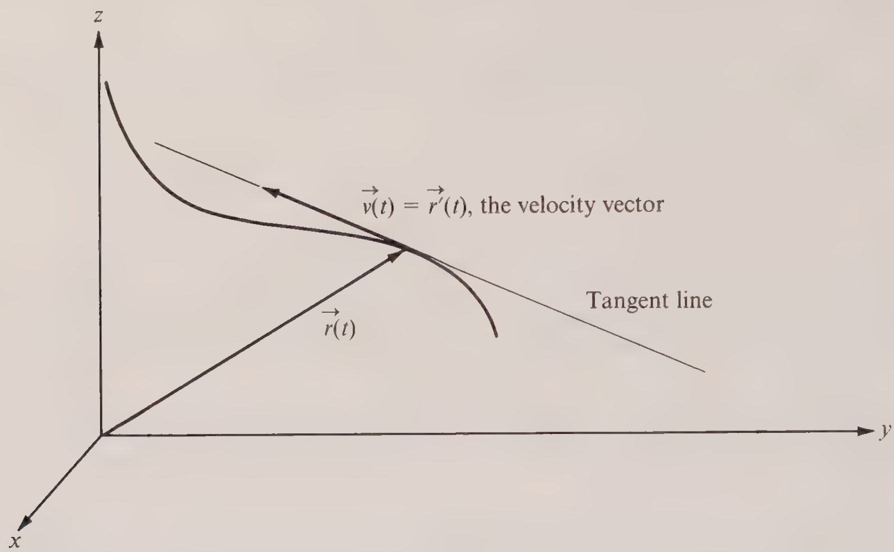


FIGURE 6.3 *The velocity vector shown tangent to the curve.*

The position $\vec{r}$, the velocity $\vec{v}$, and the acceleration $\vec{a}$ are examples of *vector-valued functions*. In general, we say that a vector-valued function $\vec{f}$ has been defined on an interval $[a, b]$ if there is a rule which associates with each t in $[a, b]$ a vector $\vec{f}(t)$. If we express each such vector in terms of its components, say we write

$$\vec{f}(t) = f_1(t)\,\vec{i} + f_2(t)\,\vec{j} + f_3(t)\,\vec{k}\,,$$

then this equation determines three real-valued functions f_1, f_2, and f_3 (called the components of $\vec{f}$) which are also defined on $[a, b]$. Therefore a description of a vector-valued function amounts to a description of three real-valued functions. Derivatives of vector-valued functions may be defined componentwise. Thus, for example, the first derivative $\vec{f}'$ is the vector-valued function given by the equation

$$\vec{f}'(t) = f_1'(t)\,\vec{i} + f_2'(t)\,\vec{j} + f_3'(t)\,\vec{k}$$

provided, of course, the derivatives of the components all exist. Higher derivatives are similarly defined.

In the work that follows we shall often have occasion to compute derivatives of vector-valued functions and it is convenient at this point to note some of the basic rules of differentiation.

Given two vector-valued functions $\vec{f}$ and $\vec{g}$, defined on a common interval $[a, b]$, we may form a new vector-valued function $\vec{h} = \vec{f} + \vec{g}$ (the *sum* of $\vec{f}$ and $\vec{g}$) as follows:

$$\vec{h}(t) = \vec{f}(t) + \vec{g}(t) \qquad \text{for each } t \text{ in } [a, b]\,.$$

It is easy to prove that the derivative of $\vec{h}$ is given by the formula

(6.6) $$\vec{h}'(t) = \vec{f}'(t) + \vec{g}'(t)\,.$$

Similarly, a given vector-valued function $\vec{f}$ may be multiplied by a real-valued function λ (λ is the Greek letter lambda) to produce a new vector-valued function $\vec{h} = \lambda\vec{f}$ which is defined as follows:

$$\vec{h}(t) = \lambda(t)\vec{f}(t) \qquad \text{for each } t \text{ in } [a, b]\,.$$

The derivative of $\vec{h}$ is given by the product formula

(6.7) $$\vec{h}'(t) = \lambda(t)\vec{f}'(t) + \lambda'(t)\vec{f}(t)\,.$$

The dot product $\vec{f}\cdot\vec{g}$ and the cross product $\vec{f}\times\vec{g}$ of two vector-valued functions are defined in a similar fashion and their derivatives may be computed by means of the following formulas:

(6.8) $$(\vec{f}\cdot\vec{g})' = \vec{f}\cdot\vec{g}\,' + \vec{f}\,'\cdot\vec{g}\,,$$

(6.9) $$(\vec{f}\times\vec{g})' = \vec{f}\times\vec{g}\,' + \vec{f}\,'\times\vec{g}\,.$$

The reader should note that the last three formulas are analogous to the usual formula for differentiating a product. Since the cross product is not commutative, one must pay attention to the order of the factors in Equation (6.9).

The operation of *composition* may be applied to combine vector functions with scalar functions. For example, if u is a real-valued (scalar) function defined on $[a, b]$ and if $\vec{f}$ is a vector-valued function defined on the range of u, the composition $\vec{g} = \vec{f}\circ u$ is a new vector-valued function defined as follows:

$$\vec{g}(t) = \vec{f}[u(t)] \qquad \text{for each } t \text{ in } [a, b]\,.$$

The chain rule for differentiating composite functions states that

(6.10) $$\vec{g}\,'(t) = u'(t)\vec{f}\,'[u(t)]$$

as in the ordinary real case.

The proofs of formulas (6.6) through (6.10) are all straightforward and are left as exercises.† As an application of (6.8) we prove the following theorem which is quite useful in some of the subsequent work.

6–1 THEOREM. If a vector-valued function f is differentiable and has a constant length on $[a, b]$, then

† For example, to prove (6.7) we write

$$\vec{h}(t) = \lambda(t)f_1(t)\,\vec{i} + \lambda(t)f_2(t)\,\vec{j} + \lambda(t)f_3(t)\,\vec{k}\,.$$

The derivative of the term involving $\vec{i}$ is $[\lambda(t)f_1'(t) + \lambda'(t)f_1(t)]\,\vec{i}$, and the derivatives of the other two terms are analogous. The terms multiplying $\lambda(t)$ have sum $\vec{f}\,'(t)$ and those multiplying $\lambda'(t)$ have sum $\vec{f}(t)$. Therefore $\vec{h}'(t)$ is given by the sum in (6.7).

$$\vec{f}'(t) \cdot \vec{f}(t) = 0 \qquad \text{for all } t \text{ on } [a, b]\,.$$

In other words, $\vec{f}'(t)$ is perpendicular to $\vec{f}(t)$ at each point.

Proof. Let $g(t) = |\vec{f}(t)|^2 = \vec{f}(t) \cdot \vec{f}(t)$. By hypothesis, g is constant and hence $g'(t) = 0$ for all t in $[a, b]$. But by (6.8) we have

$$g'(t) = \vec{f}(t) \cdot \vec{f}'(t) + \vec{f}'(t) \cdot \vec{f}(t) = 2\vec{f}'(t) \cdot \vec{f}(t)\,.$$

This proves the theorem.

Before we enter into a more detailed study of general properties of curves it may be helpful to look at some examples. In the next few sections we shall discuss some rather important curves and some of the surfaces related to them. In Section 6.23 we resume the general theory.

6.3 Exercises

Each of the vector equations in Exercises 1 through 5 describes a certain curve C in the xy-plane as t varies over the interval given. In each case: (a) Eliminate t from the corresponding parametric equations and obtain a Cartesian equation of the form $f(x, y) = 0$. Let C' denote the graph of this Cartesian equation. (b) Determine if $C = C'$. Find what restrictions, if any, must be placed on x and y so that the point (x, y) is on C' if and only if it is on C.

1. $\vec{r}(t) = t^2\,\vec{i} + 2t\,\vec{j}, \qquad 0 \le t \le 2.$
2. $\vec{r}(t) = \cos t\,\vec{i} + \sin^2 t\,\vec{j}, \qquad 0 \le t \le 2\pi.$
3. $\vec{r}(t) = \sin t\,\vec{i} + (1 - \cos 2t)\,\vec{j}, \qquad 0 \le t \le 3\pi.$
4. $\vec{r}(t) = 2e^t\,\vec{i} + 3e^t\,\vec{j}, \qquad 0 \le t \le \log 5.$
5. $\vec{r}(t) = \cosh t\,\vec{i} + \sinh t\,\vec{j}, \qquad 0 \le t \le \log 3.$

6. A curve C in the plane has Cartesian equation $f(x, y) = 0$. A line through the origin of slope t has the equation $y = tx$. This line intersects C at (x, y) if and only if x satisfies $f(x, tx) = 0$. Suppose it is possible to solve this last equation for x in terms of t, and let one solution be $x = \phi(t)$. Then the two equations $x = \phi(t)$ and $y = t\phi(t)$ may be used as parametric equations for C (or at least for a portion of C). Use this method to obtain parametric equations for the curves having the following Cartesian equations. Determine whether or not your parametric equations may be used to describe all of C.

(a) $4x^3 - y = 0.$ (b) $x^2 + y^2 - 1 = 0.$ (c) $x^3 + y^3 - xy = 0.$

7. For each of the curves described in Exercises 1 through 5, find vector equations for the velocity and acceleration vectors $\vec{v}(t)$ and $\vec{a}(t)$. Find the cosine of the angle between $\vec{v}(t)$ and $\vec{a}(t)$ for the following values of t:

(a) $t = 1$ in Exercise 1.
(b) $t = \frac{1}{2}\pi$ in Exercise 2.
(c) $t = \pi$ in Exercise 3.
(d) $t = \log 3$ in Exercise 4.
(e) $t = \log 2$ in Exercise 5.

8. The curve in the xy-plane given by the vector equation

$$\vec{r}(t) = (t^4 + 2t^2 + 1)\,i + (1 - 4t - t^4)\,\vec{j}$$

intersects the line given by the Cartesian equation $x + y = 0$. Find the cosine of the angle which the acceleration vector makes with the radius vector at the point of intersection.

9. Limits of vector-valued functions may be defined by writing $\lim_{t\to t_0} \vec{F}(t) = \vec{A}$ to mean $\lim_{t\to t_0} |\vec{F}(t) - \vec{A}| = 0$.

(a) If $\lim_{t\to t_0} \vec{F}(t) = \vec{A}$ and $\lim_{t\to t_0} \vec{G}(t) = \vec{B}$, prove that $\lim_{t\to t_0} [c\vec{F}(t)] = c\vec{A}$ and that $\lim_{t\to t_0} [\vec{F}(t) + \vec{G}(t)] = \vec{A} + \vec{B}$. [*Hint*. Use the triangle inequality.]

(b) If $\vec{F}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j} + Z(t)\,\vec{k}$ and $\vec{A} = a\,\vec{i} + b\,\vec{j} + c\,\vec{k}$, prove that $\lim_{t\to t_0} \vec{F}(t) = \vec{A}$ if and only if $\lim_{t\to t_0} X(t) = a$, $\lim_{t\to t_0} Y(t) = b$, and $\lim_{t\to t_0} Z(t) = c$.

(c) If $\lim_{t\to t_0} \vec{F}(t) = \vec{A}$ and $\lim_{t\to t_0} \vec{G}(t) = \vec{B}$, prove that $\lim_{t\to t_0} [\vec{F}(t) \cdot \vec{G}(t)] = \vec{A} \cdot \vec{B}$ and that $\lim_{t\to t_0} [\vec{F}(t) \times \vec{G}(t)] = \vec{A} \times \vec{B}$.

10. Prove the differentiation formulas in Equations (6.6) through (6.10).

11. Given constant vectors $\vec{A}$ and $\vec{B}$. If $\vec{r}(t) = e^{at}\vec{A} + e^{-at}\vec{B}$ (where a is constant), show that $\vec{r}''(t)$ and $\vec{r}(t)$ have the same direction for every t.

12. A particle moves in space so that at time t its acceleration vector $\vec{a}(t)$ is given by the formula $\vec{a}(t) = t\vec{A} + \vec{B}$, where $\vec{A}$ and $\vec{B}$ are constant vectors. Determine a vector equation for the position vector $\vec{r}(t)$ if, at time $t = 0$, the velocity is $\vec{v}_0$ and the position vector is $\vec{r}_0$.

13. A particle moves in the xy-plane with position vector $\vec{r}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j}$ and velocity vector $\vec{v}(t) = 2Y(t)\,\vec{i} - 2X(t)\,\vec{j}$. Find the components $X(t)$ and $Y(t)$ if $\vec{r}(0) = \vec{i} + 2\vec{j}$. [*Hint*. Find a second-order differential equation satisfied by X.]

14. Compute the derivative $\vec{g}'(t)$ in terms of $\vec{f}(t)$ and derivatives of $\vec{f}$ if $\vec{g}(t) = \vec{f}(t) \times \vec{f}'(t)$.

15. If $g(t) = \vec{f}(t) \cdot \vec{f}'(t) \times \vec{f}''(t)$, show that $g'(t) = \vec{f}(t) \cdot \vec{f}'(t) \times \vec{f}'''(t)$.

16. Define integrals of vector-valued functions componentwise, and prove the linearity property and the additive property.

6.4 The circle

A circle (in the xy-plane) of radius a and with center at the origin is shown in Figure 6.4. If the radius vector $\vec{r} = x\,\vec{i} + y\,j$ makes an angle θ with the positive x-axis, we have

(6.11) $$x = a\cos\theta\,, \qquad y = a\sin\theta\,.$$

These may be used as parametric equations for the circle if θ is allowed to vary over any interval† of length at least 2π, say $0 \le \theta \le 2\pi$.

If a particle starts at the point $(a, 0)$ at time $t = 0$ and moves counterclockwise around this circle with a constant angular speed, say ω (omega) radians per second, then $\theta = \omega t$ and the position vector is given by

(6.12) $$\vec{r}(t) = a\cos\omega t\,\vec{i} + a\sin\omega t\,\vec{j}\,.$$

The velocity vector for this motion is

$$\vec{v}(t) = -a\omega\sin\omega t\,\vec{i} + a\omega\cos\omega t\,\vec{j}\,.$$

Since $\vec{r}(t) \cdot \vec{v}(t) = 0$, the velocity is always perpendicular to $\vec{r}$. This also follows from Theorem 6–1 because $\vec{r}$ has constant length. This verifies that, for a circle, our definition of tangent line agrees with the one given in elementary plane geometry.

The length of the velocity vector (i.e., the *speed*) is given by

† An *arc* of the circle may be described by restricting θ to range over some interval of length less than 2π.

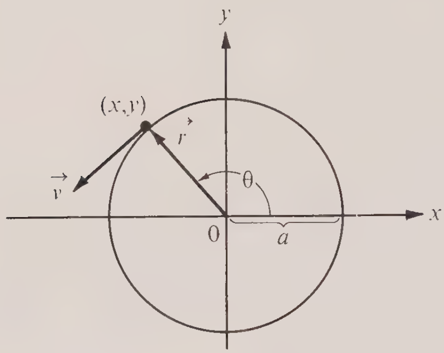


FIGURE 6.4 *A circle of radius a and center at the origin.*

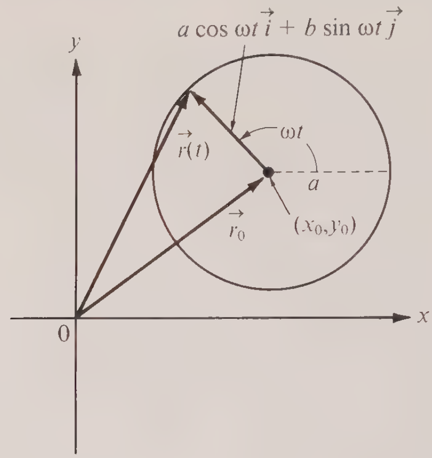


FIGURE 6.5 *A circle of radius a and center at (x_0, y_0).*

$$|\vec{v}(t)| = \sqrt{(-a\omega \sin \omega t)^2 + (a\omega \cos \omega t)^2} = a\omega\,.$$

Since this is independent of t the velocity changes only in direction and not in magnitude. This change is measured by the acceleration vector which is

$$\vec{a}(t) = \vec{v}\,'(t) = -\omega^2 a \cos \omega t\, \vec{i} - \omega^2 a \sin \omega t\, \vec{j}\,.$$

Comparing this with (6.12), we have

$$\vec{a}(t) = -\omega^2\, \vec{r}(t)\,,$$

which shows that the acceleration is always directed opposite to the radius vector. When it is drawn at the location of the particle in motion, the acceleration vector is directed toward the center of the circle. Because of this, the acceleration is called *centripetal* or "center-seeking" (a term originally proposed by Newton).

Note. If the moving particle has mass m, Newton's second law of motion states that the force acting on it (due to its acceleration) is the vector $m\vec{a}(t)$. This is called a *centripetal force* because it is directed toward the center. This force is exerted by the mechanism that confines the particle to a circular orbit. (The mechanism is a *string* in the case of a stone whirling in a slingshot, or *gravitational attraction* in the case of a satellite around the earth.) The equal and opposite reaction (due to Newton's third law), that is, the force $-m\vec{a}(t)$, is said to be *centrifugal* or "center-fleeing."

It is easy to eliminate θ from (6.11) and find a Cartesian equation for the circle. All we need do is square and add the two parametric equations to obtain

$$x^2 + y^2 = a^2\,. \tag{6.13}$$

Of course, (6.13) can also be obtained directly by using the theorem of Pythagoras.

If the center of the circle is at the point (x_0, y_0), as illustrated in Figure 6.5, this amounts to a "translation" or "displacement" of the whole curve by the vector $\vec{r}_0 = x_0\,\vec{i} + y_0\,\vec{j}$. The new position vector can be obtained merely by adding $\vec{r}_0$ to the old one. If the motion described above takes place along this translated circle its position vector is

$$\vec{r}(t) = \vec{r}_0 + (a \cos \omega t\,\vec{i} + a \sin \omega t\,\vec{j}) = (x_0 + a \cos \omega t)\,\vec{i} + (y_0 + a \sin \omega t)\,\vec{j}\,.$$

In parametric form this becomes $x = x_0 + a \cos \omega t$, $y = y_0 + a \sin \omega t$. Eliminating t as before, we obtain the Cartesian equation

$$(x - x_0)^2 + (y - y_0)^2 = a^2\,. \tag{6.14}$$

Notice that (6.14) can be derived directly from (6.13) by merely replacing x by $x - x_0$ and y by $y - y_0$. This applies not only to circles but to any space curve translated by a vector $\vec{r}_0 = x_0\,\vec{i} + y_0\,\vec{j} + z_0\,\vec{k}$. The position vector of the displaced curve may be obtained by adding $\vec{r}_0$ to the position vector of the original curve. Analytically, this is the same as replacing x, y, and z by $x - x_0$, $y - y_0$, and $z - z_0$ in the Cartesian equations.†

Although a circle is usually defined as the locus of points at a fixed distance from a given point, it has other characteristic properties as well. A line through a point P on a plane curve is called a *normal line* to the curve if it is perpendicular to the tangent line at P. It is easy to prove that circles (or arcs of circles) are the only curves whose normals at all points pass through a fixed point. In fact, if we place the origin at the fixed point and describe such a curve by a position vector $\vec{r}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j}$, then $\vec{r}(t)$ lies along the normal line through $(X(t), Y(t))$. Since the normal is perpendicular to the tangent line, we have $\vec{r}(t) \cdot \vec{r}\,'(t) = 0$ for all t in the parametric interval. But

$$\frac{d}{dt}(|\vec{r}(t)|^2) = \frac{d}{dt}[\vec{r}(t) \cdot \vec{r}(t)] = \vec{r}(t) \cdot \vec{r}\,'(t) + \vec{r}\,'(t) \cdot \vec{r}(t) = 2\vec{r}(t) \cdot \vec{r}\,'(t)\,,$$

so the condition $\vec{r}(t) \cdot \vec{r}\,'(t) = 0$ implies that $|\vec{r}(t)|^2$ is constant throughout the parametric interval.‡ If $\vec{r} \neq 0$, this means that $\vec{r}$ describes either a circle or an arc of a circle.

6.5 Exercises

1. In each case find a Cartesian equation for a circle satisfying the given conditions:
   (a) Center at $(4, -3)$, radius 6.
   (b) Center at $(1, 1)$, passes through the origin.
   (c) Center at $(3, 2)$, tangent to the x-axis.
   (d) Center at $(3, 2)$, tangent to the y-axis.
   (e) Radius 4, tangent to positive x- and y-axes.

2. By completing the squares, show that an equation of the form $x^2 + y^2 + Ax + By + C = 0$ can be written in the form

$$(x + \tfrac{1}{2}A)^2 + (y + \tfrac{1}{2}B)^2 = \tfrac{1}{4}(A^2 + B^2 - 4C)\,.$$

Show that the locus represented by this equation, if not empty, is either a single point or a circle.

† A more detailed discussion concerned with translation of curves is given in Section 6.17.

‡ Note that this proves the converse of Theorem 6–1.

Under what conditions on A, B, C does it represent a circle? When it represents a circle, determine the radius and the coordinates of the center in terms of A, B, C.

3. Find a Cartesian equation for the circle passing through the three points $(8, 1)$, $(4, 9)$, $(-10, -5)$. [*Hint.* Write an equation of the form $x^2 + y^2 + Ax + By + C = 0$ and determine three equations for the three unknowns A, B, C.]

4. Find an equation for the circle passing through the point $(2, 6)$ if its center lies on the two lines $x + 2y = 0$ and $3x - 4y = 7$.

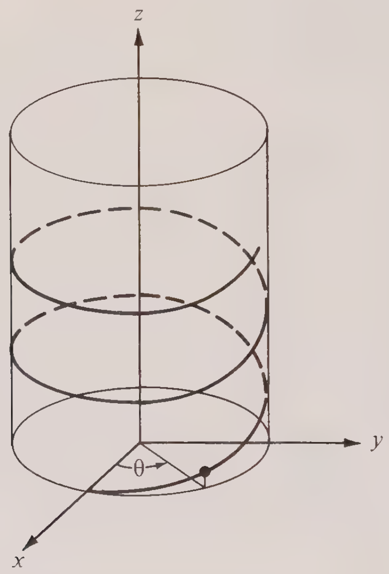


FIGURE 6.6 *A circular helix.*

5. (a) Show that the equation

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

represents a circle having the points (x_1, y_1) and (x_2, y_2) as extremities of a diameter.

(b) Interpret the equation in part (a) as the dot product of two vectors and use it to prove that an angle inscribed in a semicircle is a right angle.

6. Show that the equation $x^2 + y^2 + Cx = 0$ represents a family of circles passing through the origin, with centers on the x-axis. Show that these circles are integral curves of the first-order differential equation $x^2 - y^2 + 2xyy' = 0$.

7. Solve the differential equation $x + yy' = 0$ and show that the integral curves represent a family of circles.

8. Find a differential equation satisfied by all circles passing through the two points $(1, 0)$ and $(-1, 0)$.

9. Find a differential equation for the family of straight lines tangent to the unit circle $x^2 + y^2 = 1$. Verify that the circle is an integral curve of the equation.

10. Find a differential equation for the family of all circles of radius 1 having center on the line

$y = 4x/3$. Find (geometrically or otherwise) two straight lines that are also integral curves of the differential equation.

6.6 The circular helix

If a point revolves around the z-axis at a constant distance a from it, and simultaneously moves parallel to the z-axis in such a way that its z-component is proportional to the angle of revolution, the resulting path is called a *circular helix*. An example is shown in Figure 6.6. Using the angle θ of revolution as parameter, we have

$$x = a\cos\theta\,, \qquad y = a\sin\theta\,, \qquad z = b\theta\,, \tag{6.15}$$

where $a > 0$ and $b \neq 0$. When θ varies from 0 to 2π, the x- and y-coordinates return to their original values while z changes from 0 to $2\pi b$. The number $2\pi b$ is often referred to as the *pitch* of the helix.

If the x-, y-, and z-axes have the relative positions shown in Figure 6.6, the helix is called *right-handed* when $b > 0$ and *left-handed* when $b < 0$. (A right-handed helix is shown in the figure.) Anyone who has handled nuts and bolts with right- and left-handed threads knows that a right-handed helix cannot be superimposed on a left-handed one. This is said to be an *intrinsic property* of the helix because it does not depend on the location of the coordinate axes or the choice of parametric equations.

If a particle moves along the helix so that at time t we have $\theta = \omega t$ (where ω is a constant), the corresponding velocity and acceleration vectors are given by

$$\vec{v}(t) = -a\omega\sin\omega t\,\vec{i} + a\omega\cos\omega t\,\vec{j} + b\omega\,\vec{k}\,,$$
$$\vec{a}(t) = -\omega^2(a\cos\omega t\,\vec{i} + a\sin\omega t\,\vec{j})\,.$$

Thus, when the acceleration vector is located on the helix, it is parallel to the xy-plane and directed toward the z-axis.

If we eliminate θ from the first two equations in (6.15), we obtain the Cartesian equation $x^2 + y^2 = a^2$ which we recognize as the equation of a circle in the xy-plane. In 3-space, however, this equation represents a *surface*. A point (x, y, z) satisfies the equation if and only if its distance from the z-axis is equal to a. The locus of all such points is therefore a *right circular cylinder* of radius a with its axis along the z-axis. The helix winds around this cylinder.

6.7 Cylinders

In mathematics the word *cylinder* refers to any surface generated or swept out by a straight line moving along a plane curve and remaining parallel to a given line. The curve is called the *directrix* of the cylinder and the moving line which sweeps out the cylinder is called a *generator*. A circular cylinder is one whose directrix is a circle. If the generators are perpendicular to the plane of this circle, the surface is called a right circular cylinder. When the directrix is a straight line, the cylinder reduces to a plane.

If the directrix is in the xy-plane and has an implicit equation of the form

$$f(x, y) = 0\,, \tag{6.16}$$

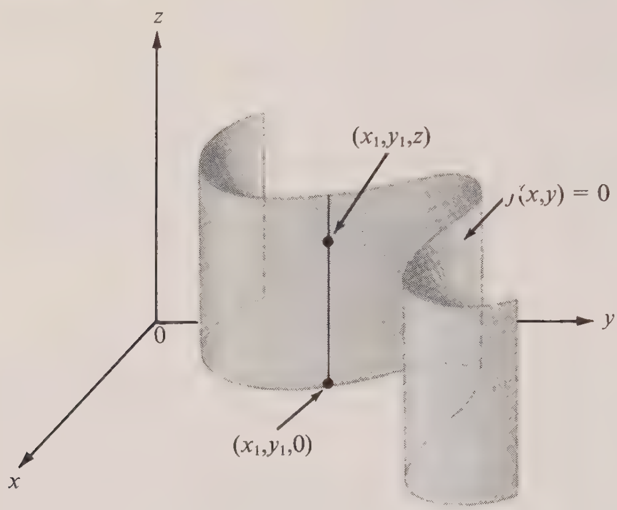


FIGURE 6.7 *A cylinder.*

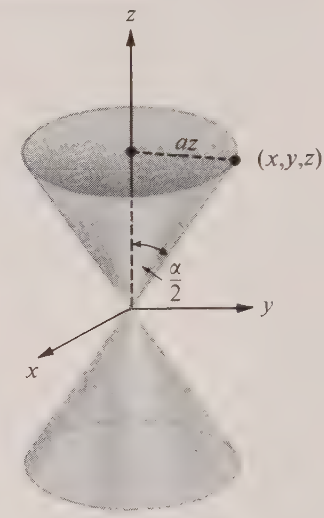


FIGURE 6.8 *A right circular cone with vertex angle α.*

it is easy to prove that the same equation describes that cylinder generated by a line moving along this curve and remaining parallel to the z-axis. In fact, if the coordinates of a particular point $(x_1, y_1, 0)$ satisfy (6.16), then so do the coordinates of the point (x_1, y_1, z) for *every* z [since z does not appear in (6.16)]. As z runs through all real numbers, the point (x_1, y_1, z) traces out that line parallel to the z-axis which passes through the point $(x_1, y_1, 0)$. (See Figure 6.7.) When $(x_1, y_1, 0)$ moves along the curve, the corresponding line (a generator) sweeps out the cylinder.

Similarly, an equation of the form $f(x, z) = 0$ (with y missing) represents a cylinder with generators parallel to the y-axis, whereas one of the form $f(y, z) = 0$ (with x missing) describes a cylinder with generators parallel to the x-axis.

Every cylinder is an example of a *ruled surface*, that is, a surface such that through each of its points there passes at least one straight line lying entirely on the surface. Another example of a ruled surface is a *cone*, which may be generated by a straight line moving along a given plane curve and passing through a fixed point (the vertex of the cone). The most common cone is the *right circular cone*, generated by a line moving along a circle, the vertex of the cone being on an axis through the center and perpendicular to the plane of the circle. Some of its properties are discussed in Section 6.9.

6.8 Exercises

1. Consider the helix described by the vector equation

$$\vec{r}(t) = a \cos \omega t \, \vec{i} + a \sin \omega t \, \vec{j} + b\omega t \, \vec{k}\,.$$

Show that the tangent line makes a constant angle with the generators of the cylinder on which the helix lies, and that the cosine of this angle is $b/\sqrt{a^2 + b^2}$.

2. (a) Referring to the helix in Exercise 1, show that the velocity $\vec{v}$ and acceleration $\vec{a}$ are vectors of constant length. (b) Show that

$$\frac{|\vec{v} \times \vec{a}|}{|\vec{v}|^3} = \frac{a}{a^2 + b^2}.$$

3. Referring to Exercise 1, let $\vec{u}(t)$ denote the unit vector $\vec{u}(t) = \sin \omega t \, \vec{i} - \cos \omega t \, \vec{j}$. Show that there are two constants A and B such that $\vec{v} \times \vec{a} = A\vec{u}(t) + B\,\vec{k}$, and express A and B in terms of a, b, and ω.

4. Find an equation for a right circular cylinder of radius a if its axis is (a) along the x-axis; (b) perpendicular to the xy-plane at the point $(2, 3)$.

5. Sketch the cylinder represented by the equation $(x - 1)^2 + y^2 = 9$. Show that the curve whose vector equation is

$$\vec{r}(t) = t\,\vec{i} + \sqrt{8 + 2t - t^2}\,\vec{j} + (t + 4)\,\vec{k}\,, \qquad 0 \le t \le 4\,,$$

lies on this cylinder. This particular curve lies in a plane. Find an equation for this plane.

6. Find an equation for a cylinder whose generators are parallel to the y-axis, given that the curve $\vec{r}(t) = 2t\,\vec{i} + t^2\,\vec{j} + t^3\,\vec{k}$ lies on this cylinder.

7. A curve C in space is the intersection of two given surfaces with Cartesian equations $f(x, y, z) = 0$ and $g(x, y, z) = 0$. Let

$$\phi(x, y, z) = f(x, y, z) + tg(x, y, z)\,,$$

where t is a constant. For each t, the equation $\phi(x, y, z) = 0$ represents a surface. As t varies over all real numbers we obtain a family of surfaces.

(a) Show that C lies on all these surfaces.

(b) Consider the special case when the two intersecting surfaces are cylinders given by the equations $x^2 + 3y^2 - 9 = 0$ and $2y^2 - 3z^2 + 1 = 0$. Find a cylinder whose generators are parallel to the y-axis and which passes through the curve C.

8. A wooden log has the shape of a solid right circular cylinder of radius r. A wedge is cut by a plane through a diameter of the base and inclined at an angle θ with the plane of the base, where $0 < \theta < \frac{1}{2}\pi$. Find the volume of the wedge.

9. The axes of two solid right circular cylinders intersect at right angles. If each cylinder has radius r, find the volume of the solid of intersection. [*Hint.* If the axes are in the xy-plane, cross sections of the solid cut by planes perpendicular to the z-axis are squares.]

10. (a) The axes of three solid right circular cylinders lie in a plane and intersect at equal angles. If each cylinder has radius r, find the volume of the solid of intersection. [*Hint.* If the axes are in the xy-plane, cross sections of the solid cut by planes perpendicular to the z-axis are regular hexagons.]

(b) Generalize part (a) for n cylinders whose axes lie in a plane and intersect at equal angles. What is the limiting value of the volume of the solid of intersection as n increases indefinitely?

6.9 The conic sections

A right circular cone with its vertex at the origin and its axis along the z-axis is shown in Figure 6.8. Every point on this cone has the property that its distance from the axis is proportional to the absolute value of its z-coordinate. If α denotes the vertex angle and if $a = \tan \frac{1}{2}\alpha$, points (x, y, z) on the upper nappe (with $z > 0$) satisfy $\sqrt{x^2 + y^2} = az$, and those on the lower nappe (with $z < 0$) satisfy $\sqrt{x^2 + y^2} = -az$. Both nappes are therefore represented by the one equation

$$x^2 + y^2 = a^2z^2\,.$$

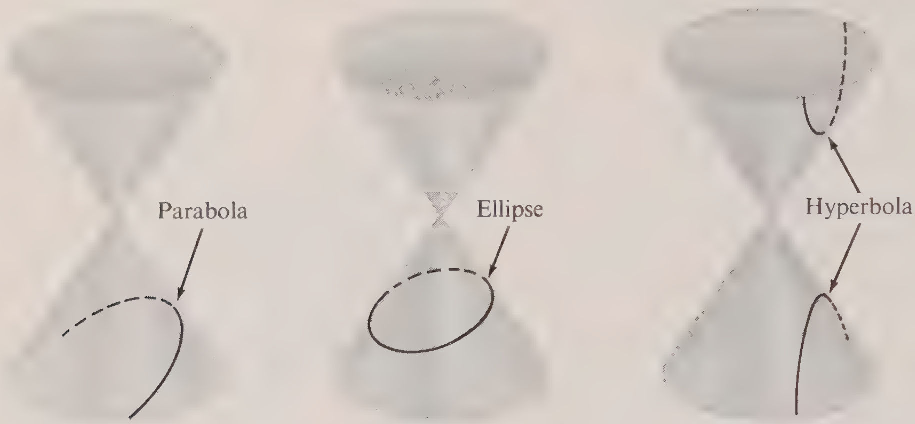


FIGURE 6.9 *The conic sections.*

The curves obtained by slicing the cone with a plane not passing through the vertex are called *conic sections*, or simply *conics*. If the cutting plane is parallel to a line of the cone through the vertex, the conic is called a *parabola*. Otherwise the intersection is called an *ellipse* or a *hyperbola*, according as the plane cuts just one or both nappes. (See Figure 6.9.) The hyperbola consists of two "branches," one on each nappe.

Many important discoveries in both pure and applied mathematics have been related to the conic sections. Appolonius' treatment of conics as early as the 3d century B.C. was one of the most profound achievements of classical Greek geometry. Nearly 2000 years later, Galileo discovered that a projectile fired horizontally from the top of a tower falls to earth along a parabolic path (if air resistance is neglected and if the motion takes place above a part of the earth that can be regarded as a flat plane). One of the turning points in the history of astronomy occurred around 1600 when Kepler suggested that all planets move in elliptical orbits. Some 80 years later, Newton was able to demonstrate

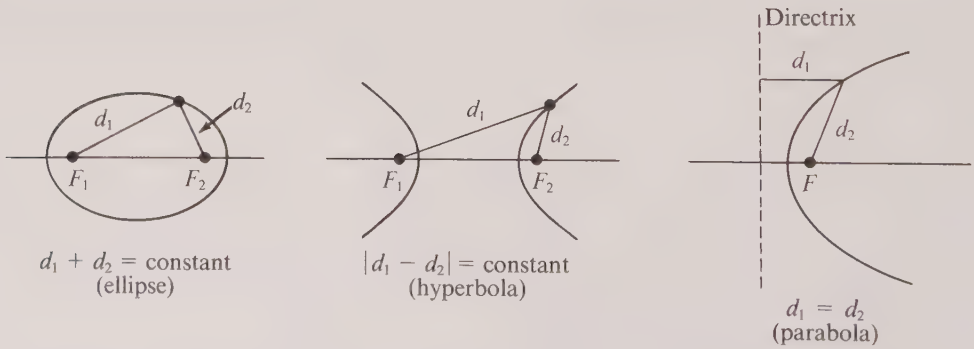


FIGURE 6.10 *Focal definitions of the conic sections.*

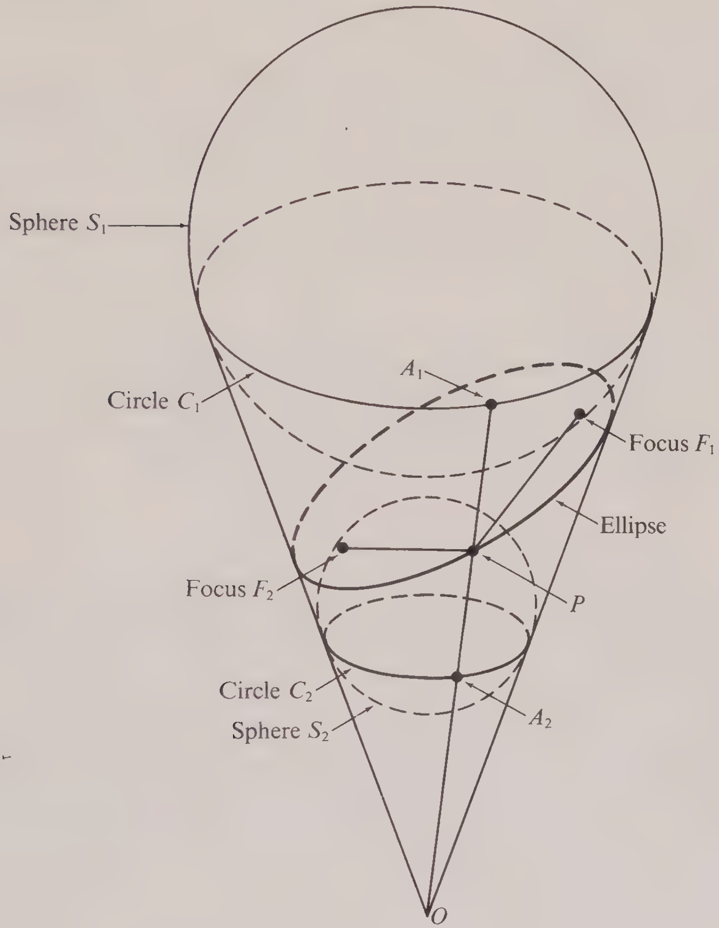


FIGURE 6.11 *The ice-cream-cone proof.*

that an elliptical planetary path implies an inverse-square law of gravitational attraction. This led Newton to formulate his famous theory of universal gravitation which has often been referred to as the greatest scientific discovery ever made. Conic sections appear not only as orbits of planets and satellites but also as trajectories of elementary atomic particles. They are used in the design of lenses and mirrors, and in architecture. These examples and many others show that the importance of the conic sections can hardly be overestimated.

There are other equivalent definitions of the conic sections. One of these refers to special points known as *foci* (singular: *focus*). An ellipse may be defined as the locus of all points in a plane the sum of whose distances d_1 and d_2 from two fixed points F_1 and F_2 (the foci) is constant. (See Figure 6.10.) If the foci coincide, the ellipse reduces to a circle. A hyperbola is the locus of all points for which the difference $d_1 - d_2$ is constant.

A parabola is the locus of all points for which the distance to a fixed point F (called the focus) is equal to the distance to a given *line* called the directrix (there is no connection with the directrix of a cylinder).

There is a very simple and elegant argument which shows that the focal property of an ellipse is a consequence of its definition as a section of a cone. This proof, which we may refer to as the "ice-cream-cone proof," was discovered in 1822 by a Belgian mathematician, G. P. Dandelin (1794–1847), and makes use of the two spheres S_1 and S_2 which are drawn so as to be tangent to the cutting plane and the cone, as illustrated in Figure 6.11. These spheres touch the cone along two parallel circles C_1 and C_2. We shall prove that the points F_1 and F_2, where the spheres contact the plane, can serve as foci of the ellipse.

Let P be an arbitrary point of the ellipse. The problem is to prove that $|\overrightarrow{PF_1}| + |\overrightarrow{PF_2}|$ is constant, that is, independent of the choice of P. For this purpose, draw that line on the cone from the vertex O to P and let A_1 and A_2 be its intersections with the circles C_1 and C_2, respectively. Then $\overrightarrow{PF_1}$ and $\overrightarrow{PA_1}$ are two tangents to S_1 from P and hence $|\overrightarrow{PF_1}| = |\overrightarrow{PA_1}|$ Similarly $|\overrightarrow{PF_2}| = |\overrightarrow{PA_2}|$, and therefore we have

$$|\overrightarrow{PF_1}| + |\overrightarrow{PF_2}| = |\overrightarrow{PA_1}| + |\overrightarrow{PA_2}| .$$

But $|\overrightarrow{PA_1}| + |\overrightarrow{PA_2}| = |\overrightarrow{A_1A_2}|$, which is the distance between the parallel circles C_1 and C_2 measured along the surface of the cone. This proves that F_1 and F_2 can serve as foci of the ellipse, as asserted.

Modifications of this proof work also for the hyperbola and the parabola. In the case of the hyperbola, the proof employs one sphere in each portion of the cone. For the parabola one sphere tangent to the cutting plane at the focus F is used. This sphere touches the cone along a circle which lies in a plane whose intersection with the cutting plane is the directrix of the parabola. With these hints the reader should be able to show that the focal properties of the hyperbola and parabola may be deduced from their definitions as sections of a cone.

6.10 The ellipse

It is easy to determine a Cartesian equation for an ellipse† if we make use of the focal definition. Given the foci F_1 and F_2, we introduce a rectangular coordinate system by taking the x-axis through the foci with the origin halfway between. Denote by $(-c, 0)$ and $(c, 0)$ the coordinates of the foci, as shown in Figure 6.12. If an arbitrary point P in the plane has coordinates (x, y), the distances d_1 and d_2 from P to the foci are given by the formulas

$$\text{(6.17)} \qquad d_1 = |\overrightarrow{PF_1}| = \sqrt{(x + c)^2 + y^2}\,, \qquad d_2 = |\overrightarrow{PF_2}| = \sqrt{(x - c)^2 + y^2}\,.$$

A point P is on the ellipse if and only if $d_1 + d_2$ is constant. Denoting this constant by $2a$, we have $d_1 + d_2 = 2a$. In order for the ellipse to have some point on it that does not lie on the segment F_1F_2, we insist on the inequality $d_1 + d_2 > 2c$ or, what is the same thing, $a > c$. The equation $d_1 + d_2 = 2a$ becomes

† Parametric equations for an ellipse are described in Exercise 16 of Section 6.11.

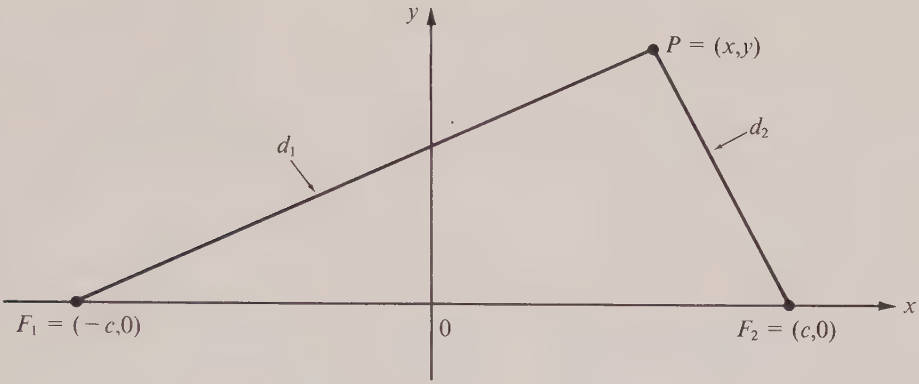


FIGURE 6.12 *The focal distances d_1 and d_2.*

$$\sqrt{(x+c)^2+y^2}+\sqrt{(x-c)^2+y^2}=2a\,.$$

This is usually simplified by transposing one of the radicals, squaring both sides, transposing the remaining radical and squaring again. These manipulations lead to the equation

$$\frac{x^2}{a^2}+\frac{y^2}{a^2-c^2}=1\,. \tag{6.18}$$

There is an alternative method of deriving (6.18) in which the algebraic manipulations are somewhat simpler. From (6.17) we find

$$d_1^2=(x+c)^2+y^2 \qquad \text{and} \qquad d_2^2=(x-c)^2+y^2\,, \tag{6.19}$$

and hence $d_1^2-d_2^2=(x+c)^2-(x-c)^2=4cx$. Since $d_1^2-d_2^2=(d_1+d_2)(d_1-d_2)$ $=2a(d_1-d_2)$ we must have $d_1-d_2=2cx/a$. If we add this to the equation $d_1+d_2=2a$, we find

$$d_1=\frac{c}{a}x+a\,, \qquad d_2=-\frac{c}{a}x+a\,. \tag{6.20}$$

These formulas express the focal distances d_1 and d_2 of a point (x, y) on the ellipse in terms of x alone. To introduce y we compute d_1^2 from (6.20) and equate this to the formula for d_1^2 in (6.19), thus obtaining (6.18).

If we let $b=\sqrt{a^2-c^2}$, Equation (6.18) becomes

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\,, \tag{6.21}$$

which is said to be an equation for the ellipse *in standard form.* We have shown that every point (x, y) on the ellipse satisfies (6.21) and it is easy to retrace our steps and show that, conversely, every point (x, y) which satisfies (6.21) must lie on the ellipse. (The proof of the converse is left as an exercise.)

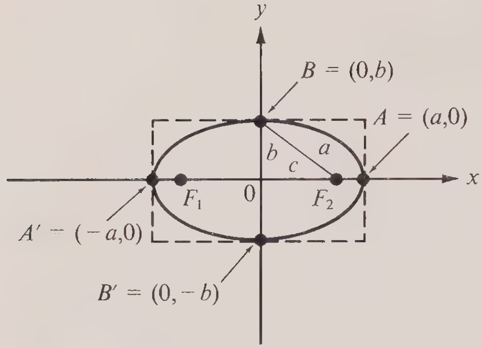


FIGURE 6.13

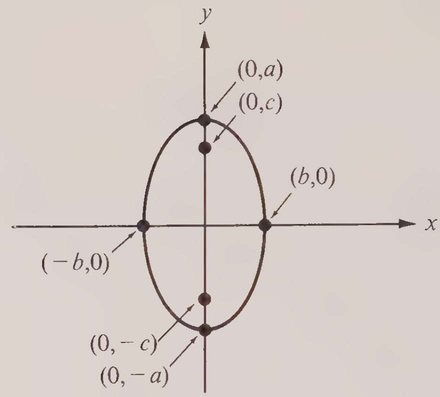


FIGURE 6.14

Much information about the shape and extent of the curve can be deduced from Equation (6.21). For example, it is clear that if a point (x, y) satisfies (6.21), so does the point $(x, -y)$ which is symmetrically located with respect to the x-axis. Similarly, the ellipse is symmetric about the y-axis because it contains $(-x, y)$ whenever it contains (x, y). Symmetry about both axes implies that $(-x, -y)$ is on the ellipse whenever (x, y) is. This property is called *symmetry about the origin*, and the origin is called the *center* of the ellipse.

If (x, y) satisfies (6.21), both x^2/a^2 and y^2/b^2 are nonnegative numbers whose sum is 1 and hence we must have

$$\frac{x^2}{a^2} \leq 1 \quad \text{and} \quad \frac{y^2}{b^2} \leq 1\,,$$

which means that all points (x, y) on the ellipse satisfy

$$-a \leq x \leq a \quad \text{and} \quad -b \leq y \leq b\,.$$

In other words, the entire ellipse lies within a rectangle of base $2a$ and height $2b$. (This rectangle is indicated by broken lines in Figure 6.13.) When $x = 0$, Equation (6.21) is satisfied by $y = b$ and $y = -b$. These are the largest and smallest values of y on the curve. Similarly, when $y = 0$, the equation is satisfied by $x = a$ and $x = -a$, which represent the largest and smallest values of x on the curve. The points $A = (a, 0)$ and $A' = (-a, 0)$ where the curve cuts the x-axis are called *vertices* of the ellipse, and the segment $A'A$ (of length $2a$) is called its *major axis*. The *minor axis* is the segment $B'B$ (of length $2b$) joining the points $B' = (0, -b)$ and $B = (0, b)$. Since $b^2 + c^2 = a^2$, the hypotenuse of right triangle BOF_2 in Figure 6.13 has length a. This relation also shows that $b < a$.

The ratio $c/a = \sqrt{1 - (b/a)^2}$ is called the *eccentricity* of the ellipse and is denoted by e. (This should not be confused with the base e of natural logarithms.) The eccentricity satisfies the inequalities $0 < e < 1$. When b is small compared to a, the foci are near the ends of the major axis, the ellipse is long and thin, and the eccentricity is near 1. As

$b \to a$ the foci move nearer the center, the ellipse becomes more like a circle, and the eccentricity approaches 0. Further properties of the eccentricity are described in some of the exercises of Section 6.11.

If the coordinate axes are drawn so the foci are on the y-axis at the points $(0, c)$ and $(0, -c)$ as shown in Figure 6.14, an argument similar to that just given leads to an equation like (6.21), except that the roles of x and y are interchanged. The standard form in this case is

$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1 . \tag{6.22}$$

The major axis (of length $2a$) and the vertices are now on the y-axis.

If the curves shown in Figures 6.13 and 6.14 are translated by a vector $\vec{r}_0 = x_0 \vec{i} + y_0 \vec{j}$, the centers move to the point (x_0, y_0) as shown in Figure 6.15 and the new Cartesian equations, obtained from (6.21) and (6.22) by replacing x by $x - x_0$ and y by $y - y_0$, become, respectively,

$$\frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} = 1 \qquad \text{and} \qquad \frac{(y - y_0)^2}{a^2} + \frac{(x - x_0)^2}{b^2} = 1 .$$

6.11 Exercises

Each of the equations in Exercises 1 through 6 represents an ellipse. Find the coordinates of the center, the foci, and the vertices, and sketch each curve. Also determine the eccentricity.

1. $\dfrac{x^2}{100} + \dfrac{y^2}{36} = 1.$
2. $\dfrac{y^2}{100} + \dfrac{x^2}{36} = 1.$
3. $\dfrac{(x - 2)^2}{16} + \dfrac{(y + 3)^2}{9} = 1.$
4. $9x^2 + 25y^2 = 25.$
5. $4y^2 + 3x^2 = 1.$
6. $\dfrac{(x + 1)^2}{16} + \dfrac{(y + 2)^2}{25} = 1.$

In each of Exercises 7 through 12, find an equation (in the appropriate standard form) for the ellipse that satisfies the conditions given. Sketch each curve.

7. Center at $(0, 0)$, one focus at $(\frac{3}{4}, 0)$, one vertex at $(1, 0)$.
8. Center at $(-3, 4)$, semiaxes of lengths 4 and 3, major axis parallel to the x-axis.
9. Same as Exercise 8, except with major axis parallel to the y-axis.
10. Vertices at $(-1, 2)$, $(-7, 2)$, minor axis of length 2.
11. Vertices at $(3, -2)$, $(13, -2)$, foci at $(4, -2)$, $(12, -2)$.
12. Center at $(2, 1)$, major axis parallel to the x-axis, the curve passing through the points $(6, 1)$ and $(2, 3)$.
13. Show that the area of the region bounded by the ellipse $x^2/a^2 + y^2/b^2 = 1$ is ab times the area of a circle of radius 1. *Note.* This statement can be proved from general properties of the integral, without performing any integrations.
14. (a) Show that the volume of the solid of revolution generated by rotating the ellipse $x^2/a^2 + y^2/b^2 = 1$ about its major axis is ab^2 times the volume of a unit sphere. *Note.* This statement can be proved from general properties of the integral, without performing any integrations.

    (b) What is the result if the ellipse is rotated about its minor axis?

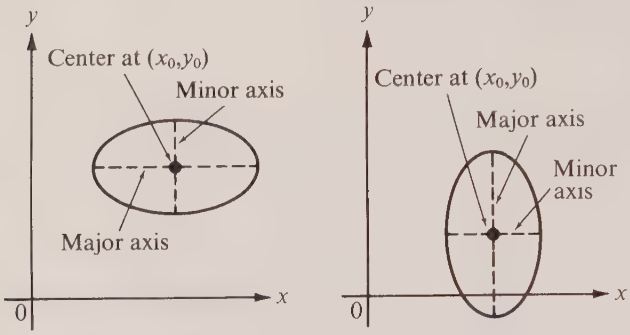


FIGURE 6.15

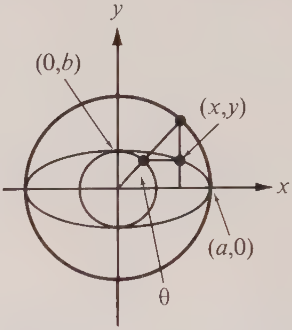


FIGURE 6.16 *The eccentric angle θ.*

15. Prove that a point (x_0, y_0) is *inside, on,* or *outside* the ellipse $x^2/a^2 + y^2/b^2 = 1$ according as $x_0^2/a^2 + y_0^2/b^2$ is *less than, equal to,* or *greater than* 1.

16. In Figure 6.16 is shown an ellipse $x^2/a^2 + y^2/b^2 = 1$ and two circles with radii a and b.

(a) Using the angle θ as parameter, show that the ellipse may be represented by the parametric equations $x = a\cos\theta$, $y = b\sin\theta$ or, what amounts to the same thing, by the vector equation $\vec{r} = a\cos\theta\,\vec{i} + b\sin\theta\,\vec{j}$. The angle θ is sometimes called the *eccentric angle* of the point (x, y).

(b) Show that the vector

$$\vec{T} = -\frac{y}{b^2}\vec{i} + \frac{x}{a^2}\vec{j}$$

is tangent to the ellipse when placed at the point (x, y) on the curve.

(c) Show that the vector

$$\vec{N} = \frac{x}{a^2}\vec{i} + \frac{y}{b^2}\vec{j}$$

is *normal* to the ellipse (perpendicular to the tangent vector) when placed at the point (x, y).

(d) If the eccentric angle of (x_0, y_0) is θ_0, show that the tangent line at (x_0, y_0) has the equation

$$\frac{x}{a}\cos\theta_0 + \frac{y}{b}\sin\theta_0 = 1 .$$

17. A particle moves around an ellipse in such a way that its eccentric angle (see Exercise 16) changes at a constant rate, say $\theta = \omega t$ at time t, where ω is a constant. If $\vec{a}(t)$ denotes the acceleration vector, show that $\vec{a}(t) = -\omega^2\vec{r}(t)$. In other words, the acceleration is centripetal, as in the case of circular motion with constant angular speed ω.

18. Show that the tangent line to the ellipse $x^2/a^2 + y^2/b^2 = 1$ at the point (x_0, y_0) has the equation $x_0x/a^2 + y_0y/b^2 = 1$. [*Hint.* Use Exercise 16(c) and write the equation of the tangent line in the form $\vec{P}\cdot\vec{N} = \vec{P}_0\cdot\vec{N}$.]

19. Let F_1 and F_2 denote the foci of an ellipse and let P be an arbitrary point on it. Show that a line perpendicular to the tangent line at P bisects the angle F_1PF_2. This *reflection property* makes the ellipse important in applications to acoustics and optics. In an elliptic mirror, light rays emanating from one focus will converge at the other.

20. Show that an ellipse is the locus of a point which moves so that the ratio of its distances from a fixed point and from a fixed line is a positive constant less than 1, and that this constant is the eccentricity of the ellipse. The fixed line is called a directrix for the ellipse.

21. (a) Prove that a similarity transformation (replacing x by tx and y by ty) carries an ellipse

with center at the origin into another ellipse with the same eccentricity. In other words, similar ellipses have the same eccentricity.

(b) Prove also the converse. That is, if two concentric ellipses have the same eccentricity and major axes on the same line, then they are related by a similarity transformation.

22. If $0 < e < 1$, show that the integral curves of the differential equation $y' = (e^2 - 1)x/y$ are ellipses with center at the origin and eccentricity e. *Note.* Since this is a homogeneous differential equation, the family of ellipses is invariant under similarity transformations. (Compare with Exercise 21.)

23. Prove that the product of the perpendicular distances from the foci of an ellipse to any tangent line is constant, this constant being the square of the length of half the minor axis.

24. Two tangent lines are drawn to the ellipse $x^2 + 4y^2 = 8$, each parallel to the line $x + 2y = 7$. Find the points of tangency.

25. A circle passes through both foci of an ellipse and is tangent to the ellipse at two points. Find the eccentricity of the ellipse.

6.12 The hyperbola

The discussion of Section 6.10 may be modified to determine Cartesian equations for the hyperbola. As before, we place the x-axis so that the foci are at the points $(-c, 0)$ and $(c, 0)$, as shown in Figure 6.12, and then we express the focal distances d_1 and d_2 by the equations in (6.17). The relation which characterizes the hyperbola now becomes $|d_1 - d_2| = 2a$, and to get at least one point on the curve that is not on the x-axis we must insist that $c > a$.

The equation $|d_1 - d_2| = 2a$ is equivalent to the statement

$$\sqrt{(x + c)^2 + y^2} - \sqrt{(x - c)^2 + y^2} = \pm 2a \,,$$

the plus or minus sign coming in to allow for the two possibilities $d_1 > d_2$ and $d_1 < d_2$. We may remove the radicals by transposing and squaring as suggested for the ellipse and thus obtain the equation

$$\frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1 \,. \tag{6.23}$$

Or, alternatively, we may use the formulas for d_1^2 and d_2^2 in (6.19) along with the equation $|d_1^2 - d_2^2| = |d_1 - d_2|(d_1 + d_2) = 2a(d_1 + d_2)$ to express the focal distances d_1 and d_2 in terms of x alone, as we did for the ellipse. In this case the formulas analogous to (6.20) are

$$d_1 = \left|\frac{c}{a}x + a\right|, \qquad d_2 = \left|\frac{c}{a}x - a\right|. \tag{6.24}$$

If we compute d_1^2 from this and equate to the formula for d_1^2 in (6.19), we are led at once to Equation (6.23).

Next we introduce $b = \sqrt{c^2 - a^2}$ and rewrite (6.23) in the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \,. \tag{6.25}$$

This equation for the hyperbola is said to be *in standard form.* All points on the hyperbola satisfy it and, conversely, every point satisfying this equation lies on the hyperbola.

If a point (x, y) satisfies (6.25), so do $(x, -y)$ and $(-x, y)$ and therefore the hyperbola is symmetric about each of the coordinate axes and hence also about the origin, its *center.* The x-axis is cut by the hyperbola at the points $A = (a, 0)$ and $A' = (-a, 0)$, called its *vertices.* The segment $A'A$ joining the vertices is known as the *transverse axis* of the hyperbola. Note that (6.25) cannot be satisfied for any y when $x = 0$ and hence the hyperbola does not intersect the y-axis.

Solving for y in terms of x in (6.25), we obtain two solutions,

$$y = \pm\frac{b}{a}\sqrt{x^2 - a^2}\,. \tag{6.26}$$

This shows that there are no points (x, y) on the curve with $x^2 < a^2$. On the other hand, any value of x for which $x > a$ or $x < -a$ leads to two values of y satisfying (6.26). Therefore the hyperbola consists of two separate branches, one lying to the right of the line $x = a$ and the other to the left of the line $x = -a$, as illustrated in Figure 6.17.

From (6.25) we may write $y^2/b^2 < x^2/a^2$, which is equivalent to the inequalities

$$-\frac{b}{a}|x| < y < \frac{b}{a}|x|\,.$$

These inequalities, along with the relation $|x| \geq a$, tell us that the entire hyperbola lies in that part of the xy-plane indicated by the shaded regions in Figure 6.17. The reader should note that the three numbers a, b, c form the sides of a right triangle with c as hypotenuse, as shown in Figure 6.17. The ratio c/a, which is always greater than 1, is called the *eccentricity* of the hyperbola. When $a = b$ the hyperbola is called *equilateral* or *rectangular.*

The two straight lines $y = bx/a$ and $y = -bx/a$ are related to the hyperbola in an interesting way. For very large positive x the number $\sqrt{x^2 - a^2}$ is approximately the same as x and hence the right-hand side of (6.26) is approximately $\pm bx/a$. As a matter of fact, it is easy to prove that the difference between $y_1 = bx/a$ and $y_2 = b\sqrt{x^2 - a^2}/a$ approaches zero as x increases without bound. This difference (illustrated in Figure 6.18) is

$$y_1 - y_2 = \frac{b}{a}(x - \sqrt{x^2 - a^2}) = \frac{b}{a}\frac{x^2 - (x^2 - a^2)}{x + \sqrt{x^2 - a^2}} = \frac{ab}{x + \sqrt{x^2 - a^2}} < \frac{ab}{x}\,.$$

As we let x increase without bound, the term $ab/x \to 0$ and this implies that the vertical distance from the hyperbola to the line $y = bx/a$ approaches zero. Because of this property, the line $y = bx/a$ is called an *asymptote* of the hyperbola. The line $y = -bx/a$ is another asymptote. The hyperbola is said to approach these lines *asymptotically.*

For each hyperbola whose standard equation is given by (6.25), there is another hyperbola with the same asymptotes whose foci lie on the y-axis at the points $(0, c)$ and $(0, -c)$, where $c = \sqrt{a^2 + b^2}$. [This is the same c as in Equation (6.23).] This new hyperbola has the standard equation

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1 \tag{6.27}$$

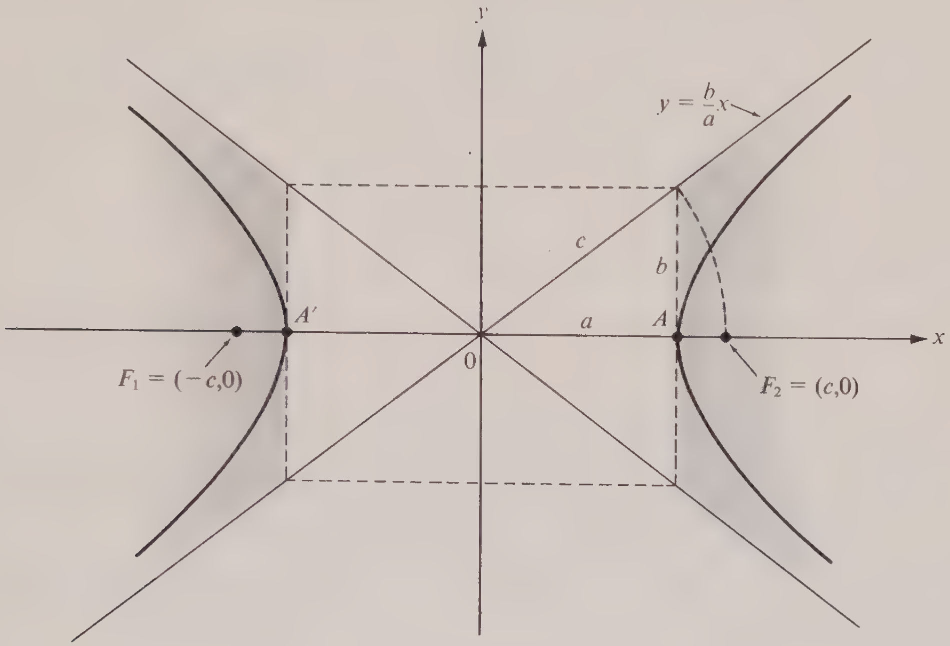


FIGURE 6.17

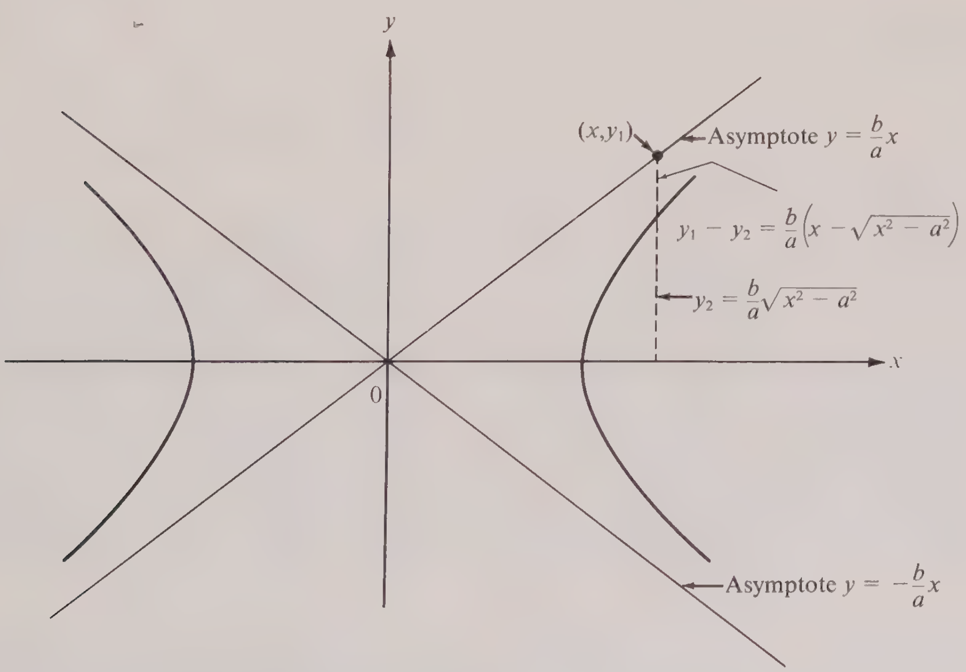


FIGURE 6.18 *Asymptotes of a hyperbola.*

and the two curves described by (6.25) and (6.27) are said to be *conjugates* of each other. Their interrelation is illustrated in Figure 6.19.

Translation of a hyperbola by a vector $\vec{r}_0 = x_0\vec{i} + y_0\vec{j}$ moves its center to (x_0, y_0). The new Cartesian equations, obtained from (6.25) and (6.27) simply by replacing x by $x - x_0$ and y by $y - y_0$, assume the form

$$\frac{(x-x_0)^2}{a^2} - \frac{(y-y_0)^2}{b^2} = 1 \qquad \text{and} \qquad \frac{(y-y_0)^2}{b^2} - \frac{(x-x_0)^2}{a^2} = 1\,.$$

The equations of the asymptotes are

$$y - y_0 = \frac{b}{a}(x - x_0) \qquad \text{and} \qquad y - y_0 = -\frac{b}{a}(x - x_0)\,.$$

We conclude this section with an interesting property relating a hyperbola to its asymptotes. Let p_1 and p_2 denote the perpendicular distances from an arbitrary point P on a hyperbola to the asymptotes, as illustrated in Figure 6.20. We shall prove that the product p_1p_2 is constant as P varies over the hyperbola.

By examining the figure, we see that $p_1 = |\vec{r}|\cos\theta$, where $\vec{r}$ is the vector $x\vec{i} + y\vec{j}$ from the origin to P, and θ is the angle between $\vec{r}$ and a line through P perpendicular to the asymptote of slope b/a, as shown. The vector $\vec{V} = b\vec{i} - a\vec{j}$ is perpendicular to this asymptote so we have $\cos\theta = \vec{V}\cdot\vec{r}/|\vec{V}|\,|\vec{r}|$ and hence

$$p_1 = \frac{\vec{V}\cdot\vec{r}}{|\vec{V}|} = \frac{bx - ay}{\sqrt{a^2 + b^2}} = \frac{bx - ay}{c}\,.$$

Similarly, we find $p_2 = (bx + ay)/c$ and so we have

$$p_1p_2 = \frac{(bx + ay)(bx - ay)}{c^2} = \frac{b^2a^2}{c^2}\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) = \frac{b^2a^2}{c^2}\,.$$

This proves that p_1p_2 is constant, as asserted.

A hyperbola is still a hyperbola if it is picked up and rotated, but we have not yet found what this does to its equation. In a later section we shall learn how an equation of a plane curve is altered by a rotation. At this point, however, we can use the foregoing property of the hyperbola to handle one particular case. Suppose we rotate the *rectangular* hyperbola $x^2/a^2 - y^2/a^2 = 1$ through $\frac{1}{4}\pi$ radians, leaving its center at the origin. In this new position the asymptotes are the coordinate axes (as shown in Figure 6.21) and the distances p_1 and p_2 from a point (x, y) on the curve to the asymptotes are $|x|$ and $|y|$, respectively. Since $b = a$ and $c = a\sqrt{2}$ in a rectangular hyperbola, the property $p_1p_2 = b^2a^2/c^2$ leads to the equation $|xy| = a^2/2$. When the branches are in the first and third quadrants, as shown in Figure 6.21, x and y have the same sign on the hyperbola and the equation is $xy = a^2/2$. The conjugate hyperbola, shown by the broken curve in Figure 6.21, is given by the equation $xy = -a^2/2$.

6.13 Exercises

1. Modify the "ice-cream-cone proof," as suggested in Section 6.9, and prove that the focal property of a hyperbola follows from its definition as a section of a cone.

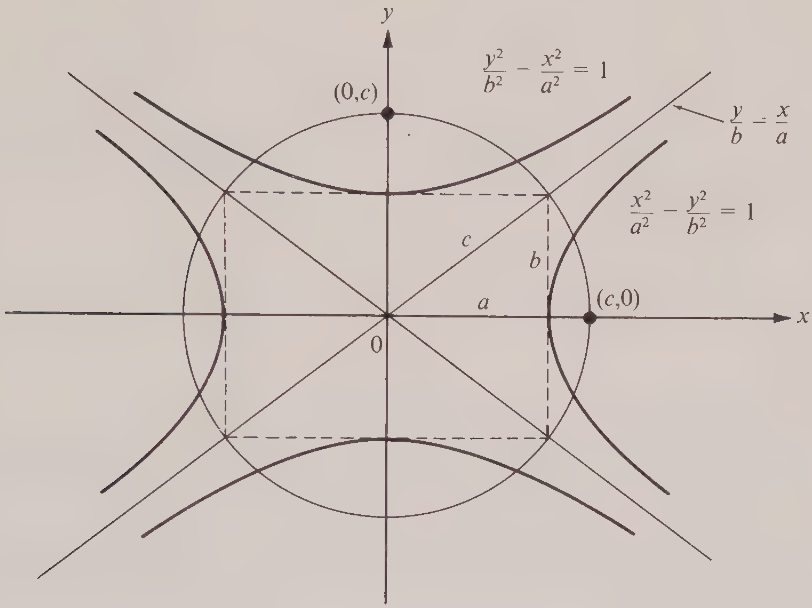


FIGURE 6.19 *Conjugate hyperbolas.*

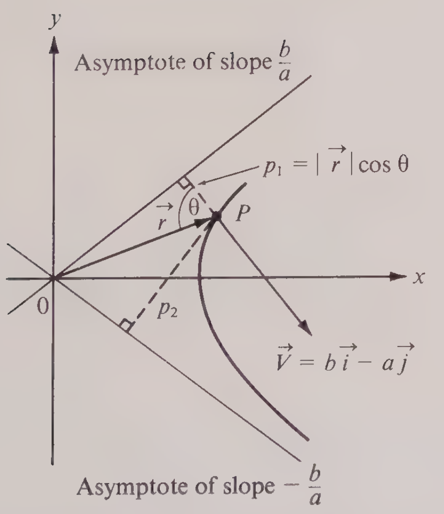


FIGURE 6.20 *Illustrating the distances p_1 and p_2 from a point P on a hyperbola to the asymptotes.*

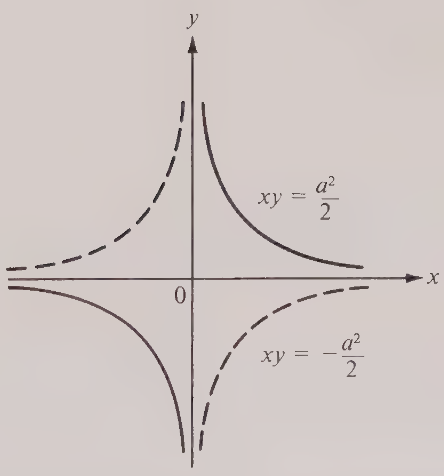
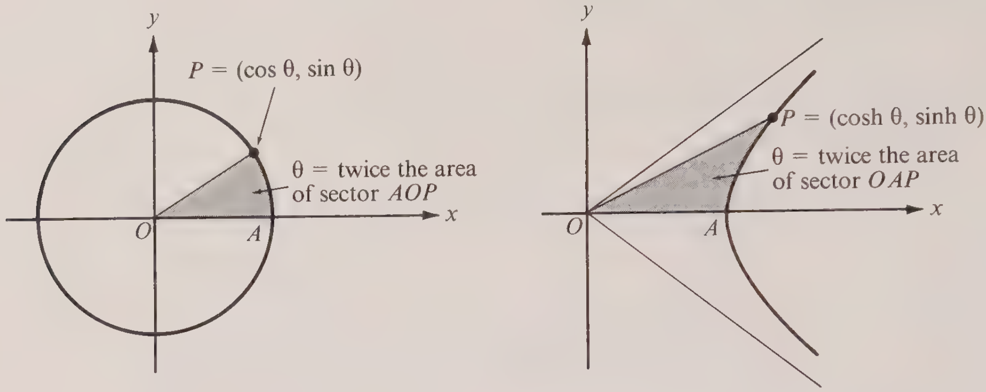


FIGURE 6.21 *Solid curve is a rectangular hyperbola, $xy = a^2/2$. Dotted curve is the conjugate hyperbola, $xy = -a^2/2$.*



(a) *Circle:* $x^2 + y^2 = 1$. (b) *Hyperbola:* $x^2 - y^2 = 1$.

FIGURE 6.22 *Analogy between parameter for a circle and that for a hyperbola.*

Each of the equations in Exercises 2 through 7 represents a hyperbola. Find the coordinates of the center, the foci, and the vertices. Sketch each curve and show the positions of the asymptotes. Also, compute the eccentricity.

2. $\dfrac{x^2}{100} - \dfrac{y^2}{64} = 1.$

3. $\dfrac{y^2}{100} - \dfrac{x^2}{64} = 1.$

4. $\dfrac{(x+3)^2}{4} - (y-3)^2 = 1.$

5. $9x^2 - 16y^2 = 144.$

6. $4x^2 - 5y^2 + 20 = 0.$

7. $\dfrac{(x-1)^2}{4} - \dfrac{(y+2)^2}{9} = 1.$

In each of Exercises 8 through 12, find an equation (in the appropriate standard form) for the hyperbola which satisfies the conditions given. Sketch each curve and show the positions of the asymptotes.

8. Center at $(0, 0)$, one focus at $(4, 0)$, one vertex at $(2, 0)$.
9. Foci at $(0, \pm\sqrt{2})$, vertices at $(0, \pm 1)$.
10. Vertices at $(\pm 2, 0)$, asymptotes $y = \pm 2x$.
11. Center at $(-1, 4)$, one focus at $(-1, 2)$, one vertex at $(-1, 3)$.
12. Center at $(2, -3)$, transverse axis parallel to one of the coordinate axes, the curve passing through $(3, -1)$ and $(-1, 0)$.
13. For what value (or values) of C will the line $3x - 2y = C$ be tangent to the hyperbola $x^2 - 3y^2 = 1$?
14. The asymptotes of a hyperbola are the lines $2x - y = 0$ and $2x + y = 0$. Find an equation for the curve if it passes through the point $(3, -5)$.
15. Prove that the perpendicular distance from a focus to an asymptote of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ is b.
16. The identity $\cosh^2\theta - \sinh^2\theta = 1$ for hyperbolic functions suggests that the hyperbola $x^2/a^2 - y^2/b^2 = 1$ may be represented by the parametric equations $x = a\cosh\theta$, $y = b\sinh\theta$, or, what amounts to the same thing, by the vector equation $\vec{r} = a\cosh\theta\,\vec{i} + b\sinh\theta\,\vec{j}$. (Compare with Exercise 16 of Section 6.11.) When $a = b = 1$ the parameter θ may be given a geometric interpretation analogous to that which holds between θ, $\sin\theta$, and $\cos\theta$ in the unit circle shown in Figure 6.22(a). Figure 6.22(b) shows one branch of the hyperbola $x^2 - y^2 = 1$.

(a) If the point P has coordinates $x = \cosh \theta$ and $y = \sinh \theta$, show that θ equals twice the area of the sector OAP shaded in the figure. [*Hint*. The area of sector OAP is $\frac{1}{2} \cosh \theta \sinh \theta - \int_1^{\cosh \theta} \sqrt{x^2 - 1}\, dx$. Introduce the substitution $x = \cosh u$ in the integral and use the identities $\sinh 2\theta = 2 \sinh \theta \cosh \theta$ and $2 \sinh^2 \theta = \cosh 2\theta - 1$.]

(b) Show that the vectors $\vec{T} = (y/b^2)\,\vec{i} + (x/a^2)\,\vec{j}$ and $\vec{N} = (x/a^2)\,\vec{i} - (y/b^2)\,\vec{j}$ are, respectively, tangent and normal to the hyperbola if placed at the point (x, y) on the curve.

17. A particle moves along a hyperbola according to the equation

$$\vec{r}(t) = a \cosh \omega t\, \vec{i} + b \sinh \omega t\, \vec{j}\,,$$

where ω is a constant. Show that the acceleration is centrifugal.

18. Prove that the tangent line at any point of a hyperbola bisects the angle formed by the lines joining this point to the foci. Interpret this as a reflection property of hyperbolic mirrors.

19. Show that the tangent line to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ at the point (x_0, y_0) is given by the equation $x_0x/a^2 - y_0y/b^2 = 1$.

20. Prove that an ellipse and a hyperbola which have the same foci intersect at right angles.

21. Show that a hyperbola is the locus of a point which moves so that the ratio of its distance from a fixed point and from a fixed line is a constant greater than 1 and that this constant is the eccentricity of the hyperbola. The fixed line is called a directrix of a hyperbola.

22. (a) Prove that a similarity transformation carries a hyperbola with center at the origin into another hyperbola with the same eccentricity. In other words, similar hyperbolas have the same eccentricity.

(b) Prove also the converse. That is, if two concentric hyperbolas have the same eccentricity and transverse axes on the same line, then they are related by a similarity transformation.

23. If $e > 1$, show that the integral curves of the differential equation $y' = (e^2 - 1)x/y$ are hyperbolas with center at the origin and eccentricity e. Note that this is the same differential equation satisfied by concentric ellipses with eccentricity $e < 1$.

24. The normal line at each point of a curve and the line from that point to the origin form an isosceles triangle whose base is on the x-axis. Show that the curve is a hyperbola. [*Hint*. Formulate the problem as a differential equation.]

25. The normal line at a point P of a curve intersects the x-axis at X and the y-axis at Y. Find the curve if each P is the mid-point of the corresponding line segment XY and if the point $(4, 5)$ is on the curve. [*Hint*. Formulate the problem as a differential equation.]

6.14 The parabola

To obtain a Cartesian equation for a parabola from its focal definition we refer to Figure 6.23 where the axes are chosen so the focus is on the x-axis at the point $(c, 0)$ and the directrix is the vertical line $x = -c$. The distances d_1 and d_2 of an arbitrary point (x, y) from the focus and directrix are given by the respective formulas

$$d_1 = \sqrt{(x - c)^2 + y^2}\,, \qquad d_2 = |x + c|\,.$$

The point (x, y) lies on the parabola if and only if $d_1 = d_2$, which implies the equation $(x - c)^2 + y^2 = (x + c)^2$. This simplifies to the so-called *standard form*:

$$y^2 = 4cx\,.$$

The point midway between the focus and directrix (the origin in Figure 6.23) is called the *vertex* of the parabola, and the line passing through the vertex and the focus is the

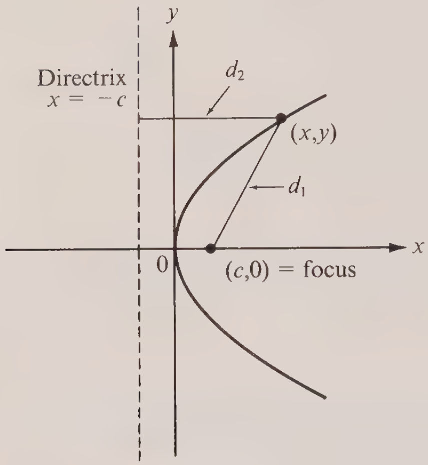


FIGURE 6.23 *The parabola* $y^2 = 4cx$.

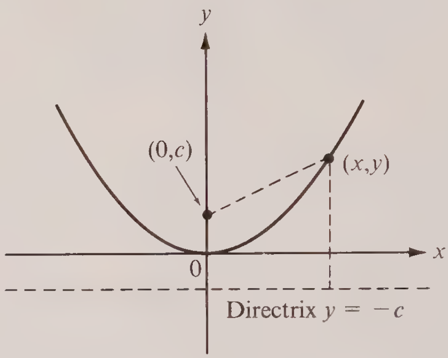


FIGURE 6.24 *The parabola* $x^2 = 4cy$.

axis of the parabola. If $c > 0$ the parabola lies to the right of the y-axis as in Figure 6.23. When $c < 0$ the curve lies to the left of the y-axis. In either case it is symmetric about its axis.

If the axes are chosen so the focus is on the y-axis at the point $(0, c)$ and if the horizontal line $y = -c$ is taken as directrix, the standard form of the Cartesian equation becomes

$$x^2 = 4cy .$$

When $c > 0$ the parabola opens upward as shown in Figure 6.24. When $c < 0$ it opens downward.

If the parabola in Figure 6.23 is translated so that its vertex is at the point (x_0, y_0), the corresponding equation becomes

$$(y - y_0)^2 = 4c(x - x_0) .$$

The focus is now at the point $(x_0 + c, y_0)$ and the directrix is the line $x = x_0 - c$. The axis of the parabola is the line $y = y_0$.

Similarly, a translation of the parabola in Figure 6.24 leads to the equation

$$(x - x_0)^2 = 4c(y - y_0) ,$$

with focus at $(x_0, y_0 + c)$. The line $y = y_0 - c$ is its directrix, the line $x = x_0$ its axis.

The reader may find it amusing to prove that a parabola does not have any asymptotes.

6.15 Exercises

1. Modify the "ice-cream-cone proof," as suggested in Section 6.9, and prove that the focal property of a parabola follows from its definition as a section of a cone.

Each of the equations in Exercises 2 through 7 represents a parabola. Find the coordinates of the vertex, an equation for the directrix, and an equation for the axis. Sketch each of the curves.

2. $y^2 = -8x$.
3. $y^2 = 3x$.
4. $(y - 1)^2 = 12x - 6$.
5. $x^2 = 6y$.
6. $x^2 + 8y = 0$.
7. $(x + 2)^2 = 4y + 9$.

In each of Exercises 8 through 13, find an equation (in appropriate standard form) for the parabola that satisfies the conditions given and sketch the curve.

8. Focus at $(0, -\frac{1}{4})$; equation of directrix, $y = \frac{1}{4}$.
9. Vertex at $(0, 0)$; equation of directrix, $x = -2$.
10. Vertex at $(-4, 3)$; focus at $(-4, 1)$.
11. Focus at $(3, -1)$; equation of directrix, $x = \frac{1}{2}$.
12. Axis is parallel to the y-axis; passes through $(0, 1)$, $(1, 0)$, and $(2, 0)$.
13. Axis is parallel to the x-axis; vertex at $(1, 3)$; passes through $(-1, -1)$.

14. Proceeding directly from the focal definition, find an equation for the parabola whose focus is the origin and whose directrix is the line $2x + y = 10$.

15. Show that the vector $\vec{T} = y\vec{i} + 2c\vec{j}$ is tangent to the parabola $y^2 = 4cx$ at the point (x, y), and that the vector $\vec{N} = 2c\vec{i} - y\vec{j}$ is perpendicular to $\vec{T}$. [*Hint*. Write a vector equation for the parabola, using y as parameter.]

16. Prove that the equation of the line of slope m that is tangent to the parabola $y^2 = 4cx$ can be written in the form $y = mx + c/m$. What are the coordinates of the point of contact?

17. (a) Solve Exercise 16 for the parabola $(y - y_0)^2 = 4c(x - x_0)$.
(b) Solve Exercise 16 for the parabola $x^2 = 4cy$ and, more generally, for the parabola $(x - x_0)^2 = 4c(y - y_0)$.

18. Prove that the equation of the line that is tangent to the parabola $y^2 = 4cx$ at the point (x_1, y_1) can be written in the form $y_1y = 2c(x + x_1)$.

19. Solve Exercise 18 for each of the parabolas described in Exercise 17.

20. The line $x - y + 4 = 0$ is tangent to the parabola $y^2 = 16x$. Find the point of contact.

21. Prove that a line perpendicular to the tangent line at a point P of a parabola bisects the angle between the line joining P to the focus and the line through P parallel to the axis. Interpret this geometric property in terms of parabolic mirrors. [*Hint*. Use Exercise 15.]

22. Determine which of the following lines are tangent to the parabola $y^2 = 16x$:

(a) $x + y + 4 = 0$;
(b) $3x - y + 2 = 0$;
(c) $2x - y - 2 = 0$;
(d) $x - y - 1 = 0$.

23. (a) A chord of length $8|c|$ is drawn perpendicular to the axis of the parabola $y^2 = 4cx$. Let P and Q be the points where the chord meets the parabola. Show that the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are perpendicular.
(b) The chord of a parabola drawn through the focus and parallel to the directrix is called the *latus rectum*. Show first that the length of the latus rectum is twice the distance from the focus to the directrix, and then show that the tangents to the parabola at both ends of the latus rectum intersect the axis of the parabola on the directrix.

24. A parabolic arch has a base of length b and altitude h. Determine the area of the region bounded by the arch and the base.

25. The region bounded by the parabola $y^2 = 8x$ and the line $x = 2$ is rotated about the x-axis. Find the volume of the solid of revolution so generated.

26. Two parabolas having the equations $y^2 = 2(x - 1)$ and $y^2 = 4(x - 2)$ enclose a plane region R.

(a) Compute the area of R by integration.

(b) Find the volume of the solid of revolution generated by revolving R about the x-axis.

(c) Same as (b), but revolve R about the y-axis.

27. Prove that a perpendicular drawn from the focus of a parabola to any tangent line meets this tangent upon the tangent at the vertex.

28. Show that the locus of the centers of a family of circles, all of which pass through a given point and are tangent to a given line, is a parabola.

29. Show that the locus of the centers of a family of circles, all of which are tangent (externally) to a given circle and also to a given straight line, is a parabola. (Exercise 28 can be considered to be a special case.)

30. A curve has the property that the part of every tangent line between the x-axis and the point of tangency is bisected by the y-axis. Find this curve if it passes through (3, 3).

31. If the normal line and ordinate are drawn at any point of a curve, they cut off a segment of length 2 on the x-axis. Find this curve if it passes through (1, 2).

32. (a) Prove that the collection of all parabolas is invariant under a similarity transformation. That is, a similarity transformation carries a parabola into a parabola.

(b) Find all the parabolas similar to $y = x^2$.

6.16 The general equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

There are two main types of problem that confront us in plane analytic geometry:

(i) A curve is described in geometric terms and we want to find a Cartesian equation which represents the curve analytically.

(ii) A Cartesian equation is given and we want to recognize the geometric configuration that it represents.

For the conic sections, problem (i) was dealt with in some detail in Sections 6.10, 6.12 and 6.14. The Cartesian equations obtained thus far for the conic sections are all obtainable as special cases of the second-degree equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0\,. \tag{6.28}$$

We turn now to the converse problem. Suppose we start with a second-degree equation of the form (6.28) and try to determine its graph. There are four possibilities:

(1) No points (x, y) satisfy (6.28). An example is the equation $x^2 + y^2 + 5 = 0$.

(2) Just one point satisfies (6.28). For example, the equation $3x^2 + 2y^2 = 0$ is satisfied only by $(x, y) = (0, 0)$.

(3) The graph consists of (one or two) straight lines. An example for one straight line is the equation $3x + 2y + 5 = 0$. Another example is the equation $y^2 = 0$. On the other hand, the equation $x^2 - 4y^2 = 0$ represents *two* straight lines since it is satisfied if either $x - 2y = 0$ or $x + 2y = 0$. Similarly, the equation $xy = 0$ is satisfied by those and only those points on the coordinate axes.

(4) The graph is a conic section. If the xy term is absent, that is, if $B = 0$ in (6.28), we may recognize the conic by completing the squares in x and y. This enables us to transform the equation to one of the standard forms discussed earlier. We illustrate this technique with three examples.

Example 1. $y^2 + 8x - 6y + 33 = 0$.

Since the equation contains y^2 but not x^2, we try to put it in the form

$$(y - y_0)^2 = 4c(x - x_0)\,. \tag{6.29}$$

We complete the square in y by writing:

$$\begin{aligned} y^2 - 6y \quad\;\; &= -8x - 33\,, \\ y^2 - 6y + 9 &= -8x - 33 + 9 = -8x - 24\,, \\ (y - 3)^2 &= -8(x + 3)\,. \end{aligned}$$

This is of the form (6.29) with $x_0 = -3$, $y_0 = 3$, $c = -2$. Therefore it represents a parabola whose axis is the line $y = 3$. The focus is at the point $(x_0 + c, y_0) = (-5, 3)$ and the directrix is the line $x = -1$. The parabola opens to the left because $c < 0$.

Example 2. $4x^2 + 9y^2 + 8x - 36y + 4 = 0$.

This equation contains both x^2 and y^2 with positive coefficients so we try to put it in the standard form for an ellipse, namely:

$$\frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} = 1\,. \tag{6.30}$$

We complete the square in both x and y as follows:

$$\begin{aligned} 4x^2 + 8x \qquad + 9y^2 - 36y \qquad &= -4\,, \\ 4(x^2 + 2x + \quad) + 9(y^2 - 4y + \quad) &= -4\,, \\ 4(x^2 + 2x + 1) + 9(y^2 - 4y + 4) &= -4 + 4 + 36\,, \\ 4(x + 1)^2 + 9(y - 2)^2 &= 36\,, \\ \frac{(x + 1)^2}{9} + \frac{(y - 2)^2}{4} &= 1\,. \end{aligned}$$

This is of the form (6.30) with $x_0 = -1$, $y_0 = 2$, $a = 3$, $b = 2$. The center of the ellipse is at $(-1, 2)$ and its major and minor axes have lengths 6 and 4, respectively.

Example 3. $3x^2 - y^2 + 20y - 148 = 0$.

The coefficients of x^2 and y^2 have opposite signs so we try to put this in one of the standard forms for a hyperbola, namely:

$$\frac{(x - x_0)^2}{a^2} - \frac{(y - y_0)^2}{b^2} = 1\,. \tag{6.31}$$

Again we complete the squares by writing:

$$\begin{aligned} 3x^2 - (y^2 - 20y + \qquad) &= 148\,, \\ 3x^2 - (y^2 - 20y + 100) &= 148 - 100\,, \\ 3x^2 - (y - 10)^2 &= 48\,, \\ \frac{x^2}{16} - \frac{(y - 10)^2}{48} &= 1\,. \end{aligned}$$

This is of the form (6.31) with $x_0 = 0$, $y_0 = 10$, $a = 4$, $b = 4\sqrt{3}$. The center of the hyperbola is at $(0, 10)$ and its asymptotes are the lines $y - 10 = \pm\sqrt{3}\,x$.

Suppose the equation in (6.28) represents a conic, and suppose that $B = 0$. If we proceed as in the foregoing examples, it is easy to see that the *type* of conic is completely determined by the two quadratic terms Ax^2 and Cy^2. If exactly one of A or C is zero, the conic is a parabola with its axis parallel to one of the coordinate axes. When both A and C are nonzero, the conic is either an ellipse or a hyperbola, depending on whether

A and C have the same or opposite signs. In other words, if (6.28) represents a conic and if $B = 0$, then the conic is a *parabola* if $AC = 0$, an *ellipse* if $AC > 0$, and a *hyperbola* if $AC < 0$. In the last two cases the first-degree terms, Dx and Ey, determine the location of the center of the conic. If $D = E = 0$, the conic has its center at the origin. Otherwise the center may be located by completing the squares as was done in the examples worked out above.

In the general case (that is, when B is not necessarily zero) the number $4AC - B^2$ determines the nature of the conic. In Section 6.18 we shall prove that if (6.28) represents a conic, then the conic is a *parabola* if $4AC - B^2 = 0$, an *ellipse* if $4AC - B^2 > 0$, and a *hyperbola* if $4AC - B^2 < 0$. The number $4AC - B^2$ is called the *discriminant* of the left-hand member of (6.28).

6.17 Translation of axes in the plane

The exact form of a Cartesian equation for a given plane curve will depend, of course, on the location of this curve relative to the coordinate axes. Experience teaches us that a judicious choice of axes may result in considerable simplification of the equation. For example, the equation of a circle contains no first-degree terms if the origin is at the center of the circle. Therefore it seems desirable to have a method for passing from one coordinate system to another.

Consider two rectangular coordinate systems in the plane. Call the first the xy-system and the second the XY-system. The second system is said to be obtained from the first by a *translation* if the corresponding axes are parallel as shown in Figure 6.25. A given point P now has two pairs of coordinates, say (x, y) and (X, Y). How are these coordinates related? Suppose the origin of the XY-system has coordinates (h, k) in the xy-system. By examining Figure 6.25, we see that we must have

$$x = X + h, \qquad y = Y + k$$

or, what is the same thing,

$$X = x - h, \qquad Y = y - k.$$

Therefore, if a curve is described in the xy-system by a Cartesian equation of the form $f(x, y) = 0$, the corresponding equation in the XY-system is simply $f(X + h, Y + k) = 0$.

For instance, in Example 2 of the foregoing section we transformed the equation $4x^2 + 9y^2 + 8x - 36y + 4 = 0$ into the form

$$\frac{(x+1)^2}{9} + \frac{(y-2)^2}{4} = 1, \tag{6.32}$$

which we recognized as the standard form of the equation of an ellipse with center at $(-1, 2)$. If we place the origin of a new coordinate system at $(-1, 2)$ by putting

$$X = x + 1, \qquad Y = y - 2,$$

Equation (6.32) is transformed into the simpler equation

$$\frac{X^2}{9} + \frac{Y^2}{4} = 1.$$

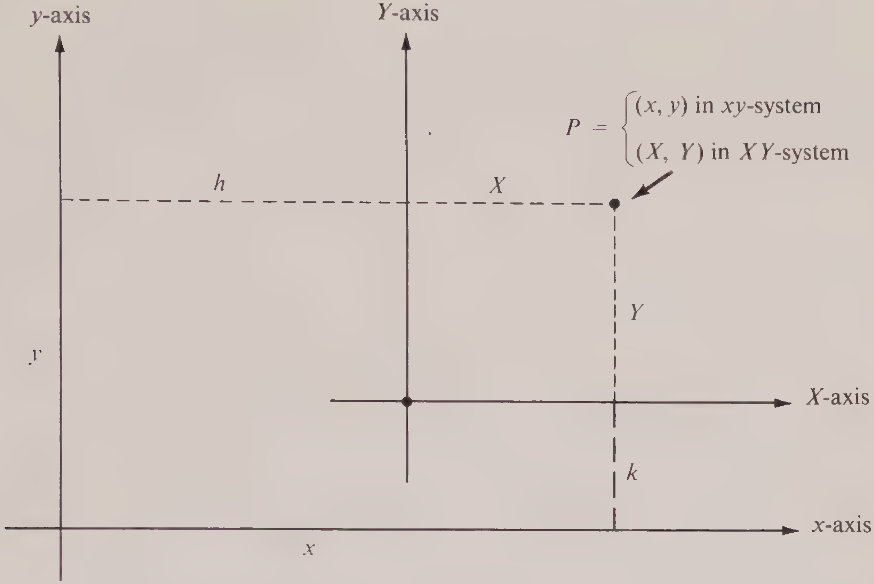


FIGURE 6.25 *Translation of axes.*

In this example we knew where to place the new axes because we recognized the curve in (6.32). If we did not have this *a priori* knowledge, we could replace x by $X + h$ and y by $Y + k$, with undetermined values of h and k, and then try to choose h and k to simplify the new equation. For example, consider the equation

$$x^2 - 4x - 2y + 10 = 0 . \tag{6.33}$$

Replacing x by $X + h$ and y by $Y + k$, we obtain

$$(X + h)^2 - 4(X + h) - 2(Y + k) + 10 = 0 .$$

Rearranging in powers of X and Y, we get

$$X^2 + (2h - 4)X - 2Y + (h^2 - 4h - 2k + 10) = 0 .$$

The first-degree term in X will disappear if we choose $h = 2$. This leads to the equation $X^2 - 2Y + (6 - 2k) = 0$. If we now choose $k = 3$, the constant term vanishes and the equation simplifies to

$$X^2 = 2Y$$

which we recognize as one of the standard equations for a parabola that opens upward and passes through the origin of the XY-system [that is, the point (2, 3) in the xy-system]. Of course, the same result can be obtained by completing the square in x in the original equation (6.33).

Note. Sometimes it is desirable to translate the curve instead of the axes. If each point of a curve $f(x, y) = 0$ is moved by a fixed vector $x_0 \vec{i} + y_0 \vec{j}$, the curve may be described in its new position by the equation $f(x - x_0, y - y_0) = 0$. In other words, we simply replace x by $x - x_0$ and y by $y - y_0$. For example, a circle of radius r and center at the origin has the equation $x^2 + y^2 = r^2$. When the circle is shifted so that its center is at (x_0, y_0), then its new equation is $(x - x_0)^2 + (y - y_0)^2 = r^2$.

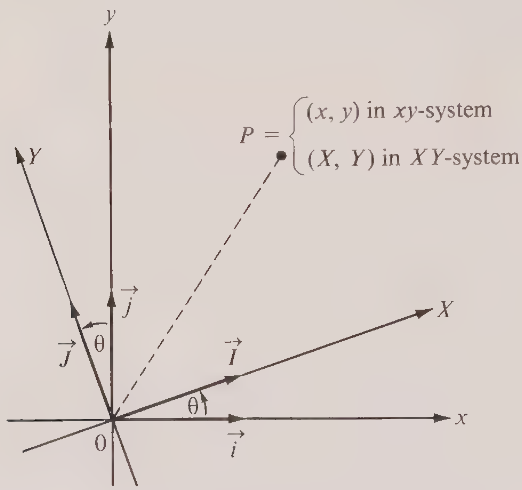


FIGURE 6.26 *Rotation of axes.*

6.18 Rotation of axes in the plane

A translation of axes moves the origin to a new position without changing the direction of the axes. We wish to consider next a *rotation* of axes in which the XY-system has the same origin as the xy-system. If θ denotes the angle between the positive x- and X-axes, we say that the XY-system is obtained from the xy-system by a rotation through the angle θ. (See Figure 6.26.)

A given point P will have two pairs of coordinates, say (x, y) and (X, Y), and the relationship between these coordinates will involve the angle θ. To find this relationship we introduce unit coordinate vectors $\vec{i}$ and $\vec{j}$ in the xy-system and $\vec{I}$ and $\vec{J}$ in the XY-system, as illustrated in Figure 6.26. Next we observe the following relations:

$$(6.34) \quad \begin{cases} \vec{i} \cdot \vec{I} = \cos\theta\,, & \vec{j} \cdot \vec{J} = \cos\theta\,, \\ \vec{i} \cdot \vec{J} = \cos\left(\theta + \frac{\pi}{2}\right) = -\sin\theta\,, & \vec{j} \cdot \vec{I} = \cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta\,. \end{cases}$$

Now let $\vec{r} = \overrightarrow{OP}$ be the radius vector from the origin to P. We then have the two expressions

(6.35) $$\vec{r} = x\vec{i} + y\vec{j}$$

and

(6.36) $$\vec{r} = X\vec{I} + Y\vec{J}\,.$$

If we form the dot product of $\vec{i}$ with $\vec{r}$ and $\vec{j}$ with $\vec{r}$, using (6.35), we get

$$x = \vec{i}\cdot\vec{r}\,, \qquad y = \vec{j}\cdot\vec{r}\,.$$

If we substitute $X\vec{I} + Y\vec{J}$ for $\vec{r}$, these equations become

$$x = \vec{i}\cdot\vec{r} = \vec{i}\cdot(X\vec{I} + Y\vec{J}) = X(\vec{i}\cdot\vec{I}) + Y(\vec{j}\cdot\vec{J})$$

and

$$y = \vec{j}\cdot\vec{r} = \vec{j}\cdot(X\vec{I} + Y\vec{J}) = X(\vec{j}\cdot\vec{I}) + Y(\vec{j}\cdot\vec{J})\,.$$

Using (6.34) to express the dot products on the right in terms of θ, we obtain the formulas

(6.37) $$\begin{cases} x = X\cos\theta - Y\sin\theta\,, \\ y = X\sin\theta + Y\cos\theta\,. \end{cases}$$

These tell us how to find x and y in terms of X and Y. If we solve these for X and Y in terms of x and y, we may derive the inverse relationship:†

(6.38) $$\begin{cases} X = x\cos\theta + y\sin\theta\,, \\ Y = -x\sin\theta + y\cos\theta\,. \end{cases}$$

As an application of the above process we shall show that a general equation of the second degree in x and y, say

(6.39) $$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0\,,$$

can always be transformed, by a suitable rotation of axes, into a new equation of second degree in X and Y in which *no* XY term is present. The graph of the new equation in X and Y may then be analyzed by the methods discussed earlier in Section 6.16.

We begin with (6.39) and substitute for x and y the expressions in (6.37). This leads to the equation

(6.40) $$\begin{aligned} &A(X\cos\theta - Y\sin\theta)^2 + B(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta) \\ &+ C(X\sin\theta + Y\cos\theta)^2 + D(X\cos\theta - Y\sin\theta) + E(X\sin\theta + Y\cos\theta) + F = 0\,. \end{aligned}$$

We want to choose θ so that the coefficient of XY will vanish. If we denote this coefficient by B', then we have

(6.41) $$\begin{aligned} B' &= -2A\sin\theta\cos\theta + B(\cos^2\theta - \sin^2\theta) + 2C\cos\theta\sin\theta \\ &= B\cos 2\theta + (C - A)\sin 2\theta. \end{aligned}$$

† Or, we may note that the xy-system is obtained from the XY-system by a rotation through the angle $2\pi - \theta$. If we replace x by X, y by Y, and θ by $2\pi - \theta$ in (6.37), we obtain (6.38).

Therefore $B' = 0$ if θ satisfies the equation

$$B \cos 2\theta = (A - C) \sin 2\theta . \tag{6.42}$$

If $B = 0$, there is no xy term in (6.39) and no rotation is necessary. If $B \neq 0$, we may write (6.42) as follows:

$$\cot 2\theta = \frac{A - C}{B} .$$

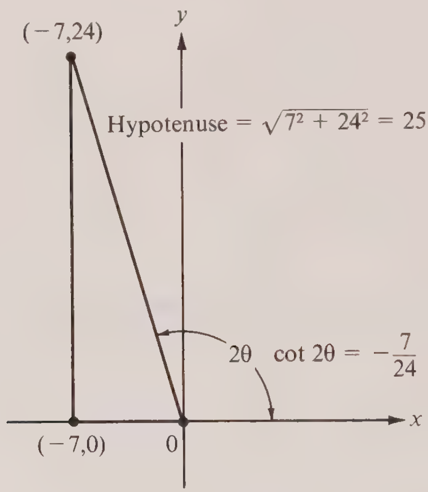


FIGURE 6.27

Since the range of the cotangent function is the set of all real numbers, this equation can always be solved for 2θ and hence for θ.

This discussion shows that every second-degree equation of the form (6.39) has for its graph one of the conic sections, or else its graph consists of one of the "degenerate" cases, that is, no points at all, a single point, or one or two straight lines. Although some of these degenerate cases may be obtained as sections of a cone by passing the cutting plane through the vertex, we shall use the word "conic" here to mean a nondegenerate conic, that is, an ellipse, hyperbola, or parabola.

Example. Consider the equation

$$9x^2 + 24xy + 16y^2 - 20x + 15y = 0 . \tag{6.43}$$

Here $A = 9$, $B = 24$, $C = 16$, and $(A - C)/B = -7/24$. This means we must choose θ so that $\cot 2\theta = -7/24$. With the help of the right triangle in Figure 6.27 we find

$$\cos 2\theta = -\frac{7}{25}, \qquad \sin \theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{4}{5}, \qquad \cos \theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{3}{5}.$$

Therefore the transformation equations in (6.37) become

$$x = \tfrac{3}{5}X - \tfrac{4}{5}Y\,, \qquad y = \tfrac{4}{5}X + \tfrac{3}{5}Y\,.$$

Substituting in (6.43) and simplifying, we obtain $Y = -X^2$. This is one of the standard equations for a parabola. Its axis is along the Y-axis and it opens downward relative to the XY-system.

Suppose we write Equation (6.40) in the form

$$A'X^2 + B'XY + C'Y^2 + D'X + E'Y + F' = 0\,. \tag{6.44}$$

We have expressed B' in terms of A, B, C in Equation (6.41). We may similarly express A' and C' and we find

$$\begin{aligned} A' &= A\cos^2\theta + \tfrac{1}{2}B\sin 2\theta + C\sin^2\theta\,,\\ C' &= A\sin^2\theta - \tfrac{1}{2}B\sin 2\theta + C\cos^2\theta\,. \end{aligned}$$

If we compute the new discriminant we find the remarkable relation

$$4A'C' - B'^2 = 4AC - B^2\,.$$

In other words, the discriminant is *invariant* under a rotation of axes. If the new equation (6.44) represents a conic and if $B' = 0$, we know that the conic is a *parabola*, *ellipse*, or *hyperbola*, according as $A'C'$ is *zero*, *positive*, or *negative*. Therefore the original equation (6.39) represents a parabola, ellipse, or hyperbola, according as $4AC - B^2$ is zero, positive, or negative. For example, the discriminant of (6.43) is $4 \cdot 9 \cdot 16 - 24^2 = 0$, so the conic is a parabola.

6.19 Exercises

In Exercises 1 through 15, identify the locus represented by the given equation. When the locus is a conic, sketch the curve and indicate the locations of important features (such as foci, vertices, center, asymptotes, etc.).

1. $3x^2 + 2y^2 - 12x + 8y + 19 = 0.$
2. $x^2 + 2x + 8y - 15 = 0.$
3. $9x^2 - 4y^2 - 54x + 45 = 0.$
4. $9y^2 - 25x^2 - 90y - 50x - 25 = 0.$
5. $x^2 - y^2 - 2x + 2y = 0.$
6. $xy + y - 2x - 2 = 0.$
7. $x^2 + y^2 - 2x - 2y + 4 = 0.$
8. $3x^2 + 2xy - y^2 + 14x + 2y + 15 = 0.$
9. $y^2 - 2xy + 2x^2 - 5 = 0.$
10. $y^2 - 2xy + 5x = 0.$
11. $y^2 - 2xy + x^2 - 5x = 0.$
12. $5x^2 - 4xy + 2y^2 - 6 = 0.$
13. $19x^2 + 4xy + 16y^2 - 212x + 104y = 356.$
14. $9x^2 + 24xy + 16y^2 - 52x + 14y = 6.$
15. $x^2 + 4xy - 2y^2 - 12 = 0.$

16. Consider the equation $2xy - 4x + 7y + c = 0$. For what value (or values) of c will its graph be a pair of lines?

Exercises 17 through 20 refer to the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0\,, \tag{6.45}$$

which is assumed to represent a (nondegenerate) conic.

17. (a) Prove that the vector $\vec{N} = (2Ax + By + D)\vec{i} + (2Cy + Bx + E)\vec{j}$ is normal to the curve (perpendicular to the tangent vector) when placed at the point (x, y) on the curve. [*Hint*. Show that for any vector representation, say $\vec{r}(t) = X(t)\vec{i} + Y(t)\vec{j}$, the vector $\vec{N}$ is perpendicular to $\vec{r}'(t)$.]

(b) Prove that the tangent line at the point (x_0, y_0) has the equation

$$Ax_0x + \tfrac{1}{2}B(x_0y + y_0x) + Cy_0y + \tfrac{1}{2}D(x + x_0) + \tfrac{1}{2}E(y + y_0) + F = 0\,.$$

18. (a) Show that the discriminant $4AC - B^2$ is invariant under translation of axes.

(b) When the conic is an ellipse with equation $Ax^2 + Bxy + Cy^2 = 1$, show that the area of the region it bounds is $2\pi/\sqrt{4AC - B^2}$.

19. (a) Show that $A + C$ is invariant under both rotation and translation of axes.

(b) If the first-degree terms are absent in Equation (6.45), show that they will also be absent in the new equation obtained by rotating the axes through any angle.

(c) Use the information in (a) and (b) to eliminate the xy term in the equation $98x^2 + 72xy + 77y^2 = 1$, without performing any calculations relating to the angle of rotation.

20. (a) Show that $D^2 + E^2 - 4F$ is invariant under a rotation of axes through any angle.

(b) When the conic is a circle, show that this invariant bears a simple relation to the radius of the circle.

6.20 Miscellaneous exercises on conic sections

1. Find an equation for the parabola whose focus is at the origin and whose directrix is the straight line $x + y + 1 = 0$.

2. Find an equation for a hyperbola which passes through the origin, given that its asymptotes are the lines $y = 2x + 1$ and $y = -2x + 3$.

3. A particle moves along the ellipse $3x^2 + y^2 = 1$ according to a vector equation of the form $\vec{r} = X(t)\vec{i} + Y(t)\vec{j}$. The motion is such that the horizontal component of the velocity vector at time t is $-Y(t)$.

(a) Does the particle move around the ellipse in a clockwise or counterclockwise direction?

(b) Show that the vertical component of the velocity vector at time t is proportional to $X(t)$ and find the factor of proportionality.

(c) How much time is required for the particle to go once around the ellipse?

4. A particle moves along the parabola $x^2 + c(y - x) = 0$ in such a way that the horizontal and vertical components of the acceleration vector are equal. If it takes T units of time to go from the point $(c, 0)$ to the point $(0, 0)$, how much time will it require to go from $(c, 0)$ to the halfway point $(c/2, c/4)$?

5. Two points P and Q are said to be symmetric with respect to a circle if P and Q are collinear with the center, if the center is not between them, and if the product of their distances from the center is equal to the square of the radius. Given that Q describes the straight line $x + 2y - 5 = 0$, find the locus of the point P symmetric to Q with respect to the circle $x^2 + y^2 = 4$.

6. (a) Given $a \neq 0$. If the two parabolas $y^2 = 4p(x - a)$ and $x^2 = 4qy$ are tangent to each other, show that the x-coordinate of the point of contact is determined by a alone.

(b) Find a condition on a, p, and q which expresses the fact that the two parabolas are tangent to each other.

7. For an arbitrary point P of the parabola $y^2 = x$ (except the vertex) let α be the angle between the tangent to the parabola and the vector $\overrightarrow{OP}$, and let θ be the angle that $\overrightarrow{OP}$ makes with the x-axis. Express α in terms of θ.

8. Two parabolas have the same point as focus and the same line as axis, but their vertices lie on opposite sides of the focus. Prove that the parabolas intersect orthogonally (i.e., their tangent lines are perpendicular at the points of intersection).

9. Consider the locus of the points P in the plane for which the distance of P from the point $(2, 3)$ is equal to the sum of the distances of P from the two coordinate axes.

(a) Show that the part of this locus which lies in the first quadrant is part of a hyperbola. Locate the asymptotes and make a sketch.

(b) Sketch the graph of the locus in the other quadrants.

10. If the equation $Ax^2 + Bxy + Cy^2 = 0$ represents two lines given by equations of the form $y = m_1x$ and $y = m_2x$, show that (a) $C \neq 0$; (b) $m_1 + m_2 = -B/C$ and $m_1m_2 = A/C$; (c) the lines are coincident if $B^2 = 4AC$; (d) the lines are perpendicular if $A + C = 0$.

Orthogonal trajectories. Two curves are said to intersect *orthogonally* at a point if their tangent lines are perpendicular at that point. A curve which intersects every member of a family of curves orthogonally is called an *orthogonal trajectory* for the family. Problems involving orthogonal trajectories are of importance in both pure and applied mathematics. For example, in the theory of fluid flow, two orthogonal systems of curves are called the *equipotential lines* and the *stream lines*, respectively. In the theory of heat they are known as *isothermal lines* and *lines of flow*.

11. Suppose a given family of curves satisfies a first-order differential equation of the form $f(x, y, y') = 0$. (a) Show that each orthogonal trajectory of this family must satisfy the differential equation $f(x, y, -1/y') = 0$. In other words, a differential equation for the orthogonal trajectories may be obtained from that of the given family by replacing y' by $-1/y'$.

(b) Show that every curve that cuts all members of the family at a constant angle θ, where $0 \leq \theta < \frac{1}{2}\pi$, must satisfy the differential equation

$$f\left(x, y, \frac{y' - \tan\theta}{1 + y' \tan\theta}\right) = 0\,.$$

12. The orthogonal trajectories of the family of all straight lines through the origin form a family of concentric circles. Prove this statement by the method suggested in Exercise 11.

In Exercises 13 through 20, use the method suggested in Exercise 11 to find the orthogonal trajectories of each of the following families. Sketch a few curves in each case.

13. $2x + 3y = C$.
14. $(x - C)^2 + y^2 = C^2$.
15. $x^2 + y^2 + 2Cy = 1$.
16. $y^2 = 4Cx$.
17. $x^2y = C$.
18. $y = Ce^{-2x}$.
19. $x^2 - y^2 = C^2$.
20. All circles through $(1, 1)$ and $(-1, -1)$.

6.21 The quadric surfaces

The three-dimensional figures that correspond to the conics in the plane are the so-called *quadric surfaces*. These surfaces are the graphs of equations of the second degree in x, y, and z.

As in the case of conics, there are certain "degeneracies" that may occur. These are essentially of four types:

(a) *no points at all;* the equation $x^2 + y^2 + z^2 + 1 = 0$ is an example.

(b) *just one point;* the equation $x^2 + 2y^2 + 3z^2 = 0$ is satisfied only by the point $(x, y, z) = (0, 0, 0)$.

(c) *a line;* the equation $x^2 + y^2 = 0$ is satisfied by those and only those points on the z-axis.

(d) *one or two planes;* the equation $z^2 = 0$ is satisfied by those and only those points that lie in the xy-plane, whereas the graph of $x^2 - y^2 = 0$ consists of the points on the two planes $x - y = 0$ and $x + y = 0$.

The simplest nondegenerate quadrics are those in which one of the coordinates x, y, or z is missing from the equation. These are *cylinders* whose generators are parallel to the axis of the missing coordinate. For example, in 3-space the equations

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\,, \qquad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\,, \qquad y^2 = 4cx$$

represent cylinders with generators parallel to the z-axis. They are called *elliptic*, *hyperbolic*, and *parabolic* cylinders, respectively, and they are illustrated in Figure 6.28.

In addition to the above, there are essentially six more types of quadric surfaces and they may be studied by examining their cross sections cut by suitably chosen planes. We list these six types below with some of the standard forms of their equations. (In the following equations, a, b, and c represent nonzero numbers.)

1. *The ellipsoid:*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \qquad (a, b, c \text{ positive})\,. \tag{6.46}$$

Putting $z = 0$, we find that this quadric intersects the xy-plane along the ellipse $x^2/a^2 + y^2/b^2 = 1$. More generally, when we put $z = k$ in (6.46) we obtain the equation

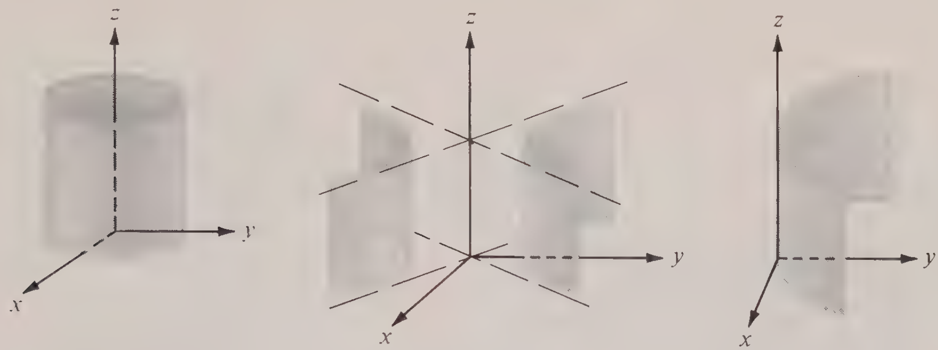
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 - \frac{k^2}{c^2}\,.$$

If $k^2 < c^2$ the plane $z = k$ cuts the quadric in an ellipse whose semiaxes have lengths $a\sqrt{1 - k^2/c^2}$ and $b\sqrt{1 - k^2/c^2}$. As k^2 increases to the value c^2 these lengths decrease to zero. When $k^2 = c^2$ the intersection is just one point, $(0, 0, k)$, and when $k^2 > c^2$ there is no intersection. A similar analysis can be made for cross sections parallel to the other coordinate planes and we conclude that an ellipsoid has the appearance of the graph shown in Figure 6.29. The numbers a, b, c (the intercepts on the coordinate axes) are the lengths of the *semiaxes* of the ellipsoid. If two of the semiaxes have equal lengths, the ellipse is called a *spheroid*. In this case it is a surface obtained by revolving an ellipse about one of its axes. A sphere is the special case $a = b = c$.

2. *The elliptic paraboloid:*

$$z = ax^2 + by^2 \qquad (a \text{ and } b \text{ both positive or both negative})\,.$$

When $z = k$ the equation becomes $ax^2 + by^2 = k$. If $k = 0$ the locus is a single point, the origin. If $k \neq 0$ this equation represents an ellipse if k has the same sign as a and b; otherwise there is no intersection. Planes parallel to the other coordinate planes intersect the surface along parabolas. When a and b are positive the parabolas open upward along the positive z-axis and the surface is like that shown in Figure 6.30. When a and b are negative the surface opens downward. When $a = b$ the horizontal cross-sections are circles and the surface is called a *paraboloid of revolution*.



Elliptic cylinder Hyperbolic cylinder Parabolic cylinder

FIGURE 6.28 *Quadric cylinders.*

3. *The hyperbolic paraboloid:*

$$z = ax^2 - by^2 \qquad (a \text{ and } b \text{ both positive or both negative}).$$

This is sometimes called a "saddle-shaped" surface. Figure 6.31 shows an example in which a and b are both positive. Cross sections cut by planes $z = k$ are hyperbolas for all values of $k \neq 0$. When $k = 0$ the intersection consists of two lines through the origin. Sections cut by planes parallel to the other coordinate planes are parabolas. When $a = b$ the x- and y-axes may be rotated through an angle $\frac{1}{4}\pi$ into new X- and Y-axes and the equation simplifies to the form $z = cXY$ for some c.

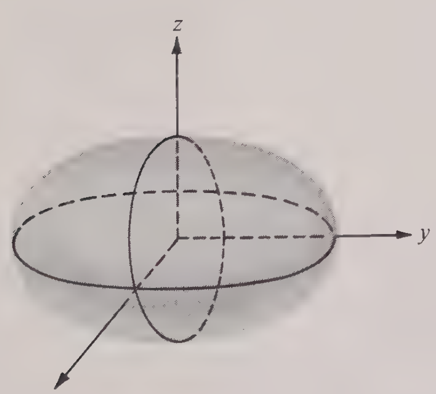


FIGURE 6.29 *Ellipsoid.*

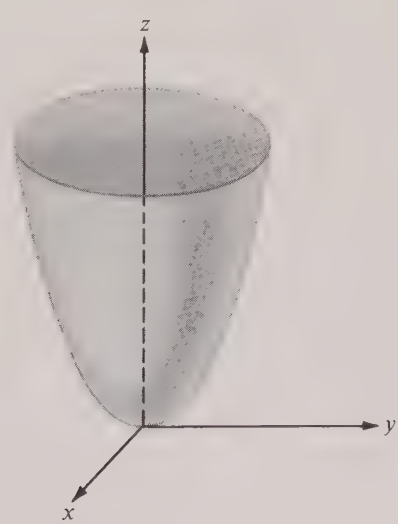


FIGURE 6.30 *Elliptic paraboloid.*

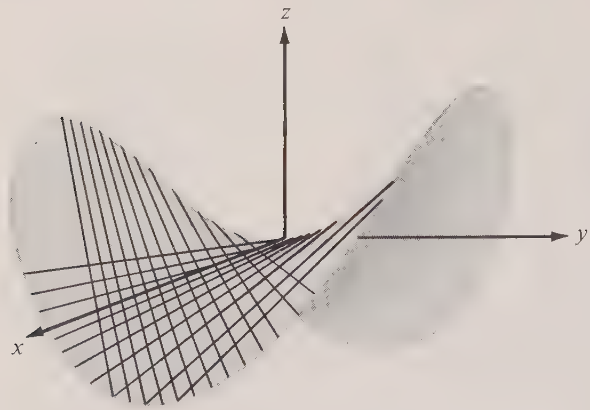


FIGURE 6.31 *Hyperbolic paraboloid.*

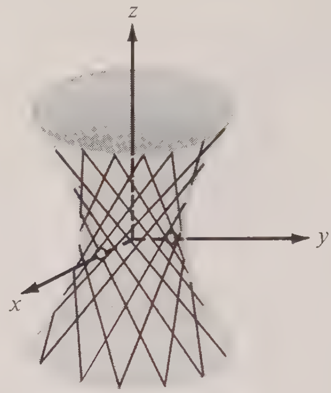


FIGURE 6.32 *Hyperboloid of one sheet.*

The hyperbolic paraboloid is a ruled surface. If a and b are both positive, planes parallel to $\sqrt{a}x = \pm\sqrt{b}y$ cut the surface along straight lines called *rulings.* Each point on the surface has a ruling passing through it. Some of them are shown in Figure 6.31.

4. *Hyperboloid of one sheet:*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \qquad (a, b, c \text{ positive})\,. \tag{6.47}$$

Cross sections by all planes $z = k$ are ellipses, and cross sections by planes $x = k$ or $y = k$ are hyperbolas. The z-axis is said to be the *axis* of this hyperboloid. An example is shown in Figure 6.32. When $a = b$ the surface is a hyperboloid of revolution.

In Exercise 18 of Section 6.22 the reader is asked to show that this, too, is a ruled surface. Some of the rulings are shown in Figure 6.32. Toy manufacturers sell kits for making interesting string models of this surface.

5. *Hyperboloid of two sheets:*

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \qquad (a, b, c \text{ positive})\,. \tag{6.48}$$

If $z = k$, where $k^2 < c^2$, there is no intersection with the surface. When $k^2 > c^2$ the plane $z = k$ cuts the surface along an ellipse and when $k^2 = c^2$ the intersection is a single point. The surface consists of two parts (two "sheets"),† one corresponding to values of $z > c$, the other to $z < -c$. Cross sections by planes $x = k$ or $y = k$ are hyperbolas. An example is shown in Figure 6.33. When $a = b$ the quadric is a surface of revolution obtained by revolving a hyperbola about an axis through its foci.

† Note that the number of "sheets" is the same as the number of minus signs appearing in each of Equations (6.47) and (6.48).

6. *The elliptic cone:*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \qquad (a, b, c \text{ positive}). \tag{6.49}$$

This is sometimes called the *asymptotic* cone of each of the hyperboloids in (6.47) and (6.48). It is related to these hyperboloids in much the same way as asymptotes are related to a hyperbola. An example is shown in Figure 6.34. Every horizontal plane $z = k$, $k \neq 0$, cuts the cone along an ellipse. Cross sections by planes $x = k$ and $y = k$ are hyperbolas. Degeneracies occur when the cutting plane passes through the vertex. That the figure is really a *cone* may be seen as follows: If a point (x_0, y_0, z_0) satisfies (6.49), so does (tx_0, ty_0, tz_0) for every real t. As t takes on all real values, the point (tx_0, ty_0, tz_0) traces out the line joining the origin to (x_0, y_0, z_0) and hence every point on this line lies on the surface. Therefore the surface may be generated by lines passing through the origin; consequently it must be a cone.

There are other equations of the second degree in x, y, and z which are not included among the forms just discussed. For example, the equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

does not appear in the above list. Nevertheless, this equation is essentially the same kind as that discussed in (6.47), the only difference being that the roles of y and z are interchanged. The corresponding surface is again a hyperboloid of one sheet but its axis is located along the y-axis, as shown in Figure 6.35.

In addition to those equations that may be reduced to one of the above types by interchanging two of the coordinates, there are other second-degree equations involving first

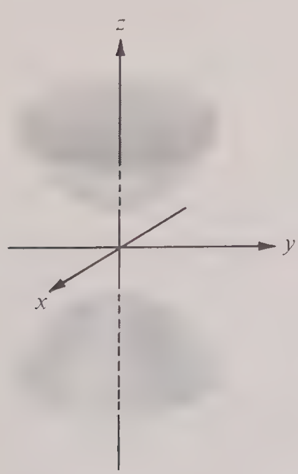


FIGURE 6.33 *Hyperboloid of two sheets.*

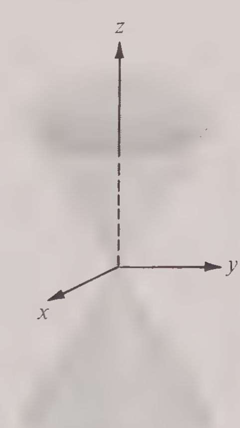


FIGURE 6.34 *An example of an elliptic cone.*

FIGURE 6.35 *Hyperboloid of one sheet.*

powers of the coordinates (but no product terms xy, yz, or xz) that can be reduced to one of the above forms by completing the squares. More general second-degree equations involving the product terms xy, yz, or xz can also be reduced to one of the forms discussed above by a suitable change of coordinate axes, but we shall not treat such examples in detail here.

6.22 Exercises

1. (a) Prove that every sphere has a Cartesian equation which may be written in the form $x^2 + y^2 + z^2 + Ax + By + Cz + D = 0$.

(b) Use (a) to show that, in general, four points in space determine a sphere.

2. Let $\vec{P}$, $\vec{P}_0$, $\vec{P}_1$ represent vectors from the origin to points P, P_0, P_1 in space.

(a) If $r > 0$, show that the equation $|\vec{P} - \vec{P}_0| = r$ represents a sphere with center at P_0 and radius r.

(b) Show that the equation $(\vec{P} - \vec{P}_0) \cdot (\vec{P} - \vec{P}_1) = 0$ represents a sphere having as a diameter the line segment joining P_0 and P_1.

3. Find equations for the two spheres of radius 5 that pass through the three points $(-2, -2, 1)$, $(3, -2, 6)$, and $(3, 3, 1)$.

4. Find an equation for the sphere which has the two planes

$$x + y + z - 3 = 0 \qquad \text{and} \qquad x + y + z - 9 = 0$$

as tangent planes, if the two planes $2x = y$ and $3x = z$ pass through the center of the sphere.

5. Given $a > b > c > 0$. Prove that the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ is cut by the sphere $x^2 + y^2 + z^2 = b^2$ along two plane curves. Find equations for the planes in which these curves lie when $a = 4$, $b = 3$, $c = 2$.

6. Given that the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ is intersected by the plane $Ax + By + Cz + D = 0$ along a curve Γ. Find an equation for the cone whose vertex is at the center of the ellipsoid and which passes through the curve Γ, if the cutting plane does not pass through the origin.

7. The locus of those points in space whose distance from the point $(2, -1, 3)$ is twice their distance from the xy-plane is a certain quadric surface. Find an equation for this surface, identify it by name, and locate its center of symmetry.

8. The curve whose vector equation is

$$\vec{r}(t) = 2\sqrt{t}\cos t\,\vec{i} + 3\sqrt{t}\sin t\,\vec{j} + \sqrt{1 - t}\,\vec{k} \qquad (0 \le t \le 1)$$

lies on a quadric surface. Find an equation for this surface and identify it.

9. Identify the quadric surface $36x^2 + 4y^2 = 9z^2 + 36$. Describe the curve of intersection of this surface with each of the following planes: $x = 1$, $y = 3$, $z = 2$.

10. Identify the quadric surface $9y^2 = 4x^2 + 144z$. Describe the curve of intersection with each of the following planes: $z = 0$, $z = -1$, $x = 3$.

Each of the equations in Exercises 11 through 16 represents a quadric surface. Identify it and locate its center of symmetry, if any.

11. $x^2 - 4y^2 - 2z^2 - 2x + 16y - 4z - 21 = 0$.
12. $x^2 + 2y^2 + 3z^2 - 2x + 4y - 12z + 9 = 0$.
13. $x^2 - y^2 - z^2 = 0$.
14. $9x^2 - 4y^2 - 36z^2 = 36$.
15. $x^2 - y^2 + z^2 - 2x + 4y = 4$.
16. $z = 2xy$.

17. Let P_0 be a fixed point in space with position vector $\vec{P}_0$ and let P be an arbitrary point in space with position vector $\vec{P}$. Describe in words and make a sketch of the locus of all points P satisfying:

(a) $\vec{P}\cdot\vec{P}_0 = 0$;
(b) $\vec{P}\cdot\vec{P}_0 = |\vec{P}_0|$;
(c) $|\vec{P}\times\vec{P}_0| = |\vec{P}_0|$;
(d) $\vec{P}\cdot\vec{P}_0 = \vec{P}\cdot\vec{P}$;
(e) $\vec{P}\cdot\vec{P}_0 = \frac{1}{2}|\vec{P}|\,|\vec{P}_0|$.

18. Let $(x_0, y_0, 0)$ be a point on the intersection of the plane $z = 0$ and the hyperboloid of one sheet given by Equation (6.47).

(a) Show that the vector equation

$$\vec{r}(t) = (x_0 + a^2y_0t)\,\vec{i} + (y_0 - b^2x_0t)\,\vec{j} + abct\,\vec{k}$$

describes a straight line which lies on the surface and passes through $(x_0, y_0, 0)$.

(b) Let (x, y, z) be an arbitrary point on the surface. Show that x_0, y_0, and t can always be found so that the line described by the vector equation in (a) passes through (x, y, z). This proves that the surface is a ruled surface.

19. (a) If t is a nonzero constant, show that any point which satisfies both the equations $x/a + z/c = t(1 + y/b)$ and $x/a - z/c = t^{-1}(1 - y/b)$ lies on a hyperboloid of one sheet. These are the equations of two planes whose intersection is a ruling of the surface.

(b) As t varies, the equations in (a) generate a family of rulings. Show that no two rulings of this family intersect.

(c) Show that another family of rulings is given by the pair of equations $x/a - z/c = s(1 + y/b)$ and $x/a + z/c = s^{-1}(1 - y/b)$, and that every member of this family intersects each member of the family in part (a).

20. (a) The plane $z = t$ cuts the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ along an ellipse. Find, in terms of t, the area bounded by this ellipse.

(b) By integrating the cross-sectional area found in (a), compute the volume bounded by the ellipsoid.

21. When a liquid is rotated about a vertical axis, show that its surface assumes the form of a paraboloid of revolution. Assume constant angular speed. [*Hint.* The resultant of the gravitational and centrifugal forces acting on a particle on the surface is normal to the surface.]

6.23 The unit tangent, the principal normal, and the osculating plane of a curve

Having discussed some of the more common examples of curves and surfaces, we turn once more to the general theory of space curves.

We recall that a curve may be described by a vector equation of the form

$$\vec{r}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j} + Z(t)\,\vec{k}\,,$$

where t is a real parameter that is allowed to range over some interval, say $[a, b]$, and X, Y, Z are real-valued functions, defined and continuous on $[a, b]$. We assume further that the functions X, Y, and Z may be differentiated as often as may be necessary at each point of $[a, b]$.

As mentioned earlier, sometimes it is convenient to think of a curve as being traced out by a moving particle which at "time" t occupies the position indicated by the terminal point of the vector $\vec{r}(t)$. With this interpretation, the derivative $\vec{r}\,'(t)$ of the position vector is called the *velocity* $\vec{v}(t)$ of the particle and it is given by the formula

$$\vec{v}(t) = X'(t)\,\vec{i} + Y'(t)\,\vec{j} + Z'(t)\,\vec{k}\,.$$

The velocity vector points in the direction of motion. (This is the direction of the *tangent* *line.*) The length of the velocity vector, $|\vec{v}(t)|$, is a scalar quantity which we denote by $v(t)$ and which we refer to as the *speed* of the particle at time t. Thus we have

$$v(t) = |\vec{v}(t)| = \sqrt{[X'(t)]^2 + [Y'(t)]^2 + [Z'(t)]^2}\,. \tag{6.50}$$

In Section 6.31 we shall formulate a definition of *arc length* which assigns, for each t, a nonnegative number $s(t)$ representing the distance the particle has traveled along the curve at time t. The distance function s which measures arc length has the property that its derivative is equal to the speed. That is, we have

$$s'(t) = v(t)\,. \tag{6.51}$$

This formula, to be proved in Section 6.33, conforms with our intuitive notion of speed as the distance per unit time being covered during the motion.

Because of (6.51), arc length may be computed by integrating the speed. Thus the distance traveled by the particle during a time interval $[t_0, t_1]$ is

$$s(t_1) - s(t_0) = \int_{t_0}^{t_1} v(t)\,dt\,. \tag{6.52}$$

If the particle begins its motion at time $t = a$, then $s(a) = 0$ and the distance covered at time t_1 is $s(t_1) = \int_a^{t_1} v(t)\,dt$.

Example. To find the length of an arc of a circle of radius R we may imagine a particle moving along the circle according to the equation $\vec{r}(t) = R\cos t\,\vec{i} + R\sin t\,\vec{j}$. The velocity vector is $\vec{v}(t) = -R\sin t\,\vec{i} + R\cos t\,j$ and the speed is $v(t) = R$. Integrating the speed over an interval of length θ, we find that the length of arc traced out is $R\theta$. In other words, the length of a circular arc is proportional to the angle it subtends; the constant of proportionality is the radius of the circle. For a unit circle, the arc length is equal to the angular measure.

It is convenient at this point to introduce the so-called *unit tangent vector* $\vec{T}$. This is another vector-valued function associated with the curve, and it is defined by the equation

$$\vec{T}(t) = \frac{1}{v(t)}\,\vec{v}(t)$$

whenever $v(t) \neq 0$. Thus, $|\vec{T}(t)| = 1$ for each t and we may write

$$\vec{v}(t) = v(t)\vec{T}(t)\,. \tag{6.53}$$

Figure 6.36 shows the position of the unit tangent vector $\vec{T}(t)$ for various values of t when it is attached to the curve. As the particle moves along the curve the corresponding vector $\vec{T}$, being of constant length, can change only in its direction. The tendency of $\vec{T}$ to change its direction is measured by its derivative, T'. Theorem 6–1 (proved at the end of Section 6.2) tells us that the derivative of a vector function of constant length is a new vector function perpendicular to the given vector function. Therefore $\vec{T}'(t)$ is perpendicular to $T(t)$ for each t.

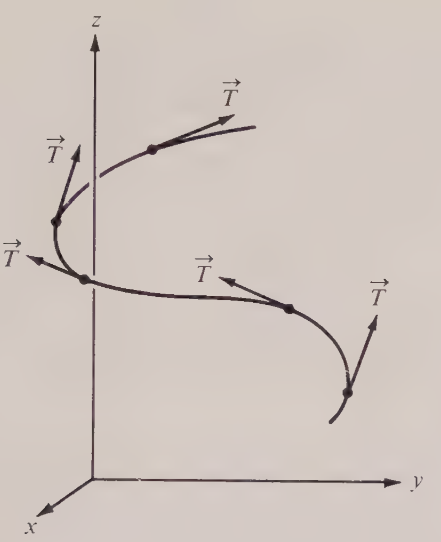


FIGURE 6.36 *The unit tangent vector $\vec{T}$.*

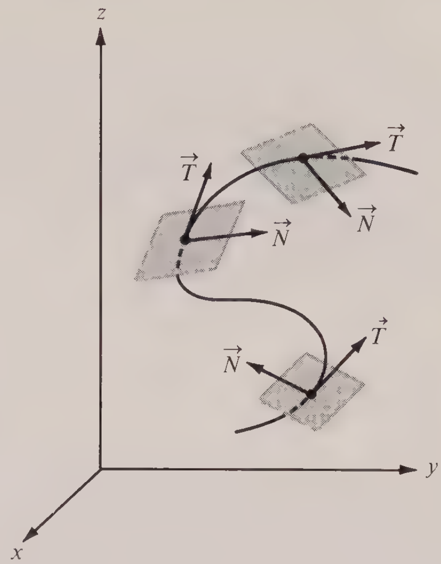


FIGURE 6.37 *The osculating plane.*

The *unit* vector having the same direction as $\vec{T}'$ is called the *principal normal* to the curve and it is denoted by $\vec{N}$. Thus $\vec{N}$ is a new vector function defined by the equation

$$\vec{N}(t) = \frac{1}{|\vec{T}'(t)|}\vec{T}'(t) \qquad \text{if} \quad \vec{T}'(t) \neq \vec{0}\,.$$

Of course, this may also be written as follows:

(6.54) $$\vec{T}'(t) = |\vec{T}'(t)|\vec{N}(t)\,.$$

When the two unit vectors $\vec{T}(t)$ and $\vec{N}(t)$ are attached to the curve they determine a plane known as the *osculating plane* of the curve. If we choose three values of t, say t_1, t_2, and t_3, and consider the plane determined by the three corresponding points on the curve, it can be shown that the position of the plane approaches the position of the osculating plane at t_1, as t_2 and t_3 approach t_1. Because of this, the osculating plane is often called the plane that "best fits the curve" at each of its points. If the curve itself is a plane curve (not a straight line), the osculating plane coincides with the plane of the curve. In general, however, the osculating plane changes with t. Examples are illustrated in Figure 6.37.

It is easy to prove that the acceleration vector $\vec{a}(t)$ (the derivative of the velocity vector) lies in the osculating plane when it is attached to the curve. For this purpose we use the expression for $\vec{v}$ given in (6.53). Differentiation gives us

$$\vec{a}(t) = \vec{v}'(t) = D[v(t)\vec{T}(t)] = v'(t)\vec{T}(t) + v(t)\vec{T}'(t)\,.$$

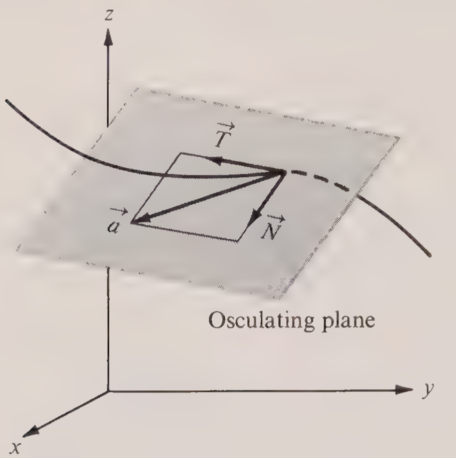


FIGURE 6.38 *The acceleration vector lies in the osculating plane.*

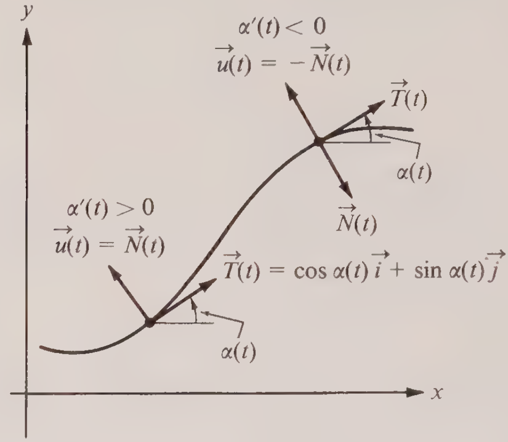


FIGURE 6.39 *The angle of inclination of the tangent vector of a plane curve.*

If we use (6.54) in the second term on the right, we obtain

$$\vec{a}(t) = v'(t)\vec{T}(t) + v(t)|\vec{T}'(t)|\vec{N}(t)\,. \tag{6.55}$$

This proves that $\vec{a}(t)$ is a linear combination of the two vectors $\vec{T}(t)$ and $\vec{N}(t)$ and hence the acceleration vector lies in the osculating plane, as asserted. An example is shown in Figure 6.38. The coefficients of $\vec{T}(t)$ and $\vec{N}(t)$ in (6.55) are called, respectively, the *tangential* and *normal components* of the acceleration. A change in speed contributes to the tangential component, whereas a change in direction contributes to the normal component.

For a plane curve, the length of $\vec{T}'(t)$ has an interesting geometric interpretation. Since $\vec{T}$ is a unit vector we may write

$$\vec{T}(t) = \cos \alpha(t)\,\vec{i} + \sin \alpha(t)\,\vec{j}\,,$$

where $\alpha(t)$ denotes the angle between the tangent vector and the positive x-axis, as shown in Figure 6.39. Differentiating, we find

$$\vec{T}'(t) = -\sin \alpha(t)\,\alpha'(t)\,\vec{i} + \cos \alpha(t)\,\alpha'(t)\,\vec{j} = \alpha'(t)\vec{u}(t)\,,$$

where $\vec{u}(t)$ is a unit vector. Therefore $|\vec{T}'(t)| = |\alpha'(t)|$ and this shows that $|\vec{T}'(t)|$ is a measure of the rate of change of the angle of inclination of the tangent vector. When the angle is increasing we have $\alpha'(t) > 0$ and hence $\vec{u}(t) = \vec{N}(t)$. When the angle is decreasing we have $\alpha'(t) < 0$ and in this case $\vec{u}(t) = -\vec{N}(t)$. The two cases are illustrated in Figure 6.39. Note that the angle of inclination of $\vec{u}(t)$ is $\alpha(t) + \frac{1}{2}\pi$ since we have

$$\vec{u}(t) = -\sin \alpha(t)\,\vec{i} + \cos \alpha(t)\,\vec{j} = \cos\left(\alpha(t) + \frac{\pi}{2}\right)\vec{i} + \sin\left(\alpha(t) + \frac{\pi}{2}\right)\vec{j}\,.$$

6.24 Curvature of a curve

For a straight line the unit tangent vector $\vec{T}$ does not change its direction and hence $\vec{T}' = \vec{0}$. If the curve is not a straight line, the derivative $\vec{T}'$ measures the tendency of the tangent to change its direction. The rate of change of the unit tangent *with respect to arc length* is called the *curvature vector* of the curve. We denote this by $d\vec{T}/ds$, where s represents arc length. The chain rule, used in conjunction with (6.51), tells us that the curvature vector $d\vec{T}/ds$ is related to the "time" derivative $\vec{T}'$ by the equation

$$\frac{d\vec{T}}{ds} = \frac{dt}{ds}\frac{d\vec{T}}{dt} = \frac{1}{s'(t)}\vec{T}'(t) = \frac{1}{v(t)}\vec{T}'(t)\,.$$

If we use (6.54) to express $\vec{T}'$ in terms of $\vec{N}$, we obtain

$$\frac{d\vec{T}}{ds} = \frac{|\vec{T}'(t)|}{v(t)}\vec{N}(t)\,, \tag{6.56}$$

which shows that the curvature vector has the same direction as the principal normal $\vec{N}(t)$. The scalar factor which multiplies $\vec{N}(t)$ in (6.56) is a nonnegative number called the *curvature* of the curve at t and it is denoted by $\kappa(t)$ (κ is the Greek letter kappa). Thus the curvature $\kappa(t)$, defined to be the *length of the curvature vector,* is given by the following formula:

$$\kappa(t) = \frac{|\vec{T}'(t)|}{v(t)}\,. \tag{6.57}$$

Equations (6.54) and (6.55) may now be written in terms of $\kappa(t)$ and they assume the following forms:

$$\vec{T}'(t) = \kappa(t)v(t)\vec{N}(t)\,,$$

$$\vec{a}(t) = v'(t)\vec{T}(t) + \kappa(t)v^2(t)\vec{N}(t)\,. \tag{6.58}$$

Example. For a circle of radius R, given by $\vec{r}(t) = R\cos t\,\vec{i} + R\sin t\,\vec{j}$, we have $\vec{v}(t) = -R\sin t\,\vec{i} + R\cos t\,\vec{j}$. Therefore $v(t) = R$, $\vec{T}(t) = -\sin t\,\vec{i} + \cos t\,\vec{j}$, and $\vec{T}'(t) = -\cos t\,\vec{i} - \sin t\,\vec{j}$. Hence $|\vec{T}'(t)| = 1$ and $\kappa(t) = 1/R$. This shows that a circle has constant curvature. The reciprocal of the curvature is the radius of the circle.

When $\kappa(t) \neq 0$, its reciprocal is called the *radius of curvature* and is denoted by $\rho(t)$ (ρ is the Greek letter rho). That circle in the osculating plane with radius $\rho(t)$ and center at the tip of the curvature vector is called the *osculating circle.* It can be shown that the osculating circle is the limiting position of circles passing through three nearby points on the curve as two of the points approach the third. Because of this property, the osculating circle is often called the circle that "best fits the curve" at each of its points.

Note. For a plane curve we have seen that $|\vec{T}'(t)| = |\alpha'(t)|$, where $\alpha(t)$ is the angle of inclination of the tangent vector, shown in Figure 6.39. Since $\alpha'(t) = d\alpha/dt = (d\alpha/ds)(ds/dt) = v(t)\,d\alpha/ds$, Equation (6.57) implies

$$\kappa(t) = \left|\frac{d\alpha}{ds}\right|. \tag{6.59}$$

In other words, the curvature of a plane curve is the absolute value of the rate of change of α with respect to arc length. It measures the change of direction per unit distance along the curve.

If $d\alpha/ds$ is a nonzero constant, say $d\alpha/ds = a$, then $\alpha = as + b$, where b is a constant, and hence $\vec{T} = \cos(as + b)\,\vec{i} + \sin(as + b)\,\vec{j}$. Integrating, we find $\vec{r} = (1/a)\sin(as + b)\,\vec{i} - (1/a)\cos(as + b)\,\vec{j} + \vec{A}$, where $\vec{A}$ is a constant vector. Therefore $|\vec{r} - \vec{A}| = 1/|a|$, so the curve is a circle (or an arc of a circle) with center at the tip of $\vec{A}$ and radius $1/|a|$. This proves that a curve of constant curvature $\kappa \neq 0$ is a circle (or an arc of a circle) with radius $1/\kappa$.

The definition in Equation (6.57) is not always the most convenient formula for computing the curvature. In practice it is usually easier to apply the formula

$$\kappa(t) = \frac{|\vec{a}(t) \times \vec{v}(t)|}{v^3(t)}\,. \tag{6.60}$$

The proof of (6.60) is quite simple. If we form the cross product $\vec{a} \times \vec{v}$ and use the formulas in (6.58) and (6.53), we find

$$\vec{a} \times \vec{v} = v'v\vec{T} \times \vec{T} + \kappa v^3\vec{N} \times \vec{T} = \kappa v^3\vec{N} \times \vec{T}\,, \tag{6.61}$$

since $\vec{T} \times \vec{T} = 0$. If we take the magnitude of both sides of (6.61) and note that $|\vec{N} \times \vec{T}| = |\vec{N}|\,|\vec{T}| \sin \frac{1}{2}\pi = 1$, we obtain $|\vec{a} \times \vec{v}| = \kappa v^3$, which is the same as (6.60). In practice it is fairly easy to compute the vectors $\vec{v}$ and $\vec{a}$ (by differentiating the position vector $\vec{r}$); hence (6.60) provides a useful method for determining the curvature.

For a straight line we have $\vec{a} \times \vec{v} = \vec{0}$ and hence the curvature is everywhere zero. A curve with a small curvature at a point has a large radius of curvature there and hence does not differ much from a straight line in the immediate vicinity of the point. Thus the curvature is a measure of the tendency of a curve to deviate from a straight line.

6.25 Exercises

In each of Exercises 1 through 5, $\vec{r}(t)$ denotes the position vector at time t for a particle moving on a space curve. For the particular value of t specified in each case, (a) express the vectors $\vec{v}$, $\vec{a}$, $\vec{T}$, $\vec{N}$ in terms of $\vec{i}, \vec{j}, \vec{k}$; (b) compute the curvature κ; (c) express the acceleration $\vec{a}$ in terms of $\vec{T}$ and $\vec{N}$.

1. $\vec{r}(t) = (3t - t^3)\,\vec{i} + 3t^2\vec{j} + (3t + t^3)\,\vec{k} \qquad (t = 2)$.
2. $\vec{r}(t) = \cos t\,\vec{i} + \sin t\,\vec{j} + e^t\,\vec{k} \qquad (t = \pi)$.
3. $\vec{r}(t) = 3t\cos t\,\vec{i} + 3t\sin t\,\vec{j} + 4t\,\vec{k} \qquad (t = 0)$.
4. $\vec{r}(t) = (t - \sin t)\,\vec{i} + (1 - \cos t)\,\vec{j} + 4\sin\frac{t}{2}\,\vec{k} \qquad (t = \pi)$.
5. $\vec{r}(t) = 3t^2\,\vec{i} + 2t^3\,\vec{j} + 3t\,\vec{k} \qquad (t = 1)$.

6. A point moves on a curve (in space) with constant speed. Show that its acceleration vector is perpendicular to the tangent vector at each point.

7. A point moves in space according to the vector equation

$$\vec{r}(t) = 4\cos t\,\vec{i} + 4\sin t\,\vec{j} + 4\cos t\,\vec{k}\,.$$

(a) Show that the path is an ellipse and find an equation for the plane in which this ellipse lies.

(b) Show that the radius of curvature is $\rho(t) = 2\sqrt{2}\,(1 + \sin^2 t)^{3/2}$.

In Exercises 8 through 11, find the length of the path traced out by a particle moving according to the given vector equation during the time interval specified in each case. Use Equation (6.52) which expresses arc length as the integral of the speed.†

8. $\vec{r}(t) = \sin t\,\vec{i} + t\vec{j} + (1 - \cos t)\,\vec{k} \qquad (0 \le t \le 2\pi)$.

9. $\vec{r}(t) = t\,\vec{i} + 3t^2\vec{j} + 6t^3\,\vec{k} \qquad (0 \le t \le 2)$.

10. $\vec{r}(t) = t\,\vec{i} + \log(\sec t)\vec{j} + \log(\sec t + \tan t)\,\vec{k} \qquad (0 \le t \le \frac{1}{4}\pi)$.

11. $\vec{r}(t) = a\cos\omega t\,\vec{i} + a\sin\omega t\,\vec{j} + b\omega t\,\vec{k} \qquad (t_0 \le t \le t_1)$.

12. A particle of unit mass with position vector $\vec{r}(t)$ at time t is moving in space under the actions of certain forces. Prove that if $\vec{r}(t) \times \vec{r}\,'(t)$ is a constant vector, the particle is moving in a plane. Hence, show that if the net force acting on the particle is always directed toward the origin, the particle must move in a plane. Is $\vec{r}(t) \times \vec{r}\,'(t)$ necessarily constant if a particle is moving in a plane?

13. For the curve whose vector equation is $\vec{r}(t) = e^t\,\vec{i} + e^{-t}\vec{j} + \sqrt{2}\,t\,\vec{k}$, show that the curvature is $\kappa(t) = \sqrt{2}/(e^t + e^{-t})^2$.

14. (a) For a plane curve described by the equation $\vec{r}(t) = X(t)\,\vec{i} + Y(t)\vec{j}$, show that the curvature is given by the formula

$$\kappa(t) = \frac{|X'(t)Y''(t) - Y'(t)X''(t)|}{\{[X'(t)]^2 + [Y'(t)]^2\}^{3/2}}\,.$$

(b) If a plane curve has the Cartesian equation $y = f(x)$, show that the curvature at the point $(x, f(x))$ is

$$\frac{|f''(x)|}{\{1 + [f'(x)]^2\}^{3/2}}\,.$$

15. If a point moves so that the velocity and acceleration vectors always have constant lengths, prove that the curvature is constant at all points of the path. Express this constant in terms of $|\vec{a}|$ and $|\vec{v}|$.

16. If two curves with Cartesian equations $y = f(x)$ and $y = g(x)$ are tangent at a point (a, b) and have the same curvature at that point, prove that $|f''(a)| = |g''(a)|$.

17. A particle moves along a plane curve with constant speed 2. The motion starts at the origin when $t = 0$ and the initial velocity $\vec{v}(0)$ is $2\,\vec{i}$. At every instant it is known that the curvature $\kappa(t) = 4t$. Find the velocity when $t = \frac{1}{4}\pi$ if the curve never goes below the x-axis.

18. (a) When a circle rolls (without slipping) along a straight line, a point on the circumference traces out a curve called a *cycloid*. If the fixed line is the x-axis and if the tracing point (x, y) is originally at the origin, show that when the circle rolls through an angle θ we have

$$x = a(\theta - \sin\theta)\,, \qquad y = a(1 - \cos\theta)\,,$$

where a is the radius of the circle. These serve as parametric equations for the cycloid.

(b) Referring to part (a), show that $dy/dx = \cot\frac{1}{2}\theta$ and deduce that the tangent line of the cycloid at (x, y) makes an angle $\frac{1}{2}(\pi - \theta)$ with the x-axis. Make a sketch and show that the tangent line passes through the highest point on the circle.

6.26 Polar coordinates

Let P be a point in the plane with rectangular coordinates $(x, y) \ne (0, 0)$. The radius vector $\vec{r} = x\,\vec{i} + y\,\vec{j}$ may be completely described by specifying its length (which we denote by r) and an angle θ which the vector makes with the positive x-axis. (An example is shown in Figure 6.40.) The two numbers r and θ are related to x and y by the equations

$$x = r\cos\theta\,, \qquad y = r\sin\theta\,. \tag{6.62}$$

† Additional exercises on arc length are given in Section 6.34.

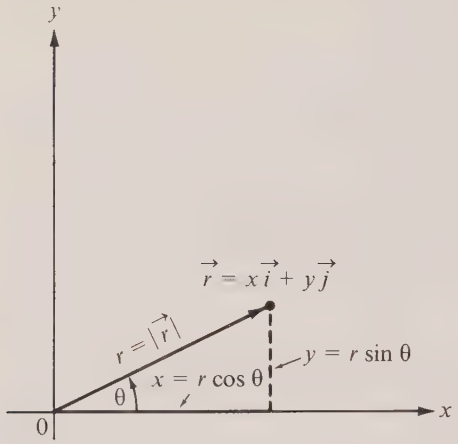


FIGURE 6.40 *Polar coordinates.*

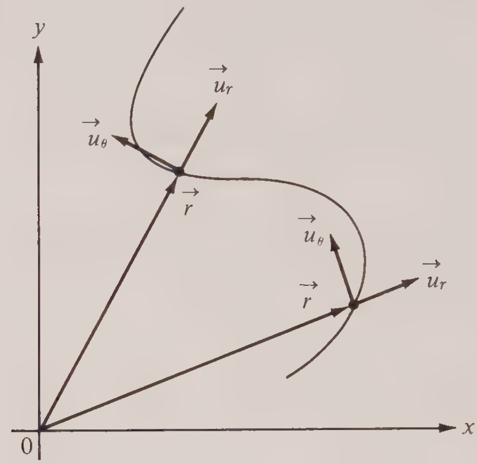


FIGURE 6.41 *The unit vectors $\vec{u}_r$ and $\vec{u}_\theta$.*

All pairs of real numbers r and θ, with $r > 0$, which satisfy these equations are called *polar coordinates* of P. The positive number r is called the *radial distance* of P, and θ is called a *polar angle.* We say *a* polar angle rather than *the* polar angle because if θ satisfies (6.62) so does $\theta + 2n\pi$, for any integer n, and each of these is a polar angle.† There is, however, only one radial distance and it is given by the equation

$$r = \sqrt{x^2 + y^2}\,.$$

When P is the origin, we assign the radial distance $r = 0$ and we agree that *any* real θ may be used as a polar angle.

A curve in the plane may be described by specifying a *polar equation*, that is, an equation of the form $f(r, \theta)$ to be satisfied by the polar coordinates of each of its points.‡ For certain curves, a polar equation may be easier to obtain and more convenient to use than a Cartesian equation. For example, the circle whose Cartesian equation is $x^2 + y^2 = R^2$ has the very simple polar equation $r = R$. Also, the graph of the equation $\theta = c$, where c is a constant, is a straight line emanating from the origin and making an angle c with the positive x-axis. The same line is also described by each of the equations $\theta = c + 2n\pi$, where n may be any integer. Further examples are discussed in the exercises in Section 6.29.

When the radius vector $\vec{r} = x\,\vec{i} + y\,\vec{j}$ is expressed in terms of polar coordinates we find

$$\vec{r} = r\cos\theta\,\vec{i} + r\sin\theta\,\vec{j} = r(\cos\theta\,\vec{i} + \sin\theta\,\vec{j})\,.$$

† Sometimes it is desirable to assign a *unique* pair of polar coordinates to the point P. This may be done by restricting θ to lie in a half-open interval of length 2π. The intervals $[0, 2\pi)$ and $(-\pi, \pi]$ are commonly used for this purpose.

‡ Since we are restricting ourselves to nonnegative values of r, only those pairs of numbers (r, θ) for which $r \geq 0$ are to be counted as part of the graph of the equation $f(r, \theta) = 0$.

The vector $\cos\theta\,\vec{i} + \sin\theta\,\vec{j}$ is a vector of unit length having the same direction as $\vec{r}$. This unit vector is usually denoted by $\vec{u}_r$ and the foregoing equation is written as follows:

(6.63) $$\vec{r} = r\vec{u}_r\,, \qquad \text{where } \vec{u}_r = \cos\theta\,\vec{i} + \sin\theta\,\vec{j}\,.$$

It is convenient to introduce also a unit vector $\vec{u}_\theta$, perpendicular to $\vec{u}_r$, which is defined as follows:

$$\vec{u}_\theta = \frac{d\vec{u}_r}{d\theta} = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}\,.$$

Note that

$$\frac{d\vec{u}_\theta}{d\theta} = -\cos\theta\,\vec{i} - \sin\theta\,\vec{j} = -\vec{u}_r\,.$$

In the study of plane curves, the two unit vectors $\vec{u}_r$ and $\vec{u}_\theta$ play the same roles in polar coordinates as the unit vectors $\vec{i}$ and $\vec{j}$ in rectangular coordinates. Figure 6.41 shows the unit vectors $\vec{u}_r$ and $\vec{u}_\theta$ attached to a curve at some of its points.

6.27 The velocity and acceleration vectors in polar coordinates

If a particle moves on a plane curve described by parametric equations, say $x = X(t)$, $y = Y(t)$, the corresponding polar coordinates r and θ may also be expressed parametrically in terms of t by two equations of the form

(6.64) $$r = f(t)\,, \qquad \theta = g(t)\,.$$

We shall derive formulas for expressing the velocity and acceleration in terms of $\vec{u}_r$ and $\vec{u}_\theta$. For the position vector we have

$$\vec{r} = r\vec{u}_r = f(t)\vec{u}_r\,.$$

Since θ depends on the parameter t, the same is true of the unit vector $\vec{u}_r$ and we must take this into account when we compute the velocity vector. Thus we have

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(r\vec{u}_r)}{dt} = \frac{dr}{dt}\vec{u}_r + r\frac{d\vec{u}_r}{dt}\,.$$

Using the chain rule, we may express $d\vec{u}_r/dt$ in terms of $\vec{u}_\theta$ by writing

(6.65) $$\frac{d\vec{u}_r}{dt} = \frac{d\theta}{dt}\frac{d\vec{u}_r}{d\theta} = \frac{d\theta}{dt}\vec{u}_\theta\,,$$

and the equation for the velocity vector becomes

(6.66) $$\vec{v} = \frac{dr}{dt}\vec{u}_r + r\frac{d\theta}{dt}\vec{u}_\theta\,.$$

In terms of the functions f and g in (6.64), this may be written as follows:

$$\vec{v} = f'(t)\vec{u}_r + f(t)g'(t)\vec{u}_\theta\,.$$

Similarly, if we differentiate both sides of (6.66), we find that the acceleration vector is given by

$$\vec{a} = \left(\frac{d^2r}{dt^2}\vec{u}_r + \frac{dr}{dt}\frac{d\vec{u}_r}{dt}\right) + \left(r\frac{d^2\theta}{dt^2}\vec{u}_\theta + \frac{dr}{dt}\frac{d\theta}{dt}\vec{u}_\theta + r\frac{d\theta}{dt}\frac{d\vec{u}_\theta}{dt}\right).$$

The derivative $d\vec{u}_r/dt$ may be expressed in terms of $\vec{u}_\theta$ by (6.65). We may similarly express the derivative of $\vec{u}_\theta$ by the equation

$$\frac{d\vec{u}_\theta}{dt} = \frac{d\theta}{dt}\frac{d\vec{u}_\theta}{d\theta} = -\frac{d\theta}{dt}\vec{u}_r .$$

This leads to the following formula for $\vec{a}$:

(6.67) $$\vec{a} = \left(\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right)\vec{u}_r + \left(r\frac{d^2\theta}{dt^2} + 2\frac{dr}{dt}\frac{d\theta}{dt}\right)\vec{u}_\theta .$$

In terms of the functions f and g in (6.64) this becomes:

$$\vec{a} = \{f''(t) - f(t)\,[g'(t)]^2\}\,\vec{u}_r + [f(t)g''(t) + 2f'(t)g'(t)]\,\vec{u}_\theta .$$

The acceleration vector is said to be *radial* if the coefficient of $\vec{u}_\theta$ in Equation (6.67) is zero. This coefficient may be expressed as follows:

$$\frac{1}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right).$$

Therefore, the acceleration is radial if and only if $r^2\,d\theta/dt$ is constant.

The product $r^2\,d\theta/dt$ has an interesting geometric interpretation. Suppose we denote by $A(t)$ the area of the region swept out by the radius vector from a fixed time t_0 to an arbitrary later time t. An example is the shaded region shown in Figure 6.42. We shall prove that the time rate of change of this area is exactly equal to $\frac{1}{2}r^2\,d\theta/dt$. That is to say, we have

(6.68) $$A'(t) = \tfrac{1}{2}r^2\frac{d\theta}{dt}.$$

To prove this, we express the radius vector $\vec{r}(t)$ in terms of rectangular coordinates, say $\vec{r}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j}$. When $X(t) = x$, let us denote the corresponding ordinate $Y(t)$ by $h(x)$. By referring to Figure 6.42, we see that the area of the sector in question is

$$A(t) = \tfrac{1}{2}X(t)Y(t) + \int_{X(t)}^{X(t_0)} h(x)\,dx - \tfrac{1}{2}X(t_0)Y(t_0).$$

Differentiation with respect to t gives us

(6.69) $$A'(t) = \tfrac{1}{2}X(t)Y'(t) + \tfrac{1}{2}X'(t)Y(t) + \frac{d}{dt}\left(\int_{X(t)}^{X(t_0)} h(x)\,dx\right).$$

To compute the derivative of the integral in (6.69) we introduce

$$F(u) = \int_u^{X(t_0)} h(x)\,dx = -\int_{X(t_0)}^u h(x)\,dx\,.$$

Then $F'(u) = -h(u)$ because of the first fundamental theorem of calculus. The integral in (6.69) is $F[X(t)]$ and hence, by the chain rule, its derivative is $F'[X(t)]\,X'(t) = -h[X(t)]\,X'(t) = -Y(t)X'(t)$. Therefore Equation (6.69) becomes

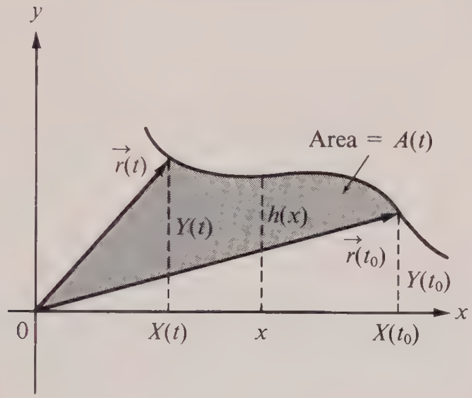


FIGURE 6.42 *A region swept out by the radius vector.*

$$A'(t) = \tfrac{1}{2}[X(t)Y'(t) - Y(t)X'(t)] = \tfrac{1}{2}X(t)^2\frac{d}{dt}\left(\frac{Y(t)}{X(t)}\right)$$

$$= \tfrac{1}{2}X(t)^2\frac{d}{dt}(\tan\theta) = \tfrac{1}{2}X(t)^2\sec^2\theta\,\frac{d\theta}{dt}\,.$$

Since $X(t) = r\cos\theta$, this equation reduces to (6.68).

As a consequence of (6.68) we may write

$$\text{(6.70)} \qquad A(t_2) - A(t_1) = \int_{t_1}^{t_2} A'(t)\,dt = \tfrac{1}{2}\int_{t_1}^{t_2} r^2\frac{d\theta}{dt}\,dt = \tfrac{1}{2}\int_{t_1}^{t_2} f^2(t)g'(t)\,dt\,,$$

where f and g are given by (6.64). This equation provides a convenient way to compute area when the curve is described in polar coordinates. In particular, if the angle θ is used as parameter, the polar equation of the curve may be written in the form $r = f(\theta)$ and (6.70) becomes

$$A(\theta_2) - A(\theta_1) = \tfrac{1}{2}\int_{\theta_1}^{\theta_2} r^2\,d\theta = \tfrac{1}{2}\int_{\theta_1}^{\theta_2} f^2(\theta)\,d\theta\,.$$

For a circle of radius a we have $f(\theta) = a$ and the integral gives $\frac{1}{2}(\theta_2 - \theta_1)a^2$ for the area of a circular sector subtending an angle $\theta_2 - \theta_1$.

The integral formula in (6.70) may also be used to determine the behavior of area under a similarity transformation. Suppose the integral in (6.70) represents the area of a region S. A similarity transformation carries each point of S with polar coordinates (r, θ) to a point with polar coordinates (kr, θ), where k is a positive constant. That is to say, the radial distances of the points in S are multiplied by a common factor k whereas the polar

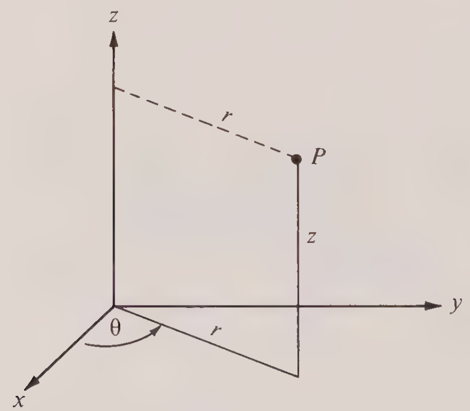


FIGURE 6.43 *Cylindrical coordinates.*

angles are unchanged. When we apply the integral formula in (6.70) to the transformed region, kS, we find that the area of kS is k^2 times that of S.

6.28 Cylindrical coordinates

If the x- and y-coordinates of a point $P = (x, y, z)$ in 3-space are replaced by polar coordinates r and θ, then the three numbers r, θ, z are called *cylindrical coordinates* for the point P. The nonnegative number r now represents the distance from the z-axis to the point P, as indicated in Figure 6.43. Those points in space for which r is constant are at a fixed distance from the z-axis and therefore lie on a circular cylinder (hence the name *cylindrical* coordinates).

To discuss *space* curves in cylindrical coordinates, the equation for the radius vector $\vec{r}$ must be replaced by one of the form

$$\vec{r} = r\vec{u}_r + Z(t)\,\vec{k}\,.$$

Corresponding formulas for the velocity and acceleration vectors are obtained by merely adding the terms $Z'(t)\,\vec{k}$ and $Z''(t)\,\vec{k}$, respectively, to the right-hand members of the two-dimensional formulas in (6.66) and (6.67).

6.29 Exercises

1. If $a > 0$, show that the polar equation $r = 2a \cos \theta$ represents a circle of radius a tangent to the y-axis at the origin. Solve this problem in two ways: (a) by transforming the Cartesian equation of this circle into polar form, and (b) by deriving the polar equation directly from the geometric properties of the circle suggested in Figure 6.44.

2. Show that the polar equation $r = 2a \sin \theta$, $a > 0$, represents a circle of radius a tangent to the x-axis at the origin. Solve this problem by each of the two methods described in Exercise 1.

3. (a) Show that a polar equation for the most general circle of radius $a > 0$ passing through the origin can be written in the form $r = 2a \cos (\theta - \alpha)$. Explain the geometric significance of α.

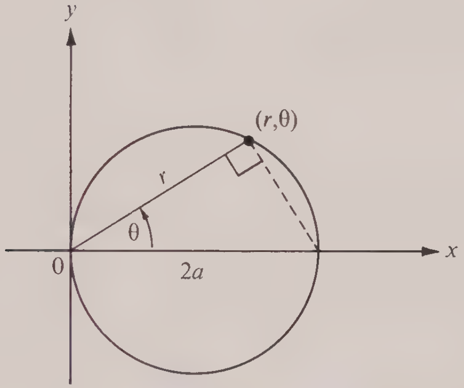


FIGURE 6.44 *The circle with polar equation* $r = 2a \cos \theta$.

(b) Find a polar equation for a circle of radius a whose center has polar coordinates (R, α). [*Hint.* Use the equation $|\vec{R} - \vec{r}|^2 = a^2$, where $\vec{R}$ and $\vec{r}$ are vectors from the origin to the points whose polar coordinates are (R, α) and (r, θ), respectively.]

4. Show that the graph of the polar equation $r = a \sin \theta + b \cos \theta$ is a circle. Express the radius and the coordinates of the center in terms of a and b.

5. A particle moves in a plane so that its position at time t has polar coordinates $r = t$, $\theta = t$. Find formulas for the velocity $\vec{v}$, the acceleration $\vec{a}$, and the curvature κ at any time t.

6. A particle moves in space so that its position at time t has cylindrical coordinates $r = t$, $\theta = t$, $z = t$.

(a) Show that the curve lies on a cone and find a Cartesian equation for this cone. (The curve is called a *conical helix.*)

(b) Find formulas for the velocity $\vec{v}$, the acceleration $\vec{a}$, and the curvature κ at time t.

(c) Find a formula for determining the angle between the tangent line to the curve and the generator of the cone at each point of the curve.

7. A particle moves in space so that its position at time t has cylindrical coordinates $r = \sin t$, $\theta = t$, $z = \log \sec t$, where $0 \le t < \frac{1}{2}\pi$.

(a) Show that the curve lies on a cylinder and find a Cartesian equation for this cylinder.

(b) Find a formula (in terms of t) for determining the angle at which this curve intersects the generators at each point.

8. The polar equation $r = e^{-\theta}$ represents a spiral in the plane. An ant crawls inward along this spiral with constant speed v.

(a) Show that $\vec{v} \cdot \vec{a} = 0$, where $\vec{v}$ and $\vec{a}$ are the ant's velocity and acceleration, respectively.

(b) The same ant can exert a force $\vec{F}$ along its direction of motion, the magnitude of this force being $|\vec{F}| = 1/r$. Find the resulting moment $\vec{M}$ about the origin. (See Exercise 23 in Section 5.18.)

In Exercises 9 through 14, R denotes the set of points in the plane with polar coordinates (r, θ), where $0 \leq r \leq f(\theta)$ and θ ranges over the indicated interval. In each case, sketch the region R and compute its area by integration.

9. $f(\theta) = \theta, \quad 0 \leq \theta \leq \dfrac{\pi}{2}.$

10. $f(\theta) = \sqrt{\cos 2\theta}, \quad 0 \leq \theta \leq \dfrac{\pi}{4}.$

11. $f(\theta) = \dfrac{1}{1 - \cos \theta}, \quad \dfrac{\pi}{4} \leq \theta \leq \dfrac{\pi}{2}.$

12. $f(\theta) = e^{\theta}, \quad 0 \leq \theta \leq \pi.$

13. $f(\theta) = 1 + \cos \theta, \quad 0 \leq \theta \leq \dfrac{\pi}{4}.$

14. $f(\theta) = \dfrac{1}{2 + \cos \theta}, \quad 0 \leq \theta \leq \dfrac{\pi}{2}.$

15. (a) If a curve is described by a vector equation in polar coordinates, show that its speed at time t is $v(t) = \sqrt{(dr/dt)^2 + r^2(d\theta/dt)^2}$.

(b) Use Equation (6.52) to find a formula for arc length in polar coordinates. What does this formula become when the curve is given by the polar equation $r = f(\theta)$?

16. The curve described by the polar equation $r = a(1 + \cos \theta)$, where $a > 0$, is called a *cardiod*. Draw a graph of the cardiod $r = 4(1 + \cos \theta)$ and use the result of Exercise 15 to compute its arc length.

17. Let ϕ denote the angle, $0 \leq \phi < \pi$, between the radius vector and the tangent line of a curve. If the curve is expressed in polar coordinates, show that $\tan \phi = r\, d\theta/dr$.

18. A missile is designed to move directly toward its target. Due to a mechanical failure, its direction in actual flight makes a fixed angle $\alpha \neq 0$ with the line from the missile to the target. Find the path if it is fired at a fixed target. Discuss how the path varies with α. Does the missile ever reach the target?

19. Sketch the curve whose polar equation is $r = \sin^2 \theta$ and show that it consists of two loops. (a) Find the area of the region enclosed by one loop of the curve. (b) Compute the length of one loop of the curve.

20. If a curve has the polar equation $r = f(\theta)$, show that its radius of curvature ρ is given by the formula $\rho = (r^2 + r'^2)^{3/2}/|r^2 - rr'' + 2r'^2|$, where $r' = f'(\theta)$ and $r'' = f''(\theta)$.

21. Prove that if a homogeneous first-order differential equation of the form $y' = f(x, y)$ is rewritten in polar coordinates, it reduces to a separable equation. Use this method to solve $y' = (y - x)/(y + x)$.

22. Due to a mechanical failure, a ground crew has lost control of a missile recently fired. It is known that the missile will proceed at a constant speed on a straight course of unknown direction. When the missile is 4 miles away, it is sighted for an instant and lost again. Immediately an anti-missile missile is fired with a constant speed three times that of the first missile. What should be the course of the second missile in order for it to overtake the first one? (Assume both missiles move in the same plane.)

23. A particle (moving in space) has velocity vector $\vec{v} = \omega \vec{k} \times \vec{r}$, where ω is a constant and $\vec{r}$ is the position vector. Prove that the particle moves along a circle with constant angular speed ω. (The angular speed is defined to be $d\theta/dt$, where θ is the polar angle at time t.)

24. A particle moves in a plane perpendicular to the z-axis. The motion takes place along a circle with center on this axis.

(a) Show that there is a vector $\vec{\omega}(t)$ parallel to the z-axis such that

$$\vec{v}(t) = \vec{\omega}(t) \times \vec{r}(t),$$

where $\vec{r}(t)$ and $\vec{v}(t)$ denote the position and velocity vectors at time t. The vector $\vec{\omega}(t)$ is called the *angular velocity* vector and its magnitude $\omega(t) = |\vec{\omega}(t)|$ is called the *angular speed*.

(b) The vector $\vec{\alpha}(t) = \vec{\omega}'(t)$ is called the *angular acceleration* vector. Show that the acceleration vector $\vec{a}(t)$ $[= \vec{v}'(t)]$ is given by the formula

$$\vec{a}(t) = [\vec{\omega}(t) \cdot \vec{r}(t)]\vec{\omega}(t) - \omega^2(t)\vec{r}(t) + \vec{\alpha}(t) \times \vec{r}(t).$$

(c) If the particle lies in the xy-plane and if the angular speed $\omega(t)$ is constant, say $\omega(t) = \omega$, prove that the acceleration vector $\vec{a}(t)$ is centripetal and that, in fact, $\vec{a}(t) = -\omega^2\vec{r}(t)$.

25. A body is said to undergo a *rigid motion* if, for every pair of particles p and q in the body, the distance $|\vec{r}_p(t) - \vec{r}_q(t)|$ is independent of t, where $\vec{r}_p(t)$ and $\vec{r}_q(t)$ denote the position vectors of p and q at time t. Prove that for a rigid body in which each particle p rotates about the z-axis we have $\vec{v}_p(t) = \vec{\omega}(t) \times \vec{r}_p(t)$, where $\vec{\omega}(t)$ is the same for each particle, and $\vec{v}_p(t)$ is the velocity of particle p.

6.30 Applications to planetary motion

By analyzing the voluminous data on planetary motion accumulated up to 1600, the German astronomer Johannes Kepler (1571–1630) tried to discover the mathematical laws governing the motions of the planets. There were six known planets at that time and, according to the Copernican theory, their orbits were thought to lie on concentric spherical shells about the sun. Kepler attempted to show that the radii of these shells were linked up with the five regular solids of geometry. He proposed an ingenious idea that the solar system was designed something like a Chinese puzzle. At the center of the system he placed the sun. Then, in succession, he arranged the six concentric spheres that can be inscribed and circumscribed around the five regular solids—the octahedron, icosahedron, dodecahedron, tetrahedron, and cube, in respective order (from inside out). The innermost sphere, inscribed in the regular octahedron, corresponded to Mercury's path. The next sphere, which circumscribed the octahedron and inscribed the icosahedron, corresponded to the orbit of Venus. Earth's orbit lay on the sphere around the icosahedron and inside the dodecahedron, and so on, the outermost sphere, containing Jupiter's orbit, being circumscribed around the cube. Although this theory seemed correct to within 5%, astronomical observations at that time were accurate to a percentage error much smaller than this and Kepler finally realized that he had to modify this theory. After much further study it occurred to him that the observed data concerning the orbits corresponded more to *elliptical* paths than the circular paths of the Copernican system. After several more years of unceasing effort, Kepler set forth three famous laws, empirically discovered, which explained all the astronomical phenomena known at that time. They may be stated as follows:

Kepler's first law: Planets move in ellipses with the sun at one focus.

Kepler's second law: The radius vector from the sun to a planet sweeps out area at a constant rate.

Kepler's third law: The square of the period† of a planet is proportional to the cube of its mean distance from the sun.

† By the *period* of a planet is meant the time required to go once around the elliptical orbit. The *mean distance* from the sun is the length of the semimajor axis of the ellipse.

The formulation of these laws from a study of astronomical tables was a remarkable achievement. Nearly 50 years later, Newton proved that all three of Kepler's laws are consequences of his own second law of motion and his celebrated universal law of gravitation. In this section we shall use vector methods to show how Kepler's laws may be deduced from Newton's.

Assume we have a fixed sun of mass M and a moving planet of mass m attracted to the sun by a force $\vec{F}$. (We neglect the influence of all other forces.) Newton's second law of motion states that

$$\vec{F} = m\vec{a}\,, \tag{6.71}$$

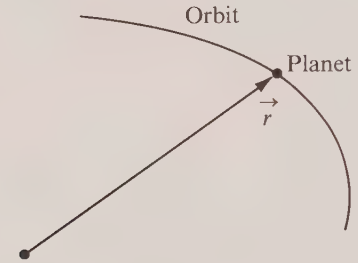


FIGURE 6.45 *The radius vector from the sun to a planet.*

where $\vec{a}$ is the acceleration vector of the moving planet. Denote by $\vec{r}$ the radius vector from the sun to the planet (as in Figure 6.45), let $r = |\vec{r}|$, and let $\vec{u}_r$ be a unit vector with the same direction as $\vec{r}$, so that $\vec{r} = r\vec{u}_r$. The universal law of gravitation states that

$$\vec{F} = -G\frac{mM}{r^2}\,\vec{u}_r\,,$$

where G is a constant. Combining this with (6.71), we obtain

$$\vec{a} = -\frac{GM}{r^2}\,\vec{u}_r\,, \tag{6.72}$$

which tells us that the acceleration is *radial*. In a moment we shall prove that the orbit lies in a plane. Once we know this, it follows at once from the results of Section 6.27 that the radius vector sweeps out area at a constant rate.

To prove that the path lies in a plane we use the fact that $\vec{r}$ and $\vec{a}$ are parallel. If we introduce the velocity vector $\vec{v} = d\vec{r}/dt$, we have

$$\vec{r}\times\vec{a} = \vec{r}\times\frac{d\vec{v}}{dt} + \vec{v}\times\vec{v} = \vec{r}\times\frac{d\vec{v}}{dt} + \frac{d\vec{r}}{dt}\times\vec{v} = \frac{d}{dt}(\vec{r}\times\vec{v})\,.$$

Since $\vec{r}\times\vec{a} = \vec{0}$ this means that $\vec{r}\times\vec{v}$ is a constant vector, say $\vec{r}\times\vec{v} = \vec{c}$. Therefore

$\vec{r} \cdot \vec{c} = 0$ and this implies that the radius vector lies in a plane perpendicular to $\vec{c}$. Since the acceleration is radial, $\vec{r}$ sweeps out area at a constant rate and this proves Kepler's second law.

It is easy to prove that this constant rate is exactly half the length of the vector $\vec{c}$. In fact, if we use polar coordinates and express the velocity in terms of $\vec{u}_r$ and $\vec{u}_\theta$ as in Equation (6.66), we find

$$(6.73) \qquad \vec{c} = \vec{r} \times \vec{v} = (r\vec{u}_r) \times \left(\frac{dr}{dt}\vec{u}_r + r\frac{d\theta}{dt}\vec{u}_\theta\right) = r^2\frac{d\theta}{dt}\vec{u}_r \times \vec{u}_\theta ,$$

and hence $|\vec{c}| = |r^2\, d\theta/dt|$. By (6.68) this is equal to $2|A'(t)|$, where $A'(t)$ is the rate at which the radius vector sweeps out area.

We shall prove next that the path is an ellipse. First of all, we form the cross product $\vec{a} \times \vec{c}$, using (6.72) and (6.73), and we find

$$\vec{a} \times \vec{c} = \left(-\frac{GM}{r^2}\vec{u}_r\right) \times \left(r^2\frac{d\theta}{dt}\vec{u}_r \times \vec{u}_\theta\right) = -GM\frac{d\theta}{dt}\vec{u}_r \times (\vec{u}_r \times \vec{u}_\theta) = GM\frac{d\theta}{dt}\vec{u}_\theta .$$

Since $\vec{a} = d\vec{v}/dt$ and $\vec{u}_\theta = d\vec{u}_r/d\theta$, the foregoing equation for $\vec{a} \times \vec{c}$ can also be written as follows:

$$\frac{d}{dt}(\vec{v} \times \vec{c}) = \frac{d}{dt}(GM\,\vec{u}_r) .$$

Integration gives us

$$(6.74) \qquad \vec{v} \times \vec{c} = GM\vec{u}_r + \vec{A} ,$$

where $\vec{A}$ is another constant vector. It is customary to write (6.74) as follows:

$$(6.75) \qquad \vec{v} \times \vec{c} = GM(\vec{u}_r + \vec{\varepsilon}) ,$$

where $GM\vec{\varepsilon} = \vec{A}$. We shall combine this with (6.73) to eliminate $\vec{v}$ and obtain an equation for r. For this purpose we dot multiply both sides of (6.73) by $\vec{c}$ and both sides of (6.75) by $\vec{r}$. Equating the two expressions for the scalar triple product $\vec{r} \cdot \vec{v} \times \vec{c}$ we are led to the equation

$$(6.76) \qquad GMr(1 + \varepsilon \cos\phi) = c^2 ,$$

where $\varepsilon = |\vec{\varepsilon}|$, $c = |\vec{c}|$, and ϕ represents the angle between the constant vector $\vec{\varepsilon}$ and the radius vector $\vec{r}$. (See Figure 6.46.) It is easy to prove that this is a polar equation of a conic section with eccentricity ε. In fact, if we let $d = c^2/(GM\varepsilon)$, then Equation (6.76) is the same as $r(1 + \varepsilon\cos\phi) = \varepsilon d$, or

$$(6.77) \qquad r = \varepsilon(d - r\cos\phi) .$$

In Figure 6.47 a line is drawn perpendicular to $\vec{\varepsilon}$ at a distance d from the sun. The distance from the planet to this line is $d - r\cos\phi$ and Equation (6.77) states that the ratio of r to this distance is the constant ε. This means that the planet moves on a conic with

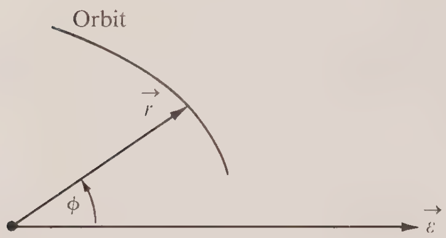


FIGURE 6.46 *The angle ϕ between the radius vector $\vec{r}$ and the constant vector ε.*

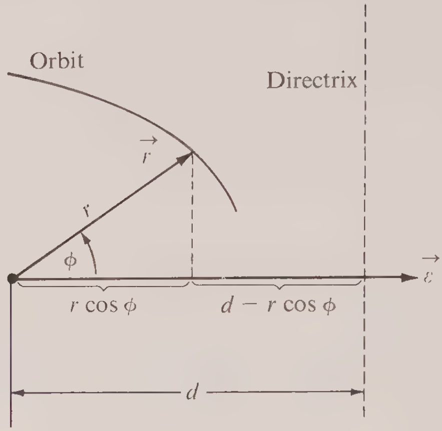


FIGURE 6.47 *The ratio $r/(d - r \cos \phi)$ is the constant ε.*

eccentricity ε, with the sun as one focus and the line at distance d as a directrix. The conic is an ellipse if $\varepsilon < 1$, a parabola if $\varepsilon = 1$, and a hyperbola if $\varepsilon > 1$. Since planets are known to move on closed paths, the orbit under consideration must be an ellipse.† This proves Kepler's first law.

Finally, we deduce Kepler's third law. Suppose the ellipse has semimajor axis a and semiminor axis b. Let T be the time it takes for the planet to go once around the ellipse. Since the radius vector sweeps out area at the rate $\frac{1}{2}c$ and since the area of the ellipse is πab, we have $\frac{1}{2}cT = \pi ab$, or $T = 2\pi ab/c$. We wish to prove that T^2 is proportional to a^3. For this purpose we express both b and c in terms of a. For b we have the equation

FIGURE 6.48 *A curve with an inscribed polygon.*

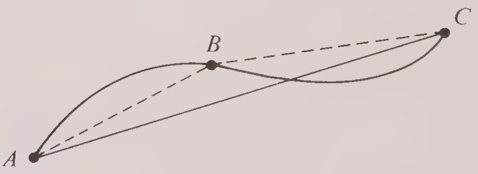


FIGURE 6.49 *The polygon ABC has a length greater than the polygon AC.*

† Parabolic and hyperbolic orbits occur when bodies enter the solar system with sufficiently high speed to escape again.

$b = a\sqrt{1 - \varepsilon^2}$. If we take the average of the values of r corresponding to $\phi = 0$ and $\phi = \pi$ in (6.77), we find $a = \varepsilon d/(1 - \varepsilon^2)$. Since $d = c^2/(GM\varepsilon)$, this implies $c = \sqrt{GMa}\sqrt{1 - \varepsilon^2}$. Therefore, we have

$$T = \frac{2\pi ab}{c} = \frac{2\pi a^2\sqrt{1 - \varepsilon^2}}{\sqrt{GMa}\sqrt{1 - \varepsilon^2}} = \frac{2\pi a^{3/2}}{\sqrt{GM}},$$

so T^2 is proportional to a^3. This proves Kepler's third law.

Supplement. Arc length.

★6.31 The definition of arc length

Various parts of calculus and analytic geometry refer to the arc length of a curve. Before we can study the properties of the length of a curve we must agree on a *definition* of arc length. The purpose of this section is to formulate such a definition. This will lead, in a natural way, to the construction of a function (called the arc-length function) which measures the length of the path traced out by a moving particle at every instant of its motion. Some of the basic properties of this function are discussed in Section 6.33. In particular, we shall prove that for most curves that arise in practice this function may be expressed as the integral of the speed.

To arrive at a definition of what we mean by the length of a curve, we proceed as though we had to measure this length with a straight yardstick. First, we mark off a number of points on the curve which we use as vertices of an inscribed polygon. (An example is shown in Figure 6.48.) Then, we measure the total length of this polygon with our yardstick and consider this as an approximation to the length of the curve. We soon observe that some polygons "fit" the curve better than others. In particular, if we start with a polygon P_1, and construct a new inscribed polygon P_2 by adding more vertices to those of P_1, it is clear that the length of P_2 will be larger than that of P_1, as suggested in Figure 6.49. In the same way we can form more and more polygons with successively larger and larger lengths.

On the other hand, our intuition tells us that the length of any inscribed polygon should not exceed that of the curve (since a straight line is the shortest path between two points). In other words, when we arrive at a definition for the length of a curve, it should be a number which is an *upper bound* to the lengths of all inscribed polygons. Therefore, it certainly seems reasonable to define the length of the curve to be the *least upper bound* of the lengths of all possible inscribed polygons.

For most curves that arise in practice, this definition gives us a useful and reasonable way to assign a length to a curve. Surprisingly enough, however, there are certain pathological cases where this definition is not applicable. There are curves for which there is *no* upper bound to the lengths of the inscribed polygons.† Therefore it becomes necessary to classify all curves into two categories: those which have a length, and those which do not. The former are called *rectifiable curves*, the others *nonrectifiable*.

† An example is the graph of the function f defined on $[0, 2\pi]$ as follows: $f(x) = x\cos(\frac{1}{2}\pi/x)$ if $x \neq 0, f(0) = 0$. It can be shown that this curve has no arc length. (See Exercise 50 in Section 9.18.)

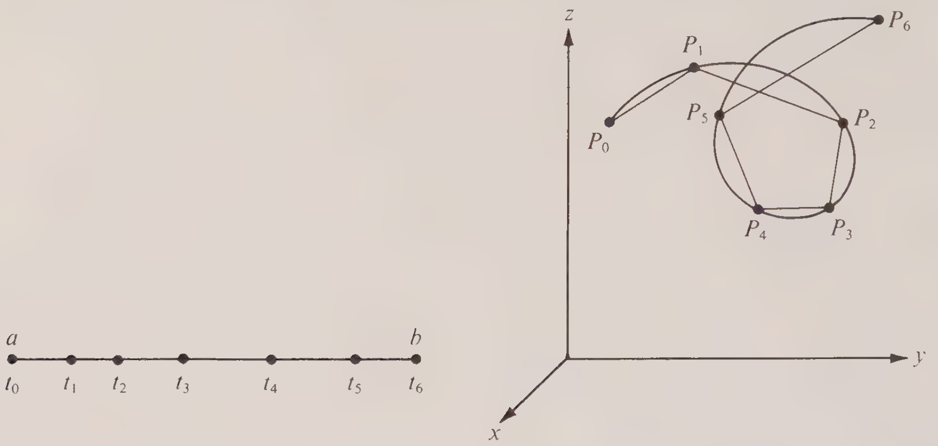


FIGURE 6.50 *A partition of* $[a, b]$ *into six subintervals and the corresponding inscribed polygon.*

Now we shall reformulate these ideas in analytic terms. For this purpose, we begin with a curve in 3-space described by the vector equation

$$\vec{r}(t) = X(t)\,\vec{i} + Y(t)\,\vec{j} + Z(t)\,\vec{k}\,, \tag{6.78}$$

and we consider that portion of the curve traced out as t varies over a parametric interval $[a, b]$. At the outset we only assume that the three functions X, Y, Z are *continuous* on $[a, b]$. (Later, we shall add further restrictions.)

Consider now any partition P of the interval $[a, b]$, say

$$P = \{t_0, t_1, \ldots, t_n\}\,, \qquad \text{where} \quad a = t_0 < t_1 < \cdots < t_n = b\,.$$

Denote by $\pi(P)$ the polygon whose vertices are the terminal points $P_0, P_1, \ldots, P_n$ of the vectors $\vec{r}(t_0), \vec{r}(t_1), \ldots, \vec{r}(t_n)$, respectively. (An example with $n = 6$ is shown in Figure 6.50.) The sides of this polygon may be described by the n vectors

$$\vec{r}(t_1) - \vec{r}(t_0)\,, \vec{r}(t_2) - \vec{r}(t_1)\,, \ldots, \vec{r}(t_n) - \vec{r}(t_{n-1})\,.$$

Therefore the length of the polygon $\pi(P)$, which we denote by $|\pi(P)|$, is the sum

$$|\pi(P)| = \sum_{k=1}^{n} |\vec{r}(t_k) - \vec{r}(t_{k-1})|\,.$$

DEFINITION. If there exists a positive number M such that

$$|\pi(P)| \leq M \tag{6.79}$$

for all partitions P of $[a, b]$, then the curve is said to be rectifiable and its arc length, denoted by $\Lambda(a, b)$, is defined to be the least upper bound of the set of all numbers $|\pi(P)|$. If there is no such M, the curve is called nonrectifiable.

Note that if an M exists satisfying (6.79), then for every partition P we have

$$|\pi(P)| \leq \Lambda(a, b) \leq M \,, \tag{6.80}$$

since the least upper bound cannot exceed any upper bound.

It is easy to prove that a curve is rectifiable whenever its velocity vector $\vec{v}$ is continuous on the parametric interval $[a, b]$. In fact, the following theorem tells us that in this case we may use the integral of the speed as an upper bound for all numbers $|\pi(P)|$.

6–2 THEOREM. Denote by $\vec{v}(t)$ the velocity vector of the curve whose vector equation is (6.78) and let $v(t) = |\vec{v}(t)|$ denote the speed. If $\vec{v}$ is continuous on the parametric interval $[a, b]$, then the curve is rectifiable and its length $\Lambda(a, b)$ satisfies the inequality

$$\Lambda(a, b) \leq \int_a^b v(t)\, dt \,. \tag{6.81}$$

Proof. For each partition P of $[a, b]$ we have

$$|\pi(P)| = \sum_{k=1}^{n} |\vec{r}(t_k) - \vec{r}(t_{k-1})| = \sum_{k=1}^{n} \left| \int_{t_{k-1}}^{t_k} \vec{r}\,'(t)\, dt \right|$$

$$= \sum_{k=1}^{n} \left| \int_{t_{k-1}}^{t_k} \vec{v}(t)\, dt \right| \leq\dagger \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} |\vec{v}(t)|\, dt = \int_a^b v(t)\, dt \,,$$

where, of course, by the integral of a vector-valued function we mean the vector obtained by integrating component by component. This shows that we have the inequality

$$|\pi(P)| \leq \int_a^b v(t)\, dt$$

for all partitions P, and hence the number $\int_a^b v(t)\, dt$ is an upper bound for the set of all numbers $|\pi(P)|$. This proves that the curve is rectifiable and, at the same time, it tells us that the length $\Lambda(a, b)$ cannot exceed the integral of the speed.

In a later section we shall prove that the inequality in (6.81) is, in fact, an *equality*. The proof of this fact will make use of the *additivity* of arc length, a property described in the next section.

★6.32 Additivity of arc length

If a rectifiable curve is cut into two pieces, the length of the whole curve is the sum of the lengths of the two parts. This is another of those "intuitively obvious" statements

† At this stage we need to know that $\left|\int_{t_{k-1}}^{t_k} \vec{v}(t)\, dt\right| \leq \int_{t_{k-1}}^{t_k} |\vec{v}(t)|\, dt$. This is an extension of a familiar property of integrals of real-valued functions. A proof for vector-valued functions is outlined in Exercise 16 of Section 6.34.

whose proof is not trivial. This property is called *additivity of arc length* and it may be expressed analytically as follows:

6–3 THEOREM. Consider a rectifiable curve of length $\Lambda(a, b)$ traced out by a vector $\vec{r}(t)$ as t varies over an interval $[a, b]$. If $a < c < b$, let C_1 and C_2 be the curves traced out by $\vec{r}(t)$ as t varies over the intervals $[a, c]$ and $[c, b]$, respectively. Then C_1 and C_2 are also rectifiable and, if $\Lambda(a, c)$ and $\Lambda(c, b)$ denote their respective lengths, we have

$$\Lambda(a, b) = \Lambda(a, c) + \Lambda(c, b) .$$

Proof. Let P_1 and P_2 be arbitrary partitions of $[a, c]$ and $[c, b]$, respectively. The points in P_1 taken together with those in P_2 give us a new partition P of $[a, b]$ for which we have

$$|\pi(P_1)| + |\pi(P_2)| = |\pi(P)| \leq \Lambda(a, b) . \tag{6.82}$$

This shows that $|\pi(P_1)|$ and $|\pi(P_2)|$ are bounded by $\Lambda(a, b)$, and hence C_1 and C_2 are rectifiable. From (6.82) we also have

$$|\pi(P_1)| \leq \Lambda(a, b) - |\pi(P_2)| .$$

Now, keep P_2 fixed and let P_1 vary over all possible partitions of $[a, c]$. Since the number $\Lambda(a, b) - |\pi(P_2)|$ is an upper bound for all numbers $|\pi(P_1)|$ it cannot be less than their least upper bound, which is $\Lambda(a, c)$. Hence we have $\Lambda(a, c) \leq \Lambda(a, b) - |\pi(P_2)|$ or, what is the same thing,

$$|\pi(P_2)| \leq \Lambda(a, b) - \Lambda(a, c) .$$

This shows that $\Lambda(a, b) - \Lambda(a, c)$ is an upper bound for all the sums $|\pi(P_2)|$, and since it cannot be less than their least upper bound, $\Lambda(c, b)$, we have $\Lambda(c, b) \leq \Lambda(a, b) - \Lambda(a, c)$. In other words, we have

$$\Lambda(a, c) + \Lambda(c, b) \leq \Lambda(a, b) . \tag{6.83}$$

Next we prove the reverse inequality. We begin with any partition P of $[a, b]$. If we adjoin the point c to P, we obtain a partition P_1 of $[a, c]$ and a partition P_2 of $[c, b]$ such that

$$|\pi(P)| \leq |\pi(P_1)| + |\pi(P_2)| \leq \Lambda(a, c) + \Lambda(c, b) .$$

This shows that $\Lambda(a, c) + \Lambda(c, b)$ is an upper bound for all numbers $|\pi(P)|$. Since this cannot be less than the least upper bound we must have

$$\Lambda(a, b) \leq \Lambda(a, c) + \Lambda(c, b) .$$

This inequality, along with (6.83), implies the additive property.

★6.33 The arc-length function

At this point we find it convenient to think of the curve as the path traced out by a particle whose position at time t is the terminal point of the vector $\vec{r}(t)$. A natural question to ask is this: How far has the particle moved along the curve at time t? To discuss this question we introduce the *arc-length function* s, defined as follows:

$$s(t) = \Lambda(a, t) \quad \text{if} \quad t > a\,, \qquad s(a) = 0\,.$$

The statement $s(a) = 0$ simply means we are assuming the motion begins when $t = a$.

The theorem on additivity enables us to derive some important properties of s. For example, we have:

6–4 THEOREM. For any rectifiable curve, the arc-length function s is monotonically increasing on $[a, b]$. That is, we have

$$s(t_1) \leq s(t_2) \quad \text{if} \quad a \leq t_1 < t_2 \leq b\,. \tag{6.84}$$

Proof. If $a \leq t_1 < t_2 \leq b$, we have

$$s(t_2) - s(t_1) = \Lambda(a, t_2) - \Lambda(a, t_1) = \Lambda(t_1, t_2)\,,$$

where the last equality comes from additivity. Since $\Lambda(t_1, t_2) \geq 0$, this proves (6.84).

Next we shall prove that the function s has a derivative at each point of the parametric interval and that this derivative is equal to the speed of the particle.

6–5 THEOREM. Let s denote the arc-length function associated with a curve and let $v(t)$ denote the speed at time t. If v is continuous on $[a, b]$, then the derivative $s'(t)$ exists for each t in (a, b) and is given by the formula

$$s'(t) = v(t)\,. \tag{6.85}$$

Note. This theorem, along with the second fundamental theorem of calculus, enables us to calculate arc length by integration. Thus we have

$$s(t_2) - s(t_1) = \int_{t_1}^{t_2} s'(t)\,dt = \int_{t_1}^{t_2} v(t)\,dt\,,$$

a result referred to earlier in Section 6.23 without proof. In particular, when $t_1 = a$ and $t_2 = b$ we obtain

$$\Lambda(a, b) = \int_a^b v(t)\,dt\,.$$

Proof. Define $F(t) = \int_a^t v(u)\,du$. We know that $F'(t) = v(t)$ because of the first fundamental theorem of calculus. We shall prove that $s'(t) = v(t)$. For this purpose we form the difference quotient

$$\left|\frac{\vec{r}(t + h) - \vec{r}(t)}{h}\right|. \tag{6.86}$$

Suppose first that $h > 0$. The line segment joining the terminal points of $\vec{r}(t)$ and $\vec{r}(t + h)$ may be thought of as a polygon approximating the arc joining these two points. Therefore, because of (6.80), we have

$$|\vec{r}(t + h) - \vec{r}(t)| \leq \Lambda(t, t + h) = s(t + h) - s(t)\,.$$

Using this in (6.86) along with the inequality (6.81) of Theorem 6–2, we have

$$\left| \frac{\vec{r}(t+h) - \vec{r}(t)}{h} \right| \leq \frac{s(t+h) - s(t)}{h} \leq \frac{1}{h} \int_t^{t+h} v(u)\, du = \frac{F(t+h) - F(t)}{h}.$$

A similar argument shows that these inequalities are also valid for $h < 0$. If we let $h \to 0$, the difference quotient on the extreme left approaches $|\vec{r}'(t)| = v(t)$ and that on the extreme right approaches $F'(t) = v(t)$. It follows that the quotient $[s(t+h) - s(t)]/h$ also approaches $v(t)$. But this means that $s'(t)$ exists and equals $v(t)$, as asserted.

★6.34 Exercises

In Exercises 1 through 5, find the length of the path traced out by a particle moving on a plane curve according to the given vector equation during the time interval specified in each case.

1. $\vec{r}(t) = a(1 - \cos t)\vec{i} + a(t - \sin t)\vec{j}, \quad 0 \leq t \leq 2\pi, \quad a > 0.$
2. $\vec{r}(t) = e^t \cos t\, \vec{i} + e^t \sin t\, \vec{j}, \quad 0 \leq t \leq 2.$
3. $\vec{r}(t) = a(\cos t + t \sin t)\vec{i} + a(\sin t - t \cos t)\vec{j}, \quad 0 \leq t \leq 2\pi, \quad a > 0.$
4. $\vec{r}(t) = \frac{c^2}{a} \cos^3 t\, \vec{i} + \frac{c^2}{b} \sin^3 t\, \vec{j}, \quad 0 \leq t \leq 2\pi, \quad c^2 = a^2 - b^2, \quad 0 < b < a.$
5. $\vec{r}(t) = a(\sinh t - t)\vec{i} + a(\cosh t - 1)\vec{j}, \quad 0 \leq t \leq T, \quad a > 0.$

6. If a function f has a continuous derivative in an interval $[a, b]$, show that the length of the graph of the equation $y = f(x)$ between $x = a$ and $x = b$ is given by the integral

$$\int_a^b \sqrt{1 + [f'(x)]^2}\, dx.$$

7. Find an integral similar to that in Exercise 6 for the length of the graph of an equation of the form $x = g(y)$, where g has a continuous derivative on an interval $[c, d]$.

8. If a curve has an equation of the form $r = f(\theta)$ in polar coordinates, where f has a continuous derivative on an interval $[a, b]$, show that the length of the graph between $\theta = a$ and $\theta = b$ is given by the integral

$$\int_a^b \sqrt{[f(\theta)]^2 + [f'(\theta)]^2}\, d\theta.$$

9. A curve has the equation $y^2 = x^3$. Find the length of the arc joining $(-1, 1)$ to $(1, 1)$.

10. Two points A and B on a unit circle with center at O determine a circular sector AOB. Prove that the arc AB has a length equal to twice the area of the sector.

11. Set up integrals for the lengths of the curves whose equations are (a) $y = e^x$, $0 \leq x \leq 1$; (b) $x = t + \log t$, $y = t - \log t$, $1 \leq t \leq e$. Show that the second length is $\sqrt{2}$ times the first one.

12. (a) Set up the integral which gives the length of the curve $y = c \cosh (x/c)$ from $x = 0$ to $x = a$ $(a > 0, c > 0)$.

(b) Show that c times the length of this curve is equal to the area of the region bounded by $y = c \cosh (x/c)$, the x-axis, the y-axis, and the line $x = a$.

(c) Evaluate this integral and find the length of the curve when $a = 2$.

13. Show that the length of the curve $y = \cosh x$ joining the points $(0, 1)$ and $(x, \cosh x)$ is $\sinh x$ if $x > 0$.

14. A nonnegative function f has the property that its ordinate set over an arbitrary interval has an area proportional to the arc length of the graph above the interval. Find f.

15. A curve is given by a polar equation $r = f(\theta)$. Find f if an arbitrary arc joining two distinct points of the curve has arc length proportional to (a) the angle subtended at the origin; (b) the

difference of the radial distances from the origin to its endpoints; (c) the area of the sector formed by the arc and the radii to its endpoints.

16. In the proof of Theorem 6–2 we used the inequality

$$\left| \int_a^b \vec{F}(t)\, dt \right| \le \int_a^b |\vec{F}(t)|\, dt \tag{6.87}$$

for integrals of vector-valued functions. The purpose of this exercise is to aid the reader to construct a proof of (6.87). For simplicity, we discuss only the two-dimensional case, since the three-dimensional case is entirely analogous.

(a) If $\vec{F} = X\vec{i} + Y\vec{j}$, show that (6.87) is equivalent to

$$\left(\int_a^b X\, dt\right)^2 + \left(\int_a^b Y\, dt\right)^2 \le \left(\int_a^b \sqrt{X^2 + Y^2}\, dt\right)^2. \tag{6.88}$$

(b) Assume that X and Y are step functions. If $P = \{t_0, t_1, \ldots, t_n\}$ is a partition of $[a, b]$ and if X and Y take the constant values a_k and b_k, respectively, on the kth open subinterval of P, show that (6.88) is equivalent to

$$\left(\sum_{k=1}^n X_k\right)^2 + \left(\sum_{k=1}^n Y_k\right)^2 \le \left(\sum_{k=1}^n \sqrt{X_k^2 + Y_k^2}\right)^2,$$

where $X_k = a_k(t_k - t_{k-1})$ and $Y_k = b_k(t_k - t_{k-1})$ or, what is the same thing,

$$\sum_{k=1}^n X_k \sum_{j=1}^n X_j + \sum_{k=1}^n Y_k \sum_{j=1}^n Y_j \le \sum_{k=1}^n \sqrt{X_k^2 + Y_k^2} \sum_{j=1}^n \sqrt{X_j^2 + Y_j^2}. \tag{6.89}$$

(c) Deduce (6.89) from the following special case of the Cauchy-Schwarz inequality:

$$X_kX_j + Y_kY_j \le \sqrt{X_k^2 + Y_k^2}\sqrt{X_j^2 + Y_j^2}.$$

(d) When you have solved parts (a), (b), and (c), you have proved (6.88) for step functions. Show how to extend the proof to more general functions. [*Hint.* Since $\left(\int_a^b X\, dt\right)^2 + \left(\int_a^b Y\, dt\right)^2 \le \left(\int_a^b |X|\, dt\right)^2 + \left(\int_a^b |Y|\, dt\right)^2$, it suffices to prove (6.88) when X and Y are nonnegative.]

17. Use the vector equation $\vec{r}(t) = a \sin t\, \vec{i} + b \cos t\, \vec{j}$, where $0 < b < a$, to show that the circumference L of an ellipse is given by the integral

$$L = 4a \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 t}\, dt,$$

where $k = \sqrt{a^2 - b^2}/a$. (The number k is the eccentricity of the ellipse.) This is a special case of an integral of the form

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 t}\, dt,$$

called an *elliptic integral of the second kind*, where $0 \le k < 1$. The numbers $E(k)$ have been tabulated for various values of k.

18. If $0 < b < 4a$, let $\vec{r}(t) = a(t - \sin t)\, \vec{i} + a(1 - \cos t)\, \vec{j} + b \sin \frac{1}{2}t\, \vec{k}$. Show that the length of the path traced out from $t = 0$ to $t = 2\pi$ is $8aE(k)$, where $E(k)$ has the meaning given in Exercise 17 and $k^2 = 1 - (b/4a)^2$.

7

THE MEAN-VALUE THEOREM AND ITS GENERALIZATIONS

7.1 The mean-value theorem for derivatives

The mean-value theorem† for derivatives holds a position of great importance in calculus. In this chapter we explain the meaning of the theorem and discuss two of its significant generalizations—*Cauchy's mean-value formula* and *Taylor's formula with remainder.* Some of the simplest applications of these theorems are also discussed here. Further applications to extremal problems and indeterminate forms are given in Chapter 8.

The mean-value theorem and its extensions are easily deduced from a special case known as *Rolle's theorem.* This is done in Section 7.4. The proof of Rolle's theorem is postponed until Section 8.1 of the next chapter where it is deduced as a simple consequence of the so-called *extreme-value theorem* for continuous functions. This latter theorem and its proof are dealt with in complete detail in Chapter 8.

Before we state the mean-value theorem in its analytic form it may be helpful to look first at its geometric significance. Each of the curves shown in Figure 7.1 is the graph of a continuous function f with a tangent line above each point of the open interval (a, b). At the point $(c, f(c))$ shown in Figure 7.1(a) the tangent line is parallel to the chord AB joining the points $A = (a, f(a))$ and $B = (b, f(b))$. In Figure 7.1(b) are two points $(c_1, f(c_1))$ and $(c_2, f(c_2))$ where the tangent line is parallel to the chord AB, and in Figure 7.1(c) there are five such points. The mean-value theorem guarantees that there will be *at least one point* with this property.

To translate this geometric property into an analytic statement, we need only observe that parallelism of two lines means equality of their slopes. Since the slope of the chord AB is the quotient $[f(b) - f(a)]/(b - a)$ and since the slope of the tangent line at c is the derivative $f'(c)$, the above assertion states that

$$\frac{f(b) - f(a)}{b - a} = f'(c) \tag{7.1}$$

for *some* c in the open interval (a, b).

† The mean-value theorem is sometimes called the "law of the mean."

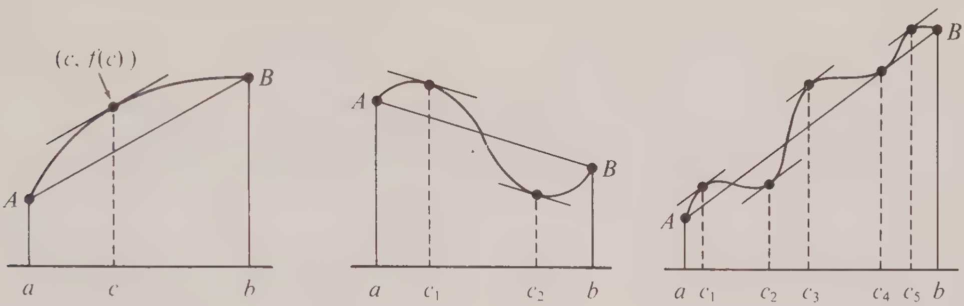


FIGURE 7.1 *Geometric significance of the mean-value theorem.*

To exhibit strong intuitive evidence for the truth of (7.1), we may think of $f(t)$ as the distance traveled by a moving particle at time t. Then the quotient on the left of (7.1) represents the *mean*, or *average* speed in the time interval $[a, b]$, and the derivative $f'(t)$ represents the instantaneous speed at time t. The equation asserts that there must be some instant when the instantaneous speed is equal to the average speed. For example, if the average speed during an automobile trip is 45 mph, then the speedometer must register 45 mph *at least once* during the trip.

The mean-value theorem may be stated formally as follows.

7–1 THEOREM. *Mean-value theorem for derivatives.* Assume $a \neq b$ and let f be a function that is continuous on the closed interval with endpoints a and b. Assume also that $f'(x)$ exists for each x between† a and b. Then there exists at least one point c between a and b such that

$$f(b) - f(a) = f'(c)(b - a) . \tag{7.2}$$

Notice that the theorem makes no assertion about the exact location of the one or more "mean values" c, except to say that they all lie *somewhere* between a and b. For some functions the position of the mean values may be specified exactly, but in most cases it is very difficult to make an accurate determination of these points. Nevertheless, the real usefulness of the theorem lies in the fact that many conclusions can be drawn from the knowledge of the mere *existence* of at least one mean value.

For example, we may use the theorem to show that

$$e^x > 1 + x \tag{7.3}$$

for all real $x \neq 0$. If we take $f(x) = e^x$ and apply Equation (7.2) with a replaced by 0 and b replaced by x, we obtain

$$e^x - 1 = e^c x ,$$

where c lies somewhere between 0 and x. To prove (7.3) it suffices to show that $e^c x > x$

† When we say x is *between* a and b we mean $a < x < b$ if $a < b$, and $b < x < a$ if $b < a$.

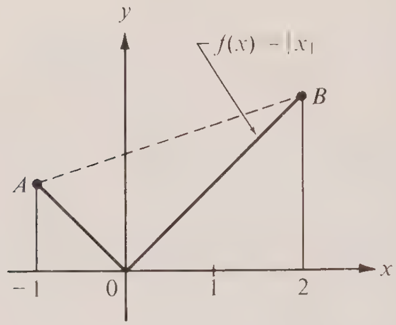


FIGURE 7.2 *A function for which the mean-value theorem is not applicable.*

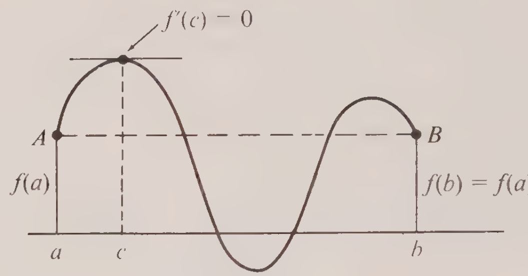


FIGURE 7.3 *Geometric interpretation of Rolle's theorem.*

for all $x \neq 0$. If $x > 0$, we have $0 < c < x$; hence $e^c > 1$ and $e^c x > x$. Therefore (7.3) is valid for positive x. If $x < 0$, we have $x < c < 0$, $e^c < 1$, and $e^c x > x$ (since x is negative). This shows that (7.3) holds for all real $x \neq 0$. The exact value of c was not required in the calculation—however we did use the fact that c was *between* 0 and x.

Note. It is important to realize that the conclusion of the mean-value theorem may fail to hold if there is any point between a and b where the derivative does not exist. For example, the function f defined by the equation $f(x) = |x|$ is continuous everywhere on the real axis and has a derivative everywhere except at 0. Figure 7.2 shows its graph above the interval $[-1, 2]$. The slope of the chord joining A and B is

$$\frac{f(2) - f(-1)}{2 - (-1)} = \frac{2 - 1}{3} = \frac{1}{3}$$

but the derivative is nowhere equal to $\frac{1}{3}$.

The mean-value theorem may be used to deduce properties of a function from a knowledge of the algebraic sign of its derivative. This is illustrated by the following theorem.

7–2 THEOREM. Let f be a function which is continuous on a closed interval $[a, b]$ and assume f has a derivative at each point of the open interval (a, b). Then we have:

(a) If $f'(x) > 0$ for every x in (a, b), f is strictly increasing on $[a, b]$.
(b) If $f'(x) < 0$ for every x in (a, b), f is strictly decreasing on $[a, b]$.
(c) If $f'(x) = 0$ for every x in (a, b), f is constant throughout $[a, b]$.

Note. If a function f is constant on a closed interval $[a, b]$, then its derivative is zero everywhere on the open interval (a, b). Part (c) of Theorem 7–2 provides the *converse* of this statement and is a result which we have used earlier without proof. (See Theorem 2–5 in Section 2.15.)

Proof. To prove (a) we must show that $f(x) < f(y)$ whenever $a \leq x < y \leq b$. Therefore, suppose $x < y$ and apply the mean-value theorem to the closed subinterval $[x, y]$.

We obtain

(7.4) $$f(y) - f(x) = f'(c)(y - x), \qquad \text{where} \quad x < c < y.$$

Since both $f'(c)$ and $y - x$ are positive, so is $f(y) - f(x)$, and this means $f(x) < f(y)$, as asserted. This proves (a), and the proof of (b) is similar. To prove (c), we use Equation (7.4) with $x = a$. Since $f'(c) = 0$ we have $f(y) = f(a)$ for every y in $[a, b]$, so f is constant on $[a, b]$.

7.2 A special case: Rolle's theorem

As we have mentioned earlier, the mean-value theorem will be deduced in Section 7.4 from one of its special cases called *Rolle's theorem* in honor of Michel Rolle (1652–1719), a French mathematician who discovered the result in 1690.

7– 3 THEOREM. *Rolle's theorem.* Let f be a function which is continuous everywhere on a closed interval $[a, b]$ and which has a derivative at each point of the open interval (a, b). Also, assume that

$$f(a) = f(b).$$

Then there is at least one interior point c of (a, b) such that

$$f'(c) = 0.$$

The geometric significance of Rolle's theorem is illustrated in Figure 7.3. The theorem simply asserts that the curve shown must have a horizontal tangent somewhere between a and b. This is a special case of the mean-value theorem because when $f(a) = f(b)$ the chord joining the points $A = (a, f(a))$ and $B = (b, f(b))$ is horizontal.

The basic idea behind the proof of Rolle's theorem is quite simple. If the function f is constant, then $f'(x) = 0$ for every x in (a, b) and we may choose *any* number c between a and b as a value for which $f'(c) = 0$. If f is *not* constant, then f takes on values larger than $f(a)$ or values smaller than $f(a)$ (or both), and the method of proof is suggested by the example shown in Figure 7.3. We simply choose for c the x-coordinate of the highest point on the graph above the chord AB (or of the lowest point below this chord if there are no points above). The tangent line at this point is horizontal and thus we have a number c between a and b for which $f'(c) = 0$, as asserted.

Although this argument is correct, it is based on two facts about the graph that may seem quite obvious when we look at any simple example but which are by no means trivial to prove for *every* function satisfying the conditions of Rolle's theorem. These two facts are: (1) *there always exists a highest point on the graph above the chord AB* (*or else a lowest point below this chord*), and (2) *the tangent line is horizontal at such a point.*

These statements describe, in geometric language, two very important theorems about real-valued functions that are expressed analytically and proved rigorously in Chapter 8. A purely analytic proof of Rolle's theorem, based on these theorems, is also given there (in Section 8.1).

7.3 Exercises

1. Show that on the parabola $y = x^2 + Ax + B$ the chord joining the points for which $x = a$ and $x = b$ is parallel to the tangent line at the point for which $x = (a + b)/2$.

2. Use Rolle's theorem to prove that the cubic equation $x^3 - 3x + b = 0$ cannot have more than one root in the interval $-1 \le x \le 1$, regardless of the value of b.

3. Define a function f as follows:

$$f(x) = \frac{3 - x^2}{2} \quad \text{if} \quad x \le 1\,, \qquad f(x) = \frac{1}{x} \quad \text{if} \quad x \ge 1\,.$$

(a) Sketch the graph of f for x in the interval $0 \le x \le 2$.

(b) Show that f satisfies the conditions of the mean-value theorem over the interval $[0, 2]$ and determine all the mean values provided by the theorem.

4. Let $f(x) = 1 - x^{2/3}$. Show that $f(1) = f(-1) = 0$, but that $f'(x)$ is never zero in the interval $[-1, 1]$. Explain how this is possible, in view of Rolle's theorem.

5. Show that $x^2 = x \sin x + \cos x$ for exactly two real values of x.

6. Let $f(x) = (x^2 - 1)e^{\alpha x}$. Show that $f'(x) = 0$ for exactly one x in the interval $(-1, 1)$ and that this x has the same sign as α.

7. Let f be a polynomial. A real number α is said to be a *zero* of f of multiplicity m if $f(x) = (x - \alpha)^m g(x)$, where $g(\alpha) \ne 0$.

(a) If f has r zeros in an interval $[a, b]$, prove that f' has at least $r - 1$ zeros, and in general, the kth derivative $f^{(k)}$ has at least $r - k$ zeros in $[a, b]$. (The zeros are to be counted as often as their multiplicity indicates.)

(b) If the kth derivative $f^{(k)}$ has *exactly* r zeros in $[a, b]$, what can you conclude about the number of zeros of f in $[a, b]$?

8. Show that the mean-value formula can be expressed in the form

$$f(x + h) = f(x) + hf'(x + \theta h)\,, \qquad \text{where} \quad 0 < \theta < 1\,.$$

Determine θ in terms of x and h when (a) $f(x) = x^2$; (b) $f(x) = x^3$. Keep x fixed, $x \ne 0$, and find the limit of θ in each case as $h \to 0$.

9. If f is continuous on $[a, b]$, let $F(x) = \int_a^x f(t)\,dt$ for $a \le x \le b$. Apply the mean-value theorem for derivatives to F and prove that

$$\int_a^b f(t)\,dt = f(c)(b - a)\,, \tag{7.5}$$

for some c between a and b. When f is positive, interpret this result geometrically by referring to areas. *Note.* The number $\int_a^b f(t)\,dt/(b - a)$ is called the *mean* or *average value* of f on $[a, b]$, and Equation (7.5) is called the *mean-value theorem for integrals.*

10. Show that the result of Exercise 9 may be expressed as follows:

$$\int_0^x f(t)\,dt = xf(\theta x)\,, \qquad \text{where} \quad 0 < \theta < 1\,,$$

if f is continuous in some closed interval containing the origin. Determine θ in terms of x when (a) $f(t) = t^n$ $(n \ge 0)$; (b) $f(t) = e^t$.

In Exercises 11 through 16, examine the sign of the derivative to determine those intervals in which the given function f is monotonic increasing and those in which it is decreasing.

11. $f(x) = x^2 - x - 1$.

12. $f(x) = x + \sin x$.

13. $f(x) = x + |\sin x|$.

14. $f(x) = \dfrac{\sqrt{x}}{x + 50}$ if $x \ge 0$.

15. $f(x) = x^2 2^{-x}$.

16. $f(x) = x^n e^{-x}$ if $n > 0,\ x \ge 0$.

17. Let I_n denote the open interval $(n\pi - \frac{1}{2}\pi, n\pi + \frac{1}{2}\pi)$, where n is an integer (positive, negative, or zero). If $\alpha \leq 1$, let $f(x) = \tan x - \alpha x$, whenever x is in any of the intervals I_n. Examine the sign of f' and show that there is exactly one x in each interval I_n such that $f(x) = 0$.

18. Let $f(x) = e^{-x} + x - 1$. Examine the sign of f' and prove that $e^{-x} > 1 - x$ whenever $x > 0$.

19. Let $f(x) = x + x^2 \sin(1/x)$ if $x \neq 0$, and let $f(0) = 0$.
 (a) Prove that $f'(0) = 1$. [*Hint.* Use the definition of $f'(0)$.]
 (b) Show that there is no interval of the form $(-h, h)$ about 0 in which f is monotonic. This shows that a positive derivative *at one point* does not necessarily mean that the function is monotonic in some neighborhood of this point.

In Exercises 20 through 25, establish the inequalities by examining the sign of the derivative of an appropriate function.

20. $e^x > 1 + x + \dfrac{x^2}{2}$ if $x > 0$.

21. $x - \dfrac{x^3}{3} < \arctan x$ if $x > 0$.

22. $\dfrac{2}{\pi} x < \sin x < x$ if $0 < x < \dfrac{\pi}{2}$.

23. $\dfrac{1}{x + \frac{1}{2}} < \log\left(1 + \dfrac{1}{x}\right) < \dfrac{1}{x}$ if $x > 0$.

24. $x - \dfrac{x^3}{6} < \sin x < x$ if $x > 0$.

25. $(x^b + y^b)^{1/b} < (x^a + y^a)^{1/a}$ if $x > 0, y > 0$, and $0 < a < b$.

Derive the inequalities in Exercises 26 through 33 by using the mean-value theorem.

26. $|\sin x - \sin y| \leq |x - y|$.

27. $|\arctan x - \arctan y| \leq |x - y|$.

28. $\dfrac{x - a}{x} < \log \dfrac{x}{a} < \dfrac{x - a}{a}$ if $0 < a < x$.

29. $ry^{r-1}(x - y) \leq x^r - y^r \leq rx^{r-1}(x - y)$ if $0 < y \leq x$ and $r > 1$.

30. $\tan x > x$ if $0 < x < \dfrac{\pi}{2}$.

31. $\dfrac{x}{1 + x^2} < \arctan x < x$ if $x > 0$.

32. $(1 + x)^\alpha < 1 + \alpha x$ if $x > 0$ or $-1 < x < 0$, and $0 < \alpha < 1$.

33. $e^a(x - a) < e^x - e^a < e^x(x - a)$ if $a < x$.

7.4 Cauchy's mean-value formula

Cauchy's generalization of the mean-value theorem deals with *two* functions defined on a common interval and, like the mean-value theorem itself, it is a result suggested by geometric considerations. As mentioned earlier, the geometric meaning of the mean-value theorem is this:

If a plane curve has a tangent line everywhere between two of its points A and B, then at least one of these tangent lines must be parallel to the chord AB.

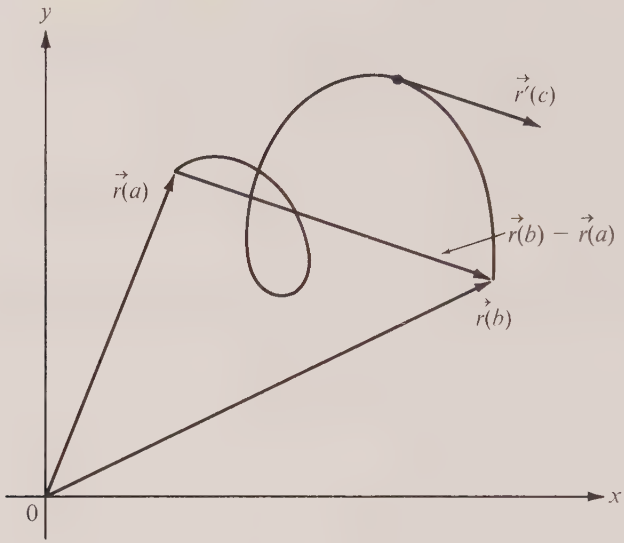


FIGURE 7.4 *Geometric motivation for Cauchy's mean-value formula.*

This geometric property holds not only when the curve is the graph of a real-valued function defined over an interval (as shown in Figure 7.1) but also for more general plane curves, such as the one shown in Figure 7.4. If a plane curve is described by a vector-valued function $\vec{r}$ defined on a parametric interval $[a, b]$, the vector joining the endpoints $\vec{r}(a)$ and $\vec{r}(b)$ is $\vec{r}(b) - \vec{r}(a)$, and for each value of t in the open interval (a, b) the derivative $\vec{r}'(t)$ is tangent to the curve. If we use Rolle's theorem, it is easy to prove that there is at least one value of t in (a, b), say $t = c$, for which the tangent vector $\vec{r}'(c)$ is parallel to the chord $\vec{r}(b) - \vec{r}(a)$, as indicated in Figure 7.4. We shall describe how this fact may be deduced from Rolle's theorem and this will lead at once to Cauchy's extension of the mean-value theorem.

Suppose the function $\vec{r}$ which describes the curve is given by

$$\vec{r}(t) = f(t)\,\vec{i} + g(t)\,\vec{j}\,, \tag{7.6}$$

where f and g are continuous on a closed interval $[a, b]$. Assume also that f and g have derivatives at each point of the open interval (a, b). Then the velocity vector is given by the equation

$$\vec{r}'(t) = f'(t)\,\vec{i} + g'(t)\,\vec{j} \qquad \text{if} \quad a < t < b$$

and we are to prove that $\vec{r}'(c)$ is parallel to $\vec{r}(b) - \vec{r}(a)$ for at least one c in the open interval (a, b). For this purpose we consider a new vector-valued function $\vec{w}$ defined by the equation

$$\vec{w}(t) = g(t)\,\vec{i} - f(t)\,\vec{j} \qquad \text{if} \quad a \le t \le b\,. \tag{7.7}$$

Since $\vec{w}(t) \cdot \vec{r}(t) = 0$ we see that each vector $\vec{w}(t)$ is perpendicular to $\vec{r}(t)$. Also, the derivative $\vec{w}'(t)$ is perpendicular to $\vec{r}'(t)$ and hence $\vec{w}'(t)$ is a vector normal to the curve. To prove that $\vec{r}'(c)$ is parallel to $\vec{r}(b) - \vec{r}(a)$ for some c it suffices to prove that $\vec{w}'(c)$ is perpendicular to $\vec{r}(b) - \vec{r}(a)$ or, what amounts to the same thing, that

$$\vec{w}'(c) \cdot [\vec{r}(b) - \vec{r}(a)] = 0 \tag{7.8}$$

for some c in (a, b). To prove this we consider the dot product

$$h(t) = \vec{w}(t) \cdot [\vec{r}(b) - \vec{r}(a)] \qquad \text{if} \quad a \le t \le b\,.$$

The real-valued function h so defined is continuous on the closed interval $[a, b]$ and its derivative $h'(t)$ exists for each t in the open interval (a, b). Also, it is easy to verify that $h(a) = h(b)$ [both are equal to $g(a)f(b) - f(a)g(b)$]. Therefore h satisfies all the conditions of Rolle's theorem and we conclude that $h'(c) = 0$ for some c in (a, b). Since $h'(t) = \vec{w}'(t) \cdot [\vec{r}(b) - \vec{r}(a)]$, this proves (7.8).

When Equation (7.8) is expressed in terms of the components of $\vec{w}$ and $\vec{r}$ it leads to an analytic statement about the two functions f and g which provides the generalization of the mean-value theorem referred to above. This extension may be stated as follows:

7–4 THEOREM. *Cauchy's mean-value formula.* Assume $a \neq b$ and let f and g be two functions, each continuous on the closed interval with endpoints a and b. Assume also that $f'(x)$ and $g'(x)$ exist for each x between a and b. Then there is at least one point c between a and b such that

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)\,. \tag{7.9}$$

Proof. If $a < b$, we define $\vec{r}$ and $\vec{w}$ by means of Equations (7.6) and (7.7), respectively. Then Equation (7.8) holds and we have

$$\vec{w}'(c) \cdot [\vec{r}(b) - \vec{r}(a)] = g'(c)\,[f(b) - f(a)] - f'(c)\,[g(b) - g(a)] = 0$$

for some c in (a, b). This is the same as Equation (7.9). If $a > b$, the result follows by interchanging the roles of a and b in the statement just proved.

Note. We have deduced Cauchy's formula as a consequence of Rolle's theorem. (The proof of Rolle's theorem appears in Section 8.1.) The mean-value theorem itself is the special case of Cauchy's formula obtained by taking $g(x) = x$.

7.5 Exercises

For each pair of functions given in Exercises 1 through 4, determine (in terms of a and b) a value of c for which Equation (7.9) is valid.

1. $f(x) = x$, $g(x) = x^2$.
2. $f(x) = x^2$, $g(x) = x^3$.
3. $f(x) = e^x$, $g(x) = e^{-x}$.
4. $f(x) = \sin x$, $g(x) = \cos x$.

5. (a) If f and g are continuous on $[a, b]$, use Cauchy's mean-value formula to prove that for some c in (a, b) we have

$$g(c) \int_a^b f(t)\,dt = f(c) \int_a^b g(t)\,dt .$$

(b) If $g(t) \neq 0$ for each t in (a, b), use part (a) to prove that

$$\int_a^b f(t)g(t)\,dt = f(c) \int_a^b g(t)\,dt \tag{7.10}$$

for some c in (a, b). This generalizes the mean-value theorem for integrals. (See Exercise 9 of Section 7.3.) Sometimes Equation (7.10) leads to a useful estimate for the integral of the product of two functions, especially if the integral of one of the factors is easy to compute. Examples are given in Exercise 6.

6. Use Equation (7.10) to establish the following inequalities:

$$\frac{1}{10\sqrt{2}} < \int_0^1 \frac{x^9}{\sqrt{1+x}}\,dx < \frac{1}{10}\,; \qquad \frac{1-e^{-100}}{200} < \int_0^{100} \frac{e^{-x}}{x+100}\,dx < \frac{1}{100}\,.$$

7. One of the following two statements is incorrect. Explain why it is wrong.

(a) The integral $\int_{2\pi}^{4\pi} (\sin t)/t\,dt > 0$ because $\int_{2\pi}^{3\pi} (\sin t)/t\,dt > \int_{3\pi}^{4\pi} |\sin t|/t\,dt$.

(b) The integral $\int_{2\pi}^{4\pi} (\sin t)/t\,dt = 0$ because, by (7.10), for some c between 2π and 4π we have

$$\int_{2\pi}^{4\pi} \frac{\sin t}{t}\,dt = \frac{1}{c} \int_{2\pi}^{4\pi} \sin t\,dt = \frac{\cos(2\pi) - \cos(4\pi)}{c} = 0\,.$$

8. If n is a positive integer, use Equation (7.10) to show that

$$\int_{\sqrt{n\pi}}^{\sqrt{(n+1)\pi}} \sin(t^2)\,dt = \frac{(-1)^n}{c}\,,$$

where $\sqrt{n\pi} < c < \sqrt{(n+1)\pi}$.

9. In this exercise we show how Equation (7.10) may be used to derive the so-called *second mean-value theorem for integrals*, which states that

$$\int_a^b f(x)g(x)\,dx = f(a) \int_a^c g(x)\,dx + f(b) \int_c^b g(x)\,dx\,, \tag{7.11}$$

for at least one c between a and b. The function g is assumed to be continuous, and f is assumed to have a derivative which is continuous and never zero on $[a, b]$. (This implies that f is monotonic on $[a, b]$.) Formula (7.11) is often used in place of (7.10) when the function g is zero at one or more points in (a, b).

(a) Let $G(x) = \int_a^x g(t)\,dt$. Use integration by parts to show that

$$\int_a^b f(x)g(x)\,dx = f(b)G(b) - \int_a^b f'(x)G(x)\,dx\,.$$

(b) Apply Equation (7.10) to the integral on the right and deduce (7.11).

10. (a) If ϕ'' is continuous and nonzero on $[a, b]$ and if there is a constant $m > 0$ such that $\phi'(t) \geq m$ for all t in $[a, b]$, use (7.11) to prove that

$$\left| \int_a^b \sin \phi(t)\,dt \right| \leq \frac{4}{m}\,.$$

[*Hint*. Multiply and divide the integrand by $\phi'(t)$.]

(b) If $a > 0$, show that $\left|\int_a^x \sin(t^2)\,dt\right| \leq 2/a$ for all $x > a$.

11. If a curve in 3-space is described by a vector-valued function $\vec{r}$ defined on a parametric interval $[a, b]$, prove that the scalar triple product $\vec{r}'(t) \cdot \vec{r}(a) \times \vec{r}(b)$ is zero for at least one t in (a, b).

Describe this result geometrically. Also, express the result in terms of the components of $\vec{r}$ and show that it includes Cauchy's mean-value formula as a special case.

7.6 Taylor's formula with remainder

Polynomials are among the simplest functions that occur in analysis. They are pleasant to work with in numerical computations because their values may be found by performing a finite number of multiplications and additions. In this section we shall use Rolle's theorem to show that certain functions, such as the exponential and trigonometric functions, can be approximated by polynomials. If the difference between a function and its polynomial approximation is sufficiently small, then we can, for practical purposes, compute with the polynomial in place of the original function.

There are many ways to approximate a given function f by polynomials, depending on what use is to be made of the approximation. Here we shall be interested in obtaining a polynomial which agrees with f and some of its derivatives at a given point. For example, the linear polynomial

$$g(x) = f(a) + f'(a)(x - a)$$

has the value $f(a)$ and the derivative $f'(a)$ at the point $(a, f(a))$. Geometrically this means that the graph of g is the tangent line of f at the point $(a, f(a))$.

If we approximate f by a quadratic polynomial P which agrees with f and its first two derivatives at a, then the graph of P will have not only the same tangent line but also the same *curvature* as f at the point $(a, f(a))$ and therefore P should provide a better approximation to f than the linear function g. This is illustrated by Figure 7.5 which shows the graph of $f(x) = e^x$ near the point (0, 1). The curve $y = e^x$ is approximated by its tangent line at (0, 1), given by $y = 1 + x$, and also by the parabola $y = 1 + x + \frac{1}{2}x^2$, which has the same curvature as the exponential curve at (0, 1).

We can hope to improve further the accuracy of the approximation by using polynomials which agree with f in the third and higher derivatives as well. It is easy to prove that such polynomials always exist. For example, suppose f has a derivative of order n at the point $x = 0$, and let us try to find a polynomial P such that

$$P(0) = f(0)\,, \quad P'(0) = f'(0)\,, \quad \ldots, \quad P^{(n)}(0) = f^{(n)}(0)\,. \tag{7.12}$$

There are $n + 1$ conditions to be satisfied so we try a polynomial of the form

$$P(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n\,, \tag{7.13}$$

with $n + 1$ coefficients to be determined. We shall use the conditions in (7.12) to determine these coefficients in succession.

First, we put $x = 0$ in (7.13) and we find $P(0) = c_0$, so $c_0 = f(0)$. Next, we differentiate both sides of (7.13) and then substitute $x = 0$ once more to find $P'(0) = c_1$; hence $c_1 = f'(0)$. If we differentiate (7.13) again and put $x = 0$, we find $P''(0) = 2c_2$, so $c_2 = f''(0)/2$. After differentiating k times, we find $P^{(k)}(0) = k!c_k$ and this gives us the formula

$$c_k = \frac{f^{(k)}(0)}{k!} \tag{7.14}$$

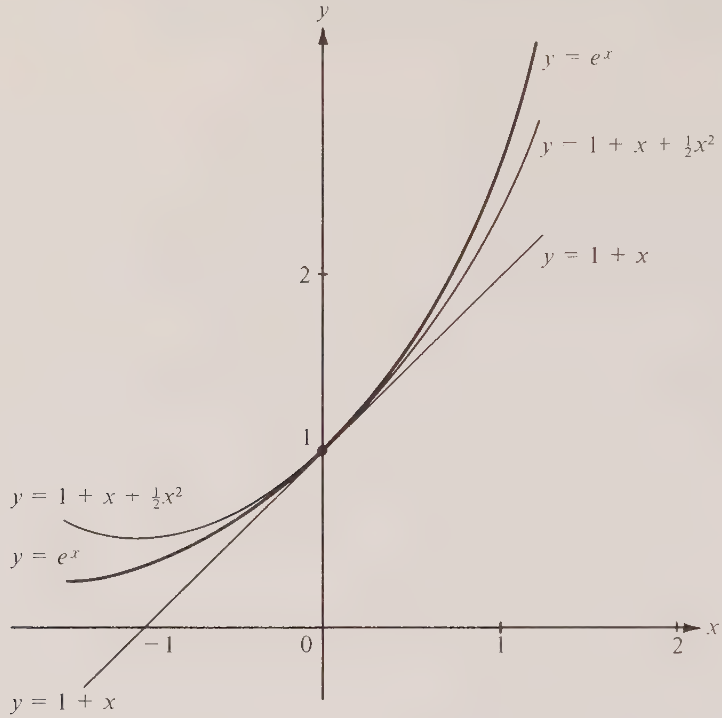


FIGURE 7.5 *Polynomial approximations to the curve $y = e^x$ near (0, 1).*

for $k = 0, 1, 2, \ldots, n$. [When $k = 0$ we interpret $f^{(0)}(0)$ to mean $f(0)$.] This argument proves that if a polynomial of degree† $\leq n$ exists which satisfies (7.12), then its coefficients are necessarily given by (7.14). Conversely, it is easy to verify that the polynomial P with coefficients given by (7.14) satisfies (7.12), and therefore we have the following theorem:

7– 5 THEOREM. Let f be a function with derivatives of order n at the point $x = 0$. Then there exists one and only one polynomial P of degree $\leq n$ which satisfies the $n + 1$ conditions

$$P(0) = f(0), \quad P'(0) = f'(0), \quad \ldots, \quad P^{(n)}(0) = f^{(n)}(0),$$

namely,

$$P(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k .$$

In the same way, we may show that there is one and only one polynomial of degree

† The degree of P will be equal to n if and only if $f^{(n)}(0) \neq 0$.

$\leq n$ which agrees with f and its first n derivatives at a point $x = x_0$. In fact, instead of (7.13), we may write P in powers of $x - x_0$ and proceed as before. If we evaluate the derivatives at x_0 in place of 0, we are led to the polynomial

$$P(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \,. \tag{7.15}$$

This is the one and only polynomial of degree $\leq n$ which satisfies the conditions

$$P(x_0) = f(x_0)\,, \quad P'(x_0) = f'(x_0)\,, \quad \ldots, \quad P^{(n)}(x_0) = f^{(n)}(x_0)\,,$$

and it is referred to as a *Taylor polynomial* in honor of the English mathematician Brook Taylor (1685–1731).

Example 1. When $f(x) = e^x$ we have $f^{(k)}(x) = e^x$ for all k, so $f^{(k)}(0) = e^0 = 1$ and the Taylor polynomial in (7.13) becomes

$$P(x) = \sum_{k=0}^{n} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}\,. \tag{7.16}$$

If we want a polynomial which agrees with f and its derivatives at the point $x_0 = 1$, we have $f^{(k)}(1) = e$ so $c_k = e/k!$ and (7.15) becomes

$$P(x) = \sum_{k=0}^{n} \frac{e}{k!}(x - 1)^k \,.$$

We turn now to a discussion of the error in the approximation of a function f by a Taylor polynomial P. The error is defined to be the difference $R(x) = f(x) - P(x)$. Thus, if f has a derivative of order n at x_0, we may write

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R(x)\,.$$

This is called *Taylor's formula*† *with remainder* $R(x)$, and it is useful whenever we can estimate the size of $R(x)$. In the next theorem we show that the remainder can be expressed in terms of the $(n + 1)$st derivative of f.

7–6 THEOREM. Assume f has a derivative of order $n + 1$ everywhere in the open interval (a, b) and that $f^{(n)}$ is continuous on the closed interval $[a, b]$. Let x and x_0 be two points of $[a, b]$, with $x \neq x_0$. Then there exists a point c between x and x_0 such that

† When Taylor's formula is applicable with $x_0 = 0$, it is usually called *Maclaurin's formula*, after Colin Maclaurin (1698–1746), a Scottish mathematician who used this formula in his *Treatise on fluxions*, published in 1742. Actually, the so-called Maclaurin formula appeared some 25 years earlier in the work of James Stirling (1692–1770).

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}. \tag{7.17}$$

In other words, the remainder in Taylor's formula is given by

$$R(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}. \tag{7.18}$$

Note. When $n = 0$ this theorem reduces to the mean-value theorem.

Although the remainder in (7.17) resembles the earlier terms, it is important to realize that the derivative $f^{(n+1)}(c)$ is evaluated at some *unknown* point c rather than at x_0, and that c depends on x and on n. This form of the remainder is especially useful when we can estimate the size of the $(n + 1)$st derivative. For example, if it is known that the $(n + 1)$st derivative is *bounded* on $[a, b]$, say $|f^{(n+1)}(t)| \leq M$ whenever $a \leq t \leq b$, then (7.18) implies the inequality

$$|R(x)| \leq \frac{M|x - x_0|^{n+1}}{(n+1)!}. \tag{7.19}$$

Examples and applications will be discussed presently. We turn now to the proof of Theorem 7–6.

Proof. Let P denote the Taylor polynomial in (7.15). Then $R(x) = f(x) - P(x)$ and, from the way P was constructed, we know that

$$R(x_0) = R'(x_0) = \cdots = R^{(n)}(x_0) = 0.$$

To prove (7.18) we keep x and x_0 *fixed* and introduce

$$g(t) = R(t) - A(t - x_0)^{n+1},$$

where $A = R(x)/(x - x_0)^{n+1}$, and t is an arbitrary point of $[a, b]$. This choice of the constant A makes $g(x) = 0$. We shall prove (7.18) by applying Rolle's theorem to the function g and to each of its first n derivatives. The first $n + 1$ derivatives of g are given by the following formulas:

$$g'(t) = R'(t) - (n+1)A(t - x_0)^n, \tag{7.20}$$

$$g''(t) = R''(t) - (n+1)nA(t - x_0)^{n-1}, \quad \ldots,$$

$$g^{(n+1)}(t) = R^{(n+1)}(t) - (n+1)!A = f^{(n+1)}(t) - (n+1)!A. \tag{7.21}$$

The last equation follows from the fact that $P^{(n+1)}(t) = 0$ for all t since P is a polynomial of degree $\leq n$.

Next we note that $g(x_0) = R(x_0) = 0$. Since we also have $g(x) = 0$, we may use Rolle's theorem to deduce that $g'(x_1) = 0$ for some x_1 between x_0 and x. From (7.20) we see that $g'(x_0) = R'(x_0) = 0$ and hence we may apply Rolle's theorem to g' to deduce that $g''(x_2) = 0$ for some x_2 between x_0 and x_1. This x_2 is also between x_0 and x. Since we also have $g''(x_0) = R''(x_0) = 0$, Rolle's theorem implies $g'''(x_3) = 0$ for some x_3 between x_0 and x_2. (This x_3 is also between x_0 and x.) In the same manner, after n applications of Rolle's theorem, we deduce that $g^{(n+1)}(c) = 0$ for some c between x_0 and x. When we use this in (7.21) we find $f^{(n+1)}(c) = (n + 1)!A$, which is the same as (7.18) since $A = R(x)/(x - x_0)^{n+1}$. This completes the proof.

Example 2. A Taylor polynomial of degree n for e^x was found in Example 1. Theorem 7–6 now enables us to write

$$e^x = \sum_{k=0}^{n} \frac{x^k}{k!} + \frac{e^c x^{n+1}}{(n+1)!}, \tag{7.22}$$

where c lies between 0 and x. This formula is valid for all real x and all integers $n \geq 0$. (The number c depends on both x and n.)

If we want to express e^x more generally in powers of $x - x_0$ we may evaluate the first n derivatives at x_0 and use (7.17), or we may simply replace x by $x - x_0$ directly in (7.22). By either method we find

$$e^x = e^{x_0} \sum_{k=0}^{n} \frac{(x - x_0)^k}{k!} + \frac{e^{c'}(x - x_0)^{n+1}}{(n+1)!},$$

where now c' lies between x_0 and x.

Polynomial approximations often enable us to obtain approximate numerical values for integrals that cannot be evaluated directly by the second fundamental theorem of calculus. A famous example is the integral

$$f(x) = \int_0^x e^{-t^2}\, dt$$

which occurs in probability theory and in many physical problems. It is known that the function f so defined is not one of the so-called *elementary functions.* That is to say, f cannot be obtained from polynomials, exponentials, logarithms, trigonometric or inverse trigonometric functions in a finite number of steps by using the operations of addition, subtraction, multiplication, division, or composition. Other examples which occur rather frequently in both theory and practice are the integrals

$$\int_0^x \frac{\sin t}{t}\, dt, \qquad \int_0^x \sin(t^2)\, dt, \qquad \int_0^x \sqrt{1 - k^2 \sin^2 t}\, dt.$$

(In the first of these it is understood that the quotient $(\sin t)/t$ is to be replaced by 1 when $t = 0$. In the third integral, k is a constant, $0 \leq k < 1$.) We conclude this section with an example which illustrates how Taylor's formula may be used to obtain an accurate estimate of the integral $\int_0^{1/2} e^{-t^2}\, dt$.

Example 3. If we use (7.22) with $n = 4$ and $x = -t^2$, we obtain

$$e^{-t^2} = 1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \frac{t^8}{4!} + r(t), \tag{7.23}$$

where $r(t) = -e^c t^{10}/5!$ and $-t^2 < c < 0$. Since $e^c < e^0 = 1$ we have $|r(t)| < (\frac{1}{2})^{10}/5! < 0.000\,009$ when $0 \le t \le \frac{1}{2}$. Integrating from 0 to $\frac{1}{2}$ we find

$$\int_0^{1/2} e^{-t^2}\,dt = \frac{1}{2} - \frac{1}{3\cdot 2^3} + \frac{1}{5\cdot 2^5\cdot 2!} - \frac{1}{7\cdot 2^7\cdot 3!} + \frac{1}{9\cdot 2^9\cdot 4!} + \theta\,,$$

where $|\theta| \le 0.000\,0045$. Rounding off to four decimals, we find $\int_0^{1/2} e^{-t^2}dt = 0.4613$.

7.7 Other forms of the remainder in Taylor's formula

The formula

$$R(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1} \tag{7.24}$$

for the remainder in Taylor's formula is due to Lagrange. The appearance of the unknown point c sometimes makes it troublesome to use the formula in practice, and other forms for the remainder have been developed which are often more convenient. One of these expresses $R(x)$ as an integral and is described in Exercise 8 of Section 7.8. Another expresses the remainder in terms of the derivative $f^{(n+1)}(x_0)$. To derive this from (7.24) we introduce the difference

$$E(x) = f^{(n+1)}(c) - f^{(n+1)}(x_0)\,,$$

and rewrite (7.24) as follows:†

$$R(x) = \frac{(x - x_0)^{n+1}}{(n+1)!}[f^{(n+1)}(x_0) + E(x)]\,.$$

This form is particularly useful when we know that $f^{(n+1)}$ is *continuous* at x_0 because, in this case, $E(x)$ is *small* when x is near x_0. In fact, if ϵ is any preassigned positive number, there exists a $\delta > 0$ such that $|f^{(n+1)}(x) - f^{(n+1)}(x_0)| < \epsilon$ whenever $0 < |x - x_0| < \delta$. Therefore, if the x in Theorem 7–6 is within a distance δ of x_0, so is the corresponding c (since c lies between x and x_0), and hence $|E(x)| < \epsilon$ for such x. This means that $E(x) \to 0$ as $x \to x_0$. Thus, when $f^{(n+1)}$ is continuous at x_0, Taylor's formula may be written in the form

$$f(x) = \sum_{k=0}^{n+1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{E(x)(x - x_0)^{n+1}}{(n+1)!}\,, \tag{7.25}$$

where $E(x) \to 0$ as $x \to x_0$.

When we are interested in using Taylor's formula for x near x_0, it is sometimes convenient to let $h = x - x_0$ and to rewrite (7.25) in powers of h as follows:

$$f(x_0 + h) = \sum_{k=0}^{n+1} \frac{f^{(k)}(x_0)}{k!}h^k + \frac{E(x_0 + h)h^{n+1}}{(n+1)!}\,, \tag{7.26}$$

† It should be noted that $E(x)$ depends on n as well as on x.

where $E(x_0 + h) \to 0$ as $h \to 0$. This tells us that, under the conditions stated, $f(x_0 + h)$ may be approximated by a polynomial in h of degree $n + 1$, and the error in this approximation is "small when compared to h^{n+1}." That is to say, the error divided by h^{n+1} tends to 0 as $h \to 0$.

Taylor's formula can be used to analyze the error made in approximating a *difference* $\Delta f(x_0; h) = f(x_0 + h) - f(x_0)$ by a *differential* $df(x_0; h) = f'(x_0)h$. In fact, if we isolate the terms in the sum in (7.26) corresponding to $k = 0$ and $k = 1$, we find

$$f(x_0 + h) - f(x_0) - f'(x_0)h = \sum_{k=2}^{n+1} \frac{f^{(k)}(x_0)}{k!} h^k + \frac{E(x_0 + h)h^{n+1}}{(n + 1)!}.$$

This shows that the difference $\Delta f(x_0; h) - df(x_0; h)$ is expressible in terms of h^2 and higher powers of h and it justifies the claims made in Section 2.19 where we asserted that $df(x_0; h)$ is a good approximation to $\Delta f(x_0; h)$ for small h.

7.8 Exercises

1. If the $(n + 1)$st derivative of f is zero throughout a closed interval $[a, b]$, prove that there exist constants $c_0, c_1, \ldots, c_n$ such that

$$f(x) = c_0 + c_1 x + \cdots + c_n x^n \qquad \text{for each } x \text{ in } [a, b].$$

2. Use Taylor's formula with remainder to derive the following polynomial approximations. In each case show that the remainder term satisfies the inequality given.

(a) $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \dfrac{x^7}{7!} + \cdots + (-1)^{n-1} \dfrac{x^{2n-1}}{(2n - 1)!} + r_n(x),$

where $|r_n(x)| \le |x|^{2n+1}/(2n + 1)!$.

(b) $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} + \cdots + (-1)^n \dfrac{x^{2n}}{(2n)!} + r_n(x),$

where $|r_n(x)| \le |x|^{2n+2}/(2n + 2)!$.

3. (a) Draw graphs of the polynomials $f_1(x) = x - x^3/3!$ and $f_2(x) = x - x^3/3! + x^5/5!$. Pay careful attention to the points where the curves cross the x-axis. Compare these graphs with that of $f(x) = \sin x$.

(b) Do the same as in part (a) for $f_1(x) = 1 - x^2/2!$, $f_2(x) = 1 - x^2/2! + x^4/4!$, and $f(x) = \cos x$.

4. (a) Derive the following identity, valid for all $t \ne -1$:

$$\frac{1}{1 + t} = 1 - t + t^2 - t^3 + \cdots + (-1)^{n-1}t^{n-1} + \frac{(-1)^n t^n}{1 + t}.$$

(b) Integrate both sides to obtain the exact formula

$$\log(1 + x) = \sum_{k=1}^{n} (-1)^{k-1} \frac{x^k}{k} + (-1)^n \int_0^x \frac{t^n}{1 + t}\, dt \qquad \text{if } |x| < 1.$$

(c) Show that

$$\left| \int_0^x \frac{t^n}{1 + t}\, dt \right| \le \frac{|x|^{n+1}}{1 - |x|} \qquad \text{if } |x| < 1.$$

5. Use an argument similar to that in Exercise 4 to obtain a polynomial approximation for $f(x) = \arctan x$ when $|x| < 1$. Make sure you give an estimate for the remainder term.

6. (a) If $0 \le x \le \frac{1}{2}$, show that $\sin x = x - x^3/3! + r(x)$, where $|r(x)| < (\frac{1}{2})^5/5!$.

(b) Use the estimate in part (a) to find an approximate value for the integral $\int_0^{\sqrt{2}/2} \sin(x^2)\,dx$. Make sure you give an estimate for the error.

7. Use the first three nonzero terms of Taylor's formula for $\sin x$ to find an approximate value for the integral $\int_0^1 (\sin x)/x\,dx$ and give an estimate for the error. [It is to be understood that the quotient $(\sin x)/x$ is equal to 1 when $x = 0$.]

8. Show that the remainder term in Taylor's formula (7.17) may be expressed as follows:

$$R(x) = \frac{1}{n!}\int_{x_0}^{x} (x - t)^n f^{(n+1)}(t)\,dt\,.$$

[*Hint.* Begin by writing $f(x) - f(x_0) = \int_{x_0}^{x} f'(t)\,dt$. Use integration by parts to obtain $f(x) = f(x_0) + f'(x_0)(x - x_0) + \int_{x_0}^{x} (x - t)f''(t)\,dt$, and repeat the process.]

7.9 The *o*- and *O*-notations

The importance of well-chosen notations in the development of mathematics was stressed in Chapter 2. There are two symbols introduced in 1909 by Landau† that are especially useful in calculations involving limits. One of these is known as the *o*-notation (the little-oh notation) and it is defined as follows:

DEFINITION OF THE LITTLE-OH NOTATION. We write

$$f(x) = o(g(x)) \qquad \text{as} \quad x \to x_0$$

whenever

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = 0\,.$$

The symbol $f(x) = o(g(x))$ is read "$f(x)$ is little-oh of $g(x)$," or "$f(x)$ is of smaller order than $g(x)$," and it is intended to convey the idea that for x near x_0, $f(x)$ is small compared to $g(x)$. For example, the relation $f(x) = o(1)$ as $x \to x_0$ means that $f(x) \to 0$ as $x \to x_0$. Also, we have $x = o(1)$, $x^2 = o(x)$, and in general, $x^m = o(x^n)$ as $x \to 0$, if $m > n$.

An equation of the form $f(x) = h(x) + o(g(x))$ is understood to mean that $f(x) - h(x) = o(g(x))$ or, in other words, $[f(x) - h(x)]/g(x) \to 0$ as $x \to x_0$. For example, $\sin x = x + o(x)$ as $x \to 0$. When the symbol $o(g(x))$ is used by itself it is understood to mean an unspecified $f(x)$ such that $f(x) = o(g(x))$.

Note. Formulas involving the *o*-notation are usually not reversible. For example, consider the following two statements:

(a) $o(x^3) = o(x^2)$ as $x \to 0$,
(b) $o(x^2) = o(x^3)$ as $x \to 0$.

† Edmund Landau (1877–1938) was a famous German mathematician who made many important contributions to mathematics. He is best known for his lucid books in analysis and in the theory of numbers.

The first one is true because it states that a relation of the form "$f(x)/x^3 \to 0$ as $x \to 0$" implies "$f(x)/x^2 \to 0$ as $x \to 0$." On the other hand, statement (b) is false because the relation "$f(x)/x^2 \to 0$ as $x \to 0$" does not necessarily imply "$f(x)/x^3 \to 0$ as $x \to 0$." This shows that transposition of terms in a correct formula involving o-symbols may lead to an incorrect formula. Some of the rules that govern algebraic manipulations with o-symbols are discussed in the exercises in the next section.

In the o-notation, Taylor's formula (7.25) can be written as follows:

$$f(x) = \sum_{k=0}^{n+1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + o((x - x_0)^{n+1}) \qquad \text{as} \quad x \to x_0 .$$

This expresses, in a brief way, the fact that the error term is small compared to $(x - x_0)^{n+1}$ when x is near x_0. In particular, when $f(x) = e^x$ and $x_0 = 0$ we may write

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n) \qquad \text{as} \quad x \to 0 .$$

The "big-oh notation" is defined as follows:

DEFINITION OF THE BIG-OH NOTATION. Let f and g be two real-valued functions defined on a set S of real numbers and assume g is nonnegative. We write

(7.27) $$f(x) = O(g(x)) \qquad \text{for } x \text{ in } S$$

if there exists a positive constant M such that

(7.28) $$|f(x)| \le Mg(x) \qquad \text{for every } x \text{ in } S .$$

The equation $f(x) = O(g(x))$ is often read as follows: "$f(x)$ is big-oh of $g(x)$" or "$f(x)$ is of the order of $g(x)$." In most applications the statement "for x in S" is omitted in Equation (7.27) because it is usually clear from the context just what values of x are to be taken into consideration. The actual value of the constant M in (7.28) is not important. To write an equation like (7.27) we need only know that *some* positive constant M exists satisfying (7.28). Note that the relation $f(x) = O(1)$ means that f is *bounded* on the set under consideration. For example, since $|\sin x| \le 1$ for all x we may write $\sin x = O(1)$ on any set S.

An equation of the form $f(x) = h(x) + O(g(x))$ is understood to mean that $f(x) - h(x) = O(g(x))$ or, in other words, that $|f(x) - h(x)| \le Mg(x)$ for some $M > 0$. When the symbol $O(g(x))$ is used by itself it stands for an unspecified $f(x)$ such that $f(x) = O(g(x))$. Formulas involving O-symbols are, like those involving o-symbols, not always reversible.

The O-notation is often used with special cases of Taylor's formula. For example, if we look at the polynomial approximation for e^{-t^2} given in Equation (7.23) we find that the remainder term is $r(t) = -e^c t^{10}/5!$, where $-t^2 < c < 0$. Since c is negative we have $e^c < 1$ and hence

$$|r(t)| < \frac{t^{10}}{5!} \qquad \text{for all real } t .$$

Therefore $r(t) = O(t^{10})$ for all t and Equation (7.23) may be written as follows:

$$e^{-t^2} = 1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \frac{t^8}{4!} + O(t^{10}) . \tag{7.29}$$

This equation, valid for all real t, tells us that we may approximate e^{-t^2} by a polynomial in t of degree 8 [the first five terms on the right of (7.29)] and that the error made in this approximation is of the order of magnitude of t^{10}. For small t the error is small compared to the other terms, but for large t the error may be rather large.

To use an equation like (7.29) to make an explicit numerical calculation as we did in Example 3 of Section 7.6, it is necessary to know something about the size of the constant M implied in the O-symbol. In this example we happen to know that we may use $M = 1/5!$. In many applications, however, an exact value for M is not needed.

We conclude this section with an example showing how Taylor's theorem and the o-notation can be used to calculate certain limits.

Example. If a and b are positive numbers, determine the limit

$$\lim_{x \to 0} \frac{a^x - b^x}{x} .$$

Solution. We cannot solve this problem by computing the limit of the numerator and denominator separately because the denominator tends to 0 and the quotient therorem on limits is not applicable. The numerator in this case also tends to 0 and the quotient is said to assume the "*indeterminate form* 0/0" as $x \to 0$. Taylor's formula and the o-notation often enable us to calculate the limit of an indeterminate form† like this one very simply. The idea is to approximate the numerator $a^x - b^x$ by a polynomial in x, then divide by x and let $x \to 0$. We could apply Taylor's formula directly to $f(x) = a^x - b^x$ but, since $a^x = e^{x \log a}$ and $b^x = e^{x \log b}$, it is simpler in this case to use the polynomial approximations already derived for the exponential function. If we begin with the linear approximation

$$e^t = 1 + t + o(t) \qquad \text{as} \quad t \to 0$$

and replace t by $x \log a$ and $x \log b$, respectively, we find

$$a^x = 1 + x \log a + o(x) \qquad \text{and} \qquad b^x = 1 + x \log b + o(x) \qquad \text{as} \quad x \to 0 .$$

Here we have used the fact that $o(x \log a) = o(x)$ and $o(x \log b) = o(x)$. If now we subtract and note that $o(x) - o(x) = o(x)$, we find $a^x - b^x = x(\log a - \log b) + o(x)$. Dividing by x and using the relation $o(x)/x = o(1)$, we obtain

$$\frac{a^x - b^x}{x} = \log \frac{a}{b} + o(1) \to \log \frac{a}{b} \qquad \text{as} \quad x \to 0 .$$

In the foregoing calculation we used a few properties of the o-symbol such as $o(x \log a) = o(x)$, $o(x) - o(x) = o(x)$, and $o(x)/x = o(1)$. These and other rules that apply when it becomes necessary to combine several terms involving the o- and O-symbols are discussed in the exercises in the next section. Further use of these symbols will be made from time to time in Chapters 8 and 9.

† A more detailed discussion of inderminate forms is given in Part II of Chapter 8.

7.10 Exercises

1. The equation $f(x) = O(1)$ implies that f is bounded. What is the meaning of the statement $f(x) = O(0)$?

2. The following equations describe, in an abbreviated form, some of the rules that apply in calculations involving the o- and O-symbols. For example, the equation in (a) means this: If $f_1(x) = O(g(x))$ and if $f_2(x) = O(g(x))$, then $f_1(x) \pm f_2(x) = O(g(x))$. This may be proved as follows: From the inequalities $|f_1(x)| \le M_1 g(x)$ and $|f_2(x)| \le M_2 g(x)$ we obtain $|f_1(x) \pm f_2(x)| \le |f_1(x)| + |f_2(x)| \le (M_1 + M_2)g(x)$.

Describe, in a similar way, the meaning of each equation in (b) through (h) and in each case prove that the assertion is correct.

(a) $O(g(x)) \pm O(g(x)) = O(g(x))$.
(b) $o(g(x)) \pm o(g(x)) = o(g(x))$.
(c) $o(cg(x)) = o(g(x)) \quad \text{if} \quad c \ne 0$.
(d) $O(cg(x)) = O(g(x)) \quad \text{if} \quad c > 0$.
(e) $o(g_1(x)) \cdot o(g_2(x)) = o(g_1(x)g_2(x))$.
(f) $O(g_1(x)) \cdot O(g_2(x)) = O(g_1(x)g_2(x))$.
(g) $o(g(x)) = O(g(x)), \qquad O(O(g(x))) = O(g(x))$.
(h) $O(o(g(x))) = o(g(x)), \qquad o(O(g(x))) = o(g(x))$.

3. (a) For $x > 1$, show that $f(x) = O(x^2)$ implies $f(x) = O(x^3)$. Show by an example that the converse is not true. In other words, for $x > 1$ we have $O(x^2) = O(x^3)$ but not $O(x^3) = O(x^2)$.
(b) For $0 < x < 1$, show that $f(x) = O(x^3)$ implies $f(x) = O(x^2)$. Is the converse true?

4. If $f(x) = O(g(x))$, prove or disprove each of the following statements:

(a) $\int_0^x f(t)\,dt = O\left(\int_0^x g(t)\,dt\right)$; (b) $f'(x) = O(g'(x))$.

5. Solve Exercise 4 when the O-symbol is replaced throughout by the o-symbol with $x \to 0$. (In part (a), assume g is positive.)

6. Use Taylor's formula or some other means to establish the following relations:
(a) $\sin x = O(|x|)$ for all x;
(b) $\dfrac{\sin x}{x} = O(1)$ for all $x \ne 0$;
(c) $\sin x = x + O(x^3)$ for $x > 0$.
(d) $\sin x = x - \dfrac{x^3}{6} + O(x^5)$ for $x > 0$.
(e) For what values of x is it true that $\sin x = x + O(x^2)$?

7. Show that $\cos x = 1 - \frac{1}{2}x^2 + o(x^3)$ as $x \to 0$. Use this to prove that $x^{-2}(1 - \cos x) \to \frac{1}{2}$ as $x \to 0$. In a similar way, find the limit of $x^{-4}(1 - \cos 2x - 2x^2)$ as $x \to 0$.

8. Given that $f(x) = x + O(x^3)$ and $g(x) = 1 + x + \frac{1}{2}x^2 + O(x^3)$ for all $x > 0$, find a quadratic polynomial approximation for the composition $g[f(x)]$, with an appropriate error term. What can you say about $g[f(x)]$ if $f(x) = x - \frac{1}{6}x^3 + O(x^5)$ and $g(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4)$?

9. Solve Exercise 8 when the O-symbol is replaced throughout by the o-symbol with $x \to 0$.

10. Prove that:

(a) $\dfrac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + O(x^{n+1}) \quad \text{if} \quad 0 < x \le \frac{1}{2}$.

(b) $\dfrac{1}{1+f(x)} = 1 + O(x) \quad \text{if} \quad |f(x)| \le \frac{1}{2} \quad \text{and} \quad f(x) = O(x) \quad \text{for} \quad x > 0$.

(c) $\dfrac{1}{1+f(x)} = 1 - f(x) + o(f(x)) \quad \text{if} \quad f(x) = o(1) \quad \text{as} \quad x \to 0$.

11. Use Taylor's formula and the o-notation with $x \to 0$ to prove that:

(a) $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} + o(x^6)$;
(b) $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} + o(x^5)$;
(c) $\log(1 + x) = x - \frac{1}{2}x^2 + o(x^2)$;
(d) $\tan x = x + \dfrac{x^3}{3} + \dfrac{2x^5}{15} + o(x^6)$.

12. Compute the limits of the following as $x \to 0$:

(a) $x^{-2}\left(\dfrac{\sin x}{x} - 1\right)$;

(b) $\dfrac{\tan 2x}{\sin 3x}$;

(c) $\dfrac{\log (1 + x)}{e^{2x} - 1}$;

(d) $\dfrac{1 - \cos^2 x}{x \tan x}$.

[*Hint*. Use Exercise 11.]

8

APPLICATIONS OF THE MEAN-VALUE THEOREM

Part I. Extremum Problems

8.1 Extreme values of functions. Proof of Rolle's theorem

Many natural questions in both pure and applied mathematics are concerned with a maximum or minimum property of some sort. For example, how should a rocket be designed so that it offers the least possible air resistance during its flight? What is the shape of the box of largest volume that can be formed from a given rectangular piece of material? What is the shortest path joining two points on a given surface? Questions of this type are called *extremum problems* and some of them can be attacked systematically with the use of differential calculus. As a matter of fact, the rudiments of differential calculus were first developed when Fermat tried to find general methods for answering questions about maxima and minima.

Before an extremum problem can be studied with the use of calculus, it must be formulated in terms of some real-valued function. In this section we shall introduce some of the definitions and theorems concerned with extreme values of functions of one real variable. Useful criteria for determining maxima or minima of such functions will be given here and in Section 8.3. The theory of extrema for functions of several variables is treated in Volume II.

Two of the theorems concerning extrema of functions of one variable were referred to previously in Section 7.2 when we discussed the idea behind the proof of Rolle's theorem. One of these states that a function f which is continuous everywhere on a closed interval $[a, b]$ must have a maximum somewhere in the interval. The other states that if the maximum occurs at an interior point, say at $x = c$, and if the derivative $f'(c)$ exists there, then $f'(c)$ is necessarily equal to zero. These theorems will be proved in this chapter, but before they can be discussed intelligently we must define first what we mean by the maximum of a function.

Actually, there are two different uses of the word "maximum" in calculus, and they are distinguished by means of the two prefixes *absolute* and *relative*. Their exact definitions are as follows:

DEFINITION OF ABSOLUTE MAXIMUM. Let f be a real-valued function defined on a set S

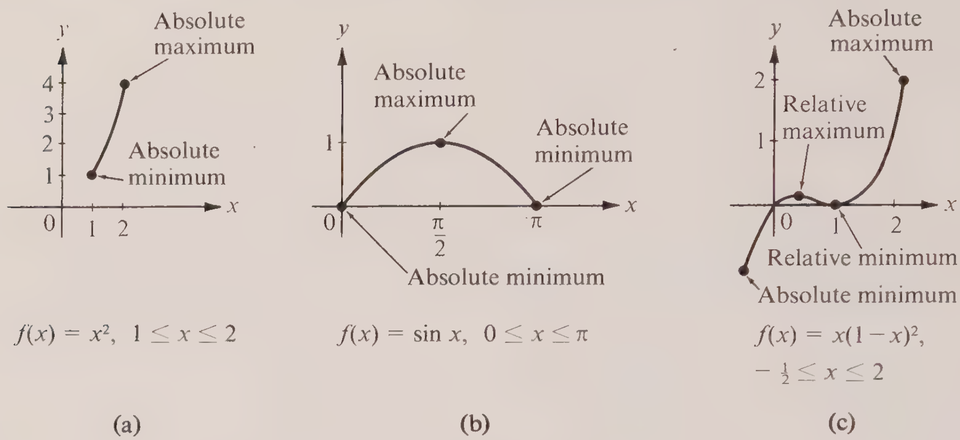


FIGURE 8.1 *Extrema of functions.*

of real numbers. Then f is said to have an absolute maximum on the set S if there is at least one point c in S such that

$$f(x) \leq f(c) \qquad \text{for all } x \text{ in } S\,.$$

The number $f(c)$ is called the absolute maximum value of f on S. [The concept of absolute minimum may be similarly defined if we use the inequality $f(x) \geq f(c)$.]

Examples of functions having absolute maxima and minima are illustrated in Figure 8.1.

DEFINITION OF RELATIVE MAXIMUM. Let f be a real-valued function defined on a set S of real numbers, and suppose c is a point in S. Then f is said to have a relative maximum at c if there is some open interval (a, b) containing c such that

$$f(x) \leq f(c) \qquad \text{for all } x \text{ which lie both in } S \text{ and in } (a, b)\,.$$

(The concept of relative minimum is similarly defined.)

In other words, a relative maximum at c is the absolute maximum in some neighborhood of c, although this need not be the absolute maximum on the whole of S. [An example is shown in Figure 8.1(c).] Of course, every absolute maximum is, in particular, a relative maximum.

Note. The terms *local* and *global* are sometimes used instead of *relative* and *absolute*. In general, a property of a set S which is described in terms of what happens in the immediate vicinity of a point in S is referred to as a *local property*. Thus the possession of a relative maximum is a local property. On the other hand, a property which involves the set as a whole is called a *global property*. Hence the possession of an absolute maximum on a set is a global property.

DEFINITION OF EXTREMUM. A number which is either a relative maximum or a relative minimum of a function f is called an extreme value or an extremum of f.

These concepts are illustrated in Figure 8.1. In Figure 8.1(a) the set S is the closed interval $[1, 2]$ and $f(x) = x^2$. The absolute minimum of f on S is $f(1) = 1$ and the absolute maximum is $f(2) = 4$.

In Figure 8.1(b), S is the closed interval $[0, \pi]$ and $f(x) = \sin x$. The absolute minimum, which occurs at both endpoints of the interval, is 0. The absolute maximum is $f(\frac{1}{2}\pi) = 1$.

In Figure 8.1(c), $S = [-\frac{1}{2}, 2]$ and $f(x) = x(x - 1)^2$. The absolute minimum occurs at $x = -\frac{1}{2}$ and is $f(-\frac{1}{2}) = -9/8$. The absolute maximum occurs at $x = 2$ and is $f(2) = 2$. At $x = \frac{1}{3}$ the function has a relative maximum, $f(\frac{1}{3}) = 4/27$, and at $x = 1$ there is a relative minimum, $f(1) = 0$.

A function may or may not have an absolute maximum or minimum on a given set. To illustrate, suppose we modify the example shown in Figure 8.1(a) by replacing the closed interval $S = [1, 2]$ by the *open* interval $T = (1, 2)$, and suppose we let $f(x) = x^2$ as before. Although we have $f(x) < 4$ for every x in T, the number 4 does not qualify as the absolute maximum value of f on the set T because there is no point c *in* T where $f(c) = 4$. Also, no number less than 4 can qualify as a maximum. (The reader should verify this.) Therefore f has *no* absolute maximum on T. Similarly, it is easy to verify that f has no absolute minimum on T.

Another example of a function with no absolute maximum is shown in Figure 8.2. The set S is the half-open interval $(0, 2]$ and the function f is defined on S by the equation

$$f(x) = \frac{1}{x} \quad \text{if} \quad 0 < x \leq 2 .$$

Although this function has an absolute minimum at $x = 2$ there is no absolute maximum in this case. This function fails to have a maximum because it is unbounded on S.

The two foregoing examples show that the property of possessing an absolute maximum or minimum on a set depends not only on the nature of the function but also on the set under consideration.

Now we are ready to state the first of the two important theorems referred to above.

8–1 THEOREM. *Extreme-value theorem for continuous functions.* Let f be a function which is continuous everywhere on a closed interval $[a, b]$. Then f has an absolute maximum and an absolute minimum on $[a, b]$.

In other words, if a function f is continuous on a *closed interval* $[a, b]$ there will always be at least one point x_1 in $[a, b]$ such that $f(x_1)$ is the maximum value of f in $[a, b]$ and at least one point x_2 in $[a, b]$ such that $f(x_2)$ is the minimum value of f in $[a, b]$. The foregoing examples show why it is necessary to assume that the interval $[a, b]$ is *closed.* The conclusion does not necessarily hold for functions continuous on an open interval or on a half-open interval. The proof of the extreme-value theorem makes use of the least-upper-bound axiom for the real-number system and is given in Section 8.8.

The second basic theorem concerned with extreme values is the following:

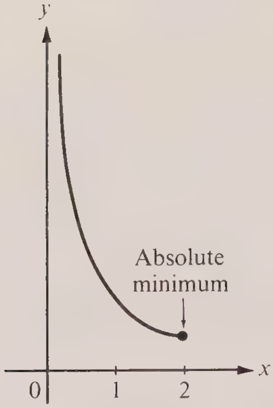


FIGURE 8.2 *A function with no absolute maximum:* $f(x) = 1/x, \quad 0 < x \leq 2.$

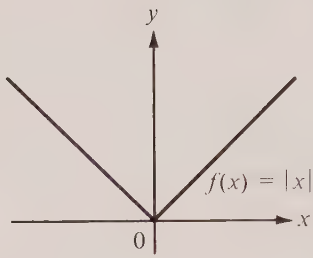


FIGURE 8.3 *There is an extremum at 0, but $f'(0)$ does not exist.*

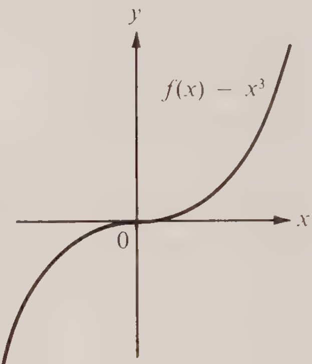


FIGURE 8.4 *Here $f'(0)$ equals 0 but there is no extremum at 0.*

8– 2 THEOREM. *Vanishing of the derivative at an extremum.* Let f be defined on an interval (a, b), and assume that f has a relative maximum or a relative minimum at an interior point c of (a, b). If the derivative $f'(c)$ exists at c, then $f'(c) = 0$.

Before we discuss the proof of this theorem we wish to emphasize one or two important points related to it. First of all, the existence of an extremum at c does not necessarily imply $f'(c) = 0$. The example $f(x) = |x|$, shown in Figure 8.3, has a relative minimum at $x = 0$ but, of course, there is no derivative at 0. Theorem 8–2 *assumes the existence* of the derivative at c. It also assumes that c is an *interior point* of (a, b). In the example shown in Figure 8.1(a) we have $f(x) = x^2$ over the closed interval $[1, 2]$. This function takes on its extreme values at the endpoints and $f'(x)$ is never zero in $[1, 2]$.

Next we note that the converse of Theorem 8–2 is not true. If we know that $f'(c) = 0$ at an interior point, it does not necessarily follow that f has an extremum at c. For example, consider the case in which $f(x) = x^3$ and $c = 0$. (See Figure 8.4.) Here we have $f'(0) = 0$ but f is increasing in every interval containing 0 and hence there is no extremum at 0. Theorem 8–2 tells us only that the extrema at interior points at which the derivative exists are to be found *among* the roots of the equation $f'(x) = 0$.

If a function f is continuous on a closed interval $[a, b]$ and has a derivative at each interior point, then the only places where extrema can occur are:

(i) at the endpoints a and b,
(ii) at those interior points x where $f'(x) = 0$.

Points of type (ii) are called *stationary values* of f. The example in Figure 8.4 shows that the condition $f'(c) = 0$ is, by itself, not enough to guarantee an extremum at a stationary value c. Therefore, to decide whether there is a maximum or a minimum (or neither) at a stationary value c we need more information about f. Sometimes the nature of the stationary value may be determined from the algebraic sign of the derivative near c.

For example, suppose there is a closed subinterval $[x_1, x_2]$ of (a, b) in which c is the *only* stationary value, say $a < x_1 < c < x_2 < b$, and $f'(c) = 0$. If in this interval the derivative changes sign from positive to negative or from negative to positive, then there definitely is an extremum at c. In fact, if $f'(x) > 0$ for $x < c$ and $f'(x) < 0$ for $x > c$, then f is strictly increasing on the interval $[x_1, c]$ and strictly decreasing on $[c, x_2]$, as shown in Figure 8.5(a), and there is a relative maximum at c. On the other hand, if the derivative changes from negative to positive as shown in Figure 8.5(b), then there is a relative minimum at c. If $f'(x)$ has the same sign on both sides of c, then f is strictly monotonic on $[x_1, x_2]$ and there is no extremum at c. The example in Figure 8.4 has a positive derivative on both sides of c.

The proof of Theorem 8–2 is based on a simple principle which states that if a derivative $f'(c)$ is not zero, then the difference quotient $[f(x) - f(c)]/(x - c)$ has the same sign as $f'(c)$ if x is sufficiently near to c. This principle will be used again in some of the later work and it is convenient to state and prove it here as a formal theorem. Then we shall show how Theorem 8–2 follows easily from this principle.

8–3 THEOREM. Let f be defined on an open interval (a, b) and assume $f'(c)$ exists and is nonzero at some point c in (a, b). Then there is an open subinterval $(c - \delta, c + \delta)$ of (a, b) in which the difference quotient

$$\frac{f(x) - f(c)}{x - c}$$

has the same sign as $f'(c)$. In particular, if $f'(c) > 0$, then in this subinterval we have

$$f(x) > f(c) \quad \text{if} \quad x > c \qquad \text{and} \qquad f(x) < f(c) \quad \text{if} \quad x < c\,.$$

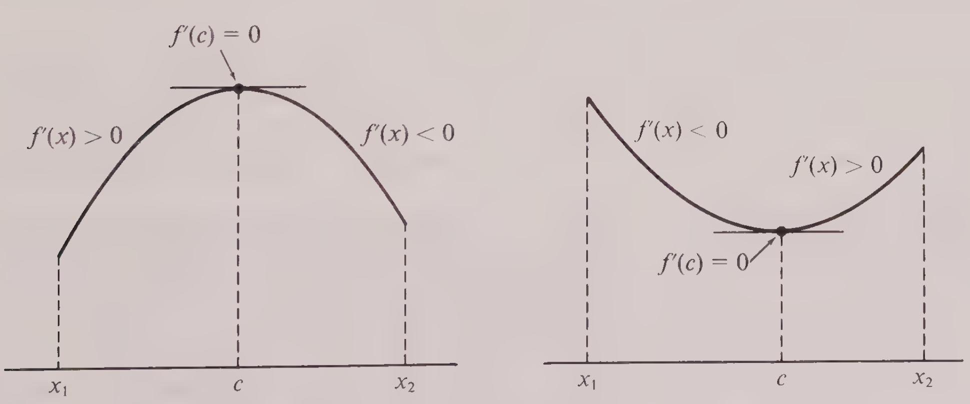


FIGURE 8.5 *An extremum occurs when the derivative changes sign.*

Proof. Since $f'(c)$ exists we have

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c) .$$

This means that for every preassigned positive number ϵ there corresponds another positive number δ (depending on ϵ) such that

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \epsilon \tag{8.1}$$

for all x satisfying $c - \delta < x < c + \delta$. In particular, there is a δ that corresponds to the choice $\epsilon = \frac{1}{2}|f'(c)|$. [This ϵ is positive because $f'(c) \neq 0$.] For this δ the inequality in (8.1) states that

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \tfrac{1}{2}|f'(c)| \qquad \text{whenever} \quad c - \delta < x < c + \delta .$$

If $f'(c) > 0$ this implies

$$\frac{1}{2}f'(c) < \frac{f(x) - f(c)}{x - c} < \frac{3}{2}f'(c) ,$$

which proves that the difference quotient is also positive for each x in $(c - \delta, c + \delta)$. If $f'(c) < 0$ we find, in a similar way, that the difference quotient is negative for x in $(c - \delta, c + \delta)$, and this proves Theorem 8–3.

Proof of Theorem 8–2. We are given that f has an extremum at an interior point c of (a, b) and that $f'(c)$ exists, and we are to prove that $f'(c) = 0$. We shall do this by showing that each of the inequalities $f'(c) > 0$ and $f'(c) < 0$ leads to a contradiction.

If $f'(c) > 0$, then by Theorem 8–3 there is an interval $(c - \delta, c + \delta)$ about c in which $f(x) < f(c)$ when $x < c$ and $f(x) > f(c)$ when $x > c$. This contradicts the assumption that f has an extremum at c. Hence the inequality $f'(c) > 0$ is impossible. A similar argument shows that we cannot have $f'(c) < 0$. Therefore $f'(c) = 0$, as asserted.

Proof of Rolle's theorem as a consequence of Theorems 8–1 and 8–2. In the hypothesis of Rolle's theorem we are given a function f which is continuous everywhere on a closed interval $[a, b]$ and which has a derivative at each point of the open interval (a, b). In addition, we are given that $f(a) = f(b)$. The conclusion is that $f'(c) = 0$ for at least one c satisfying $a < c < b$.

The proof can be presented in two ways. One is based on the idea already discussed in Section 7.2 and it makes use of Theorems 8–1 and 8–2. We shall leave it as an exercise for the reader to translate this idea into an analytic proof. There is another way of formulating the proof, and it may be instructive to describe it here.

We assume that $f'(x) \neq 0$ for every x in the open interval (a, b) and we arrive at a contradiction as follows: By the extreme-value theorem, f must take on its absolute maximum M and its absolute minimum m somewhere in the closed interval $[a, b]$. Theorem 8–2 tells us that neither extreme value can be taken on at any interior point (otherwise the derivative would vanish there). Hence both extreme values are taken on at the end-points a and b. But since $f(a) = f(b)$, this means that $m = M$ and hence f is constant on $[a, b]$. This contradicts the fact that $f'(x) \neq 0$ for all x interior to (a, b). It follows that $f'(c) = 0$ for at least one c satisfying $a < c < b$. This proves that Rolle's theorem is a consequence of Theorems 8–1 and 8–2. When we prove Theorem 8–1 in Section 8.8 the proof of Rolle's theorem will be complete in all details.

8.2 Exercises

In Exercises 1 through 8, (a) find all points c such that $f'(c) = 0$; (b) examine the sign of f' and determine those intervals in which f is monotonic; (c) locate the points (if any) where f has a relative maximum and where f has a relative minimum; (d) make a sketch of the graph of f.

1. $f(x) = (x - 1)^2(x + 2)$.
2. $f(x) = x^3 - 6x^2 + 9x + 5$.
3. $f(x) = 2 + (x - 1)^4$.
4. $f(x) = x/(1 + x^2)$.
5. $f(x) = \sin x - \cos x$.
6. $f(x) = \sin x - 3 \cos (x/3)$.
7. $f(x) = xe^{-x}$.
8. $f(x) = x + 1/x \quad$ if $\quad x \neq 0$.

In Exercises 9 through 16, find the absolute maximum and minimum values of f (whenever they exist) in the designated intervals.

9. $f(x) = 2^x, \qquad [-1, 5]$.
10. $f(x) = \sqrt{5 - 4x}, \qquad [-1, 1]$.
11. $f(x) = \dfrac{x^2 + 100}{x^2 - 25}, \qquad [-1, 3]$.
12. $f(x) = x + \dfrac{1}{x}, \qquad \left[\dfrac{1}{10}, 10\right]$.
13. $f(x) = x^2 - 3x + 2, \qquad [-3, 10]$.
14. $f(x) = |x^2 - 3x + 2|, \qquad [-3, 10]$.
15. $f(x) = \dfrac{\log^2 x}{x}, \qquad [1, 3]$.
16. $f(x) = e^x \sin x, \qquad [0, 2\pi]$.

In Exercises 17 through 20, find the local maxima and minima (when they exist) of the given functions.

17. $f(x) = \dfrac{10}{1 + \sin^2 x}$.
18. $f(x) = |x|e^{-|x-1|}$.
19. $f(x) = e^{-x} \displaystyle\sum_{k=0}^{n} \frac{x^k}{k!}$.
20. $f(x) = \arctan x - \log (1 + x^2)^{1/2}$.

8.3 Other tests for extrema at a stationary value

As we have already mentioned, the condition $f'(c) = 0$ by itself does not guarantee an extremum at a stationary value c. In Section 8.1 we learned how to determine the nature of a stationary value c by examining the algebraic sign of the derivative at points near c. There are other tests that involve *only* the stationary value c but they require a knowledge of the higher derivatives at c. The simplest of these is described in the next theorem.

8–4 THEOREM. *Second-derivative test for an extremum at a stationary value.* Assume f has a derivative everywhere in an open interval (a, b) and suppose c is a station-

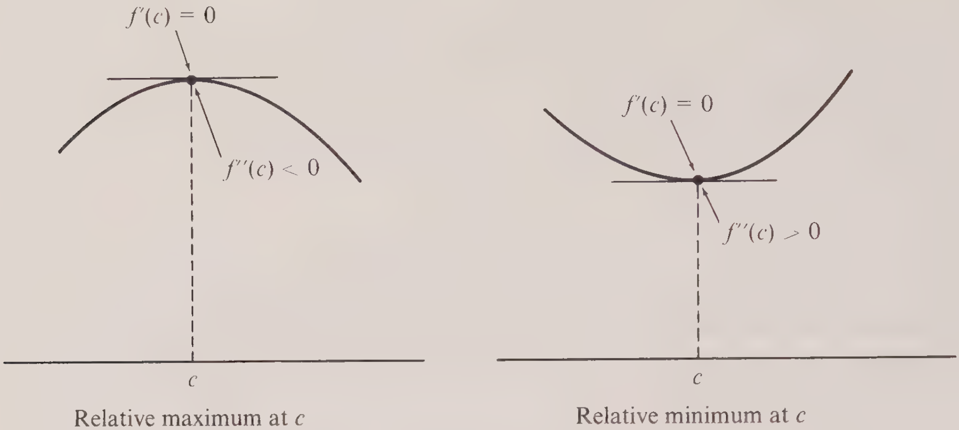


FIGURE 8.6 *The second-derivative test.*

ary value [that is, $a < c < b$ and $f'(c) = 0$]. Assume also that the second derivative $f''(c)$ exists and is different from zero. Then:
(a) If $f''(c) > 0$, f has a relative minimum at c.
(b) If $f''(c) < 0$, f has a relative maximum at c.

The two cases are illustrated in Figure 8.6.

Proof. Let $g(x) = f'(x)$ and consider case (a). The hypothesis $f''(c) > 0$ means that $g'(c) > 0$. Therefore, by Theorem 8–3 (applied to g), there is some interval $(c - \delta, c + \delta)$ in which we have

$$g(x) < g(c) \quad \text{if} \quad x < c \qquad \text{and} \qquad g(x) > g(c) \quad \text{if} \quad x > c\,,$$

or, in other words, since $g(c) = 0$, we have

$$f'(x) < 0 \quad \text{if} \quad c - \delta < x < c \qquad \text{and} \qquad f'(x) > 0 \quad \text{if} \quad c < x < c + \delta\,.$$

This knowledge of the sign of f' implies that f has a relative minimum at c, as asserted. This proves part (a), and the proof of (b) is similar.

In general, no conclusion regarding extrema can be drawn from just the knowledge that $f''(c) = 0$ at a stationary value. Each of the three examples

$$f(x) = x^3\,, \qquad g(x) = x^4\,, \qquad h(x) = -x^4$$

has a stationary value at $x = 0$ and each has second derivative zero at 0. However, f has no extremum at 0, g has a relative minimum at 0, and h has a relative maximum at 0.

When $f''(c) = 0$, further analysis requires a knowledge of the higher derivatives at c. If we use Taylor's formula with remainder, a sufficient condition may be given as described in the following theorem.

8–5 THEOREM. *nth-derivative test.* Suppose f has an nth derivative $f^{(n)}$ everywhere on an open interval (a, b), and assume that for some c in (a, b) we have

$$f'(c) = f''(c) = \cdots = f^{(n-1)}(c) = 0\,, \qquad \text{but} \qquad f^{(n)}(c) \neq 0\,.$$

Assume also that $f^{(n)}$ is continuous at c. Then for n even, f has a local minimum at c if $f^{(n)}(c) > 0$, and a local maximum at c if $f^{(n)}(c) < 0$. If n is odd, there is neither a local maximum nor a local minimum at c.

Proof. Since $f^{(n)}(c) \neq 0$, and since $f^{(n)}$ is continuous at c, there is an open interval $(c - \delta, c + \delta)$ about c in which the derivative $f^{(n)}(x)$ has the same sign as $f^{(n)}(c)$. (See Theorem 2–9 of Section 2.30.) By Taylor's formula, for every x in $(c - \delta, c + \delta)$, $x \neq c$, we can write

$$f(x) - f(c) = \frac{f^{(n)}(x_1)}{n!}(x - c)^n\,, \tag{8.2}$$

where x_1 is between x and c; this x_1 is also in $(c - \delta, c + \delta)$.

If n is *even*, the right-hand side of (8.2) has the same sign as $f^{(n)}(c)$, which means that $f(x) > f(c)$ if $f^{(n)}(c) > 0$ and $f(x) < f(c)$ if $f^{(n)}(c) < 0$. This proves the theorem for n even.

If n is *odd*, the right-hand side of (8.2) has the same sign as $f^{(n)}(c)$ if $x > c$ but the opposite sign if $x < c$. Thus, if $f^{(n)}(c) > 0$ we have $f(x) > f(c)$ when $x > c$, but $f(x) < f(c)$ when $x < c$, and hence there is no extremum at c. A similar argument shows there is no extremum at c if $f^{(n)}(c) < 0$. This proves the theorem for n odd.

8.4 Worked examples of extremum problems

In this section we shall illustrate how some of the foregoing theorems on extrema may be applied in a variety of problems.

Example 1: Method of least squares. Given n real numbers $a_1, \ldots, a_n$, determine the value of x for which the sum

$$\sum_{k=1}^{n} (x - a_k)^2$$

is a minimum.

Solution. Denote the sum in question by $f(x)$. Then

$$f'(x) = \sum_{k=1}^{n} 2(x - a_k) = 2nx - 2\sum_{k=1}^{n} a_k = 2n(x - \bar{a})\,,$$

where $\bar{a}$ denotes the arithmetic mean of $a_1, \ldots, a_n$; that is,

$$\bar{a} = \frac{1}{n}\sum_{k=1}^{n} a_k\,.$$

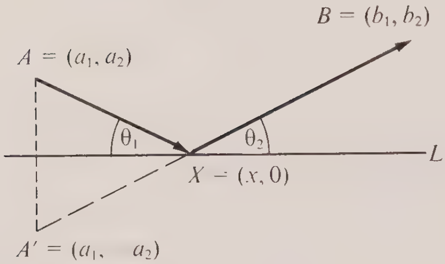


FIGURE 8.7 *Example 3.*

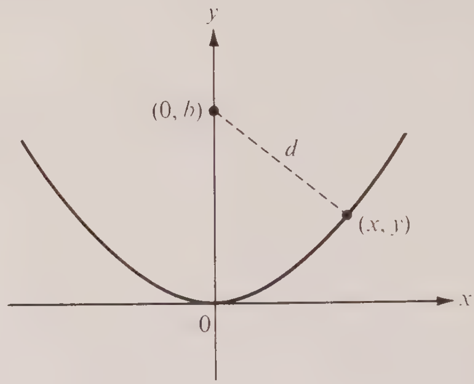


FIGURE 8.8 *Example 4.*

The only stationary value of f is $x = \bar{a}$. Since $f'(x) < 0$ for every $x < \bar{a}$ and $f'(x) > 0$ for every $x > \bar{a}$, it follows (by Theorem 7–2) that $f(x) > f(\bar{a})$ for all $x \neq \bar{a}$. Therefore, $x = \bar{a}$ is the required value. The result of this example is the starting point for the so-called "method of least squares."

Example 2. Prove that, among all rectangles having a given perimeter, the square is the one with maximum area.

Solution. Let p denote the given perimeter, and let x and y denote the sides of a general rectangle. Then x and y are related by the equation

$$2x + 2y = p\,. \tag{8.3}$$

The quantity to be maximized is the area xy, subject to the restriction (8.3). Solving (8.3) for y, we get $y = \frac{1}{2}p - x$ and this allows us to express the area in terms of x alone. The problem then amounts to finding the maximum of the function A given by

$$A(x) = x(\tfrac{1}{2}p - x) = \tfrac{1}{2}px - x^2\,.$$

Although this equation defines $A(x)$ for all real x, the conditions of the problem require that we consider only those x in the interval $0 \leq x \leq \frac{1}{2}p$. At the endpoints we have $A(0) = A(\frac{1}{2}p) = 0$ which is clearly not the maximum area. Therefore the maximum occurs in the interior at a stationary value. To find all stationary values we compute the derivative $A'(x) = \frac{1}{2}p - 2x$. This is zero only when $x = \frac{1}{4}p$. From what was just said we know that the absolute maximum of A on $[0, \frac{1}{2}p]$ must occur when $x = \frac{1}{4}p$. [We can also verify this by examining the second derivative which has the constant value $A''(x) = -2$.] The corresponding value of y is also $\frac{1}{4}p$, and hence the maximizing rectangle is a square.

Example 3: Fermat's reflection principle. Given two points A and B on the same side of a line L (all in the same plane), determine that point X on L for which the broken line AXB shall have minimum length.

Solution. We pass the x-axis through L and assign coordinates (a_1, a_2) and (b_1, b_2) to A and B, respectively, as shown in Figure 8.7. (There is no loss in generality if we assume $b_1 \geq a_1$.) If X has coordinates $(x, 0)$, the problem is to minimize the function f for which

$$f(x) = |\overrightarrow{AX}| + |\overrightarrow{XB}| = \sqrt{(x - a_1)^2 + a_2^2} + \sqrt{(b_1 - x)^2 + b_2^2}.$$

The derivative is

$$(8.4) \qquad f'(x) = \frac{x - a_1}{\sqrt{(x - a_1)^2 + a_2^2}} - \frac{b_1 - x}{\sqrt{(b_1 - x)^2 + b_2^2}} = \frac{x - a_1}{|\overrightarrow{AX}|} - \frac{b_1 - x}{|\overrightarrow{XB}|}.$$

If θ_1 denotes the angle between the vectors $\overrightarrow{AX}$ and $\vec{i}$, and if θ_2 denotes the angle between $\overrightarrow{XB}$ and $\vec{i}$, where $0 < \theta_1 < \pi$ and $0 < \theta_2 < \pi$, then we have

$$\cos\theta_1 = \frac{\overrightarrow{AX}\cdot\vec{i}}{|\overrightarrow{AX}|} = \frac{x - a_1}{|\overrightarrow{AX}|} \qquad \text{and} \qquad \cos\theta_2 = \frac{\overrightarrow{XB}\cdot\vec{i}}{|\overrightarrow{XB}|} = \frac{b_1 - x}{|\overrightarrow{XB}|}.$$

Therefore Equation (8.4) becomes $f'(x) = \cos\theta_1 - \cos\theta_2$, and hence f has a stationary value if and only if $\cos\theta_1 = \cos\theta_2$. Since the angles θ_1 and θ_2 are restricted to the open interval $(0, \pi)$, their cosines are equal if and only if $\theta_1 = \theta_2$.

Let x_0 denote the corresponding stationary value. Since $\cos\theta_1$ increases as x increases, and since $\cos\theta_2$ decreases as x increases, the derivative f' is strictly increasing over the whole real axis. Since $f'(x_0) = 0$ it follows that $f(x) > f(x_0)$ for all $x \neq x_0$, and hence f has its absolute minimum at x_0. Therefore the desired path is that for which θ_1 (the angle of "incidence") is equal to θ_2 (the angle of "reflection").

The point $X = (x_0, 0)$ so determined can be located geometrically as follows: Let A' denote the mirror image of A [that is, the point $(a_1, -a_2)$ in Figure 8.7]. The line segment $A'B$ intersects L at the desired point X. This geometric argument was known to the ancient Greeks.

Example 4. Find the shortest distance from a given point $(0, b)$ on the y-axis to the parabola $x^2 = 4y$. (The number b may have any real value.)

Solution. The parabola is shown in Figure 8.8. The quantity to be minimized is the distance d, where

$$d = \sqrt{x^2 + (y - b)^2},$$

subject to the restriction $x^2 = 4y$. It is clear from the figure that when b is *negative* the minimum distance is $|b|$. As the point $(0, b)$ moves upward along the positive y-axis, the minimum is b until the point reaches a certain critical position, above which the minimum is $< b$. The exact location of this critical position will now be determined.

First of all, we observe that the point (x, y) that minimizes d also minimizes d^2. (This observation enables us to avoid differentiation of square roots.) At this stage we may express d^2 in terms of x alone or else in terms of y alone. We shall express d^2 in terms of y and leave it as an exercise for the reader to carry out the calculations when d^2 is expressed in terms of x.

Therefore the function f to be minimized is given by the formula

$$f(y) = d^2 = 4y + (y - b)^2.$$

Although $f(y)$ is defined for all real y, the nature of the problem requires that we seek the minimum only among those $y \geq 0$. The derivative, given by $f'(y) = 4 + 2(y - b)$, is zero only when $y = b - 2$. When $b < 2$ this leads to a negative stationary value y which is excluded by the restriction $y \geq 0$. In other words, if $b < 2$ the minimum does not occur at a stationary value. In fact, when $b < 2$ we see that $f'(y) > 0$ when $y \geq 0$, and hence f is strictly increasing for $y \geq 0$. Therefore the absolute minimum occurs at the endpoint $y = 0$. The corresponding minimum d is $\sqrt{b^2} = |b|$.

If $b \geq 2$ there is a legitimate stationary value at $y = b - 2$. Since $f''(y) = 2$ for all y, the derivative f' is increasing and hence the *absolute minimum* of f occurs at this stationary value. The minimum d is $\sqrt{4(b - 2) + 4} = 2\sqrt{b - 1}$. Thus we have shown that the minimum distance is $|b|$ if $b < 2$, and is $2\sqrt{b - 1}$ if $b \geq 2$. (The value $b = 2$ is the critical value referred to above.)

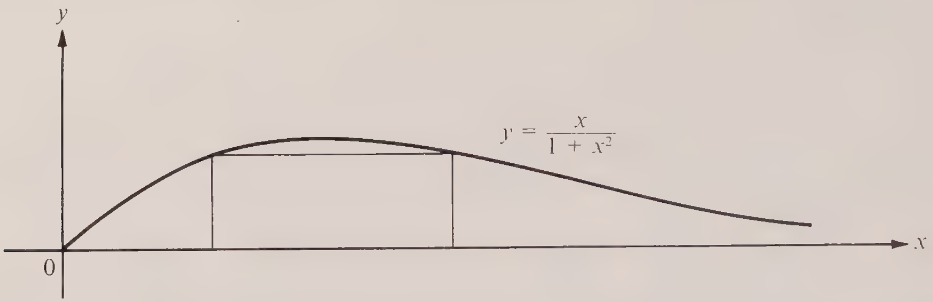


FIGURE 8.9 *Exercise 11.*

8.5 Exercises

1. Find those points on the hyperbola $x^2 - y^2 = 1$ nearest to the point $(0, 1)$.

2. Find the rectangle of largest area that can be inscribed in a semicircle, the lower base being on the diameter.

3. Find the trapezoid of largest area that can be inscribed in a semicircle, the lower base being on the diameter.

4. Prove that, among all rectangles of a given area, the square has the smallest perimeter.

5. An open box is made from a rectangular piece of material by removing equal squares at each corner and turning up the sides. Find the dimensions of the box of largest volume that can be made in this manner if the material has sides (a) 10 and 10; (b) 12 and 18.

6. If a and b are the legs of a right triangle whose hypotenuse is 1, find the largest value of $2a + b$.

7. An "hourglass" is formed by rotating the curve $y = e^{2x}$ about the line $y = x$. How large a sphere can be passed through the constriction?

8. A log 12 feet long has the shape of a frustum of a right circular cone with diameters 4 feet and $(4 + h)$ feet at its ends, where $h \geq 0$. Determine, as a function of h, the volume of the largest right circular cylinder that can be cut from the log, if its axis coincides with that of the log. Discuss the case $h = 1$.

9. For what value of the positive constant a will the maximal value of $f(x) = x^a e^{2a-x}$ over the nonnegative real axis be as small as possible?

10. Let α be the angle $(0 \leq \alpha \leq \frac{1}{2}\pi)$ between the radius vector $\overrightarrow{OP}$ and the tangent line at a point P of the parabola $y = x^2$. Study the behavior of $\tan \alpha$ as P moves to the right along the curve, starting from the origin. In particular, find $\tan \alpha$ when $P = (1, 1)$ and find the maximum value of $\tan \alpha$.

11. A cylinder is obtained by revolving a rectangle about the x-axis, the base of the rectangle lying on the x-axis and the entire rectangle lying in the region between the curve $y = x/(x^2 + 1)$ and the x-axis. (See Figure 8.9.) Find the maximum possible volume of the cylinder.

12. The lower right-hand corner of a page is folded over so as to reach the leftmost edge. (See Figure 8.10.) If the width of the page is six inches, find the minimum length of the crease. What angle will this minimal crease make with the rightmost edge of the page? Assume the page is long enough to prevent the crease reaching the top of the page.

13. A truck is to be driven 300 miles on a freeway at a constant speed of x miles per hour. Speed laws require $30 \le x \le 60$. Assume that fuel costs 30 cents per gallon and is consumed at the rate of $2 + x^2/600$ gallons per hour. If the driver's wages are D dollars per hour and if he obeys all speed laws, find the most economical speed and the cost of the trip if (a) $D = 0$, (b) $D = 1$, (c) $D = 2$, (d) $D = 3$, (e) $D = 4$.

14. Given the parabola $x^2 = 4cy$ and a fixed point (a, b) with $a^2 \le 4bc$. Consider the chords of the parabola which pass through this fixed point. Among all these chords find the one whose projection on the x-axis has the smallest length. Compute this minimal length.

15. (a) An isosceles triangle is inscribed in a circle of radius r as shown in Figure 8.11. If the angle 2α at the apex is restricted to lie between 0 and $\frac{1}{2}\pi$, find the largest value and the smallest value of the perimeter of the triangle. Give full details of your reasoning.

(b) What is the radius of the smallest circular disk large enough to cover *every* isosceles triangle of a given perimeter L? Give full details of your reasoning.

16. Water flows from a hemispherical basin through an orifice at the bottom. Let y denote the height of the water above the orifice and let V denote the corresponding volume of water remaining at time t. A reasonable physical assumption is that dV/dt is proportional to $\sqrt{y}$. (See Example 5 of Section 4.10.) Show that the level of the water falls most slowly when the depth is two-thirds the radius of the basin.

17. A window is to be made in the form of a rectangle surmounted by a semicircle with diameter equal to the base of the rectangle. The rectangular portion is to be of clear glass, and the semicircular portion is to be of a colored glass admitting only half as much light per square foot as the clear glass. The total perimeter of the window frame is to be a fixed length P. Find, in terms of P, the dimensions of the window which will admit the most light.

18. Let $f(x) = (ax + b)/(cx + d)$ if $cx + d \ne 0$. If the constants a, b, c, d are chosen so that f is not a constant, show that f has no local extrema.

19. Assume that $|f''(x)| \le m$ for each x in the interval $[0, a]$, and assume that f takes on its largest value at an interior point of this interval. Show that $|f'(0)| + |f'(a)| \le am$.

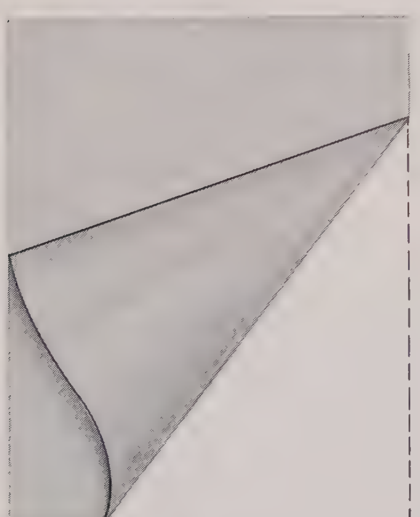


FIGURE 8.10 *Exercise 12.*

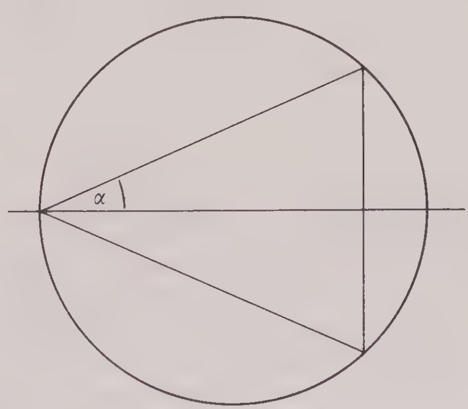
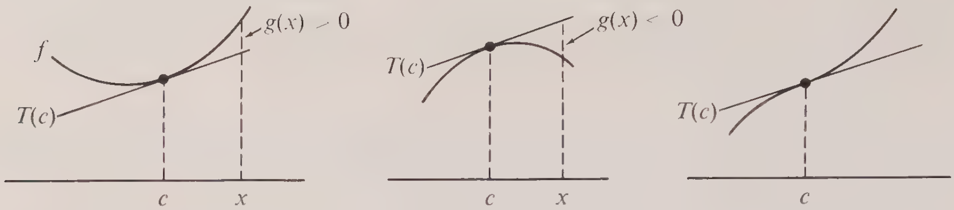


FIGURE 8.11 *Exercise 15.*



Concave upward Concave downward Point of inflection

FIGURE 8.12

20. Given a function f which satisfies the differential equation

$$xf''(x) + 3x[f'(x)]^2 = 1 - e^{-x}$$

for all real x. (Do not attempt to solve this differential equation.)

(a) If f has an extremum at a point $c \neq 0$, show that this extremum is a minimum.
(b) If f has an extremum at 0, is it a maximum or a minimum? Justify your conclusion.
(c) Find the *smallest* constant A such that $f(x) \leq Ax^2$ for all $x \geq 0$.

8.6 The significance of the sign of the second derivative. Concavity

At a stationary value c of a function f the sign of the second derivative $f''(c)$ tells us the kind of extremum that occurs at c. There is a relative maximum at c if $f''(c) < 0$ and a relative minimum if $f''(c) > 0$. The two cases have been illustrated in Figure 8.6 in Section 8.3. Notice that in the first case the graph near c lies below the tangent line at the point $(c, f(c))$. We shall prove in a moment that this geometric property holds not only at the stationary points of f but at any point where the second derivative is negative.

At a point $(c, f(c))$ where the derivative $f'(c)$ exists the graph of f has a tangent line whose equation may be written as follows:

$$y = f(c) + f'(c)(x - c).$$

Let us denote this tangent line by $T(c)$. To determine whether the graph of f is above or below $T(c)$ we introduce

$$g(x) = f(x) - [f(c) + f'(c)(x - c)], \tag{8.5}$$

which measures the difference between the height of the curve and the height of the tangent at x. Thus the graph of f is above $T(c)$ if $g(x) > 0$, and below $T(c)$ if $g(x) < 0$. At the point c itself we have $g(c) = 0$. The graph is said to be *concave upward* at c if $g(x) > 0$ for all $x \neq c$ in some open interval about c. If $g(x) < 0$ for all $x \neq c$ in some open interval about c, the graph is called *concave downward* at c. The point c is called a *point of inflection* of the graph if, in some open interval about c, $g(x)$ is positive on one side of c and negative on the other side of c. Examples are shown in Figure 8.12. Notice that the graph crosses its tangent at a point of inflection.

The nature of the concavity may be determined by the sign of the second derivative as described in the following theorem.

8–6 THEOREM. Assume $f'(x)$ exists for each x in some open interval $(c - h, c + h)$ about c, and suppose $f''(c)$ also exists. Then we have:

(a) If $f''(c) > 0$, the graph of f is concave upward at c.
(b) If $f''(c) < 0$, the graph of f is concave downward at c.
(c) If c is a point of inflection, we must have $f''(c) = 0$.

Note. The condition $f''(c) = 0$ does not necessarily mean that there is a point of inflection at c. For example, when $f(x) = x^4$ we have $f''(0) = 0$ but the graph is concave upward at c.

Proof. The difference $g(x)$ in Equation (8.5) may be written as follows:

$$g(x) = [f(x) - f(c)] - f'(c)(x - c)\,. \tag{8.6}$$

By the mean-value theorem we have $f(x) - f(c) = f'(x_1)(x - c)$, where x_1 lies between x and c. Hence (8.6) becomes

$$g(x) = [f'(x_1) - f'(c)]\,(x - c)\,. \tag{8.7}$$

Suppose now that $f''(c) > 0$. Then by Theorem 8–3 (applied to f') there is a neighborhood of c in which $f'(x) > f'(c)$ if $x > c$ and $f'(x) < f'(c)$ if $x < c$. If x is in this neighborhood, so is the point x_1 in (8.7) and hence the two factors on the right of (8.7) have the same sign. Therefore $g(x) > 0$ for each x in this neighborhood and this means that the curve is concave upward at c. This proves part (a), and the proof of (b) is similar. Part (c) is a consequence of parts (a) and (b).

Some of the results discussed thus far in this chapter may be used as aids in curve tracing. In drawing the graph of an equation $y = f(x)$, we should first determine the domain of f [the set of x for which $f(x)$ is defined] and, if it is easy to do so, we should find the range of f [the set of values that can be taken on by f]. A knowledge of the domain and range gives us an idea of the extent of the curve since it specifies a portion of the xy-plane in which the entire curve must lie. Then it is a good idea to try to locate those points (if any) where the curve crosses the coordinate axes. These are called *intercepts* of the graph. The y-intercept is simply the point $(0, f(0))$ (assuming 0 is in the domain of f) and the x-intercepts are those points $(x, 0)$ for which $f(x) = 0$. Finding the x-intercepts may be extremely difficult in practice and we may have to be content with approximate values only.

We should also try to determine the relative maxima and minima and the points of inflection. Here it is helpful to remember that the only possible candidates for stationary values and points of inflection are the roots of the equations $f'(x) = 0$ and $f''(x) = 0$, respectively. It may be useful to determine intervals in which f is monotonic by examining the sign of f' and to determine intervals of concavity by studying the sign of f''. Special attention should be paid to those points where f' or f'' may not exist.

The behavior of $f(x)$ in the neighborhood of certain points is sometimes easy to determine by inspection. For example, if $f(x) = x + 1/x$ $(x \neq 0)$, then for x near 0 the term x is small compared to $1/x$ and the curve behaves like the hyperbola $y = 1/x$. (See Figure 8.13.) On the other hand, for very large x (positive or negative) the term $1/x$ is small compared to x and the curve behaves very much like the line $y = x$. In fact, in this example the line $y = x$ is an asymptote of the curve. One should, of course, look for

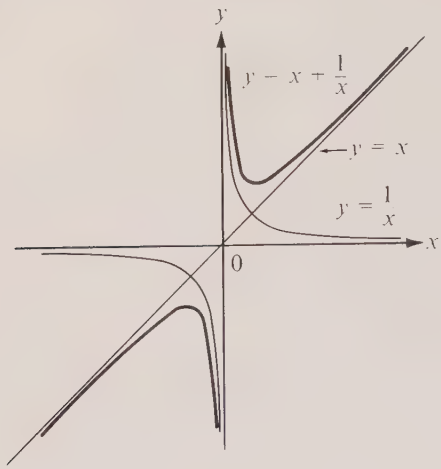


FIGURE 8.13 *Graph of $f(x) = x + 1/x$.*

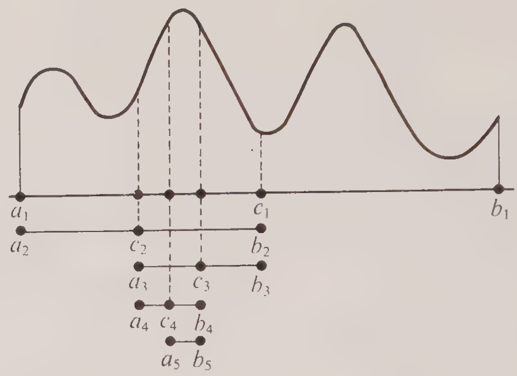


FIGURE 8.14 *Proof of extreme-value theorem for continuous functions.*

asymptotes. Vertical asymptotes often occur among those values of x where $f(x)$ is undefined. For example, the vertical lines $x = \pm 5$ are asymptotes for the graph of $f(x) = 1/(x^2 - 25)$. If a nonvertical line with an equation $y = mx + b$ is suspected to be an asymptote for $y = f(x)$, we then consider the difference $f(x) - (mx + b)$. The line is an asymptote if and only if this difference tends to 0 as x takes on arbitrarily large positive values or if it tends to 0 as x takes arbitrarily large negative values.

8.7 Exercises

In Exercises 1 through 8, find the intervals of concavity and locate the points of inflection (if any exist). Also sketch the graph of f, showing essential features.

1. $f(x) = x^3 - 4x$.
2. $f(x) = \arctan x$.
3. $f(x) = e^{-x^2}$.
4. $f(x) = \log(1 + x^2)$.
5. $f(x) = \sqrt{1 + x^2}$.
6. $f(x) = x \sin(\log x), \quad x > 0$.
7. $f(x) = x\sqrt{2 - x}, \quad x \le 2$.
8. $f(x) = (x^2 - 4)/(x^2 - 9), \quad x^2 \ne 9$.

9. Assume $f''(x) > 0$ for each x in an open interval (a, b). Prove that for every pair of numbers x_1 and x_2 in (a, b) we have

$$f(t_1x_1 + t_2x_2) < t_1f(x_1) + t_2f(x_2)$$

whenever $t_1 > 0$, $t_2 > 0$, and $t_1 + t_2 = 1$. [*Hint.* Interpret geometrically.]

10. (a) Show that each of the following functions has a positive second derivative $f''(x)$ if $x > 0$:

$$f(x) = x^n \quad (n > 1); \qquad f(x) = e^x; \qquad f(x) = x \log x .$$

(b) Use the results of part (a) together with Exercise 9 to derive the following inequalities:

$$\left(\frac{x+y}{2}\right)^n < \frac{x^n + y^n}{2} \qquad \text{if} \quad x > 0\,, y > 0\,, x \neq y\,, n > 1\,;$$

$$e^{(x+y)/2} < \frac{e^x + e^y}{2} \qquad \text{if} \quad x \neq y\,;$$

$$(x + y) \log \frac{x+y}{2} < x \log x + y \log y \qquad \text{if} \quad x > 0\,, y > 0\,, x \neq y\,.$$

★8.8 Proof of the extreme-value theorem for continuous functions

This section is devoted to a proof of the extreme-value theorem for continuous functions, described in Section 8.1.

We are given a function f which is continuous on a closed interval $[a, b]$ and we are to prove that f has an absolute maximum and an absolute minimum on $[a, b]$. It suffices to prove the theorem only for the maximum—the result for the minimum then follows as a consequence because a minimum of f is a maximum of $-f$.

The proof involves the construction of a collection of closed intervals $I_1 = [a_1, b_1]$, $I_2 = [a_2, b_2], \ldots, I_n = [a_n, b_n], \ldots$, where each interval I_{n+1} is either the left or right half of I_n. The endpoints of these intervals are determined successively as follows: First of all, we take $a_1 = a$ and $b_1 = b$. To determine a_{n+1} and b_{n+1} in terms of a_n and b_n, we denote by c_n the mid-point of I_n [that is, $c_n = \frac{1}{2}(a_n + b_n)$] and we examine the function f over the two closed subintervals $[a_n, c_n]$ and $[c_n, b_n]$. If there is a number x in $[a_n, c_n]$ such that $f(t) \leq f(x)$ for all t in $[c_n, b_n]$ we let $a_{n+1} = a_n$ and $b_{n+1} = c_n$. Otherwise we let $a_{n+1} = c_n$ and $b_{n+1} = b_n$. (Examples are shown in Figure 8.14.) In the second case we note that for each t in $[a_n, c_n]$ there is at least one x in $[c_n, b_n]$ such that $f(t) < f(x)$. Now we note that the intervals $I_1, I_2, \ldots$ so constructed have the following properties:

(a) The length of I_{n+1} is

$$b_{n+1} - a_{n+1} = \tfrac{1}{2}(b_n - a_n) = \tfrac{1}{4}(b_{n-1} - a_{n-1}) = \cdots = \frac{b-a}{2^n}\,.$$

(b) No value of $f(t)$ for t outside I_{n+1} exceeds *all* the values of $f(x)$ for x in I_{n+1}.

Let A denote the set of numbers $a_1, a_2, \ldots, a_n, \ldots$ (the leftmost endpoints of $I_1, I_2, \ldots, I_n, \ldots$) and let α denote the least upper bound of A. (The existence of α is guaranteed by Axiom 10 for the real-number system.) We shall prove that α is in $[a, b]$ and that $f(\alpha)$ is the absolute maximum of f on $[a, b]$.

Each b_n, being greater than every a_m, is an upper bound for the set A and hence cannot be less than α, the *least* upper bound. That is, $\alpha \leq b_n$ for every n. Also, $a_n \leq \alpha$ for every n so α lies in every interval I_n and hence, in particular, α lies in $[a, b]$. We wish to prove that $f(x) \leq f(\alpha)$ for every x in $[a, b]$.

Suppose this were not true. Then there would be a point β in $[a, b]$ with $f(\beta) > f(\alpha)$. Let $\epsilon = \frac{1}{2}[f(\beta) - f(\alpha)]$. Then $\epsilon > 0$ and

$$f(\beta) = f(\alpha) + 2\epsilon\,. \tag{8.8}$$

Because of the continuity of f at α, for this ϵ there is a $\delta > 0$ such that

$$f(x) < f(\alpha) + \epsilon \tag{8.9}$$

for every x which lies in the interval $(\alpha - \delta, \alpha + \delta)$ and in $[a, b]$. We can assume also that δ is small enough so that β is not in $(\alpha - \delta, \alpha + \delta)$. Because of property (a), the interval I_{n+1} will lie inside $(\alpha - \delta, \alpha + \delta)$ when n is taken large enough so that $(b - a)/2^n < \delta$. But then, by (8.8) and (8.9), we see that $f(\beta)$ exceeds $f(x)$ for each x in I_{n+1}, contradicting property (b). This contradiction shows that there is no β in $[a, b]$ with $f(\beta) > f(\alpha)$. Hence $f(x) \leq f(\alpha)$ for all x in $[a, b]$, so $f(\alpha)$ is the absolute maximum of f on $[a, b]$. This completes the proof of the extreme-value theorem for continuous functions.

Part II. Indeterminate Forms

8.9 L'Hôpital's rule for the indeterminate form 0/0

In the remainder of this chapter and also in Chapter 9 we shall discuss various extensions of the limit concept. Chapter 9 deals primarily with a limit process that extends the operation of addition from *finite* collections of numbers to *infinite* collections, and it provides an introduction to the subject known as the *theory of infinite series.* In the latter part of Chapter 9 another limit process is introduced which extends the theory of integration from finite intervals to infinite intervals. A somewhat different aspect of the limit concept, known as the *theory of indeterminate forms*, constitutes the subject matter of the rest of this chapter. Some of the consequences of this theory are useful in the study of infinite series. The proofs of the principal results concerning indeterminate forms make use of the mean-value theorem and its extensions.

The subject of indeterminate forms was touched on briefly at the end of Section 7.9 where we used Taylor's theorem and the o-notation to show that

$$\lim_{x \to 0} \frac{a^x - b^x}{x} = \log \frac{a}{b} \qquad (a > 0,\ b > 0)\,. \tag{8.10}$$

The general problem we want to deal with now is that of evaluating the limit of a quotient $f(x)/g(x)$ when both $f(x) \to 0$ and $g(x) \to 0$ as $x \to a$. The limit relation in (8.10) is one example of such a problem. Another is the limit formula

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

encountered earlier in Chapter 2. The quotient theorem on limits tells us that if we have

$$\lim_{x \to a} f(x) = A \qquad \text{and} \qquad \lim_{x \to a} g(x) = B\,,$$

then we also have

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{A}{B}, \qquad \textit{provided } B \neq 0\,. \tag{8.11}$$

If $B = 0$, the theorem gives no information about the behavior of $f(x)/g(x)$. In fact, the

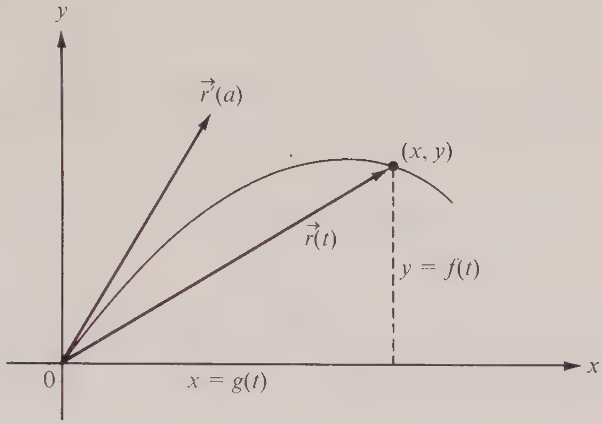


FIGURE 8.15 *The slope of $\vec{r}(t)$ approaches that of $\vec{r}'(a)$ as $t \to a$.*

quotient $f(x)/g(x)$ may or may not tend to a limit as $x \to a$. For example, if $A \neq 0$ and $B = 0$ the quotient $f(x)/g(x)$ takes on arbitrarily large values as $x \to a$, and $f(x)/g(x)$ has no finite limit. But if *both* A and B are zero, it is quite possible that the quotient $f(x)/g(x)$ *does* approach a finite limit, as in the case of $(\sin x)/x$. In fact, if c is any given real number, it is easy to construct examples where the limit of the quotient is c. For example, if $c \neq 0$ we have

$$\lim_{x \to 0} \frac{\sin cx}{x} = \lim_{x \to 0} \left(c \cdot \frac{\sin cx}{cx} \right) = c \cdot \lim_{x \to 0} \frac{\sin cx}{cx} = c\,.$$

A similar example with limit 0 is

$$\lim_{x \to 0} \frac{\sin^2 x}{x} = \lim_{x \to 0} \left(\sin x \, \frac{\sin x}{x} \right) = \left(\lim_{x \to 0} \sin x \right)\left(\lim_{x \to 0} \frac{\sin x}{x} \right) = 0 \cdot 1 = 0\,.$$

These examples show that there is no natural value we can assign to the "quotient" A/B in (8.11) when *both* A and B are zero. For this reason the quotient $f(x)/g(x)$ is said to assume the "indeterminate form 0/0" as $x \to a$.

One way to attack problems on indeterminate forms is to obtain polynomial approximations to $f(x)$ and $g(x)$ as we did in treating the example in Equation (8.10). In this section we discuss another method, known as *L'Hôpital's rule*,† which is more efficient and has wider application than the use of Taylor's formula. The basic idea of the method is to study the quotient of derivatives $f'(x)/g'(x)$ and thereby try to deduce information about $f(x)/g(x)$. This idea is suggested by simple geometric considerations. Figure 8.15

† In 1696, Guillaume François Antoine de L'Hôpital (1661–1704) wrote the first textbook on differential calculus. This work appeared in many editions and played a significant role in the popularization of the subject. Much of the content of the book, including the method known as "L'Hôpital's rule," was based on the earlier work of Johann Bernoulli, one of L'Hôpital's teachers.

shows a curve with vector equation $\vec{r}(t) = g(t)\,\vec{i} + f(t)\,\vec{j}$ which passes through the origin when $t = a$. [This means $f(a) = g(a) = 0$.] As $t \to a$, the radius vector $\vec{r}(t) \to \vec{0}$, but its position approaches that of the tangent vector $\vec{r}'(a)$. If the derivative $\vec{r}'$ is continuous at a, then $\vec{r}'(t) \to \vec{r}'(a)$ as $t \to a$ and this suggests that the *slopes* of the vectors $\vec{r}(t)$ and $\vec{r}'(t)$ should have the same limit as $t \to a$. In other words, the following equation should be true when $f(a) = g(a) = 0$ and when f' and g' have continuous derivatives at a:

(8.12)
$$\lim_{t\to a} \frac{f(t)}{g(t)} = \lim_{t\to a} \frac{f'(t)}{g'(t)}\,.$$

We shall show presently that Equation (8.12) is valid even when the derivatives $f'(a)$ and $g'(a)$ do not exist, provided only that the quotient $f'(t)/g'(t)$ tends to a finite limit as $t \to a$. In fact, we have the following theorem.

8–7 THEOREM. *L'Hôpital's rule for 0/0.* Assume f and g have derivatives $f'(x)$ and $g'(x)$ at each point x of an open interval (a, b), and suppose that

(8.13)
$$\lim_{x\to a+} f(x) = 0 \qquad \text{and} \qquad \lim_{x\to a+} g(x) = 0\,.$$

Assume also that $g'(x) \neq 0$ for each x in (a, b). If the limit

(8.14)
$$\lim_{x\to a+} \frac{f'(x)}{g'(x)}$$

exists and has the value L, say, then the limit

(8.15)
$$\lim_{x\to a+} \frac{f(x)}{g(x)}$$

also exists and has the value L.

Note that no assumption is made about f, g, or their derivatives *at* the point $x = a$. Note also that the limits in (8.13), (8.14), and (8.15) are "right-handed." That is to say, we let $x \to a$ through values greater than a, and we have indicated this by writing $x \to a+$. There is, of course, a similar theorem in which the hypotheses are satisfied in some open interval of the form (b, a) and all the limits are "left-handed" (indicated by writing $x \to a-$, which means that we let $x \to a$ through values less than a). Also, by combining the two "one-sided" theorems there follows a "two-sided" result of the same kind in which $x \to a$ in an unrestricted fashion.

Before we discuss the proof of Theorem 8–7 we shall illustrate the use of this theorem in a number of examples.

Example 1. We shall use L'Hôpital's rule to obtain the familiar formula

(8.16)
$$\lim_{x\to 0} \frac{\sin x}{x} = 1\,.$$

Here $f(x) = \sin x$ and $g(x) = x$. The quotient of derivatives is $f'(x)/g'(x) = (\cos x)/1$ and this tends to 1 as $x \to 0$. By Theorem 8–7, the limit in (8.16) also exists and equals 1.

Example 2. To determine the limit

$$\lim_{x\to 0} \frac{x - \tan x}{x - \sin x}$$

by L'Hôpital's rule, we let $f(x) = x - \tan x$, $g(x) = x - \sin x$, and we find

$$\frac{f'(x)}{g'(x)} = \frac{1 - \sec^2 x}{1 - \cos x}. \tag{8.17}$$

Although this, too, assumes the form 0/0 as $x \to 0$, we may remove the indeterminacy at this stage by algebraic means. If we write

$$1 - \sec^2 x = 1 - \frac{1}{\cos^2 x} = \frac{\cos^2 x - 1}{\cos^2 x} = -\frac{(1 + \cos x)(1 - \cos x)}{\cos^2 x},$$

the quotient in (8.17) becomes

$$\frac{f'(x)}{g'(x)} = -\frac{1 + \cos x}{\cos^2 x},$$

and this approaches -2 as $x \to 0$. Notice that the indeterminacy disappeared when we canceled the common factor $1 - \cos x$. Canceling common factors usually tends to simplify the work in problems of this kind.

When the quotient of derivatives $f'(x)/g'(x)$ also assumes the indeterminate form 0/0 we may try L'Hôpital's rule again. In the next example, the indeterminacy is removed after two applications of the rule.

Example 3. For any real number c we have

$$\lim_{x\to 1} \frac{x^c - cx + c - 1}{(x - 1)^2} = \lim_{x\to 1} \frac{cx^{c-1} - c}{2(x - 1)} = \lim_{x\to 1} \frac{c(c - 1)x^{c-2}}{2} = \frac{c(c - 1)}{2}.$$

In this sequence of equations it is understood that the existence of each limit implies that of the preceding and also their equality.

The next example shows that L'Hôpital's rule is not infallible.

Example 4. Let $f(x) = e^{-1/x}$ if $x \neq 0$, and let $g(x) = x$. The quotient $f(x)/g(x)$ assumes the indeterminate form 0/0 as $x \to 0+$, and one application of L'Hôpital's rule leads to the quotient

$$\frac{f'(x)}{g'(x)} = \frac{(1/x^2)e^{-1/x}}{1} = \frac{e^{-1/x}}{x^2}.$$

This, too, is indeterminate as $x \to 0+$, and if we differentiate numerator and denominator we obtain $(1/x^2)e^{-1/x}/(2x) = e^{-1/x}/(2x^3)$. After n steps we are led to the quotient $e^{-1/x}/(n!x^{n+1})$, so the indeterminacy never disappears by this method.

Example 5. When using L'Hôpital's rule repeatedly, some care is needed to make certain that the quotient under consideration actually assumes an indeterminate form. A common type of error is illustrated by the following calculation:

$$\lim_{x\to 1} \frac{3x^2 - 2x - 1}{x^2 - x} = \lim_{x\to 1} \frac{6x - 2}{2x - 1} = \lim_{x\to 1} \frac{6}{2} = 3.$$

The first step is correct but the second is not. The quotient $(6x - 2)/(2x - 1)$ is not indeterminate as $x \to 1$. The correct limit, 4, is obtained by substituting 1 for x in $(6x - 2)/(2x - 1)$.

Example 6. Sometimes the work can be shortened by a change of variable. For example, we could apply L'Hôpital's rule directly to calculate the limit

$$\lim_{x \to 0+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}},$$

but we may avoid differentiation of square roots by writing $t = \sqrt{x}$ and noting that

$$\lim_{x \to 0+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}} = \lim_{t \to 0+} \frac{t}{1 - e^{2t}} = \lim_{t \to 0+} \frac{1}{-2e^{2t}} = -\frac{1}{2}.$$

We turn now to the proof of Theorem 8–7.

Proof. We make use of Cauchy's mean-value formula (Theorem 7–4 of Section 7.4) applied to a closed interval having a as its left endpoint. Since the functions f and g may not be defined at a, we introduce two new functions that *are* defined there. Let

(8.18)
$$F(x) = f(x) \quad \text{if} \quad x \neq a, \qquad F(a) = 0,$$
$$G(x) = g(x) \quad \text{if} \quad x \neq a, \qquad G(a) = 0.$$

Because of (8.13), both F and G are continuous at a. In fact, if $a < x < b$, both functions F and G are continuous on the *closed interval* $[a, x]$ and have derivatives everywhere in the *open interval* (a, x). Therefore Cauchy's formula is applicable to the interval $[a, x]$ and we obtain

$$[F(x) - F(a)]\, G'(c) = [G(x) - G(a)]\, F'(c),$$

where c is some point satisfying $a < c < x$. If we use (8.18), this becomes

(8.19)
$$f(x)g'(c) = g(x)f'(c).$$

Now $g'(c) \neq 0$ [since, by hypothesis, g' is never zero in (a, b)] and also $g(x) \neq 0$. In fact, if we had $g(x) = 0$ then we would have $G(x) = G(a) = 0$ and, by Rolle's theorem, there would be a point x_1 between a and x where $G'(x_1) = 0$, contradicting the hypothesis that g' is never zero in (a, b). Therefore we may divide by $g'(c)$ and $g(x)$ in (8.19) to obtain

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}.$$

As $x \to a$, the point $c \to a$ (since $a < c < x$) and the quotient on the right approaches L [by (8.14)]. Hence $f(x)/g(x)$ also approaches L and the theorem is proved.

8.10 Exercises

Evaluate the limits in Exercises 1 through 23. The letters a and b denote positive constants.

1. $\displaystyle\lim_{x \to 0} \frac{\sin ax}{\sin bx}$.
2. $\displaystyle\lim_{x \to 2} \frac{3x^2 + 2x - 16}{x^2 - x - 2}$.
3. $\displaystyle\lim_{x \to 0} \frac{\log(\cos ax)}{\log(\cos bx)}$.
4. $\displaystyle\lim_{x \to 0} \frac{\sin x - x}{x^3}$.

5. $\lim_{x\to 0+} \frac{x - \sin x}{(x \sin x)^{3/2}}$.

6. $\lim_{x\to 0} \frac{\sin x}{\arctan x}$.

7. $\lim_{x\to 0} \frac{a^x - 1}{b^x - 1}$, $\quad b \neq 1$.

8. $\lim_{x\to 1} \frac{\log x}{x^2 + x - 2}$.

9. $\lim_{x\to 0} \frac{1 - \cos x^2}{x^2 \sin x^2}$.

10. $\lim_{x\to 0} \frac{x(e^x + 1) - 2(e^x - 1)}{x^3}$.

11. $\lim_{x\to 0} \frac{\log(1 + x) - x}{1 - \cos x}$.

12. $\lim_{x\to \frac{1}{2}\pi} \frac{\cos x}{x - \frac{1}{2}\pi}$.

13. $\lim_{x\to 1} \frac{[\sin(\pi/2x)](\log x)}{(x^3 + 5)(x - 1)}$.

14. $\lim_{x\to 0} \frac{\cosh x - \cos x}{x^2}$.

15. $\lim_{x\to 0} \frac{a^x - a^{\sin x}}{x^3}$.

16. $\lim_{x\to 0} \frac{\cos(\sin x) - \cos x}{x^4}$.

17. $\lim_{x\to a+} \frac{\sqrt{x} - \sqrt{a} + \sqrt{x - a}}{\sqrt{x^2 - a^2}}$.

18. $\lim_{x\to 0} \frac{3 \tan 4x - 12 \tan x}{3 \sin 4x - 12 \sin x}$.

19. $\lim_{x\to 1+} \frac{x^x - x}{1 - x + \log x}$.

20. $\lim_{x\to 0} \frac{\arcsin 2x - 2 \arcsin x}{x^3}$.

21. $\lim_{x\to 0} \frac{x \cot x - 1}{x^2}$.

22. $\lim_{x\to 1} \frac{\sum_{k=1}^{n} x^k - n}{x - 1}$.

23. $\lim_{x\to 0+} \frac{1}{x\sqrt{x}}\left(a \arctan \frac{\sqrt{x}}{a} - b \arctan \frac{\sqrt{x}}{b}\right)$.

24. For what value of the constant a will $x^{-2}(e^{ax} - e^x - x)$ tend to a finite limit as $x \to 0$? What is the value of this limit?

25. Find constants a and b such that

$$\lim_{x\to 0} \frac{1}{bx - \sin x} \int_0^x \frac{t^2\,dt}{\sqrt{a + t}} = 1 .$$

8.11 The symbols $+\infty$ and $-\infty$. Extension of L'Hôpital's rule

L'Hôpital's rule may be extended in several ways. First of all, we may wish to consider the quotient $f(x)/g(x)$ as x increases without bound. It is convenient to have a short descriptive symbolism to express the fact that we are allowing x to increase indefinitely. For this purpose mathematicians use the special symbol $+\infty$, called "plus infinity." Although we shall not attach any meaning to the symbol $+\infty$ *by itself*, we shall give precise definitions of various statements involving this symbol.

One of these statements is written as follows:

$$\lim_{x\to +\infty} f(x) = A ,$$

and is read "The limit of $f(x)$, as x tends to plus infinity, is A." The idea we are trying to express here is that the function values $f(x)$ can be made arbitrarily close to the real number A by taking x large enough. To make this statement mathematically precise we must explain what is meant by "arbitrarily close" and by "large enough." This is done by means of the following definition:

DEFINITION. The symbolism

$$\lim_{x\to +\infty} f(x) = A$$

means that for every number $\epsilon > 0$, there is another number $M > 0$ (which may depend on ϵ) such that

$$|f(x) - A| < \epsilon \quad \text{whenever} \quad x > M .$$

In actual practice, calculations involving limits as $x \to +\infty$ may be reduced to a more familiar case. We simply replace x by $1/t$ (that is, let $t = 1/x$) and note that $t \to 0$ through positive values as $x \to +\infty$. More precisely, we introduce a new function F, where

$$F(t) = f\left(\frac{1}{t}\right) \quad \text{if} \quad t \neq 0 , \tag{8.20}$$

and simply observe that the two statements

$$\lim_{x\to+\infty} f(x) = A \quad \text{and} \quad \lim_{t\to0+} F(t) = A$$

mean exactly the same thing. The proof of this equivalence requires only the definitions of the two limit symbols and is left as an exercise.

When we are interested in the behavior of $f(x)$ for large *negative* x we introduce the symbol $-\infty$ ("minus infinity") and write

$$\lim_{x\to-\infty} f(x) = A$$

to mean: For every $\epsilon > 0$ there is an $M > 0$ such that

$$|f(x) - A| < \epsilon \quad \text{whenever} \quad x < -M .$$

If F is defined by (8.20) it is easy to verify that the two statements

$$\lim_{x\to-\infty} f(x) = A \quad \text{and} \quad \lim_{t\to0-} F(t) = A$$

are equivalent.

In view of the above remarks, it is not surprising to find that all the usual rules for calculating with limits (as stated in Theorem 2–1 of Section 2.8) also apply to limits as $x \to \pm\infty$. The same is true of L'Hôpital's rule which may be extended as follows:

8–8 THEOREM. Assume that f and g have derivatives $f'(x)$ and $g'(x)$ for all x greater than a certain fixed $M > 0$. Suppose that

$$\lim_{x\to+\infty} f(x) = 0 \quad \text{and} \quad \lim_{x\to+\infty} g(x) = 0 ,$$

and that $g'(x) \neq 0$ for $x > M$. If $f'(x)/g'(x)$ tends to a limit as $x \to +\infty$, then $f(x)/g(x)$ also tends to a limit and the two limits are equal. In other words,

$$\lim_{x\to+\infty} \frac{f'(x)}{g'(x)} = L \quad \text{implies} \quad \lim_{x\to+\infty} \frac{f(x)}{g(x)} = L . \tag{8.21}$$

Proof. Let $F(t) = f(1/t)$ and $G(t) = g(1/t)$. Then $f(x)/g(x) = F(t)/G(t)$ if $t = 1/x$, and $t \to 0+$ as $x \to +\infty$. Since $F(t)/G(t)$ assumes the indeterminate form $0/0$ as $t \to 0+$, we examine the quotient of derivatives $F'(t)/G'(t)$. By the chain rule we have

$$F'(t) = \frac{-1}{t^2} f'\left(\frac{1}{t}\right) \quad \text{and} \quad G'(t) = \frac{-1}{t^2} g'\left(\frac{1}{t}\right).$$

Also, $G'(t) \neq 0$ if $0 < t < 1/M$. When $x = 1/t$ and $x > M$, we have $F'(t)/G'(t) = f'(x)/g'(x)$ since the common factor $-1/t^2$ cancels. Therefore, if $f'(x)/g'(x) \to L$ as $x \to +\infty$, then $F'(t)/G'(t) \to L$ as $t \to 0+$ and hence, by Theorem 8–7, $F(t)/G(t) \to L$. Since $F(t)/G(t) = f(x)/g(x)$ this proves (8.21).

There is, of course, a result analogous to Theorem 8–8 in which we consider limits as $x \to -\infty$.

8.12 Infinite limits

In the foregoing section we used the notation $x \to +\infty$ to convey the idea that x takes on arbitrarily large positive values. The symbol $+\infty$ is also used when we want to suggest that a function may be taking on arbitrarily large function values. Thus we write

$$\lim_{x \to a} f(x) = +\infty \tag{8.22}$$

or, alternatively,

$$f(x) \to +\infty \quad \text{as} \quad x \to a \tag{8.23}$$

to indicate that $f(x)$ can be made arbitrarily large by taking x sufficiently close to a. The precise meaning of these symbols is given in the following definition:

DEFINITION. The symbolism in (8.22) or in (8.23) means that for every positive number M (no matter how large) there corresponds another positive number δ (which may depend on M) such that

$$f(x) > M \quad \text{whenever} \quad 0 < |x - a| < \delta .$$

If $f(x) > M$ whenever $0 < x - a < \delta$ we write

$$\lim_{x \to a+} f(x) = +\infty$$

and we say that $f(x)$ tends to plus infinity as x approaches *a from the right*. If $f(x) > M$ whenever $0 < a - x < \delta$, we write

$$\lim_{x \to a-} f(x) = +\infty$$

and we say that $f(x)$ tends to plus infinity as x approaches *a from the left*.

The symbols

$$\lim_{x \to a} f(x) = -\infty, \quad \lim_{x \to a+} f(x) = -\infty, \quad \text{and} \quad \lim_{x \to a-} f(x) = -\infty$$

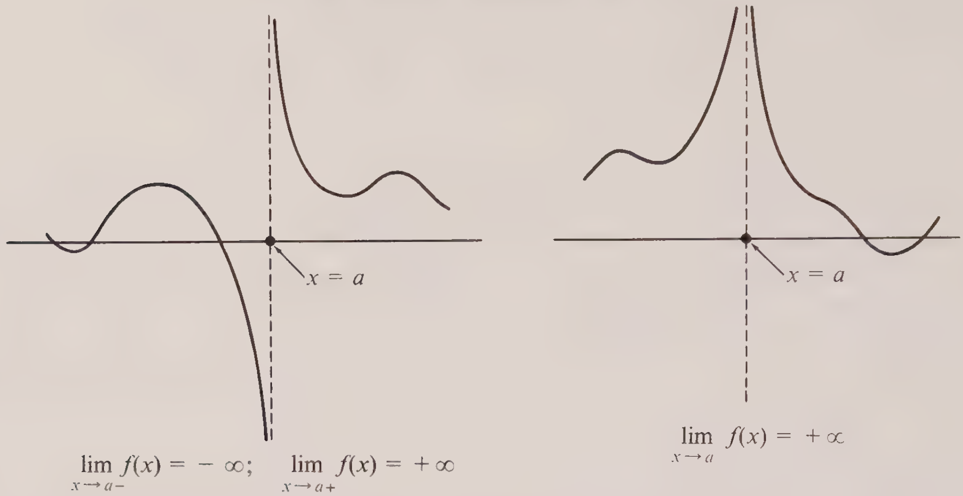


FIGURE 8.16 *Infinite limits.*

are similarly defined, the only difference being that we replace $f(x) > M$ by $f(x) < -M$. Examples are shown in Figure 8.16.

It is also convenient to extend the definitions of these symbols further to cover the cases when $x \to \pm\infty$. Thus, for example, we write

$$\lim_{x \to +\infty} f(x) = +\infty$$

if, for every positive number M, there exists another positive number X such that $f(x) > M$ whenever $x > X$.

The reader should have no difficulty in formulating similar definitions for the symbols

$$\lim_{x \to -\infty} f(x) = +\infty\,, \qquad \lim_{x \to +\infty} f(x) = -\infty\,, \qquad \text{and} \qquad \lim_{x \to -\infty} f(x) = -\infty\,.$$

Examples. In Chapter 3 we proved that the logarithm function is unbounded on the positive real axis. We may express this fact briefly by writing

(8.24)
$$\lim_{x \to +\infty} \log x = +\infty\,.$$

We also proved in Chapter 3 that $\log x < 0$ when $0 < x < 1$ and that the logarithm has no lower bound in the interval (0, 1). Therefore we may also write $\lim_{x \to 0+} \log x = -\infty$.

From the relation that holds between the logarithm and the exponential function it is easy to prove that

(8.25)
$$\lim_{x \to +\infty} e^x = +\infty \qquad \text{and} \qquad \lim_{x \to -\infty} e^x = 0 \qquad \left(\text{or} \quad \lim_{x \to +\infty} e^{-x} = 0\right).$$

Using these results it is not difficult to show that for $\alpha > 0$ we have

(8.26)
$$\lim_{x \to +\infty} x^\alpha = +\infty \qquad \text{and} \qquad \lim_{x \to +\infty} \frac{1}{x^\alpha} = 0\,.$$

The idea is to write $x^\alpha = e^{\alpha \log x}$ and use (8.25) together with (8.24). The formulas in (8.25) also give us the relations

$$\lim_{x\to 0-} e^{-1/x} = +\infty \qquad \text{and} \qquad \lim_{x\to 0+} e^{-1/x} = 0\,.$$

The proofs of these statements make good exercises for testing a reader's understanding of limit symbols involving $\pm\infty$.

8.13 The indeterminate form ∞/∞

Infinite limits lead to new types of indeterminate forms. For example, we may have a quotient $f(x)/g(x)$ where both $f(x)\to+\infty$ and $g(x)\to+\infty$ as $x\to a$ (or as $x\to\pm\infty$). In this case we say that the quotient $f(x)/g(x)$ assumes the indeterminate form ∞/∞ as $x\to a$ (or as $x\to\pm\infty$). Thus, if $f(x) = \log x$ and $g(x) = x^\alpha$ ($\alpha > 0$), the relations (8.24) and (8.26) show that $f(x)/g(x)$ assumes the indeterminate form ∞/∞ as $x\to+\infty$. In this example it is easy to prove that the quotient $f(x)/g(x)$ tends to 0. That is, we have

$$\lim_{x\to+\infty} \frac{\log x}{x^\alpha} = 0 \qquad \text{if} \quad \alpha > 0\,. \tag{8.27}$$

A simple and interesting proof of (8.27) may be given directly from the definition of the logarithm as an integral. If $c > 0$ and $t \ge 1$ we have $t^{-1} \le t^{c-1}$. Hence, if $x > 1$, we may write

$$0 < \log x = \int_1^x \frac{1}{t}\,dt \le \int_1^x t^{c-1}\,dt = \frac{x^c - 1}{c} < \frac{x^c}{c}\,.$$

Therefore

$$0 < \frac{\log x}{x^\alpha} < \frac{x^{c-\alpha}}{c} \qquad \text{for every } c > 0\,.$$

If we choose $c = \frac{1}{2}\alpha$ and let $x\to+\infty$ we obtain (8.27).

Note. The inequality $0 < \log x < x^c/c$ allows us to write

$$\log x = O(x^c) \qquad \text{for} \quad x > 1\,,$$

where c is any positive number. Also, with a natural extension of the o-notation, the relation in (8.27) can be written as follows:

$$\log x = o(x^\alpha) \qquad \text{as} \quad x\to+\infty\,,$$

where α is any positive number. This latter relation is often described by saying that $\log x$ tends to infinity "more slowly" than any positive power of x as $x\to+\infty$.

There are various extensions of L'Hôpital's rule that often help to determine the behavior of a quotient $f(x)/g(x)$ when it assumes the indeterminate form ∞/∞. We shall state just one version which is a direct extension of Theorem 8–7.

8–9 THEOREM. Assume f and g have derivatives $f'(x)$ and $g'(x)$ at each point of an open interval (a, b), and suppose that

$$\lim_{x\to a+} f(x) = +\infty \qquad \text{and} \qquad \lim_{x\to a+} g(x) = +\infty\,.$$

Assume also that $g'(x) \neq 0$ for each x in (a, b). If $f'(x)/g'(x)$ tends to a limit L as $x \to a+$ (where L is "finite"), then $f(x)/g(x)$ also tends to L as $x \to a+$.

Other variants of the theorem exist where $x \to a-$ or where $x \to \pm\infty$. The theorem holds also in all these cases when $L = \pm\infty$. A simple geometric idea underlies the proof in all these cases and it is discussed in detail in Section 8.17.

Example 1. Theorem 8–9 leads to another proof of (8.27). If we put $f(x) = \log x$ and $g(x) = x^\alpha$, the derivatives are $f'(x) = 1/x$, $g'(x) = \alpha x^{\alpha-1}$, and hence

$$\frac{f'(x)}{g'(x)} = \frac{x^{-1}}{\alpha x^{\alpha-1}} = \frac{1}{\alpha x^\alpha} \to 0 \quad \text{as} \quad x \to +\infty .$$

Therefore

$$\lim_{x\to+\infty} \frac{\log x}{x^\alpha} = 0 \quad \text{when} \quad \alpha > 0 ,$$

as asserted in (8.27).

Example 2. Consider the following limit:

$$\lim_{x\to+\infty} \frac{\log(\log x)}{(\log x)^\alpha}, \quad \text{where} \quad \alpha > 0 .$$

Although this can be handled by L'Hôpital's rule, it is easier to introduce $t = \log x$ and write the quotient in the form $(\log t)/t^\alpha$. Since $t \to +\infty$ as $x \to +\infty$, Equation (8.27) tells us that the required limit is zero. This example illustrates once more that a straightforward application of L'Hôpital's rule is not necessarily the simplest method for dealing with an indeterminate form.

Example 3. In Example 4 of Section 8.9 we showed that the behavior of $e^{-1/x}/x$ for x near 0 could not be decided by any number of applications of L'Hôpital's rule for 0/0. However, if we write $t = 1/x$, this quotient becomes t/e^t and it assumes the indeterminate form ∞/∞ as $t \to +\infty$. One application of L'Hôpital's rule now gives us

(8.28) $$\lim_{t\to+\infty} \frac{t}{e^t} = \lim_{t\to+\infty} \frac{1}{e^t} = 0 .$$

Therefore $e^{-1/x}/x \to 0$ as $x \to 0+$ or, in other words, $e^{-1/x} = o(x)$ as $x \to 0+$.

8.14 Exercises

Evaluate the limits in Exercises 1 through 12. The letters a and b denote positive constants.

1. $\displaystyle\lim_{x\to0} \frac{e^{-1/x^2}}{x^{1000}}$.
2. $\displaystyle\lim_{x\to+\infty} \frac{\sin(1/x)}{\arctan(1/x)}$.
3. $\displaystyle\lim_{x\to\frac{1}{2}\pi} \frac{\tan 3x}{\tan x}$.
4. $\displaystyle\lim_{x\to+\infty} \frac{\log(a + be^x)}{\sqrt{a + bx^2}}$.
5. $\displaystyle\lim_{x\to+\infty} x^4\left(\cos\frac{1}{x} - 1 + \frac{1}{2x^2}\right)$.
6. $\displaystyle\lim_{x\to\pi} \frac{\log(\sin x)}{\log(\sin 2x)}$.
7. $\displaystyle\lim_{x\to\frac{1}{2}-} \frac{\log(1 - 2x)}{\tan \pi x}$.
8. $\displaystyle\lim_{x\to+\infty} \frac{\cosh(x + 1)}{e^x}$.
9. $\displaystyle\lim_{x\to+\infty} \frac{a^x}{x^b}, \quad a > 1$.
10. $\displaystyle\lim_{x\to\frac{1}{2}\pi} \frac{\tan x - 5}{\sec x + 4}$.

11. $\lim_{x\to 0+} \frac{1}{\sqrt{x}}\left(\frac{1}{\sin x} - \frac{1}{x}\right)$.

12. $\lim_{x\to+\infty} x^{1/4} \sin(1/\sqrt{x})$.

13. Without attempting to evaluate the integral, find the value of the following limit:

$$\lim_{p\to+\infty} \frac{1}{p} \int_0^p e^{x^2-p^2}(x^2+1)\,dx\,.$$

Notice that the integral $\int_0^p e^{x^2}(x^2+1)\,dx \to +\infty$ as $p \to +\infty$, since its integrand is ≥ 1.

14. For a certain value of c the limit

$$\lim_{x\to+\infty} x^c e^{-2x} \int_0^x e^{2t}\sqrt{3t^2+1}\,dt$$

exists, is finite, and is not zero. Determine this c and find the value of the limit. (Do not attempt to evaluate the integral.)

8.15 Other indeterminate forms

There are other indeterminate forms besides $0/0$ and ∞/∞. Among these is $\infty - \infty$, which occurs when we have a difference $f(x) - g(x)$ in which both $f(x) \to +\infty$ and $g(x) \to +\infty$ or both $f(x) \to -\infty$ and $g(x) \to -\infty$. Other indeterminate forms, denoted by the symbols $0 \cdot \infty$, 0^0, ∞^0, 0^∞, and 1^∞, are illustrated by the examples given below. In each of these examples, algebraic manipulation enables us to reduce the problem to an indeterminate form of the type $0/0$ or ∞/∞ which may be handled by L'Hôpital's rule, or to a form that may be treated by some other method.

Example 1: ($\infty - \infty$). Prove that

$$\lim_{x\to 0}\left(\cot x - \frac{1}{x}\right) = 0\,. \tag{8.29}$$

Solution. In this case we write

$$\cot x - \frac{1}{x} = \frac{\cos x}{\sin x} - \frac{1}{x} = \frac{x\cos x - \sin x}{x \sin x}\,,$$

which leads to the indeterminate form $0/0$. This may be handled with one application of L'Hôpital's rule. Putting

$$f(x) = x\cos x - \sin x\,, \qquad g(x) = x \sin x\,,$$

we find

$$f'(x) = -x\sin x\,, \qquad g'(x) = \sin x + x\cos x\,,$$

and hence

$$\frac{f'(x)}{g'(x)} = \frac{-x\sin x}{\sin x + x\cos x} = \frac{-x}{1 + [x/(\sin x)]\cos x} \to 0 \quad \text{as} \quad x \to 0\,.$$

This proves (8.29).

Another example leading to the indeterminate form $\infty - \infty$ was encountered in Chapter 6 where it was shown that the straight line $y = (b/a)x$ is an asymptote of the hyperbola $x^2/a^2 - y^2/b^2 = 1$. The indeterminacy in this case was removed entirely by algebraic means.

Example 2: $(0 \cdot \infty)$. Prove that

(8.30) $$\lim_{x \to 0+} x^{\alpha} \log x = 0 \qquad \text{for each fixed } \alpha > 0\,.$$

Solution. Writing $t = 1/x$, we find $x^{\alpha} \log x = -(\log t)/t^{\alpha}$ and, by (8.27), this tends to 0 as $t \to +\infty$.

This example can also be treated directly by L'Hôpital's rule, writing $x^{\alpha} \log x = f(x)/g(x)$, where $f(x) = \log x$ and $g(x) = x^{-\alpha}$. The quotient assumes the indeterminate form ∞/∞ as $x \to 0+$ and we find

$$\frac{f'(x)}{g'(x)} = \frac{x^{-1}}{-\alpha x^{-\alpha-1}} = -\frac{x^{\alpha}}{\alpha} \to 0 \qquad \text{as} \quad x \to 0\,.$$

Example 3: $(0 \cdot \infty)$. Prove that

(8.31) $$\lim_{x \to +\infty} x^{a} e^{-bx} = 0 \qquad \text{if} \quad a > 0 \quad \text{and} \quad b > 0\,.$$

Solution. We convert this to the form ∞/∞ by writing $x^{a}e^{-bx} = f(x)/g(x)$, where $f(x) = x^{a}$ and $g(x) = e^{bx}$. The quotients of derivatives are

$$\frac{f'(x)}{g'(x)} = \frac{ax^{a-1}}{be^{bx}}\,, \qquad \frac{f''(x)}{g''(x)} = \frac{a(a-1)x^{a-2}}{b^{2}e^{bx}}\,,$$

and so on. It is clear that after a finite number of steps the exponent of x in the numerator becomes 0 or negative and hence the limit is 0, as asserted.

Example 3 tells us that $x^{a} = o(e^{bx})$ as $x \to +\infty$, where $a > 0$ and $b > 0$. In other words, no matter how large a may be and no matter how small b may be (if both are *positive*), x^{a} tends to infinity more slowly than e^{bx} as $x \to +\infty$. [Equation (8.28) is a special case.]

An interesting consequence of (8.31) is obtained by putting $y = e^{x}$. Then $x = \log y$ and we have

$$\frac{x^{a}}{e^{bx}} = \frac{(\log y)^{a}}{y^{b}}\,.$$

Since $y \to +\infty$ as $x \to +\infty$, using (8.31) we obtain the following limit relation:

(8.32) $$\lim_{y \to +\infty} \frac{(\log y)^{a}}{y^{b}} = 0 \qquad \text{if} \quad a > 0 \quad \text{and} \quad b > 0\,.$$

That is to say, $(\log y)^{a} = o(y^{b})$ as $y \to +\infty$. In other words, no matter how large a may be and no matter how small b may be (if both are *positive*), $(\log y)^{a}$ tends to infinity more slowly than y^{b} as $y \to +\infty$. [Equation (8.27) is a special case.]

Example 4: (0^{0}). Show that

$$\lim_{x \to 0+} x^{x} = 1\,.$$

Solution. Since $x^{x} = e^{x \log x}$, by continuity of the exponential function we have

$$\lim_{x \to 0+} x^{x} = \exp\,(\lim_{x \to 0+} x \log x)\,,$$

if the last limit exists. But by Example 2 we know that $x \log x \to 0$ as $x \to 0+$, and hence $x^{x} \to e^{0} = 1$.

Example 5: (∞^0). Show that

$$\lim_{x\to+\infty} x^{1/x} = 1\,.$$

Solution. We write $x^{1/x} = e^{(1/x)\log x}$ and the problem amounts to determining

$$\lim_{x\to+\infty} \frac{\log x}{x}\,.$$

By (8.27) this limit is 0, and hence $x^{1/x} \to e^0 = 1$.

Example 6: (0^∞). Prove that

$$\lim_{x\to 0+} x^{\log x} = +\infty\,.$$

Solution. $x^{\log x} = e^{(\log x)\log x} = e^{\log^2 x} \to +\infty$ as $x \to 0+$. A similar argument shows that

$$\lim_{x\to 0+} x^{\log^2 x} = 0\,.$$

Example 7: (1^∞). The classical example of this indeterminate form is

$$\lim_{x\to 0} (1+x)^{a/x}\,,$$

where a is any nonzero real number. [When $a = 0$ there is no indeterminacy since $(1+x)^0 = 1$ for small $x \neq 0$.] We write

$$(1+x)^{a/x} = e^{(a/x)\log(1+x)}$$

and the problem reduces to the computation of the following limit:

$$\lim_{x\to 0} \frac{a\log(1+x)}{x}\,.$$

This assumes the indeterminate form 0/0 and one application of L'Hôpital's rule yields

$$\lim_{x\to 0} \frac{a\log(1+x)}{x} = \lim_{x\to 0} \frac{a/(1+x)}{1} = a\,.$$

Therefore $e^{(a/x)\log(1+x)} \to e^a$ as $x \to 0$. In other words, we have shown that

(8.33) $$\lim_{x\to 0} (1+x)^{a/x} = e^a$$

for every real $a \neq 0$. This also holds for $a = 0$, since both sides are equal to 1, so (8.33) holds for all real a. Sometimes this limit relation is taken as the starting point for the theory of the exponential function.

An example related to (8.33) is the limit

(8.34) $$\lim_{x\to 0} (1+ax)^{1/x} = e^a$$

which may be reduced to (8.33) by writing $y = ax$ when $a \neq 0$. Similarly, replacing x by $1/x$ in (8.33) and (8.34), respectively, we obtain the formulas

$$\lim_{x\to+\infty} \left(1+\frac{1}{x}\right)^{ax} = e^a \quad\text{and}\quad \lim_{x\to+\infty} \left(1+\frac{a}{x}\right)^{x} = e^a\,,$$

both of which are valid for all real a.

Note. Examples 4 through 7 were all of the type $f(x)^{g(x)}$. These are usually dealt with by writing

$$f(x)^{g(x)} = e^{g(x)\log f(x)}$$

and then treating the exponent $g(x)\log f(x)$ by one of the methods discussed earlier.

8.16 Exercises

Evaluate the limits in Exercises 1 through 20.

1. $\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{e^x - 1}\right)$.
2. $\lim_{x\to 1}\left(\frac{1}{\log x} - \frac{1}{x-1}\right)$.
3. $\lim_{x\to +\infty}(x^2 - \sqrt{x^4 - x^2 + 1})$.
4. $\lim_{x\to 0+}\left[\frac{\log x}{(1+x)^2} - \log\left(\frac{x}{1+x}\right)\right]$.
5. $\lim_{x\to 1-}(\log x)\log(1-x)$.
6. $\lim_{x\to 0+} x^{(x^x - 1)}$.
7. $\lim_{x\to 0+}[x^{(x^x)} - 1]$.
8. $\lim_{x\to 0-}(1 - 2^x)^{\sin x}$.
9. $\lim_{x\to 0+} x^{1/\log x}$.
10. $\lim_{x\to 0}(\cot x)^{\sin x}$.
11. $\lim_{x\to \frac{1}{4}\pi}(\tan x)^{\tan 2x}$.
12. $\lim_{x\to 0+}\left(\log\frac{1}{x}\right)^x$.
13. $\lim_{x\to 0}(x + e^{2x})^{1/x}$.
14. $\lim_{x\to 0+} x^{e/(1+\log x)}$.
15. $\lim_{x\to 1} x^{1/(1-x)}$.
16. $\lim_{x\to 1}(2-x)^{\tan(\pi x/2)}$.
17. $\lim_{x\to 0}\frac{(1+x)^{1/x} - e}{x}$.
18. $\lim_{x\to 0}\left(\frac{(1+x)^{1/x}}{e}\right)^{1/x}$.
19. $\lim_{x\to 0}\left(\frac{\arcsin x}{x}\right)^{1/x^2}$.
20. $\lim_{x\to 0}\left(\frac{1}{\log(x + \sqrt{1+x^2})} - \frac{1}{\log(1+x)}\right)$.

★8.17 Geometric proof of L'Hôpital's rule for ∞/∞

L'Hôpital's rule for ∞/∞ (Theorem 8-9) states that, if $f(x)\to +\infty$ and $g(x)\to +\infty$ as $x\to a$, then the relation $f'(x)/g'(x)\to L$ as $x\to a$ implies $f(x)/g(x)\to L$ as $x\to a$. Here a may be finite or infinite and L may be finite or infinite. In this section we shall discuss the proof of this result in the case $a = +\infty$ and L finite. The other cases involve essentially the same ideas and they need not be discussed in detail.

The hypotheses we make are

$$\lim_{x\to+\infty} f(x) = +\infty, \qquad \lim_{x\to+\infty} g(x) = +\infty, \tag{8.35}$$

$$\lim_{x\to+\infty}\frac{f'(x)}{g'(x)} = L, \tag{8.36}$$

and $g'(x)\neq 0$ for all sufficiently large x. We are to prove that

$$\lim_{x\to+\infty}\frac{f(x)}{g(x)} = L. \tag{8.37}$$

We shall give a simple geometric proof† which makes use of the geometric interpretation of Cauchy's mean-value formula. For this purpose we introduce the curve whose vector equation is $\vec{r}(t) = g(t)\,\vec{i} + f(t)\,\vec{j}$. For each t we let $P(t)$ denote the tip of $\vec{r}(t)$, that is, the point $(g(t), f(t))$ on the curve. The hypothesis in (8.36) can be expressed as follows: If ϵ is any preassigned positive number, then there exists a number t_ϵ such that $t \geq t_\epsilon$ implies

$$L - \epsilon < \frac{f'(t)}{g'(t)} < L + \epsilon. \tag{8.38}$$

In other words, for all $t \geq t_\epsilon$ the slope of the curve at $P(t)$ lies between $L - \epsilon$ and $L + \epsilon$. We shall show that for some T we have

$$L - \epsilon < \frac{f(t)}{g(t)} < L + \epsilon \tag{8.39}$$

whenever $t > T$, and this will prove (8.37). Geometrically, (8.39) means that the radius vector $\vec{r}(t)$ has slope between $L - \epsilon$ and $L + \epsilon$ for all $t > T$.

We prove first a geometric property which we state as a lemma. (A *lemma* is an auxiliary theorem that is used only in the demonstration of some other theorem.)

LEMMA. Let Q be any point in the plane and consider the wedge-shaped region, with vertex Q, bounded by two lines of slopes $L - \epsilon$ and $L + \epsilon$ which meet at Q.‡ (An example is shown in Figure 8.17.) If, for some $T_1 \geq t_\epsilon$, a point $P(T_1)$ of the curve lies inside or on the boundary of this wedge, then all points $P(T_2)$ with $T_2 > T_1$ and $g(T_2) > g(T_1)$ also lie inside this wedge. [The condition $g(T_2) > g(T_1)$ means that $P(T_2)$ lies to the right of $P(T_1)$.]

We prove the lemma by contradiction. Suppose some point $P(T_2)$ with $T_2 > T_1$ and $g(T_2) > g(T_1)$ is outside or on the boundary of the wedge. (An example is shown in Figure 8.17.) Then the chord joining $P(T_2)$ to $P(T_1)$ has a slope which is $\geq L + \epsilon$ or $\leq L - \epsilon$. By Cauchy's extension of the mean-value theorem, for some c between T_1 and T_2 the tangent line at $P(c)$ has the same slope as this chord, contradicting (8.38). This contradiction proves the lemma.

Now we shall use the lemma to prove (8.37). Let $t_{\epsilon/2}$ be such that

$$L - \frac{\epsilon}{2} < \frac{f'(t)}{g'(t)} < L + \frac{\epsilon}{2}$$

for all $t \geq t_{\epsilon/2}$, and consider the narrower wedge whose vertex is at $P(t_{\epsilon/2})$ and whose boundaries are the lines with slopes $L - \frac{1}{2}\epsilon$ and $L + \frac{1}{2}\epsilon$. [We can assume also that $t_{\epsilon/2} \geq t_\epsilon$.] Since $g(t) \to +\infty$ as $t \to +\infty$, there exist points $P(t)$ with $t > t_{\epsilon/2}$ such that $g(t) > g(t_{\epsilon/2})$. Using the lemma (with ϵ replaced by $\epsilon/2$), we find that each such point is

† There is no difficulty in reformulating this proof entirely in analytic terms. However, the geometric argument is more transparent.

‡ More precisely, if $Q = (x_0, y_0)$ the wedge in question is the set of all points (x, y) such that $x > x_0$ and $(L - \epsilon)(x - x_0) < y - y_0 < (L + \epsilon)(x - x_0)$.

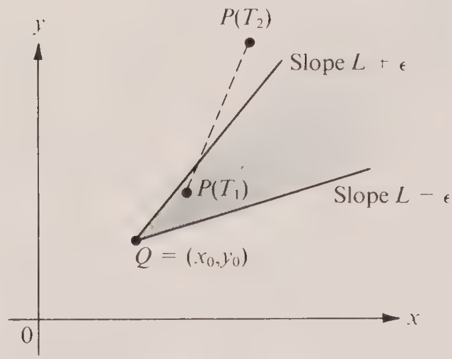


FIGURE 8.17 *A wedge-shaped region bounded by two lines of slopes $L - \epsilon$ and $L + \epsilon$ which meet at Q.*

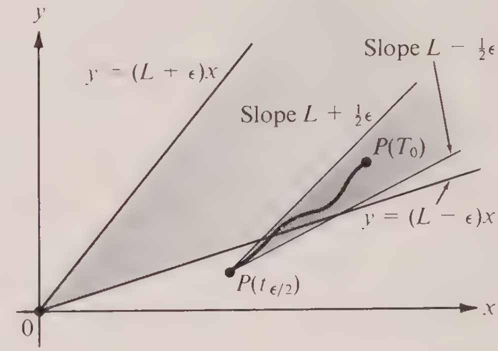


FIGURE 8.18 *The curve eventually enters the wedge formed by the lines $y = (L \pm \epsilon)x$.*

in this wedge. Since the boundaries of this wedge have slopes $L \pm \frac{1}{2}\epsilon$, all but at most a finite portion of this wedge must lie inside the larger wedge with vertex at O formed by the lines $y = (L \pm \epsilon)x$. (See Figure 8.18.) Therefore, since $g(t) \to +\infty$ as $t \to +\infty$, the curve must eventually enter the larger wedge. Let us say $P(T_0)$ is in the larger wedge, where $T_0 \geq t_{\epsilon/2}$. Since $g(t) \to +\infty$ as $t \to +\infty$, there exists a $T \geq T_0$ such that $g(t) > g(T_0)$ for all $t \geq T$. By the lemma, $P(t)$ is in the larger wedge for each such t and this means that (8.39) holds for all $t \geq T$. As we have already noted, this implies (8.37).

8.18 Miscellaneous review exercises

1. Let $f(x) = x^4 + cx^2 + 1$, where c is a real constant. Find where f is increasing, where it is decreasing, and locate the extreme values. Using the information obtained, draw the graph of f when $c = 18$ and when $c = -18$.

2. Let $f(x) = 2 \sec x - \tan x$ if $0 \leq x \leq \pi$, $x \neq \frac{1}{2}\pi$. Find where f is positive, where it is negative, where it is increasing, where it is decreasing, and locate the local maxima and minima. Draw the graph, showing in particular the behavior as $x \to \frac{1}{2}\pi$.

3. Let

$$f(x) = \int_0^x \frac{t^2(1 - 2t)}{1 + t^6}\, dt\,.$$

(Do not attempt to evaluate this integral.)

(a) For what x does $f(x)$ reach a maximum? Explain why it is a maximum and not a minimum.

(b) Prove that $f(x) \leq \frac{1}{96}$ for all real x.

4. Determine $g'(0)$ if $g(x) = (x^2 + 2x + 3)f(x)$, given that $f(0) = 5$ and $\lim_{x\to 0} [f(x) - 5]/x = 4$.

5. Given a function f whose derivative always lies between -1 and $+1$. Show that there exists at most one value of $x > \frac{1}{2}$ for which $f(x) = x^2$. Give full details of your reasoning.

6. Given that the line $y = c$ intersects the parabola $y = x^2$ at two points. Find the radius of

the circle passing through these two points and through the vertex of the parabola. The radius you determine depends on c. What happens to this radius as $c \to 0$?

7. Let V be one of the two vertices of a hyperbola whose semitransverse axis has length a and whose eccentricity $c/a = 2$. Let P be a point on the curve on the same branch as V and let $r = |\overrightarrow{VP}|$. Denote by A the area of the region bounded by the hyperbola and the line segment VP.

(a) Place the coordinate axes in a convenient position and write an equation for the hyperbola.

(b) Express the area A as an integral and, without attempting to evaluate this integral, show that Ar^{-3} tends to a limit as the point P tends to V. Find this limit.

8. (a) Let P be a point on the parabola $y = x^2$. Let Q be the point of intersection of the normal line at P with the y-axis. What is the limiting position of Q as P tends to the y-axis?

(b) Solve the same problem for the curve $y = f(x)$, where $f'(0) = 0$.

9. (a) Determine the limit of the quotient

$$\frac{(\sin 4x)(\sin 3x)}{x \sin 2x}$$

as $x \to 0$ and also as $x \to \frac{1}{2}\pi$.

(b) For what values of the constants a and b is

$$\lim_{x \to 0} (x^{-3} \sin 3x + ax^{-2} + b) = 0\,?$$

10. Given the following information about a function f:

$$\lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} f'(x) = \lim_{x \to +\infty} f''(x) = +\infty\,; \qquad \lim_{x \to +\infty} \frac{xf'''(x)}{f''(x)} = 1\,.$$

Determine the value of the limit $\lim_{x \to +\infty} xf'(x)/f(x)$.

11. Given a function f defined at each point of the open interval $(-1, 1)$, assume that the derivative $f'(x)$ exists everywhere in $(-1, 1)$, *except possibly* at 0. For each of the following statements about $f'(0)$ either give a proof or exhibit a counterexample:

Statement 1. If f is continuous at 0, then $f'(0)$ exists.

Statement 2. If $\lim_{x \to 0} f'(x)$ exists, then $f'(0)$ exists. (Note that we do not assume continuity of f at 0 in this case.)

Statement 3. If f is continuous at 0 and if $\lim_{x \to 0} f'(x)$ exists, then $f'(0)$ exists.

12. Let $y = f(x)$ be that solution of the differential equation

$$y' = \frac{2y^2 + x}{3y^2 + 5}$$

which satisfies the initial condition $f(0) = 0$. (Do not attempt to solve this differential equation.)

(a) The differential equation shows that $f'(0) = 0$. Discuss whether f has a relative maximum or minimum or neither at 0.

(b) Notice that $f'(x) \geq 0$ for each $x \geq 0$ and that $f'(x) \geq 2/3$ for each $x \geq 10/3$. Exhibit two positive numbers a and b such that $f(x) > ax - b$ for each $x \geq 10/3$.

(c) Show that $x/y^2 \to 0$ as $x \to +\infty$. Give full details of your reasoning.

(d) Show that y/x tends to a finite limit as $x \to +\infty$ and determine this limit.

9

SEQUENCES, INFINITE SERIES, IMPROPER INTEGRALS

9.1 Zeno's paradox

The principal subject matter of this chapter had its beginning nearly 2400 years ago when the Greek philosopher Zeno of Elea (495–435 B.C.) precipitated a crisis in ancient mathematics by setting forth a number of ingenious paradoxes. One of these, often called the *racecourse paradox*, may be described as follows:

> A runner can never reach the end of a racecourse because he must cover half of any distance before he covers the whole. That is to say, having covered the first half he still has the second half before him. When half of this is covered, one-fourth yet remains. When half of this one-fourth is covered, there remains one-eighth, and so on, *ad infinitum*.

Zeno was talking, of course, about an idealized situation in which the runner is to be thought of as a particle or point moving from one end of a line segment to the other. To analyze Zeno's argument in more detail, let us assume that the runner starts at the point marked 1 in Figure 9.1 and runs toward the goal marked 0. The positions labeled $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, etc., indicate the fraction of the course yet to be covered when these points are reached. These fractions (each of which is half the previous one) subdivide the whole course into an endless number of smaller portions. Since a positive amount of time is required to cover each portion separately, it seems reasonable to assert that the time required for the whole course must be the sum total of all these amounts. Zeno argued that a sum of an endless number of positive time intervals cannot possibly be finite and hence the goal cannot be reached in a finite length of time. Thus we have a paradox—a result obtained by reasoning that is in direct conflict with our physical intuition.

Consider now another example which arises from a different source but which contains the same mathematical ingredients as Zeno's paradox. Suppose a philanthropic foundation agrees to make annual gifts forever according to the following plan: The first year the gift will be one million dollars, the second year one-half million, the third year one-fourth million, and so on, each payment after the first being half the previous year's grant. If we argue as Zeno did, that the sum of an endless number of positive quantities cannot possibly be finite, then we must conclude that the foundation will need unlimited

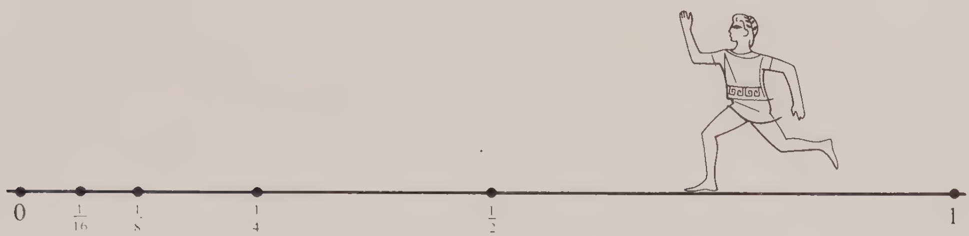


FIGURE 9.1 *The racecourse paradox.*

resources to fulfill its pledge. However, this conclusion is false. As we shall see presently, a fund totaling two million dollars will suffice to continue this plan indefinitely.

Zeno's assertion that an endless number of positive quantities cannot have a finite sum was rejected 2000 years later when the theory of infinite series was created. In the 17th and 18th centuries mathematicians began to realize that it *is* possible to extend the ideas of ordinary addition from *finite* collections of numbers to *infinite* collections so that sometimes infinitely many positive numbers have a finite "sum." To see how this extension might come about and to get an idea of some of the difficulties that might be encountered in making the extension, let us analyze Zeno's paradox in more detail.

Suppose the aforementioned runner travels at a *constant speed* and suppose it takes him T minutes to cover the first half of the course. The next quarter of the course will take $T/2$ minutes, the next eighth will take $T/4$ minutes, and, in general, the portion from $1/2^n$ to $1/2^{n+1}$ will take $T/2^n$ minutes. The "sum" of all these time intervals may be indicated symbolically by writing the following expression:

$$T + \frac{T}{2} + \frac{T}{4} + \cdots + \frac{T}{2^n} + \cdots. \tag{9.1}$$

This is an example of what is known as an *infinite series* and the problem here is to decide whether there is some reasonable way to assign a number which may be called the *sum* of this series.

Our physical experience tells us that a runner who travels at a constant speed should reach his goal in twice the time it takes for him to reach the halfway point. Since it takes T minutes to cover half the course, it should require $2T$ minutes for the whole course. This line of reasoning strongly suggests that we should assign the "sum" $2T$ to the series in (9.1) and it leads us to expect that the equation

$$T + \frac{T}{2} + \frac{T}{4} + \cdots + \frac{T}{2^n} + \cdots = 2T \tag{9.2}$$

should be "true" in some sense.

The theory of infinite series tells us exactly how to interpret this equation. The idea is this: First we add a *finite number* of the terms, say the first n, and denote their sum by s_n. Thus we have

$$s_n = T + \frac{T}{2} + \frac{T}{4} + \cdots + \frac{T}{2^{n-1}}. \tag{9.3}$$

This is called the nth *partial sum* of the series. Now we study the behavior of s_n as n takes larger and larger values. In particular, we try to determine whether the partial sums s_n approach a finite limit as n increases without bound.

In this example it is easy to see that $2T$ is the limiting value of the partial sums. In fact, if we calculate a few of these partial sums we find

$$s_1 = T, \qquad s_2 = T + \frac{T}{2} = \frac{3}{2}T, \qquad s_3 = T + \frac{T}{2} + \frac{T}{4} = \frac{7}{4}T,$$
$$s_4 = T + \frac{T}{2} + \frac{T}{4} + \frac{T}{8} = \frac{15}{8}T.$$

Now, observe that these results may be expressed as follows:

$$s_1 = (2 - 1)T, \qquad s_2 = (2 - \tfrac{1}{2})T, \qquad s_3 = (2 - \tfrac{1}{4})T, \qquad s_4 = (2 - \tfrac{1}{8})T.$$

This leads us to conjecture the following general formula:

$$s_n = \left(2 - \frac{1}{2^{n-1}}\right)T \quad \text{for all positive integers } n. \tag{9.4}$$

Formula (9.4) is easily verified by induction. Since $1/2^{n-1} \to 0$ as n increases indefinitely, this shows that $s_n \to 2T$. Therefore Equation (9.2) is "true" if we interpret it to mean that $2T$ is the *limit* of the partial sums s_n. This limit process seems to invalidate Zeno's objection that the sum of an infinite number of time intervals can never be finite.

Exactly the same reasoning explains why a fund of two million dollars will suffice to guarantee the pledge of the philanthropic foundation mentioned earlier. In fact, if we let T represent the amount of the first year's gift (one million dollars) then the nth partial sum s_n in (9.3) represents the total amount paid out after n years, and Equation (9.4) shows that $s_n < 2T$. Therefore the total amount contributed never reaches two million dollars. Since s_n becomes arbitrarily close to $2T$ as n increases indefinitely, any amount less than two million dollars will eventually be exhausted, so two million dollars is the minimum fund that suffices to guarantee the pledge forever.

Now we shall give an argument which lends considerable support to Zeno's point of view. Suppose we make a small but important change in the foregoing analysis of the racecourse paradox. Instead of assuming that the speed of the runner is constant, let us suppose that his speed gradually decreases in such a way that he requires T minutes to go from 1 to 1/2, $T/2$ minutes to go from 1/2 to 1/4, $T/3$ minutes to go from 1/4 to 1/8, and, in general, T/n minutes to go from $1/2^{n-1}$ to $1/2^n$. The "total time" for the course may now be represented by the following infinite series:

$$T + \frac{T}{2} + \frac{T}{3} + \cdots + \frac{T}{n} + \cdots. \tag{9.5}$$

In this case our physical experience does not suggest any natural or obvious "sum" to assign to this series and hence we must rely entirely on mathematical analysis to deal with this example.

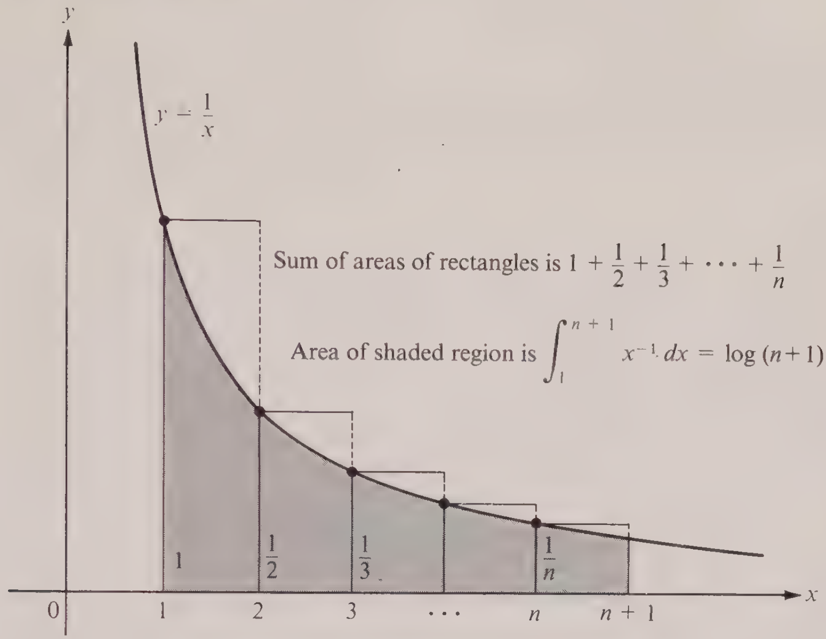


FIGURE 9.2 *Geometric meaning of the inequality* $1 + 1/2 + \cdots + 1/n \geq \log(n+1)$.

Let us proceed as before and introduce the partial sums s_n. That is, let

$$s_n = T + \frac{T}{2} + \frac{T}{3} + \cdots + \frac{T}{n}. \tag{9.6}$$

Our object is to decide what happens to s_n for larger and larger values of n. These partial sums are not as easy to study as those in (9.3) because there is no simple formula analogous to (9.4) for simplifying the expression on the right of (9.6). Nevertheless it is easy to obtain an *estimate* for the size of s_n if we compare the partial sum with an appropriate integral.

Figure 9.2 shows part of the hyperbola $y = 1/x$. (The scale is distorted along the y-axis.) The rectangles shown there have a total area equal to the sum

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}. \tag{9.7}$$

The area of the region under the hyperbola and above the interval $[1, n + 1]$ is $\int_1^{n+1} x^{-1}\, dx = \log(n + 1)$. Since this area cannot exceed the sum of the areas of the rectangles, we have the inequality†

† To prove (9.8) analytically, let $f(x) = 1/x$ if $1 \leq x \leq n + 1$, and let $t(x) = 1/k$ if $k \leq x < k + 1$ for every integer $k \geq 1$. Then $t(x) \geq f(x)$ for all x in $[1, n + 1]$; hence $\int_1^{n+1} t(x)\, dx \geq \int_1^{n+1} f(x)\, dx$. Integrating, we obtain (9.8).

$$(9.8) \qquad 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \geq \log (n + 1).$$

Multiplying both sides by T we obtain $s_n \geq T \log (n + 1)$. In other words, if the runner's speed decreases in the manner described above, the time required to reach the point $1/2^n$ is at least $T \log (n + 1)$ minutes. Since $\log (n + 1)$ increases without bound as n increases, we must agree with Zeno and conclude that the runner cannot reach his goal in any finite time.

In the same way we could change the plan of the philanthropic foundation so it agrees to contribute T dollars the first year, $T/2$ dollars the second year, $T/3$ dollars the third year and, in general, T/n dollars the nth year. Under this modified plan, the foundation would have contributed a total exceeding $T \log (n + 1)$ dollars at the end of n years and therefore would require unlimited resources to continue the plan indefinitely, even if the initial gift T were only one dollar.

The general theory of infinite series makes a distinction between series like (9.1) whose partial sums tend to a finite limit, and those like (9.5) whose partial sums have no finite limit. The former are called *convergent*, the latter *divergent*. Early investigators in the field paid little or no attention to questions of convergence or divergence. They treated infinite series as though they were ordinary finite sums, subject to the usual laws of algebra, not realizing that these laws cannot be universally extended to infinite series. Therefore it is not surprising that some of the results they obtained were later shown to be incorrect. Fortunately, many of the early pioneers possessed unusual intuition and skill which prevented them from arriving at too many false conclusions, even though they could not justify all their methods. Foremost among these men was Leonard Euler who discovered one beautiful formula after another and at the same time used infinite series as a unifying idea to bring together many branches of mathematics, hitherto unrelated. The great quantity of Euler's work that has survived the test of history is a tribute to his remarkable instinct for what is mathematically correct.

The widespread use of infinite series began late in the 17th century, nearly fifty years before Euler was born, and coincided with the early development of the integral calculus. Nicholas Mercator (1620–1687) and William Brouncker (1620–1684) discovered an infinite series for the logarithm in 1668 while attempting to calculate the area of a hyperbolic segment. Shortly thereafter, Newton discovered the *binomial series*. This discovery proved to be a landmark in the history of mathematics. A special case of the binomial series is the now-familiar *binomial theorem* which states that

$$(1 + x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k ,$$

where x is an arbitrary real number, n is a nonnegative integer, and $\binom{n}{k}$ is the binomial coefficient. Newton found that this formula could be extended from *integer* values of the exponent n to arbitrary *real* values of n by replacing the finite sum on the right by a suitable infinite series, although he gave no proof of this fact. Actually, a careful treatment of the binomial series raises some rather delicate questions of convergence that could not have been answered in Newton's time.

Shortly after Euler's death in 1783 the flood of new discoveries began to recede and the formal period in the history of series came to a close. A new and more critical period began in 1812 when Gauss published a celebrated memoir which contained, for the first time in history, a thorough and rigorous treatment of the convergence of a particular infinite series. A few years later Cauchy introduced an analytic definition of the limit concept in his treatise *Cours d'analyse algébrique* (published in 1821) and laid the foundations of the modern theory of convergence and divergence. The rudiments of that theory are discussed in the sections that follow.

9.2 Sequences

In everyday usage of the English language the words "sequence" and "series" are synonyms and they are used to suggest a succession of things or events arranged in some order. In mathematics these words have special technical meanings. The word "sequence" is employed as in the common use of the term to convey the idea of a set of things arranged in order, but the word "series" is used in a somewhat different sense. The concept of a sequence will be discussed in this section, and series will be defined in Section 9.5.

If for every positive integer n there is associated a real number a_n, then the ordered set

$$a_1, a_2, a_3, \ldots, a_n, \ldots$$

is said to define an infinite sequence. The important thing here is that each member of the set has been labeled with an integer so that we may speak of the *first term* a_1, the *second term* a_2, and, in general, the *nth term* a_n. Each term a_n has a successor a_{n+1} and hence there is no "last" term.

The most common examples of sequences can be constructed if we give some rule or formula for describing the nth term. Thus, for example, the formula $a_n = 1/n$ defines a sequence whose first five terms are

$$1, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{1}{5}.$$

Sometimes two or more formulas may be employed as, for example,

$$a_{2n-1} = 1, \qquad a_{2n} = 2n^2,$$

the first few terms in this case being

$$1, 2, 1, 8, 1, 18, 1, 32, 1.$$

Another common way to define a sequence is by a set of instructions which explains how to carry on after a given start. Thus we may have

$$a_1 = a_2 = 1, \qquad a_{n+1} = a_n + a_{n-1} \qquad \text{for} \quad n \geq 2.$$

This particular rule is known as a *recursion* formula and it defines a famous sequence whose terms are called the *Fibonacci*† *numbers.* The first few terms are

$$1, 1, 2, 3, 5, 8, 13, 21, 34.$$

† Fibonacci, also known as Leonardo of Pisa (*circa* 1175–1250), encountered this sequence in a problem concerning the offspring of rabbits.

In any sequence the essential thing is that there be some function f defined on the positive integers such that $f(n)$ is the nth term of the sequence for each $n = 1, 2, 3, \ldots$. In fact, this is probably the most convenient way to state a technical definition of sequence.

DEFINITION. A function f whose domain is the set of all positive integers $1, 2, 3, \ldots$ is called an infinite sequence. The function value $f(n)$ is called the nth term of the sequence.

The *range* of the function (that is, the set of function values) is usually displayed by writing the terms in order, thus:

$$f(1), f(2), f(3), \ldots, f(n), \ldots.$$

For brevity, the notation $\{f(n)\}$ is used to denote the sequence whose nth term is $f(n)$. Very often the dependence on n is denoted by using subscripts and we write a_n, s_n, x_n, u_n, or something similar instead of $f(n)$.

The main question we are concerned with here is to decide whether or not the terms $f(n)$ tend to a finite limit as n increases indefinitely. To treat this problem we must extend the limit concept to sequences. This is done as follows:

DEFINITION. A sequence $\{f(n)\}$ is said to have a limit L if, for every positive number ϵ, there is another positive number N (which may depend on ϵ) such that

$$|f(n) - L| < \epsilon \qquad \text{for all } n \geq N.$$

In this case we say the sequence $\{f(n)\}$ converges to L and we write

$$\lim_{n\to\infty} f(n) = L.$$

A sequence which does not converge is called divergent.

It is clear that any function defined for all positive real x may be used to construct a sequence by restricting x to take only *integer* values. This explains the strong analogy between the definition just given and the one in Section 8.11 for more general functions. The analogy carries over to *infinite limits* as well and we leave it for the reader to define the symbols

$$\lim_{n\to\infty} f(n) = +\infty \qquad \text{and} \qquad \lim_{n\to\infty} f(n) = -\infty$$

as was done in Section 8.12.

The phrase "convergent sequence" is used only for a sequence whose limit is *finite*. Sequences with limit $+\infty$ or $-\infty$ are said to diverge. There are, of course, divergent sequences that do not tend to $+\infty$ or $-\infty$. Examples are defined by the following formulas:

$$f(n) = (-1)^n, \qquad f(n) = \sin\frac{n\pi}{2}, \qquad f(n) = (-1)^n\left(1 + \frac{1}{n}\right).$$

The basic rules for dealing with limits of sums, products, etc. also hold for limits of convergent sequences. The reader should have no difficulty in formulating these theorems for himself. Their proofs are somewhat similar to those given in Section 2.29.

The convergence or divergence of many sequences may be determined by using properties of familiar functions that are defined for all positive x. We mention a few important examples whose limits may be found directly or by using some of the results derived in Chapter 8:

(9.9) $$\lim_{n\to\infty} \frac{1}{n^\alpha} = 0 \quad \text{if} \quad \alpha > 0 .$$

(9.10) $$\lim_{n\to\infty} x^n = 0 \quad \text{if} \quad |x| < 1 .$$

(9.11) $$\lim_{n\to\infty} \frac{(\log n)^a}{n^b} = 0 \qquad \text{for all } a > 0,\, b > 0 .$$

(9.12) $$\lim_{n\to\infty} n^{1/n} = 1 .$$

(9.13) $$\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = e^a \qquad \text{for all real } a .$$

9.3 Monotonic sequences

A sequence $\{f(n)\}$ is said to be *increasing* if

$$f(n) \le f(n + 1) \qquad \text{for all } n \ge 1 .$$

We indicate this briefly by writing $f(n)\nearrow$. If, on the other hand, we have

$$f(n) \ge f(n + 1) \qquad \text{for all } n \ge 1 ,$$

we call the sequence *decreasing* and write $f(n)\searrow$. A sequence is called *monotonic* if it is increasing or if it is decreasing.

Monotonic sequences are pleasant to work with because their convergence or divergence is particularly easy to determine. In fact, we have the following simple criterion:

9–1 THEOREM. A monotonic sequence converges if and only if it is bounded.†

Proof. It is clear that an unbounded sequence cannot converge. Therefore all we need to prove is that a bounded monotonic sequence must converge.

Assume $f(n)\nearrow$ and let L denote the least upper bound of the set of function values. (Since the sequence is bounded it has a least upper bound by Axiom 10 of the real-number system.) Then $f(n) \le L$ for all n, and we shall prove that the sequence converges to L.

Choose any positive number ϵ. Since $L - \epsilon$ cannot be an upper bound for *all* numbers $f(n)$ we must have $L - \epsilon < f(N)$ for some N. (This N may depend on ϵ.) If $n \ge N$ we have $f(N) \le f(n)$ since $f(n)\nearrow$. Hence we have $L - \epsilon < f(n) \le L$ for all $n \ge N$, as illustrated in Figure 9.3. From these inequalities we find

$$0 \le L - f(n) < \epsilon \qquad \text{for all } n \ge N$$

† A sequence $\{f(n)\}$ is called *bounded* if there exists a positive number M such that $|f(n)| \le M$ for all n. A sequence that is not bounded is called *unbounded*.

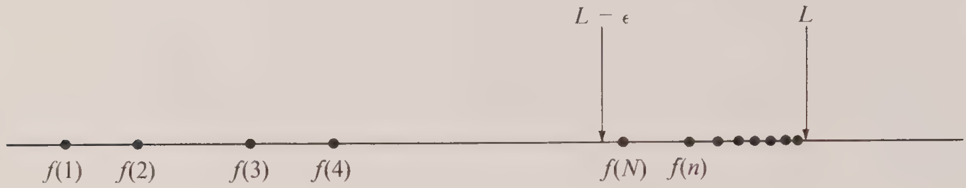


FIGURE 9.3 *A bounded increasing sequence converges to its least upper bound.*

and this means that the sequence converges to L, as asserted.

If $f(n)\searrow$ the proof is similar, the limit in this case being the greatest lower bound of the set of function values.

9.4 Exercises

In Exercises 1 through 18, a sequence $\{f(n)\}$ is defined by the formula given. In each case (a) determine whether the sequence converges or diverges and (b) find the limit of each convergent sequence. In some cases it may be helpful to replace the integer n by an arbitrary positive real x and to study the resulting function of x by the methods of Chapter 8. You may use formulas (9.9) through (9.13) listed at the end of Section 9.2.

1. $f(n) = \dfrac{n}{n+1} - \dfrac{n+1}{n}$.
2. $f(n) = \dfrac{n^2}{n+1} - \dfrac{n^2+1}{n}$.
3. $f(n) = \cos \dfrac{n\pi}{2}$.
4. $f(n) = \dfrac{n^2 + 3n - 2}{5n^2}$.
5. $f(n) = \dfrac{n}{2^n}$.
6. $f(n) = 1 + (-1)^n$.
7. $f(n) = \dfrac{1 + (-1)^n}{n}$.
8. $f(n) = \dfrac{(-1)^n}{n} + \dfrac{1 + (-1)^n}{2}$.
9. $f(n) = 2^{1/n}$.
10. $f(n) = n^{(-1)^n}$.
11. $f(n) = \dfrac{n^{2/3}\sin(n!)}{n+1}$.
12. $f(n) = \dfrac{3^n + (-2)^n}{3^{n+1} + (-2)^{n+1}}$.
13. $f(n) = \sqrt{n+1} - \sqrt{n}$.
14. $f(n) = na^n$, where $|a| < 1$.
15. $f(n) = \dfrac{\log_a n}{n}$, $a > 1$.
16. $f(n) = \dfrac{100{,}000n}{1 + n^2}$.
17. $f(n) = \left(1 + \dfrac{2}{n}\right)^n$.
18. $f(n) = 1 + \dfrac{n}{n+1} \cos \dfrac{n\pi}{2}$.

Each of the sequences $\{a_n\}$ in Exercises 19 through 24 is convergent. Therefore, for every preassigned $\epsilon > 0$, there exists an integer N (depending on ϵ) such that $|a_n - L| < \epsilon$ if $n \geq N$, where $L = \lim_{n\to\infty} a_n$. In each case determine a value of N that is suitable for each of the following values of ϵ: $\epsilon = 1, 0.1, 0.01, 0.001, 0.0001$.

19. $a_n = \dfrac{1}{n}$.

20. $a_n = \dfrac{n}{n+1}$.

21. $a_n = \dfrac{(-1)^{n+1}}{n}$.

22. $a_n = \dfrac{1}{n!}$.

23. $a_n = \dfrac{2n}{n^3+1}$.

24. $a_n = (-1)^n\left(\dfrac{9}{10}\right)^n$.

25. Prove that a sequence cannot converge to two different limits.

26. Assume $\lim_{n\to\infty} a_n = 0$. Use the definition of limit to prove that $\lim_{n\to\infty} a_n^2 = 0$.

27. If $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, use the definition of limit to prove that we have $\lim_{n\to\infty} (a_n + b_n) = A + B$, and $\lim_{n\to\infty} (ca_n) = cA$, where c is a constant.

28. From the results of Exercises 26 and 27 prove that if $\lim_{n\to\infty} a_n = A$ then $\lim_{n\to\infty} a_n^2 = A^2$. Then use the identity $2a_nb_n = (a_n + b_n)^2 - a_n^2 - b_n^2$ to prove that $\lim_{n\to\infty} (a_nb_n) = AB$ if $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$.

29. If α is a real number and n a nonnegative integer, the binomial coefficient $\binom{\alpha}{n}$ is defined by the equation

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-n+1)}{n!}.$$

(a) When $\alpha = -\frac{1}{2}$, show that

$$\binom{\alpha}{1} = -\tfrac{1}{2}, \quad \binom{\alpha}{2} = \frac{3}{8}, \quad \binom{\alpha}{3} = -\frac{5}{16}, \quad \binom{\alpha}{4} = \frac{35}{128}, \quad \binom{\alpha}{5} = -\frac{63}{256}.$$

(b) Let $a_n = (-1)^n\binom{-\frac{1}{2}}{n}$. Prove that $a_n > 0$ and that $a_{n+1} < a_n$.

9.5 Infinite series

From a given sequence of real numbers we can always generate a *new* sequence by adding together successive terms. Thus if the given sequence has the terms

$$a_1, a_2, \ldots, a_n, \ldots,$$

we may form, in succession, the "partial sums"

$$s_1 = a_1, \qquad s_2 = a_1 + a_2, \qquad s_3 = a_1 + a_2 + a_3,$$

and so on, the partial sum s_n of the first n terms being defined as follows:

(9.14)
$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k.$$

The sequence $\{s_n\}$ of partial sums is called an *infinite series*, or simply a *series*, and is also denoted by the following symbols:

(9.15)
$$a_1 + a_2 + a_3 + \cdots, \qquad a_1 + a_2 + \cdots + a_n + \cdots, \qquad \sum_{k=1}^{\infty} a_k.$$

For example, the series $\sum_{k=1}^{\infty} 1/k$ represents the sequence $\{s_n\}$ for which

$$s_n = \sum_{k=1}^{n} \frac{1}{k}.$$

The symbols in (9.15) are intended to remind us that the sequence of partial sums $\{s_n\}$ is obtained from the sequence $\{a_n\}$ by addition of successive terms.

If there is a real number S such that

$$\lim_{n\to\infty} s_n = S$$

we say that the series $\sum_{k=1}^{\infty} a_k$ is *convergent* and has the *sum* S, in which case we write

$$\sum_{k=1}^{\infty} a_k = S.$$

If $\{s_n\}$ diverges, we say that the series $\sum_{k=1}^{\infty} a_k$ *diverges* and has no sum.

Example 1. In the discussion of Zeno's paradox we showed that the partial sums s_n of the series $\sum_{k=1}^{\infty} 1/k$ satisfy the inequality

$$s_n = \sum_{k=1}^{n} \frac{1}{k} \geq \log(n+1).$$

Since $\log(n+1) \to \infty$ as $n \to \infty$, the same is true of s_n and hence the series $\sum_{k=1}^{\infty} 1/k$ diverges. This series is called the *harmonic series.*

Example 2. In the discussion of Zeno's paradox we also encountered the partial sums of the series $1 + \frac{1}{2} + \frac{1}{4} + \cdots$, given by the formula

$$\sum_{k=1}^{n} \frac{1}{2^{k-1}} = 2 - \frac{1}{2^{n-1}},$$

which is easily proved by induction. As $n \to \infty$ these partial sums approach the limit 2, hence the series converges and has sum 2. We may indicate this by writing

(9.16) $$1 + \tfrac{1}{2} + \tfrac{1}{4} + \cdots = 2.$$

The reader should realize that the word "sum" is used here in a very special sense. The sum of a convergent series is not obtained by ordinary addition but rather as the *limit of the sequence of partial sums.* Also, the reader should note that for a convergent series the symbol $\sum_{k=1}^{\infty} a_k$ is used to denote both the *series* and its *sum*, even though the two are conceptually distinct. The sum represents a *number* and it is not capable of being convergent or divergent. Once the distinction between a series and its sum has been realized, the use of one symbol to represent both should cause no confusion.

As in the case of finite summation notation, the letter k used in the symbol $\sum_{k=1}^{\infty} a_k$ is a "dummy index" and may be replaced by any other convenient symbol. The letters n, m, and r are commonly used for this purpose. Sometimes it is desirable to start the summation from $k = 0$ or from $k = 2$ or from some other value of k. Thus, for example, the series in (9.16) could be written as $\sum_{k=0}^{\infty} 1/2^k$. In general, if $p \geq 0$ we define the symbol $\sum_{k=p}^{\infty} a_k$ to mean the same as $\sum_{k=1}^{\infty} b_k$, where $b_k = a_{p+k-1}$. Thus $b_1 = a_p$, $b_2 = a_{p+1}$, etc. When there is no danger of confusion or when the starting point is unimportant, we write Σa_k instead of $\sum_{k=p}^{\infty} a_k$.

It is easy to prove that the two series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=p}^{\infty} a_k$ both converge or both diverge. Suppose we let $s_n = a_1 + \cdots + a_n$ and $t_n = a_p + a_{p+1} + \cdots + a_{p+n-1}$. If $p = 0$ we have $t_{n+1} = a_0 + s_n$, so if $s_n \to S$ as $n \to \infty$ then $t_n \to a_0 + S$ and, conversely, if $t_n \to T$ as $n \to \infty$ then $s_n \to T - a_0$. Therefore both series converge or both diverge when $p = 0$. The same holds true if $p \geq 1$. For $p = 1$ we have $s_n = t_n$ and for $p > 1$ we have $t_n = s_{n+p-1} - s_{p-1}$ and again it follows that the sequences $\{s_n\}$ and $\{t_n\}$ both converge or both diverge. This is often described by saying that a finite number of terms may be omitted or added at the beginning of a series without affecting its convergence or divergence.

9.6 The linearity property of convergent series

Ordinary finite sums have the following important properties:

$$\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k \qquad \text{(additive property)} \tag{9.17}$$

and

$$\sum_{k=1}^{n} (ca_k) = c \sum_{k=1}^{n} a_k \qquad \text{(homogeneous property)}\,. \tag{9.18}$$

The next theorem provides a natural extension of these properties to convergent infinite series and thereby justifies many algebraic manipulations in which convergent series are treated as though they were finite sums. Both additivity and homogeneity may be combined into one property called *linearity* which may be described as follows:

9–2 THEOREM. Let Σa_n and Σb_n be convergent infinite series and let α and β be constants. Then the series $\Sigma\, (\alpha a_n + \beta b_n)$ also converges and its sum is given by the equation

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n\,. \tag{9.19}$$

Proof. Using (9.17) and (9.18), we may write

$$\sum_{k=1}^{n} (\alpha a_k + \beta b_k) = \alpha \sum_{k=1}^{n} a_k + \beta \sum_{k=1}^{n} b_k\,. \tag{9.20}$$

When $n \to \infty$ the first term on the right of (9.20) tends to $\alpha \sum_{k=1}^{\infty} a_k$ and the second term tends to $\beta \sum_{k=1}^{\infty} b_k$. Therefore the left-hand side tends to their sum, and this proves that the series $\Sigma\, (\alpha a_k + \beta b_k)$ converges to the sum indicated by (9.19).

Theorem 9–2 has an interesting corollary which is often used to establish the divergence of a series.

9–3 THEOREM. If Σa_n converges and if Σb_n diverges, then $\Sigma\, (a_n + b_n)$ diverges.

Proof. Since $b_n = (a_n + b_n) - a_n$, and since Σa_n converges, Theorem 9–2 tells us that convergence of $\Sigma\,(a_n + b_n)$ implies convergence of Σb_n. Therefore $\Sigma\,(a_n + b_n)$ cannot converge if Σb_n diverges.

Example. The series $\Sigma(1/k + 1/2^k)$ diverges because $\Sigma 1/k$ diverges and $\Sigma 1/2^k$ converges.

If Σa_n and Σb_n are *both* divergent, the series $\Sigma\,(a_n + b_n)$ may or may not converge. For example, when $a_n = b_n = 1$ for all n, then $\Sigma\,(a_n + b_n)$ diverges. But when $a_n = 1$ and $b_n = -1$ for all n, then $\Sigma\,(a_n + b_n)$ converges.

9.7 Telescoping series

Another important property of finite sums is the so-called telescoping property which states that

(9.21)
$$\sum_{k=1}^{n} (b_k - b_{k+1}) = b_1 - b_{n+1}\,.$$

When we try to extend this property to infinite series we are led to consider those series Σa_n for which each term a_n may be expressed as a difference of the form

(9.22)
$$a_n = b_n - b_{n+1}\,.$$

These series are known as *telescoping series* and their behavior is characterized by the following theorem.

9– 4 THEOREM. Let $\{a_n\}$ and $\{b_n\}$ be two sequences such that

(9.23)
$$a_n = b_n - b_{n+1} \qquad \text{for } n = 1, 2, 3, \ldots.$$

Then the series Σa_n converges if and only if the sequence $\{b_n\}$ converges, in which case we have

(9.24)
$$\sum_{n=1}^{\infty} a_n = b_1 - L\,, \qquad \text{where} \quad L = \lim_{n\to\infty} b_n\,.$$

Proof. Let s_n denote the nth partial sum of Σa_n. Then we have

$$s_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n} (b_k - b_{k+1}) = b_1 - b_{n+1}\,,$$

because of (9.21). Therefore both sequences $\{s_n\}$ and $\{b_n\}$ converge or both diverge. Moreover, if $b_n \to L$ as $n \to \infty$ then $s_n \to b_1 - L$, and this proves (9.24).

Note. Every series is telescoping because we can always satisfy (9.22) if we first choose b_1 to be arbitrary and then choose $b_{n+1} = b_1 - s_n$ for $n \geq 1$, where $s_n = a_1 + \cdots + a_n$.

Example 1. Let $a_n = 1/(n^2 + n)$. Then we have

$$a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

and hence (9.23) holds with $b_n = 1/n$. Since $b_1 = 1$ and $L = 0$ we obtain

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1 .$$

Example 2. If x is not a negative integer, we have the decomposition

$$\frac{1}{(n+x)(n+x+1)(n+x+2)} = \frac{1}{2}\left(\frac{1}{(n+x)(n+x+1)} - \frac{1}{(n+x+1)(n+x+2)}\right)$$

for each integer $n \geq 1$. Therefore, by the telescoping property, the following series converges and has the sum indicated:

$$\sum_{n=1}^{\infty} \frac{1}{(n+x)(n+x+1)(n+x+2)} = \frac{1}{2(x+1)(x+2)} .$$

Example 3. Since $\log [n/(n+1)] = \log n - \log (n+1)$, and since $\log n \to \infty$ as $n \to \infty$, the series $\sum \log [n/(n+1)]$ diverges.

Note. Telescoping series illustrate an important difference between finite sums and infinite series. If we write (9.21) in extended form, it becomes

$$(b_1 - b_2) + (b_2 - b_3) + \cdots + (b_n - b_{n+1}) = b_1 - b_{n+1}$$

which can be verified by merely removing parentheses and canceling. Suppose now we perform the same operations on the infinite series

$$(b_1 - b_2) + (b_2 - b_3) + (b_3 - b_4) + \cdots .$$

We leave b_1, cancel b_2, cancel b_3, and so on. For each $n > 1$, at some stage we cancel b_n. Thus every b_n cancels with the exception of b_1. This leads us to the conclusion that the sum of the series is b_1. Because of Theorem 9-4 this conclusion is false unless $\lim_{n\to\infty} b_n = 0$. This shows that parentheses cannot always be removed in an infinite series as they can in a finite sum. (See also Exercise 24 in Section 9.9.)

9.8 The geometric series

The telescoping property of finite sums may be used to study a very important example known as the *geometric series*. This series is generated by successive addition of the terms in a geometric progression and has the form $\sum x^n$, where the nth term x^n is the nth power† of a fixed real number x.

Let s_n denote the nth partial sum of this series, so that

$$s_n = 1 + x + x^2 + \cdots + x^{n-1} .$$

If $x = 1$, each term on the right is 1 and $s_n = n$. In this case the series diverges since $s_n \to \infty$ as $n \to \infty$. If $x \neq 1$ we may simplify the sum for s_n by writing

$$(1 - x)s_n = (1 - x)\sum_{k=0}^{n-1} x^k = \sum_{k=0}^{n-1} (x^k - x^{k+1}) = 1 - x^n ,$$

† It is convenient to start this series with $n = 0$, with the understanding that the initial term, x^0, is equal to 1.

since the last sum telescopes. Dividing by $1 - x$, we obtain the formula

$$s_n = \frac{1 - x^n}{1 - x} = \frac{1}{1 - x} - \frac{x^n}{1 - x} \quad \text{if} \quad x \neq 1 .$$

This shows that the behavior of s_n for large n depends entirely on the behavior of x^n. When $|x| < 1$, then $x^n \to 0$ as $n \to \infty$ and the series converges to the sum $1/(1 - x)$. If $|x| > 1$ or if $x = -1$, the term x^n does not tend to a finite limit and the series diverges. Therefore we have proved the following theorem:

9- 5 THEOREM. If $|x| < 1$, the geometric series $\sum_{n=0}^{\infty} x^n$ converges and has sum $1/(1 - x)$. That is to say, we have

$$1 + x + x^2 + \cdots + x^n + \cdots = \frac{1}{1 - x} \quad \text{if} \quad |x| < 1 . \tag{9.25}$$

If $|x| \geq 1$, the series diverges.

The geometric series, with $|x| < 1$, is one of those rare examples whose sum we are able to determine by finding first a simple formula for its partial sums. (A special case with $x = \frac{1}{2}$ was encountered in Section 9.1 in connection with Zeno's paradox.) The real importance of this series lies in the fact that it may be used as a starting point for determining the sums of a large number of other interesting series. For example, if we assume $|x| < 1$ and replace x by x^2 in (9.25), we obtain the formula

$$1 + x^2 + x^4 + \cdots + x^{2n} + \cdots = \frac{1}{1 - x^2} \quad \text{if} \quad |x| < 1 . \tag{9.26}$$

Notice that this series contains those terms of (9.25) with *even* exponents. To find the sum of the odd powers alone we need only multiply both sides of (9.26) by x to obtain

$$x + x^3 + x^5 + \cdots + x^{2n+1} + \cdots = \frac{x}{1 - x^2} \quad \text{if} \quad |x| < 1 . \tag{9.27}$$

If we replace x by $-x$ in (9.25), we find

$$1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots = \frac{1}{1 + x} \quad \text{if} \quad |x| < 1 . \tag{9.28}$$

Replacing x by x^2 in (9.28), we find

$$1 - x^2 + x^4 - x^6 + \cdots + (-1)^n x^{2n} + \cdots = \frac{1}{1 + x^2} \quad \text{if} \quad |x| < 1 . \tag{9.29}$$

Multiplying both sides of (9.29) by x, we obtain

$$x - x^3 + x^5 - x^7 + \cdots + (-1)^n x^{2n+1} + \cdots = \frac{x}{1 + x^2} \quad \text{if} \quad |x| < 1 . \tag{9.30}$$

If we replace x by $2x$ in (9.26) we find

$$1 + 4x^2 + 16x^4 + \cdots + 4^n x^{2n} + \cdots = \frac{1}{1 - 4x^2},$$

which is valid if $|2x| < 1$ or, what is the same thing, if $|x| < \frac{1}{2}$. It is clear that many other examples may be constructed by similar means.

All these series have the special form

$$\sum_{n=0}^{\infty} a_n x^n$$

and are known as *power series.* The numbers $a_0, a_1, a_2, \ldots$ are called *coefficients* of the power series. The geometric series is an example with all the coefficients equal to 1. We shall find later, when we discuss the general theory of power series, that it is permissible to differentiate and to integrate both sides of each of the equations (9.25) through (9.30), treating the left-hand members as though they were ordinary finite sums. These operations lead to many remarkable new formulas. For example, differentiation of (9.25) gives us

(9.31) $$1 + 2x + 3x^2 + \cdots + nx^{n-1} + \cdots = \frac{1}{(1 - x)^2} \quad \text{if} \quad |x| < 1,$$

whereas integration of (9.28) yields the interesting formula

(9.32) $$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{(-1)^n x^{n+1}}{n + 1} + \cdots = \log(1 + x)$$

which expresses the logarithm as a power series. This is the discovery of Mercator and Brouncker (1668) that we mentioned earlier. Although each of the equations (9.25) through (9.31) is valid for x in the open interval $-1 < x < +1$, it turns out that the logarithmic series in (9.32) is valid at the endpoint $x = +1$ as well.

Another important example, which may be obtained by integration of (9.29), is the following power-series expansion for the inverse tangent, discovered in 1671 by James Gregory (1638–1675):

(9.33) $$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + \frac{(-1)^n x^{2n+1}}{2n + 1} + \cdots = \arctan x.$$

Gregory's series converges and represents the inverse tangent function everywhere in the closed interval $-1 \le x \le +1$.

Many of the other elementary functions of calculus, such as the sine, cosine, and exponential, may also be represented by power series. This is not too surprising, in view of Taylor's formula which tells us that any function may be approximated by a polynomial in x of degree $\le n$ if it has derivatives of order $n + 1$ in some neighborhood of the origin. In the examples given above, the partial sums of the power series are precisely these approximating polynomials. When a function f has derivatives of *every* order in a neighborhood of the origin then for every positive integer n Taylor's formula leads to an equation of the form

(9.34) $$f(x) = \sum_{k=0}^{n} a_k x^k + R_n(x),$$

where the finite sum $\sum_{k=0}^{n} a_k x^k$ is an approximating polynomial of degree $\leq n$ and $R_n(x)$ is the error for this approximation. If, now, we keep x fixed and let n increase without bound in (9.34), the approximating polynomials give rise to a power series, namely $\sum_{k=0}^{\infty} a_k x^k$, where each coefficient a_k is determined as follows:

$$a_k = \frac{f^{(k)}(0)}{k!}.$$

If, for some x, the remainder $R_n(x)$ tends to 0 as $n \to \infty$, then for this x we have

$$\lim_{n\to\infty} \sum_{k=0}^{n} a_k x^k = \sum_{k=0}^{\infty} a_k x^k = f(x),$$

or, in other words, the power series in question converges to $f(x)$. If x is a point for which $R_n(x)$ does not tend to 0 as $n \to \infty$, then the partial sums will not approach $f(x)$. Conditions on f for guaranteeing that $R_n(x) \to 0$ will be discussed later in Section 9.22.

To lay a better foundation for the general theory of power series, we turn next to certain general questions related to the convergence and divergence of arbitrary series. We shall return to the subject of power series in Section 9.19.

9.9 Exercises

Each of the series in Exercises 1 through 10 is a telescoping series, or a geometric series, or some related series whose partial sums may be simplified. In each case prove that the series converges and has the sum indicated.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}.$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{2}{3^{n-1}} = 3.$

3. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^2-1} = \frac{3}{4}.$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n+3^n}{6^n} = \frac{3}{2}.$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n^2+n}} = 1.$

6. $\displaystyle\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)(n+3)} = \frac{1}{4}.$

7. $\displaystyle\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2} = 1.$

8. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n+n^2+n}{2^{n+1}n(n+1)} = 1.$

9. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(2n+1)}{n(n+1)} = 1.$

10. $\displaystyle\sum_{n=2}^{\infty} \frac{\log\,[(1+1/n)^n(1+n)]}{(\log n^n)[\log\,(n+1)^{n+1}]} = \log_2 \sqrt{e}.$

Power series for $\log\,(1 + x)$ and arc tan x were obtained in Section 9.8 by performing various operations on the geometric series. In a similar manner, without attempting to justify the steps, obtain the formulas in Exercises 11 through 19. They are all valid at least for $|x| < 1$. (The theoretical justification is provided in Section 9.21.)

11. $\displaystyle\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$.

12. $\displaystyle\sum_{n=1}^{\infty} n^2x^n = \frac{x^2+x}{(1-x)^3}$.

13. $\displaystyle\sum_{n=1}^{\infty} n^3x^n = \frac{x^3+4x^2+x}{(1-x)^4}$.

14. $\displaystyle\sum_{n=1}^{\infty} n^4x^n = \frac{x^4+11x^3+11x^2+x}{(1-x)^5}$.

15. $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n} = \log\frac{1}{1-x}$.

16. $\displaystyle\sum_{n=1}^{\infty} \frac{x^{2n-1}}{2n-1} = \tfrac{1}{2}\log\frac{1+x}{1-x}$.

17. $\displaystyle\sum_{n=0}^{\infty} (n+1)x^n = \frac{1}{(1-x)^2}$.

18. $\displaystyle\sum_{n=0}^{\infty} \frac{(n+1)(n+2)}{2!}x^n = \frac{1}{(1-x)^3}$.

19. $\displaystyle\sum_{n=0}^{\infty} \frac{(n+1)(n+2)(n+3)}{3!}x^n = \frac{1}{(1-x)^4}$.

20. The results of Exercises 11 through 14 suggest that there exists a general formula of the form

$$\sum_{n=1}^{\infty} n^k x^n = \frac{P_k(x)}{(1-x)^{k+1}},$$

where $P_k(x)$ is a polynomial of degree k, the term of lowest degree being x and that of highest degree being x^k. Prove this by induction, without attempting to justify the formal manipulations with the series.

21. The results of Exercises 17 through 19 suggest the more general formula

$$\sum_{n=0}^{\infty} \binom{n+k}{k} x^n = \frac{1}{(1-x)^k}, \qquad \text{where} \quad \binom{n+k}{k} = \frac{(n+1)(n+2)\cdots(n+k)}{k!}.$$

Prove this by induction, without attempting to justify the formal manipulations with the series.

22. Given that $\sum_{n=0}^{\infty} x^n/n! = e^x$ for all x, find the sums of the following series, assuming it is permissible to operate on infinite series as though they were finite sums.

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{n-1}{n!}$. (b) $\displaystyle\sum_{n=2}^{\infty} \frac{n+1}{n!}$. (c) $\displaystyle\sum_{n=2}^{\infty} \frac{(n-1)(n+1)}{n!}$.

23. (a) Given that $\sum_{n=0}^{\infty} x^n/n! = e^x$ for all x, show that

$$\sum_{n=1}^{\infty} \frac{n^2x^n}{n!} = (x^2+x)e^x,$$

assuming it is permissible to operate on these series as though they were finite sums.

(b) The sum of the series $\sum_{n=1}^{\infty} n^3/n!$ is ke, where k is a positive integer. Find the value of k. Do not attempt to justify formal manipulations.

24. Two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are called *identical* if $a_n = b_n$ for each $n \geq 1$. For example, the series

$$0+0+0+\cdots \qquad \text{and} \qquad (1-1)+(1-1)+(1-1)+\cdots$$

are identical, but the series

$$1+1+1+\cdots \qquad \text{and} \qquad 1+0+1+0+1+0+\cdots$$

are not identical. Determine whether or not the series are identical in each of the following pairs:

(a) $1 - 1 + 1 - 1 + \cdots$ and $(2 - 1) - (3 - 2) + (4 - 3) - (5 - 4) + \cdots$.
(b) $1 - 1 + 1 - 1 + \cdots$ and $(1 - 1) + (1 - 1) + (1 - 1) + (1 - 1) + \cdots$.
(c) $1 - 1 + 1 - 1 + \cdots$ and $1 + (-1 + 1) + (-1 + 1) + (-1 + 1) + \cdots$.
(d) $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ and $1 + (1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{4}) + (\frac{1}{4} - \frac{1}{8}) + \cdots$.

25. (a) Use (9.26) to prove that

$$1 + 0 + x^2 + 0 + x^4 + \cdots = \frac{1}{1 - x^2} \quad \text{if} \quad |x| < 1 .$$

Note that, according to the definition given in Exercise 24, this series is not identical to the one in (9.26) if $x \neq 0$.

(b) Apply Theorem 9–2 to the result in part (a) and to (9.25) to deduce (9.27).
(c) Show that Theorem 9–2 when applied directly to (9.25) and (9.26) does not yield (9.27). Instead, it yields the formula $\sum_{n=1}^{\infty} (x^n - x^{2n}) = x/(1 - x^2)$, valid for $|x| < 1$.

★9.10 Exercises on decimal expansions

Decimal representations of real numbers were introduced in Section 1.16. It was shown there that every positive real x has a decimal representation of the form

$$x = a_0 \,.\, a_1a_2a_3 \cdots ,$$

where $0 \leq a_k \leq 9$ for each $k \geq 1$. The number x is related to the digits $a_0, a_1, a_2, \ldots$ by the inequalities

$$a_0 + \frac{a_1}{10} + \cdots + \frac{a_n}{10^n} \leq x < a_0 + \frac{a_1}{10} + \cdots + \frac{a_{n-1}}{10^{n-1}} + \frac{a_n + 1}{10^n} . \tag{9.35}$$

If we let $s_n = \sum_{k=0}^{n} a_k/10^k$, and if we subtract s_n from each member of (9.35), we obtain

$$0 \leq x - s_n < 10^{-n} .$$

This shows that $s_n \to x$ as $n \to \infty$, and hence x is given by the convergent series

$$x = \sum_{k=0}^{\infty} \frac{a_k}{10^k} . \tag{9.36}$$

Each of the infinite decimal expansions in Exercises 1 through 5 is understood to be repeated indefinitely as suggested. In each case, express the decimal as an infinite series, find the sum of the series, and thereby express x as a quotient of two integers.

1. $x = 0.4444 \cdots$.
2. $x = 0.51515151 \cdots$.
3. $x = 2.02020202 \cdots$.
4. $x = 0.123123123123 \cdots$.
5. $x = 0.142857142857142857142857 \cdots$.

6. Prove that every repeating decimal represents a rational number.

7. If a number has a decimal expansion which ends in zeros, such as $\frac{1}{8} = 0.1250000 \cdots$, then this number can also be written as a decimal which ends in nines if we decrease the last nonzero digit by one unit. For example, $\frac{1}{8} = 0.1249999 \cdots$. Use infinite series to prove this statement.

The decimal representation in (9.36) may be generalized by replacing the integer 10 by any other integer $b > 1$. If $x > 0$, let a_0 denote the greatest integer in x; assuming that $a_0, a_1, \ldots, a_{n-1}$ have been defined, let a_n denote the largest integer such that

$$\sum_{k=0}^{n} \frac{a_k}{b^k} \leq x .$$

The following exercises refer to the sequence of integers $a_0, a_1, a_2, \ldots$ so obtained.

8. Show that $0 \leq a_k \leq b - 1$ for each $k \geq 1$.
9. Describe a geometric method for obtaining the numbers $a_0, a_1, a_2, \ldots$.
10. Show that the series $\sum_{k=0}^{\infty} a_k/b^k$ converges and has sum x. This provides a decimal expansion of x in the scale of b. Important special cases, other than $b = 10$, are the *binary scale*, $b = 2$, and the *duodecimal scale*, $b = 12$.

9.11 Tests for convergence. Comparison tests

In theory, the convergence or divergence of a particular series Σa_n is decided by examining its partial sums s_n to see whether or not they tend to a finite limit as $n \to \infty$. In some special cases, such as the geometric series, the sums defining s_n may be simplified to the point where it becomes a simple matter to determine their behavior for large n. However, in the majority of cases there is no nice formula for simplifying s_n and the convergence or divergence may be rather difficult to establish in a straightforward manner. Early investigators in the subject, notably Cauchy and his contemporaries, realized this difficulty and they developed a number of "convergence tests" that bypassed the need for an explicit knowledge of the partial sums. A few of the simplest and most useful of these tests will be discussed in this section, but first we want to make some general remarks about the nature of these tests.

Convergence tests may be broadly classified into three categories: (i) *sufficient* conditions; (ii) *necessary* conditions; (iii) *necessary and sufficient* conditions. A test of type (i) may be expressed symbolically as follows:

"If C is satisfied, then Σa_n converges,"

where C stands for the condition in question. Tests of type (ii) have the form

"If Σa_n converges, then C is satisfied,"

whereas those of type (iii) may be written thus:

"Σa_n converges if and only if C is satisfied."

We shall see presently that there are tests of type (ii) that are not of type (i) (and vice versa). Beginners often use such tests incorrectly by failing to realize the difference between a *necessary* condition and a *sufficient* condition. Therefore the reader should make an effort to keep this distinction in mind when using a particular test in practice.

The simplest of all convergence tests gives a *necessary* condition for convergence and may be stated as follows:

9–6 THEOREM. If the series Σa_n converges, then

$$\lim_{n\to\infty} a_n = 0 . \tag{9.37}$$

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$. Then $a_n = s_n - s_{n-1}$. As $n \to \infty$, both s_n and s_{n-1} tend to the same limit and hence $a_n \to 0$. This proves the theorem.

This is an example of a test of type (ii) which is not of type (i). Condition (9.37) is *not* sufficient for convergence. For example, when $a_n = 1/n$, the condition $a_n \to 0$ is satisfied but the series $\Sigma 1/n$ diverges. The real usefulness of this test is that it gives us a *sufficient* condition for *divergence*. That is, if the terms a_n of a series Σa_n do *not* tend to zero, then the series must diverge. This statement is logically equivalent to Theorem 9–6.

In the remainder of this section we shall be concerned primarily with series having *nonnegative terms*, that is, series of the form Σa_n, where each $a_n \geq 0$. Since the partial sums of such series are monotonic increasing, we may use Theorem 9–1 to obtain the following *necessary and sufficient condition* for convergence.

9–7 THEOREM. Assume that $a_n \geq 0$ for each $n \geq 1$. Then the series Σa_n converges if and only if the sequence of its partial sums is bounded above.

If the partial sums are bounded above by a number M, say, then the sum of the series cannot exceed M.

Example 1. Theorem 9–7 may be used to establish the convergence of the series $\Sigma_{n=1}^{\infty} 1/n!$. We estimate the partial sums from above by using the inequality

$$\frac{1}{k!} \leq \frac{1}{2^{k-1}}$$

which is obviously true for all $k \geq 1$ since $k!$ consists of $k - 1$ factors, each ≥ 2. Therefore

$$\sum_{k=1}^{n} \frac{1}{k!} \leq \sum_{k=1}^{n} \frac{1}{2^{k-1}} = \sum_{k=0}^{n-1} (\tfrac{1}{2})^k \leq \sum_{k=0}^{\infty} (\tfrac{1}{2})^k = 2,$$

the last series being a geometric series. The series $\Sigma_{n=1}^{\infty} 1/n!$ is therefore convergent and has a sum ≤ 2. We shall see later that the sum of this series is $e - 1$, where e is the Euler number.

The convergence of the foregoing example was established by comparing the terms of the given series with those of a series known to converge. This idea may be pursued further to yield a number of tests known as *comparison tests.*

9–8 THEOREM. *Comparison test.* Assume $a_n \geq 0$ and $b_n \geq 0$ for all $n \geq 1$. If there exists a positive constant c such that

$$a_n \leq cb_n \tag{9.38}$$

for all n, then convergence of Σb_n implies convergence of Σa_n.

Note. The conclusion may also be formulated as follows: "Divergence of Σa_n implies divergence of Σb_n." This statement is logically equivalent to Theorem 9–8. When the inequality (9.38) is satisfied we say that the series Σb_n *dominates* the series Σa_n.

Proof. Let $s_n = a_1 + \cdots + a_n$, $t_n = b_1 + \cdots + b_n$. Then (9.38) implies $s_n \leq ct_n$. If Σb_n converges its partial sums are bounded, say by M. Then $s_n \leq cM$ and hence Σa_n is also convergent since its partial sums are bounded by cM. This completes the proof.

Omitting a finite number of terms at the beginning of a series does not affect its convergence or divergence. Therefore Theorem 9–8 still holds true if the inequality (9.38) is valid only for all $n \geq N$ for some N.

9–9 THEOREM. *Limit comparison test.* Assume that $a_n > 0$ and $b_n > 0$ for all $n \geq 1$, and suppose that

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1 . \tag{9.39}$$

Then Σa_n converges if and only if Σb_n converges.

Proof. There exists an N such that $n \geq N$ implies $\frac{1}{2} < a_n/b_n < \frac{3}{2}$. Therefore $b_n < 2a_n$ and $a_n < \frac{3}{2}b_n$ for all $n \geq N$, and the theorem follows by applying Theorem 9–8 twice.

Note that Theorem 9–9 also holds if $\lim_{n\to\infty} a_n/b_n = c$, provided that $c > 0$, because we then have $\lim_{n\to\infty} a_n/(cb_n) = 1$ and we may compare Σa_n with $\Sigma(cb_n)$. However, if $\lim_{n\to\infty} a_n/b_n = 0$ we conclude only that convergence of Σb_n implies convergence of Σa_n.

Two sequences $\{a_n\}$ and $\{b_n\}$ are said to be *asymptotically equal* if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1 .$$

This relation is often indicated symbolically by writing

$$a_n \sim b_n \qquad \text{as} \quad n \to \infty . \tag{9.40}$$

The notation $a_n \sim b_n$ is read "a_n is asymptotically equal to b_n" and it is intended to suggest that a_n and b_n behave in essentially the same way for large n. Using this terminology, we may state the limit comparison test as follows:

> *Two series Σa_n and Σb_n with terms that are positive and asymptotically equal converge together or they diverge together.*

The O-notation (introduced in Section 7.9) may be immediately adapted to sequences by considering functions defined on the positive integers. Thus if $\{a_n\}$ and $\{b_n\}$ are two sequences with $b_n \geq 0$, the notation

$$a_n = O(b_n) \qquad \text{for } n \text{ in } S$$

means that there is a constant $M > 0$ such that $|a_n| \leq Mb_n$ for each n in S. In practice, S is usually the set of all positive integers or else the set of all integers $\geq N$ for some $N > 0$ and the reference "for n in S" is ordinarily omitted.

The comparison test may be restated in terms of the O-notation as follows: If $a_n \geq 0$ and $b_n \geq 0$ and if $a_n = O(b_n)$, then convergence of Σb_n implies convergence of Σa_n. If both relations $a_n = O(b_n)$ and $b_n = O(a_n)$ hold simultaneously, then we say that the two sequences $\{a_n\}$ and $\{b_n\}$ have *the same order of magnitude*. In this case both series Σa_n and Σb_n converge or else both diverge. In particular, two nonnegative sequences $\{a_n\}$ and

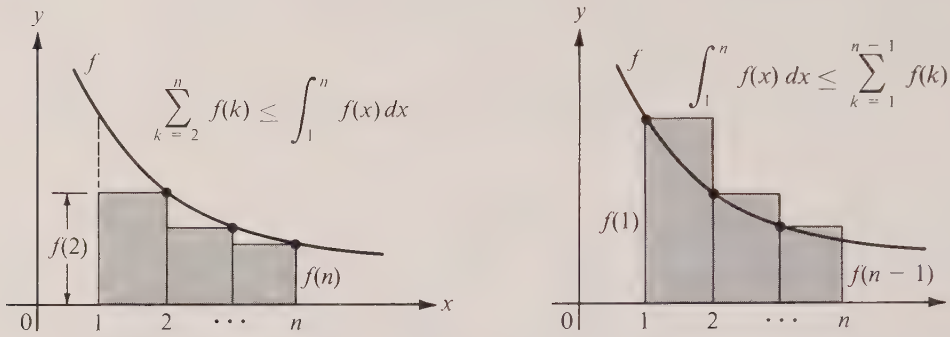


FIGURE 9.4 *Proof of the integral test.*

$\{b_n\}$ have the same order of magnitude if $a_n \sim b_n$ or, more generally, if $a_n \sim cb_n$ for some $c > 0$.

Example 2. In Example 1 of Section 9.7 we proved that $\Sigma 1/(n^2 + n)$ is a convergent telescoping series. If we use this as a comparison series it follows that $\Sigma 1/n^2$ is convergent, since $1/n^2 \sim 1/(n^2 + n)$ as $n \to \infty$. Also, $\Sigma 1/n^2$ dominates $\Sigma 1/n^s$ for $s \ge 2$, and therefore $\Sigma 1/n^s$ converges for every real $s \ge 2$. We shall prove in the next section that this series also converges for every $s > 1$. Its sum, denoted by $\zeta(s)$ (ζ is the Greek letter zeta), defines an important function in analysis known as the *Riemann zeta-function:*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad \text{if} \quad s > 1 .$$

Euler discovered many beautiful formulas involving $\zeta(s)$. In particular, he found that $\zeta(2) = \pi^2/6$, a result which is not easy to derive at this stage.

Example 3. Since $\Sigma 1/n$ diverges, every series having positive terms asymptotically equal to $1/n$ must also diverge. For example, this is true of the two series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+10)}} \quad \text{and} \quad \sum_{n=1}^{\infty} \sin \frac{1}{n} .$$

The relation $\sin 1/n \sim 1/n$ follows from the fact that $(\sin x)/x \to 1$ as $x \to 0$.

9.12 The integral test

To use comparison tests effectively, we must have at our disposal some examples of series of known behavior. The geometric series and the zeta-function are useful for this purpose. New examples can be obtained very simply by applying the so-called *integral test*, first proved by Cauchy in 1837.

9–10 THEOREM. *Integral test.* Let f be a positive decreasing function, defined for all real $x \ge 1$. For each $n \ge 1$, let

$$s_n = \sum_{k=1}^{n} f(k) \qquad \text{and} \qquad t_n = \int_1^n f(x)\,dx\,.$$

Then both sequences $\{s_n\}$ and $\{t_n\}$ converge or both diverge.

Proof. By comparing f with appropriate step functions as suggested in Figure 9.4, we obtain the inequalities†

$$\sum_{k=2}^{n} f(k) \le \int_1^n f(x)\,dx \le \sum_{k=1}^{n-1} f(k)\,, \tag{9.41}$$

or $s_n - f(1) \le t_n \le s_{n-1}$. Since both sequences $\{s_n\}$ and $\{t_n\}$ are monotonic increasing, these inequalities show that both are bounded above or both are unbounded. Therefore both sequences converge or both diverge, as asserted.

Example 1. The integral test enables us to prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^s} \qquad \text{converges if and only if} \quad s > 1\,.$$

Taking $f(x) = x^{-s}$, we have

$$t_n = \int_1^n \frac{1}{x^s}\,dx = \begin{cases} \dfrac{n^{1-s} - 1}{1 - s} & \text{if} \quad s \ne 1\,, \\ \log n & \text{if} \quad s = 1\,. \end{cases}$$

When $s > 1$ the term $n^{1-s} \to 0$ as $n \to \infty$ and hence $\{t_n\}$ converges. By the integral test, this implies convergence of the series for $s > 1$.

When $s \le 1$, then $t_n \to \infty$ as $n \to \infty$ and the series diverges. The special case $s = 1$ (the *harmonic series*) was discussed earlier in Section 9.5. Its divergence was known to Leibniz.

Example 2. The same method may be used to prove that

$$\sum_{n=2}^{\infty} \frac{1}{n\,(\log n)^s} \qquad \text{converges if and only if} \quad s > 1\,.$$

(We start the sum with $n = 2$ to avoid n for which $\log n$ may be zero).

The corresponding integral in this case is

$$t_n = \int_2^n \frac{1}{x(\log x)^s}\,dx = \begin{cases} \dfrac{(\log n)^{1-s} - (\log 2)^{1-s}}{1 - s} & \text{if} \quad s \ne 1\,, \\ \log(\log n) - \log(\log 2) & \text{if} \quad s = 1\,. \end{cases}$$

Thus $\{t_n\}$ converges if and only if $s > 1$ and hence, by the integral test, the same holds true for the series in question.

9.13 Exercises

Test the following series for convergence or divergence. In each case give a reason for your decision.

† For each integer $k \ge 1$, let $s(x) = f(k+1)$ if $k < x \le k+1$ and let $t(x) = f(k)$ if $k \le x < k+1$. Also, define $s(1) = 0$. Then, on the interval $[1, n]$, s and t are step functions such that $s(x) \le f(x) \le t(x)$. Integration of these inequalities yields (9.41).

1. $\displaystyle\sum_{n=1}^{\infty} \frac{n}{(4n-3)(4n-1)}$.

2. $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{2n-1}\log(4n+1)}{n(n+1)}$.

3. $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{2^n}$.

4. $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{2^n}$.

5. $\displaystyle\sum_{n=1}^{\infty} \frac{|\sin nx|}{n^2}$.

6. $\displaystyle\sum_{n=1}^{\infty} \frac{2+(-1)^n}{2^n}$.

7. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{(n+2)!}$.

8. $\displaystyle\sum_{n=2}^{\infty} \frac{\log n}{n\sqrt{n+1}}$.

9. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$.

10. $\displaystyle\sum_{n=1}^{\infty} \frac{1+\sqrt{n}}{(n+1)^3-1}$.

11. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{(\log n)^s}$.

12. $\displaystyle\sum_{n=1}^{\infty} \frac{|a_n|}{10^n}$, $\quad |a_n| < 10$.

13. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1000n+1}$.

14. $\displaystyle\sum_{n=1}^{\infty} \frac{n\cos^2(n\pi/3)}{2^n}$.

15. $\displaystyle\sum_{n=3}^{\infty} \frac{1}{n\log n\,(\log\log n)^s}$.

16. $\displaystyle\sum_{n=1}^{\infty} ne^{-n^2}$.

17. $\displaystyle\sum_{n=1}^{\infty} \int_0^{1/n} \frac{\sqrt{x}}{1+x^2}\,dx$.

18. $\displaystyle\sum_{n=1}^{\infty} \int_n^{n+1} e^{-\sqrt{x}}\,dx$.

19. Assume f is a nonnegative increasing function defined for all $x \geq 1$. Use the method suggested by the proof of the integral test to show that

$$\sum_{k=1}^{n-1} f(k) \leq \int_1^n f(x)\,dx \leq \sum_{k=2}^{n} f(k).$$

Take $f(x) = \log x$ and deduce the inequalities

$$e\,n^n e^{-n} < n! < (n+1)^{n+1} e^{-n}.$$

These give a rough estimate of the order of magnitude of $n!$. A more exact estimate is provided by *Stirling's formula* which states that

$$\sqrt{2\pi}\, n^{n+1/2} e^{-n} < n! < \sqrt{2\pi}\, n^{n+1/2} e^{-n}\left(1+\frac{1}{4n}\right).$$

This is discussed more fully in Volume II.

9.14 The root test and the ratio test

Using the geometric series Σx^n as a comparison series, Cauchy developed two useful tests known as the *root test* and the *ratio test*.

If Σa_n is a series whose terms (from some point on) satisfy an inequality of the form

$$0 \leq a_n \leq x^n, \quad \text{where} \quad 0 < x < 1, \tag{9.42}$$

a direct application of the comparison test (Theorem 9–8) tells us that Σa_n converges. The inequalities in (9.42) are equivalent to

$$0 \leq a_n^{1/n} \leq x \,, \tag{9.43}$$

hence the name *root test.*

If the sequence $\{a_n^{1/n}\}$ is convergent the test may be restated in a somewhat more useful form that makes no reference to the number x.

9–11 THEOREM. *Root test.* Let Σa_n be a series of nonnegative terms such that

$$a_n^{1/n} \to R \qquad \text{as} \quad n \to \infty \,.$$

(a) If $R < 1$, the series converges.
(b) If $R > 1$, the series diverges.
(c) If $R = 1$, the test is inconclusive.

Proof. Assume $R < 1$ and choose x so that $R < x < 1$. Then (9.43) must be satisfied for all $n \geq N$ for some N. Hence Σa_n converges by the comparison test. This proves (a).

To prove (b), we observe that $R > 1$ implies $a_n > 1$ for infinitely many values of n and hence a_n cannot tend to 0. Therefore, by Theorem 9–6, Σa_n diverges. This proves (b).

To prove (c), consider the two examples in which $a_n = 1/n$ and $a_n = 1/n^2$. In both cases $R = 1$ since $n^{1/n} \to 1$ as $n \to \infty$ [see Equation (9.12) of Section 9.2], but $\Sigma 1/n$ diverges whereas $\Sigma 1/n^2$ converges.

Example 1. The root test makes it easy to determine the convergence of the series $\Sigma_{n=3}^{\infty} (\log n)^{-n}$ since

$$a_n^{1/n} = \frac{1}{\log n} \to 0 \qquad \text{as} \quad n \to \infty \,.$$

Example 2. Applying the root test to $\Sigma[n/(n+1)]^{n^2}$, we find

$$a_n^{1/n} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n} \to \frac{1}{e} \qquad \text{as} \quad n \to \infty$$

by Equation (9.13) of Section 9.2. Since $1/e < 1$, the series converges.

A slightly different use of the comparison test yields the ratio test.

9–12 THEOREM. *Ratio test.* Let Σa_n be a series of positive terms such that

$$\frac{a_{n+1}}{a_n} \to L \qquad \text{as} \quad n \to \infty \,.$$

(a) If $L < 1$, the series converges.
(b) If $L > 1$, the series diverges.
(c) If $L = 1$, the test is inconclusive.

Proof. Assume $L < 1$ and choose x so that $L < x < 1$. Then there must be an N such that $a_{n+1}/a_n < x$ for all $n \geq N$. This implies

$$\frac{a_{n+1}}{x^{n+1}} < \frac{a_n}{x^n} \qquad \text{for all } n \geq N .$$

In other words, the sequence $\{a_n/x^n\}$ is decreasing for $n \geq N$. In particular, when $n \geq N$ we must have $a_n/x^n \leq a_N/x^N$, or, in other words,

$$a_n \leq cx^n , \qquad \text{where} \quad c = \frac{a_N}{x^N} .$$

Therefore Σa_n is dominated by the convergent series Σx^n. This proves (a).

To prove (b), we simply observe that $L > 1$ implies $a_{n+1} > a_n$ for all $n \geq N$ for some N, and hence a_n cannot approach 0.

Finally, (c) is proved by using the same examples as in Theorem 9–11.

Warning. If the test ratio a_{n+1}/a_n is always less than 1, it does not necessarily follow that the *limit* L will be less than 1. For example, the harmonic series, which diverges, has test ratio $n/(n + 1)$ which is always less than 1 but the limit L equals 1. On the other hand, for divergence it is sufficient that the test ratio be greater than 1 for all sufficiently large n because for such n we have $a_{n+1} > a_n$ and a_n cannot approach 0.

Example 3. We may establish the convergence of the series $\Sigma n!/n^n$ by the ratio test. The ratio of consecutive terms is

$$\frac{a_{n+1}}{a_n} = \frac{(n + 1)!}{(n + 1)^{n+1}} \cdot \frac{n^n}{n!} = \left(\frac{n}{n + 1}\right)^n = \frac{1}{(1 + 1/n)^n} \rightarrow \frac{1}{e} \qquad \text{as} \quad n \rightarrow \infty ,$$

by formula (9.13) of Section 9.2. Since $1/e < 1$ the series converges. In particular, this implies that the general term of the series tends to 0; that is,

$$\frac{n!}{n^n} \rightarrow 0 \qquad \text{as} \quad n \rightarrow \infty . \tag{9.44}$$

This is often described by saying that n^n "grows faster" than $n!$ for large n. Also, with a natural extension of the o-notation, we can write (9.44) as follows: $n! = o(n^n)$ as $n \rightarrow \infty$.

Note. The relation (9.44) may also be proved directly by writing

$$\frac{n!}{n^n} = \frac{1}{n} \cdot \frac{2}{n} \cdot \cdots \cdot \frac{k}{n} \cdot \frac{k + 1}{n} \cdot \cdots \cdot \frac{n}{n} ,$$

where $k = n/2$ if n is even, and $k = (n - 1)/2$ if n is odd. If $n \geq 2$, the product of the first k factors on the right does not exceed $(\frac{1}{2})^k$, and each of the remaining factors does not exceed 1. Since $(\frac{1}{2})^k \rightarrow 0$ as $n \rightarrow \infty$, this proves (9.44).

The reader should realize that both the root test and the ratio test are, in reality, special cases of the comparison test. In both tests when we have case (a) convergence is deduced from the fact that the series in question can be dominated by a suitable geometric series Σx^n. The usefulness of these tests in practice is that a knowledge of a particular comparison series Σx^n is not explicitly required. Further convergence tests may be deduced by using the comparison test in other ways. Two important examples known as *Raabe's test* and *Gauss' test* are described in Exercises 16 and 17 of Section 9.15. These are often helpful when the ratio test fails.

9.15 Exercises

Test the following series for convergence or divergence and give a reason for your decision in each case.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$.

2. $\displaystyle\sum_{n=1}^{\infty} \frac{(n!)^2}{2^{n^2}}$.

3. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$.

4. $\displaystyle\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$.

5. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{3^n}$.

6. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{2^{2n}}$.

7. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{(\log n)^{1/n}}$.

8. $\displaystyle\sum_{n=1}^{\infty} (\sqrt[n]{n} - 1)^n$.

9. $\displaystyle\sum_{n=1}^{\infty} e^{-n^2}$.

10. $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{n} - e^{-n^2}\right)$.

11. $\displaystyle\sum_{n=1}^{\infty} \frac{(1000)^n}{n!}$.

12. $\displaystyle\sum_{n=1}^{\infty} \frac{n^{n+1/n}}{(n + 1/n)^n}$.

13. $\displaystyle\sum_{n=1}^{\infty} \frac{n^3[\sqrt{2} + (-1)^n]^n}{3^n}$.

14. $\displaystyle\sum_{n=1}^{\infty} r^n \, |\sin nx|, \qquad r > 0.$

15. Let $\{a_n\}$ and $\{b_n\}$ be two sequences with $a_n > 0$ and $b_n > 0$ for all $n \geq N$, and let $c_n = b_n - b_{n+1}a_{n+1}/a_n$. Prove that:

(a) If there is a positive constant r such that $c_n \geq r > 0$ for all $n \geq N$, then $\sum a_n$ converges. [*Hint*. Show that $\sum_{k=N}^{n} a_k \leq a_N b_N / r$.]

(b) If $c_n \leq 0$ for $n \geq N$ and if $\sum 1/b_n$ diverges, then $\sum a_n$ diverges. [*Hint*. Show that $\sum a_n$ dominates $\sum 1/b_n$.]

16. Let $\sum a_n$ be a series of positive terms. Prove *Raabe's test*: If there is an $r > 0$ and an $N \geq 1$ such that

$$\frac{a_{n+1}}{a_n} \leq 1 - \frac{1}{n} - \frac{r}{n} \qquad \text{for all } n \geq N,$$

then $\sum a_n$ converges. The series $\sum a_n$ diverges if

$$\frac{a_{n+1}}{a_n} \geq 1 - \frac{1}{n} \qquad \text{for all } n \geq N.$$

[*Hint*. Use Exercise 15 with $b_{n+1} = n$.]

17. Let $\sum a_n$ be a series of positive terms. Prove *Gauss' test*: If there is an $N \geq 1$ and an $s > 1$ such that

$$\frac{a_{n+1}}{a_n} = 1 - \frac{A}{n} + O\left(\frac{1}{n^s}\right) \qquad \text{for } n \geq N,$$

then $\sum a_n$ converges if $A > 1$ and diverges if $A \leq 1$. [*Hint*. If $A \neq 1$, use Exercise 16. If $A = 1$, use Exercise 15 with $b_{n+1} = n \log n$.]

18. Use Gauss' test (in Exercise 17) to prove that the series

$$\sum_{n=1}^{\infty}\left(\frac{1\cdot 3\cdot 5\cdot\cdots\cdot(2n-1)}{2\cdot 4\cdot 6\cdot\cdots\cdot(2n)}\right)^k$$

converges if $k > 2$ and diverges if $k \le 2$. For this example the ratio test fails.

9.16 Alternating series

Up to now we have been concerned largely with series of nonnegative terms. We wish to turn our attention next to series whose terms may be positive or negative. The simplest examples occur when the terms alternate in sign. These are called *alternating series* and they have the form

$$\sum_{n=1}^{\infty}(-1)^{n-1}a_n = a_1 - a_2 + a_3 - a_4 + \cdots + (-1)^{n-1}a_n + \cdots, \tag{9.45}$$

where each $a_n > 0$.

Examples of alternating series were known to many early investigators. We have already mentioned the logarithmic series

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n-1}\frac{x^n}{n} + \cdots.$$

As we shall prove later on, this series converges and has the sum $\log(1 + x)$ whenever $-1 < x \le 1$. For positive x it is an alternating series. In particular, when $x = 1$ we obtain the formula

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{(-1)^{n-1}}{n} + \cdots, \tag{9.46}$$

which tells us that the alternating harmonic series has the sum log 2. This result is of special interest in view of the fact that the harmonic series $\Sigma 1/n$ diverges.

Closely related to (9.46) is the interesting formula

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots + \frac{(-1)^{n-1}}{2n-1} + \cdots, \tag{9.47}$$

discovered by James Gregory in 1671. Leibniz rediscovered this result in 1673 while computing the area of a unit circle.

Both series in (9.46) and in (9.47) are alternating series of the form (9.45) in which the sequence $\{a_n\}$ decreases monotonically to zero. Leibniz noticed, in 1705, that this simple property of the a_n implies the convergence of *any* alternating series.

9–13 THEOREM. *Leibniz's rule.* If $\{a_n\}$ is a monotonic decreasing sequence with limit 0, then the alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ converges.† If S denotes its sum and s_n its nth partial sum, we also have the inequalities

† The series also converges if the monotonicity of $\{a_n\}$ holds only for all n greater than some N.

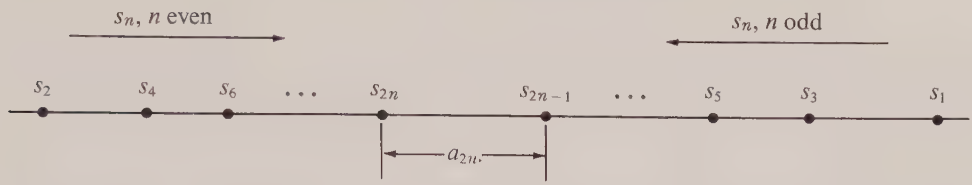


FIGURE 9.5 *Proof of Leibniz's rule for alternating series.*

$$0 < (-1)^n(S - s_n) < a_{n+1} \qquad \text{for each } n \geq 1 . \tag{9.48}$$

The inequalities in (9.48) provide a useful way to estimate the error in approximating the sum S by any partial sum s_n. The first inequality tells us that the error, $S - s_n$, has the sign $(-1)^n$, which is the same as the sign of the first neglected term, $(-1)^n a_{n+1}$. The second inequality states that the absolute value of this error is less than that of the first neglected term. Section 9.26 contains a nice example which shows how the inequalities in (9.48) may be used in practice to compute the decimal expansion of π to a high degree of accuracy. In another example in Section 9.26 we show how (9.48) may be used to prove that e is irrational.

Proof. The idea of the proof of Leibniz's rule is quite simple and is illustrated in Figure 9.5. The partial sums s_{2n} (consisting of an even number of terms) form an increasing sequence because $s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} > 0$. Similarly, the partial sums s_{2n-1} form a decreasing sequence. Both sequences are bounded below by s_2 and above by s_1. Therefore each sequence $\{s_{2n}\}$ and $\{s_{2n-1}\}$, being monotonic and bounded, converges to a limit, say $s_{2n} \to S'$, and $s_{2n-1} \to S''$. But $S' = S''$ because

$$S' - S'' = \lim_{n\to\infty} s_{2n} - \lim_{n\to\infty} s_{2n-1} = \lim_{n\to\infty} (s_{2n} - s_{2n-1}) = \lim_{n\to\infty} (-a_{2n}) = 0 .$$

If we denote this common limit by S, it is clear that the series converges and has sum S.

To derive the inequalities in (9.48) we argue as follows: Since $s_{2n} \nearrow$ and $s_{2n-1} \searrow$ we have

$$s_{2n} < s_{2n+2} \leq S \qquad \text{and} \qquad S \leq s_{2n+1} < s_{2n-1} \qquad \text{for all } n \geq 1 .$$

Therefore we have the inequalities

$$0 < S - s_{2n} \leq s_{2n+1} - s_{2n} = a_{2n+1} \qquad \text{and} \qquad 0 < s_{2n-1} - S \leq s_{2n-1} - s_{2n} = a_{2n} ,$$

which, taken together, yield (9.48). This completes the proof.

Example 1. Since $1/n \searrow$ and $1/n \to 0$ as $n \to \infty$, the convergence of the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ is an immediate consequence of Leibniz's rule. The sum of this series is computed below in Example 4.

Example 2. The alternating series $\sum(-1)^n (\log n)/n$ converges. To prove this using Leibniz's rule we must show that $(\log n)/n \to 0$ as $n \to \infty$ and that $(\log n)/n \searrow$. The first statement follows from Equation (9.11) of Section 9.2. To prove the second statement we note that the function f for which

$$f(x) = \frac{\log x}{x} \quad \text{when} \quad x > 0$$

has the derivative $f'(x) = (1 - \log x)/x^2$. When $x > e$ this is negative and f is monotonic decreasing. In particular, $f(n + 1) < f(n)$ for $n \geq 3$.

Example 3. An important limit relation may be derived as a consequence of Leibniz's rule. Let

$$a_1 = 1\,, \qquad a_2 = \int_1^2 \frac{dx}{x}\,, \qquad a_3 = \frac{1}{2}\,, \qquad a_4 = \int_2^3 \frac{dx}{x}\,, \qquad \ldots,$$

where, in general,

$$a_{2n-1} = \frac{1}{n} \qquad \text{and} \qquad a_{2n} = \int_n^{n+1} \frac{dx}{x} \qquad \text{for} \quad n = 1, 2, 3, \ldots.$$

It is easy to verify that $a_n \to 0$ as $n \to \infty$ and that $a_n \searrow$. Hence the series $\sum(-1)^{n-1}a_n$ converges. Denote its sum by C and its nth partial sum by s_n. The $(2n - 1)$st partial sum may be expressed as follows:

$$\begin{aligned} s_{2n-1} &= 1 - \int_1^2 \frac{dx}{x} + \frac{1}{2} - \int_2^3 \frac{dx}{x} + \cdots + \frac{1}{n-1} - \int_{n-1}^n \frac{dx}{x} + \frac{1}{n} \\ &= 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \int_1^n \frac{dx}{x} = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n\,. \end{aligned}$$

Since $s_{2n-1} \to C$ as $n \to \infty$ we obtain the following limit formula:

$$\lim_{n\to\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n\right) = C\,. \tag{9.49}$$

The number C defined by this limit is called *Euler's constant* (sometimes denoted by γ). Like π and e, this number appears in many analytic formulas. Its value, correct to ten decimals, is 0.5772156649. An interesting problem, unsolved to this time, is to decide whether Euler's constant is rational or irrational.

The existence of the limit in (9.49) proves that the partial sums of the harmonic series are asymptotically equal to $\log n$. That is, we have

$$1 + \frac{1}{2} + \cdots + \frac{1}{n} \sim \log n \qquad \text{as} \quad n \to \infty\,, \tag{9.50}$$

because

$$\frac{1 + 1/2 + \cdots + 1/n}{\log n} - 1 = \frac{1 + 1/2 + \cdots + 1/n - \log n}{\log n} \to 0 \qquad \text{as} \quad n \to \infty\,.$$

The relation (9.50) not only explains why the harmonic series $\sum 1/n$ diverges but it also gives us some concrete idea of the rate of growth of its partial sums. In the next example we use (9.49) to prove that the alternating harmonic series has the sum $\log 2$.

Example 4. Let $s_m = \sum_{k=1}^m (-1)^{k-1}/k$. We know that s_m tends to a limit as $m \to \infty$, and we shall prove now that this limit is $\log 2$. When m is even, say $m = 2n$, we may separate the positive and negative terms to obtain

$$s_{2n} = \sum_{k=1}^{n} \frac{1}{2k-1} - \sum_{k=1}^{n} \frac{1}{2k} = \left(\sum_{k=1}^{2n} \frac{1}{k} - \sum_{k=1}^{n} \frac{1}{2k}\right) - \sum_{k=1}^{n} \frac{1}{2k}$$

$$= \sum_{k=1}^{2n} \frac{1}{k} - \sum_{k=1}^{n} \frac{1}{k} = \log 2 + \left[\sum_{k=1}^{2n} \frac{1}{k} - \log 2n\right] - \left[\sum_{k=1}^{n} \frac{1}{k} - \log n\right],$$

where in the last step we have used the identity $\log 2 = \log 2n - \log n$. By (9.49) each term in square brackets approaches Euler's constant as $n \to \infty$, and hence $s_{2n} \to \log 2 + C - C = \log 2$. This proves that the sum of the alternating harmonic series is log 2.

9.17 Conditional and absolute convergence

Although the alternating harmonic series $\Sigma(-1)^{n-1}/n$ is convergent, the series obtained by replacing each term by its absolute value is divergent. This shows that, in general, convergence of Σa_n does not imply convergence of $\Sigma\, |a_n|$. In the other direction we have the following theorem:

9–14 THEOREM. Assume $\Sigma\, |a_n|$ converges. Then Σa_n also converges and we have

$$\left|\sum_{n=1}^{\infty} a_n\right| \leq \sum_{n=1}^{\infty} |a_n| \, . \tag{9.51}$$

Proof. Let $b_n = a_n + |a_n|$. We shall prove that Σb_n converges. It then follows (by Theorem 9–2) that Σa_n converges because $a_n = b_n - |a_n|$.

Since b_n is either 0 or $2|a_n|$ we have $0 \leq b_n \leq 2|a_n|$ and hence $\Sigma |a_n|$ dominates Σb_n. Therefore Σb_n converges and, as already mentioned, this implies convergence of Σa_n. To derive (9.51) we note that $\left|\Sigma_{k=1}^{n} a_k\right| \leq \Sigma_{k=1}^{n} |a_k|$ and then we let $n \to \infty$.

DEFINITION. A series Σa_n is called absolutely convergent if $\Sigma\, |a_n|$ converges. It is called conditionally convergent if Σa_n converges but $\Sigma\, |a_n|$ diverges.

If Σa_n and Σb_n are absolutely convergent, then so is the series $\Sigma(\alpha a_n + \beta b_n)$ for every choice of α and β. This follows at once from the inequalities

$$\sum_{n=1}^{M} |\alpha a_n + \beta b_n| \leq |\alpha| \sum_{n=1}^{M} |a_n| + |\beta| \sum_{n=1}^{M} |b_n| \leq |\alpha| \sum_{n=1}^{\infty} |a_n| + |\beta| \sum_{n=1}^{\infty} |b_n| \, ,$$

which show that the partial sums of $\Sigma\, |\alpha a_n + \beta b_n|$ are bounded.

All the convergence tests of the earlier sections that were developed for series of nonnegative terms may be used to test *absolute* convergence of a series with arbitrary terms. More intricate tests exist† for determining convergence when the series might not converge absolutely, but we shall not discuss any of them here. Instead, we shall describe a result of a rather different nature.

Suppose a given series Σa_n has infinitely many positive terms and infinitely many negative terms. We can form two new series by considering the positive terms alone, and

† See Section 12–12 of the author's *Mathematical Analysis*, Addison-Wesley Publishing Co., Reading, Mass., 1957.

also the negative terms alone. These two series are related to the given series in a manner which may be described as follows:

9–15 THEOREM. Let Σa_n be a given series and define

$$p_n = \frac{|a_n| + a_n}{2}, \qquad q_n = \frac{|a_n| - a_n}{2} \qquad \text{for each } n \geq 1. \tag{9.52}$$

Then the following statements hold:

(a) If Σa_n is conditionally convergent, both Σp_n and Σq_n diverge.

(b) If $\Sigma\, |a_n|$ converges, both Σp_n and Σq_n converge and we have

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} p_n - \sum_{n=1}^{\infty} q_n .$$

Note. If $a_n \geq 0$, then $p_n = a_n$ and $q_n = 0$. If $a_n \leq 0$, then $p_n = 0$ and $q_n = -a_n$.

Proof. To prove part (a) we note that $\Sigma \frac{1}{2} a_n$ converges and $\Sigma \frac{1}{2}|a_n|$ diverges. Therefore, by Theorem 9–3, Σp_n diverges. In the same way, we find that Σq_n diverges. To prove (b) we use (9.52) along with the linearity property (Theorem 9-2).

9.18 Exercises

In Exercises 1 through 32, determine convergence or divergence of the given series. In case of convergence, determine whether the series converges absolutely or conditionally.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$.

2. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}}{n + 100}$.

3. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}$.

4. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(\frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n)}\right)^3$.

5. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n(n-1)/2}}{2^n}$.

6. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(\frac{2n + 100}{3n + 1}\right)^n$.

7. $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n} + (-1)^n}$.

8. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[n]{n}}$.

9. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{1 + n^2}$.

10. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\log (e^n + e^{-n})}$.

11. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n \log^2 (n + 1)}$.

12. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\log (1 + 1/n)}$.

13. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n n^{37}}{(n + 1)!}$.

14. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \int_n^{n+1} \frac{e^{-x}}{x}\, dx$.

15. $\displaystyle\sum_{n=1}^{\infty} \sin (\log n)$.

16. $\displaystyle\sum_{n=1}^{\infty} \log \left(n \sin \frac{1}{n}\right)$.

17. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(1 - n \sin \frac{1}{n}\right)$.

18. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(1 - \cos \frac{1}{n}\right)$.

19. $\sum_{n=1}^{\infty} (-1)^n \arctan \frac{1}{2n+1}$.

20. $\sum_{n=1}^{\infty} (-1)^n \left(\frac{\pi}{2} - \arctan (\log n)\right)$.

21. $\sum_{n=1}^{\infty} \log \left(1 + \frac{1}{|\sin n|}\right)$.

22. $\sum_{n=2}^{\infty} \sin \left(n\pi + \frac{1}{\log n}\right)$.

23. $\sum_{n=1}^{\infty} \frac{1}{n(1 + 1/2 + \cdots + 1/n)}$.

24. $\sum_{n=1}^{\infty} (-1)^n \left[e - \left(1 + \frac{1}{n}\right)^n\right]$.

25. $\sum_{n=2}^{\infty} \frac{(-1)^n}{(n + (-1)^n)^s}$.

26. $\sum_{n=1}^{\infty} (-1)^{n(n-1)/2} \left(\frac{n^{100}}{2^n}\right)$.

27. $\sum_{n=1}^{\infty} a_n$, where $a_n = \begin{cases} 1/n & \text{if } n \text{ is a square,} \\ 1/n^2 & \text{otherwise.} \end{cases}$

28. $\sum_{n=1}^{\infty} a_n$, where $a_n = \begin{cases} 1/n^2 & \text{if } n \text{ is odd,} \\ -1/n & \text{if } n \text{ is even.} \end{cases}$

29. $\sum_{n=1}^{\infty} \left(\sin \frac{1}{n}\right)^{3/2}$.

30. $\sum_{n=1}^{\infty} \frac{\sin (1/n)}{n}$.

31. $\sum_{n=1}^{\infty} \left(1 - n \sin \frac{1}{n}\right)$.

32. $\sum_{n=1}^{\infty} \frac{1 - n \sin (1/n)}{n}$.

In Exercises 33 through 44, determine the set of real x for which the given series converges.

33. $\sum_{n=1}^{\infty} n^n x^n$.

34. $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$.

35. $\sum_{n=1}^{\infty} \left(\frac{x}{2x+1}\right)^n$.

36. $\sum_{n=1}^{\infty} \frac{n}{n+1} \left(\frac{x}{2x+1}\right)^n$.

37. $\sum_{n=1}^{\infty} \frac{(-1)^n}{x+n}$.

38. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^n \sin^{2n} x}{n}$.

39. $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}} \log \frac{2n+1}{n}$.

40. $\sum_{n=1}^{\infty} \frac{(2x+3)^n}{n \log (n+1)}$.

41. $\sum_{n=1}^{\infty} \left(1 + \frac{1}{5n+1}\right)^{n^2} x^{17n}$.

42. $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \left(\frac{1-x}{1+x}\right)^n$.

43. $\sum_{n=1}^{\infty} \frac{2^n \sin^n x}{n^2}$.

44. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} (x-1)^n}{n}$.

45. If $a_n > 0$ and Σa_n converges, prove that $\Sigma 1/a_n$ diverges.

46. If $\Sigma |a_n|$ converges, prove that Σa_n^2 converges. Give a counterexample in which Σa_n^2 converges but $\Sigma |a_n|$ diverges.

47. Given a convergent series Σa_n, where each $a_n \geq 0$. Prove that $\Sigma \sqrt{a_n} n^{-p}$ converges if $p > \frac{1}{2}$. Give a counterexample for $p = \frac{1}{2}$.

48. Prove or disprove the following statements:

(a) If Σa_n converges absolutely, then so does $\Sigma a_n^2/(1 + a_n^2)$.

(b) If Σa_n converges absolutely, and if no $a_n = -1$, then $\Sigma a_n/(1 + a_n)$ converges absolutely.

49. (a) If p and q are fixed integers, $p \geq q \geq 1$, show that

$$\lim_{n \to \infty} \sum_{k=qn}^{pn} \frac{1}{k} = \log \frac{p}{q}.$$

(b) The following series is a rearrangement of the alternating harmonic series in which there appear, alternately, three positive terms followed by two negative terms:

$$1 + \tfrac{1}{3} + \tfrac{1}{5} - \tfrac{1}{2} - \tfrac{1}{4} + \tfrac{1}{7} + \tfrac{1}{9} + \tfrac{1}{11} - \tfrac{1}{6} - \tfrac{1}{8} + + + - - \cdots.$$

Show that the series converges and has sum $\log 2 + \frac{1}{2}\log \frac{3}{2}$. [*Hint.* Consider the partial sum s_{5n} and use part (a).]

(c) Rearrange the alternating harmonic series, writing alternately p positive terms followed by q negative terms. Then use part (a) to show that this rearranged series converges and has sum $\log 2 + \frac{1}{2}\log (p/q)$.

50. Consider the plane curve whose vector equation is $\vec{r}(t) = t\vec{i} + f(t)\vec{j}$, where

$$f(t) = t \cos\left(\frac{\pi}{2t}\right) \quad \text{if } t \neq 0, \qquad f(0) = 0.$$

Consider the following partition of the interval [0, 1]:

$$P = \left\{0, \frac{1}{2n}, \frac{1}{2n-1}, \ldots, \frac{1}{3}, \frac{1}{2}, 1\right\}.$$

(a) Show that the corresponding inscribed polygon $\pi(P)$ has length

$$|\pi(P)| > 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2n}.$$

(b) Use part (a) to deduce that this curve has no arc length. (For the definition of arc length, see Section 6.31.)

9.19 Power series. Interval of convergence

An infinite series of the form

$$(9.53) \qquad \sum_{n=0}^{\infty} a_n(x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_n(x - x_0)^n + \cdots$$

is called a power series in $x - x_0$. The numbers $a_0, a_1, a_2, \ldots$ are called its *coefficients.* The convergence or divergence of (9.53) depends both on the coefficients and on the numbers x and x_0. Later in this section we shall prove that with each power series there is associated an interval, called the *interval of convergence*, such that the series converges absolutely for every x interior to this interval and diverges for every x outside this interval. This interval is of the form $[x_0 - r, x_0 + r]$, symmetrically located about the point x_0. (See Figure 9.6.) The nonnegative number r is called the *radius of convergence.* In extreme cases the interval may shrink to the single point x_0, in which case $r = 0$, or it may consist of the whole real axis, in which case we say that r is $+\infty$.

At the endpoints $x_0 - r$ and $x_0 + r$ the behavior of the series cannot be predicted in advance. There may be convergence at none, one, or both of the endpoints, as we shall see presently.

For many power series that occur in practice, the interval of convergence can be determined by using either the ratio test or the root test, as in the following examples.

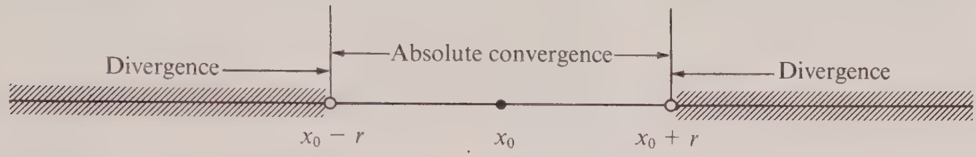


FIGURE 9.6 *The interval of convergence of a power series.*

Example 1. To find the interval of convergence of the series $\Sigma x^n/n!$, we apply the ratio test. If $x \neq 0$, the ratio of consecutive terms has absolute value

$$\left|\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}\right| = \frac{|x|}{n+1}.$$

Since this ratio tends to 0 as $n \to \infty$ we conclude that the series converges absolutely for all real $x \neq 0$. It also converges for $x = 0$, and therefore the interval of convergence is the whole real axis.

Since the general term of a convergent series must tend to 0, the result of the foregoing example proves that

$$\lim_{n\to\infty} \frac{x^n}{n!} = 0 \tag{9.54}$$

for every real x. That is, $n!$ "grows faster" than the nth power of any fixed real number x as $n \to \infty$. This may also be shown directly by the following argument: If $0 \leq |x| \leq 1$, Equation (9.54) holds trivially because $|x|^n/n! \leq 1/n!$. If $|x| \geq 1$, we may write

$$\frac{|x|^n}{n!} = \frac{|x|}{1} \cdot \frac{|x|}{2} \cdot \cdots \cdot \frac{|x|}{k} \cdot \frac{|x|}{k+1} \cdot \cdots \cdot \frac{|x|}{n} \leq \frac{|x|^k}{k!}\left(\frac{|x|}{k+1}\right)^{n-k},$$

where $k < n$. If we choose k to be the greatest integer $\leq |x|$, then $|x| < k + 1$ and the last factor tends to 0 as $n \to \infty$. This proves that (9.54) holds for all real k.

Example 2. To test the series $\Sigma n^2 3^n x^n$, we use the root test. We have

$$(n^2 3^n |x|^n)^{1/n} = 3|x| n^{2/n} \to 3|x| \qquad \text{as} \quad n \to \infty,$$

since $n^{2/n} = (n^{1/n})^2$ and $n^{1/n} \to 1$ as $n \to \infty$. [See Equation (9.12) of Section 9.2.] Therefore the series converges absolutely if $|x| < \frac{1}{3}$ and diverges if $|x| > \frac{1}{3}$. The radius of convergence is $r = \frac{1}{3}$. This series diverges at both endpoints $x = \pm\frac{1}{3}$.

Example 3. For each of the series $\Sigma x^n/n$ and $\Sigma x^n/n^2$, the ratio test tells us that the radius of convergence is 1. The first series converges at the endpoint $x = -1$ and diverges at $x = 1$. The second series converges at both endpoints.

We conclude this section with a proof that every power series $\Sigma a_n x^n$ has associated with it an interval of convergence. The proof is based on a preliminary result which we state as a lemma.

LEMMA. Assume the power series $\Sigma a_n x^n$ converges for a particular $x \neq 0$, say for $x = x_1$. Then the series converges absolutely for every x satisfying $|x| < |x_1|$.

Proof. Since $\Sigma a_n x_1^n$ converges, its general term tends to 0 as $n \to \infty$. In particular, $|a_n x_1^n| < 1$ from some point on, say for $n \geq N$. If $|x| < |x_1|$ and $n \geq N$, we have

$$|a_n x^n| = |a_n x_1^n| \left|\frac{x}{x_1}\right|^n < \left|\frac{x}{x_1}\right|^n = t^n, \qquad \text{where} \quad t = \left|\frac{x}{x_1}\right|.$$

Since $0 < t < 1$, the series $\Sigma\, |a_n x^n|$ is dominated by the convergent geometric series Σt^n. This proves the lemma.

9–16 THEOREM. Assume that the power series $\Sigma a_n x^n$ converges for at least one $x \neq 0$, say for $x = x_1$, and that it diverges for at least one x, say for $x = x_2$. Then there exists a positive real number r such that the series converges absolutely if $|x| < r$ and diverges if $|x| > r$.

Proof. Let S denote the set of all real numbers $x \neq 0$ for which the power series converges. The set S is not empty since, by hypothesis, it contains x_1. Also, no number in S can exceed $|x_2|$ (because of the lemma). Hence $|x_2|$ is an upper bound for S. Therefore S is a nonempty set of real numbers that is bounded above, and hence, by the least-upper-bound axiom, there is a number which is the *least upper bound* for S. Call this number r. It is clear that $r > 0$ since $r \geq |x_1|$. Because of the lemma, no number in S can have an absolute value exceeding r. Therefore the series diverges if $|x| > r$. We shall prove that the series converges *absolutely* if $|x| < r$. If $|x| < r$, there is a positive number x' *in* S such that $|x| < x' < r$. By the lemma, $\Sigma a_n x^n$ converges absolutely. This completes the proof.

In proving Theorem 9–16, we have shown that the series converges *absolutely* in the open interval $(-r, r)$, and that it diverges outside the closed interval $[-r, r]$. The actual set of points where the series converges is either the open interval $(-r, r)$, or the closed interval $[-r, r]$ or one of the half-open intervals $[-r, r)$ or $(-r, r]$. Examples illustrating all four possibilities are:

$$\sum_{n=1}^{\infty} x^n, \qquad \sum_{n=1}^{\infty} \frac{x^n}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{x^n}{n}, \qquad \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}.$$

The set of points where a power series converges absolutely is either the open interval $(-r, r)$ or the closed interval $[-r, r]$.

There is, of course, a corresponding theorem for power series in $x - x_0$ which may be deduced from the case just treated by introducing the translation $X = x - x_0$. The interval of convergence is then symmetrically located about the point x_0 as shown in Figure 9.6.

9.20 Exercises

In Exercises 1 through 15, determine the interval of convergence of the given power series. If the interval is finite, test for convergence at both endpoints. The letters a and b denote positive real numbers.

1. $\sum_{n=0}^{\infty} \frac{(x+3)^n}{(n+1)2^n}$.
2. $\sum_{n=0}^{\infty} \frac{n(x-2)^n}{3^n(n+1)}$.
3. $\sum_{n=0}^{\infty} \frac{(-1)^n(x+1)^n}{n^2+1}$.
4. $\sum_{n=1}^{\infty} \frac{n!x^n}{n^n}$.
5. $\sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n} x^{2n}}{2n}$.
6. $\sum_{n=1}^{\infty} [1-(-2)^n]x^n$.
7. $\sum_{n=0}^{\infty} (\sin an)x^n$.
8. $\sum_{n=0}^{\infty} (\sinh an)x^n$.
9. $\sum_{n=0}^{\infty} a^{n^2}x^n, \qquad a < 1$.
10. $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2} x^n$.
11. $\sum_{n=1}^{\infty} \left(\frac{a^n}{n}+\frac{b^n}{n^2}\right) x^n$.
12. $\sum_{n=1}^{\infty} \frac{x^n}{a^n+b^n}$.
13. $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n$.
14. $\sum_{n=1}^{\infty} \frac{3^{\sqrt{n}}x^n}{\sqrt{n^2+1}}$.
15. $\sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n)}\right)^3 \left(\frac{x-1}{2}\right)^n$.

9.21 Properties of functions having power-series expansions

Every power series defines a function f whose value at each x in the interval of convergence is given by an expression of the form

$$f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n . \tag{9.55}$$

The series is said to *represent the function* f in the interval of convergence and it is called the *power-series expansion* of f about x_0.

There are two basic problems about power-series expansions that we shall be concerned with here:

(1) Given the coefficients $a_0, a_1, a_2, \ldots$, to find properties of the function f defined by Equation (9.55).

(2) Given a function f, to find whether or not it may be represented by a power series.

It turns out that only rather special functions possess power-series expansions. Nevertheless, the class of such functions includes most examples that arise in practice and hence their study is of great importance.

To see what kinds of functions may be represented by power series, we turn our attention to problem (1). The study may be carried out for the special case in which $x_0 = 0$, since the more general case can always be reduced to this by a translation of the form $X = x - x_0$. Assume, then, that f is defined by a power-series expansion of the form

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + \cdots . \tag{9.56}$$

We shall prove that f is differentiable in the interval of convergence and that $f'(x)$ may be obtained by differentiating the series in (9.56) term by term. For this purpose we derive some preliminary results which we state as lemmas.

LEMMA 1. The differentiated series, that is, the series

$$\sum_{n=1}^{\infty} na_n x^{n-1}, \tag{9.57}$$

has the same radius of convergence as the series $\Sigma a_n x^n$.

Proof. Assume the series $\Sigma a_n x^n$ in (9.56) converges absolutely in an interval $(-r, r)$, where $r > 0$. We prove first that (9.57) also converges absolutely in this interval. Choose any positive x such that $0 < x < r$, and let h be a small positive number such that $0 < x < x + h < r$. Then the series for $f(x)$ and for $f(x + h)$ are each absolutely convergent. Hence we may write

$$\frac{f(x+h) - f(x)}{h} = \sum_{n=0}^{\infty} a_n \frac{(x+h)^n - x^n}{h}. \tag{9.58}$$

The series on the right is absolutely convergent since it is a linear combination of absolutely convergent series. Now we apply the mean-value theorem to write

$$(x + h)^n - x^n = hnc_n^{n-1},$$

where $x < c_n < x + h$. Hence the series in (9.58) is identical to the series

$$\sum_{n=1}^{\infty} na_n c_n^{n-1} \tag{9.59}$$

which must be absolutely convergent, since that in (9.58) is. The series (9.59) is no longer a power series but it dominates† the power series (9.57), so the series (9.57) must be absolutely convergent for this x. This proves that the radius of convergence of (9.57) is at least as large as the radius of convergence of (9.56). On the other hand, the radius of convergence of (9.57) cannot exceed that of (9.56) because (9.57) dominates (9.56). This proves Lemma 1.

If we apply Lemma 1 to the power series in (9.57), we obtain as a corollary:

LEMMA 2. The power series $\Sigma n(n - 1)a_n x^{n-2}$ has the same radius of convergence as the series $\Sigma a_n x^n$.

Now we are ready to prove the theorem on term-by-term differentiation.

9–17 THEOREM. If the power-series expansion $f(x) = \Sigma_{n=0}^{\infty} a_n x^n$ is valid in an interval of the form $(-r, r)$, then for every x inside this interval the derivative $f'(x)$ exists and is given by the power-series expansion

† That is, $|na_n x^{n-1}| \leq |na_n c_n^{n-1}|$ since $x^{n-1} < c_n^{n-1}$.

(9.60)
$$f'(x) = \sum_{n=1}^{\infty} na_n x^{n-1}.$$

Proof. First we choose any x in $(-r, r)$ and then we choose x_1 so that $0 \le |x| < |x_1| < r$. Let F be defined by the series

$$F(x) = \sum_{n=1}^{\infty} na_n x^{n-1}.$$

This series converges absolutely since $|x| < r$ (by Lemma 1), and our aim is to prove that $f'(x)$ exists and equals $F(x)$. For this purpose we choose $h \neq 0$ so that $0 \le |x + h| < |x_1|$. Proceeding as in the proof of Lemma 1, we have

(9.61)
$$F(x) - \frac{f(x+h) - f(x)}{h} = \sum_{n=2}^{\infty} na_n(x^{n-1} - c_n^{n-1}),$$

where each number c_n lies between x and $x + h$. This series converges absolutely since it is the difference of two absolutely convergent series. Now we apply the mean-value theorem again to obtain

$$x^{n-1} - c_n^{n-1} = (n-1)t_n^{n-2}(x - c_n),$$

where each number t_n lies between x and c_n. Hence we have

(9.62)
$$F(x) - \frac{f(x+h) - f(x)}{h} = \sum_{n=2}^{\infty} n(n-1)a_n t_n^{n-2}(x - c_n),$$

the series on the right of (9.62) being absolutely convergent since it is identical to the one in (9.61). But $|x - c_n| < |h|$ and $|t_n| < |x_1|$ so that we have

(9.63)
$$0 \le \left| F(x) - \frac{f(x+h) - f(x)}{h} \right| \le |h| \sum_{n=2}^{\infty} n(n-1)|a_n|\,|x_1|^{n-2},$$

the series in (9.63) being convergent by Lemma 2. The relation (9.63) is valid for any $h \neq 0$ satisfying the inequality $0 \le |x + h| < |x_1|$. Since this inequality is valid for all sufficiently small $|h|$ we may let $h \to 0$ in (9.63) to deduce that $f'(x)$ exists and equals $F(x)$. This proves the theorem.

As a corollary of Theorem 9–17 we conclude that f has derivatives of *every* order and they may be obtained by repeated differentiation, term by term, of the power series in (9.56). If we compute each of these derivatives and then let $x = 0$, we find that the nth coefficient a_n in (9.56) is given by the formula

(9.64)
$$a_n = \frac{f^{(n)}(0)}{n!} \qquad \text{for } n = 1, 2, 3, \ldots.$$

This formula also holds for $n = 0$ if we interpret $f^{(0)}(0)$ to mean $f(0)$.

When the function f has an expansion in powers of $x - x_0$ rather than in powers of x, the nth derivatives in (9.64) must be evaluated at x_0 instead of at 0, and the expansion has the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n .$$

Incidentally, this property provides a *uniqueness theorem* for power-series expansions. It shows that if two functions f and g agree in some neighborhood of the point x_0 and if both are representable by power series in $x - x_0$, then these series must be equal term by term, since the coefficients are given in terms of the functions and their derivatives at x_0.

As a further corollary of Theorem 9–17 we shall prove that a power series may be integrated term by term within its interval of convergence.

9–18 THEOREM. Assume the power-series expansion $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is valid in an open interval of the form $(-r, r)$. Then the function f is integrable on every closed subinterval of $(-r, r)$ and its integral may be computed by integrating the power series term by term. In particular, for every x in $(-r, r)$ we have

$$\int_0^x f(t)\, dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} . \tag{9.65}$$

The integrated series has the same radius of convergence as $\Sigma a_n x^n$.

Proof. Let G be the function defined by the integrated series, that is, let $G(x) = \Sigma a_n x^{n+1}/(n+1)$, and let R denote the radius of convergence of this series. Since the series for f dominates that for G, we have $R \geq r$ and hence R is *positive*. By Lemma 1, the series for f has the same radius of convergence as that for G. By Theorem 9–17, we have $G'(x) = f(x)$ for each x in $(-r, r)$. Since f has a derivative in $(-r, r)$, it follows that f is continuous and therefore integrable on every closed subinterval of $(-r, r)$. Applying the second fundamental theorem of calculus, we have $\int_0^x f(t)\, dt = G(x) - G(0) = G(x)$. This proves (9.65).

The theorems of this section justify the formal manipulations of Section 9.8 where we obtained various power-series expansions using term-by-term differentiation and integration of the geometric series. In particular, these theorems establish the validity of the expansions for $\log(1 + x)$ and arc tan x given in formulas (9.32) and (9.33), whenever x is in the *open* interval $-1 < x < 1$.

Note. Although term-by-term differentiation and integration of power series is always permissible, this is not necessarily true for series expansions other than power series. In Section 9.30 an example is given in which term-by-term integration leads to an incorrect result.

9.22 The Taylor's series generated by a function

We turn now to the second problem raised at the beginning of the foregoing section. Suppose we are given a real-valued function f defined in some open interval about a point x_0, and suppose f has derivatives of every order in this interval. Then we can certainly *form* the power series

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n, \tag{9.66}$$

using the numbers $a_n = f^{(n)}(x_0)/n!$ as coefficients. This is called the *Taylor's series about* x_0 *generated by* f. We now ask two questions: Does this series converge for any x other than $x = x_0$? If so, is its sum equal to $f(x)$? Surprisingly enough, the answer to both questions is, in general, "no." The series in (9.66) may or may not converge for $x \neq x_0$ and, if it does converge, its sum may or may not be $f(x)$. (An example where the series converges to a sum different from $f(x)$ is given in Exercise 13 in Section 9.23.)

A necessary and sufficient condition for answering both questions in the affirmative can be given by using Taylor's formula with remainder, which provides a *finite* expansion of the form

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n)}(c)(x - x_0)^n}{n!}, \tag{9.67}$$

where c is some point between x and x_0. The point c depends on the choices of x, x_0, and n, and the finite sum on the right of (9.67) is the nth partial sum of the series in (9.66). This series will converge to $f(x)$ if and only if the remainder term tends to 0 as $n \to \infty$, that is, whenever

$$\lim_{n\to\infty} \frac{f^{(n)}(c)}{n!}(x - x_0)^n = 0. \tag{9.68}$$

In practice it may be difficult to verify (9.68) because of the unknown position of c. In some cases, however, a suitable upper bound for the size of $f^{(n)}(c)$ can be obtained and the limit can be shown to be zero. In particular, the following theorem gives a *sufficient* condition for a function to be representable by its Taylor's series.

9–19 THEOREM. Assume f has derivatives of every order in an interval of the form $(x_0 - r, x_0 + r)$ and assume that there is a constant M (which may depend on x_0) such that

$$|f^{(n)}(x)| \leq M \qquad \text{for all } x \text{ in } (x_0 - r, x_0 + r) \tag{9.69}$$

and all $n \geq N$ (for some N). Then for each x in $(x_0 - r, x_0 + r)$ the Taylor's series of f about x_0 converges to $f(x)$. That is, we have

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

Proof. For each x in $(x_0 - r, x_0 + r)$ we have an expansion of the form (9.67) with a remainder term which, for $n \geq N$, may be estimated as follows:

$$\left|\frac{f^{(n)}(c)(x-x_0)^n}{n!}\right| \le \frac{M|x-x_0|^n}{n!} = \frac{Ma^n}{n!}, \qquad \text{where} \quad a = |x - x_0| .$$

But $a^n/n! \to 0$ as $n \to \infty$ [see Equation (9.54) following Example 1 in Section 9.19], and hence the limit in (9.68) is also zero. This proves the theorem.

Examples. This criterion may be used to prove that the following power-series expansions are valid for every real x:

(9.70)
$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^{n-1}\frac{x^{2n-1}}{(2n-1)!} + \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n\frac{x^{2n}}{(2n)!} + \cdots.$$

In the case of the exponential function we note that (9.69) holds with $M = e^r$ when $x_0 = 0$. Therefore Theorem 9–19 establishes the validity of (9.70) in every interval of the form $(-r, r)$. Since r is arbitrary, this proves that (9.70) is valid for all real x. The series expansions for the sine and cosine are similarly established. For these functions, inequality (9.69) holds with $M = 1$ for all x.

The foregoing power-series expansions for the sine and cosine can be used as the starting point for a completely analytic treatment of the trigonometric functions. If we use these series as *definitions* of the sine and cosine, it is possible to derive all the familiar algebraic and analytic properties of the trigonometric functions from these series alone. For example, the series immediately give us the formulas

$$\sin 0 = 0, \qquad \cos 0 = 1, \qquad \sin(-x) = -\sin x, \qquad \cos(-x) = \cos x,$$
$$D\sin x = \cos x, \qquad D\cos x = -\sin x.$$

The addition formulas may be derived by the following simple device: Let u and v be new functions defined by the equations

$$u(x) = \sin(x+a) - \sin x\cos a - \cos x \sin a,$$
$$v(x) = \cos(x+a) - \cos x\cos a + \sin x \sin a,$$

where a is a fixed real number, and let $f(x) = [u(x)]^2 + [v(x)]^2$. Then it is easy to verify that $u'(x) = v(x)$ and $v'(x) = -u(x)$, and so $f'(x) = 0$ for all x. Therefore f is a constant and, since $f(0) = 0$, we must have $f(x) = 0$ for all x. This implies $u(x) = v(x) = 0$ for all x or, in other words,

$$\sin(x+a) = \sin x\cos a + \cos x \sin a,$$
$$\cos(x+a) = \cos x\cos a - \sin x \sin a.$$

The number π may be introduced as the smallest positive x such that $\sin x = 0$ (such an x can be shown to exist) and then it can be shown that the sine and cosine are periodic with period 2π, that $\sin(\frac{1}{2}\pi) = 1$, and that $\cos(\frac{1}{2}\pi) = 0$. The details, which we shall not present here, may be found in the book *Theory and Application of Infinite Series* by K. Knopp (New York: Hafner, 1951).

Another useful sufficient condition for convergence of a Taylor's series was formulated by the Russian mathematician Sergei N. Bernstein (1880–). We simply state the result here without proof.†

9–20 THEOREM OF BERNSTEIN. Assume f and all its derivatives are nonnegative in a closed interval $[a, b]$. That is, assume

$$f(x) \geq 0 \qquad \text{and} \qquad f^{(n)}(x) \geq 0$$

for each x in $[a, b]$ and every $n = 1, 2, 3, \ldots$. Let x_0 be a point in the open interval (a, b). Then for every x in $[a, b]$ such that $|x - x_0| \leq b - x_0$, the Taylor's series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

converges to $f(x)$.

To illustrate how Bernstein's theorem may be applied in practice, we shall use it to show that the following expansion, known as the *binomial series*, is valid for every x in the interval $-1 < x < 1$ and every real α:

$$(9.71)\quad (1 + x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \qquad \text{where} \quad \binom{\alpha}{n} = \frac{\alpha(\alpha - 1)\cdots(\alpha - n + 1)}{n!}.$$

Bernstein's theorem is not *directly* applicable in this case because the derivatives of $(1 + x)^\alpha$ are not always nonnegative. However we can use Bernstein's theorem if we argue as follows: Let

$$f(x) = (1 - x)^{-c}, \qquad \text{where} \quad c > 0 \quad \text{and} \quad x < 1.$$

Then

$$f'(x) = c(1 - x)^{-c-1}, \quad \ldots, \quad f^{(n)}(x) = c(c + 1)\cdots(c + n - 1)(1 - x)^{-c-n},$$

and hence $f^{(n)}(x) \geq 0$ for each n, provided $x < 1$. Applying Bernstein's theorem with $x_0 = 0$, $a = -b$, $0 < b < 1$, we find that the expansion

$$\frac{1}{(1 - x)^c} = \sum_{k=0}^{\infty} \binom{-c}{k} (-x)^k$$

is valid for $-1 < x < 1$ and every $c > 0$. If we replace c by $-\alpha$ and x by $-x$, this proves

† A proof may be found on pp. 418–420 of the author's *Mathematical Analysis, ibid.*

that (9.71) is valid for $\alpha < 0$ and $-1 < x < 1$. Equation (9.71) is also true for $\alpha = 0$ because both sides reduce to 1 in this case.

To extend the validity of (9.71) to *every* real α, we use successive integration. For example, integration of (9.71) yields

$$\frac{(1+x)^{\alpha+1}}{\alpha+1} - \frac{1}{\alpha+1} = \sum_{n=0}^{\infty} \binom{\alpha}{n} \frac{x^{n+1}}{n+1} \qquad \text{if} \quad -1 < x < 1, \quad \alpha \le 0, \quad \alpha \ne -1.$$

Multiplying through by $\alpha + 1$ and writing β for $\alpha + 1$, we obtain

$$(1+x)^{\beta} = \sum_{n=0}^{\infty} \binom{\beta}{n} x^n \qquad \text{if} \quad -1 < x < 1 \qquad \text{and} \quad \beta \le 1\,.$$

This provides the extension of (9.71) from $\alpha \le 0$ to $\alpha \le 1$. Repeating the argument as often as is necessary, we may show that (9.71) is valid for any real α. Another method of establishing (9.71), without using Bernstein's theorem, is outlined in Exercise 12 of Section 9.25.

Note. When α is a positive integer, say $\alpha = m$, then the binomial coefficient $\binom{m}{n} = 0$ for $n > m$ and (9.71) reduces to a *finite sum* (the familiar *binomial theorem*).

9.23 Exercises

Each of the functions in Exercises 1 through 12 has a power-series representation in powers of x. Assume the existence of the expansion, verify that the coefficients have the form given, and show that the series converges for the values of x indicated. The expansions given earlier in the text may be used whenever it is convenient to do so.

1. $a^x = \displaystyle\sum_{n=0}^{\infty} \frac{(\log a)^n}{n!} x^n, \qquad a > 0 \qquad$ (all x). [*Hint.* $a^x = e^{x \log a}$.]

2. $\sinh x = \displaystyle\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} \qquad$ (all x).

3. $\sin^2 x = \displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^{2n-1}}{(2n)!} x^{2n} \qquad$ (all x). [*Hint.* $\cos 2x = 1 - 2\sin^2 x$.]

4. $\dfrac{1}{2-x} = \displaystyle\sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}} \qquad$ ($|x| < 2$).

5. $e^{-x^2} = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} \qquad$ (all x).

6. $\sin^3 x = \dfrac{3}{4}\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^{2n}-1}{(2n+1)!} x^{2n+1} \qquad$ (all x).

7. $\log \sqrt{\dfrac{1+x}{1-x}} = \displaystyle\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \qquad$ ($|x| < 1$).

8. $\dfrac{x}{1 + x - 2x^2} = \dfrac{1}{3} \displaystyle\sum_{n=1}^{\infty} [1 - (-2)^n]x^n \qquad (|x| < \tfrac{1}{2}).$

[*Hint.* $3x/(1 + x - 2x^2) = 1/(1 - x) - 1/(1 + 2x)$.]

9. $\dfrac{12 - 5x}{6 - 5x - x^2} = \displaystyle\sum_{n=0}^{\infty} \left(1 + \frac{(-1)^n}{6^n}\right) x^n \qquad (|x| < 1).$

10. $\dfrac{1}{x^2 + x + 1} = \dfrac{2}{\sqrt{3}} \displaystyle\sum_{n=0}^{\infty} \sin \frac{2\pi(n + 1)}{3} x^n \qquad (|x| < 1).$

[*Hint.* $x^3 - 1 = (x - 1)(x^2 + x + 1)$.]

11. $\dfrac{x}{(1 - x)(1 - x^2)} = \dfrac{1}{2} \displaystyle\sum_{n=1}^{\infty} \left(n + \frac{1 - (-1)^n}{2}\right) x^n \qquad (|x| < 1).$

12. $\arcsin x = x + \displaystyle\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n - 1)}{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n)} \frac{x^{2n+1}}{2n + 1} \qquad (|x| < 1).$

[*Hint.* Integrate the binomial series for $(1 - x^2)^{-1/2}$.]

13. Let $f(x) = e^{-1/x^2}$ if $x \neq 0$, and let $f(0) = 0$.
(a) Show that f has derivatives of every order everywhere on the real axis.
(b) Show that $f^{(n)}(0) = 0$ for all $n \geq 1$. This example shows that the Taylor's series generated by f about the point 0 converges everywhere on the real axis, but that it represents f *only* at the origin.

9.24 Applications of power series to differential equations

Power series sometimes enable us to obtain solutions of differential equations when other methods fail. An example will illustrate some of the ideas and techniques involved.

Consider the second-order differential equation

$$(1 - x^2)y'' = -2y \,. \tag{9.72}$$

Assume there exists a solution, say $y = f(x)$, which may be represented by a power-series expansion in some neighborhood of the origin, say

$$y = \sum_{n=0}^{\infty} a_n x^n \,. \tag{9.73}$$

The first thing we do is determine the coefficients $a_0, a_1, a_2, \ldots$.

One way to proceed is this: Differentiating (9.73) twice, we obtain

$$y'' = \sum_{n=2}^{\infty} n(n - 1)a_n x^{n-2} \,.$$

Multiplying by $1 - x^2$, we find

$$\begin{aligned} (1 - x^2)y'' &= \sum_{n=2}^{\infty} n(n - 1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n - 1)a_n x^n \\ &= \sum_{n=0}^{\infty} (n + 2)(n + 1)a_{n+2} x^n - \sum_{n=0}^{\infty} n(n - 1)a_n x^n \end{aligned}$$

$$(9.74) \qquad = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} - n(n-1)a_n]x^n.$$

Substituting each of the series (9.73) and (9.74) in the differential equation, we obtain an equation involving two power series, valid in some neighborhood of the origin. By the uniqueness theorem, these power series must be equal term by term. Therefore we may equate coefficients of x^n and obtain the relation

$$(n+2)(n+1)a_{n+2} - n(n-1)a_n = -2a_n$$

or, what amounts to the same thing,

$$a_{n+2} = \frac{n^2 - n - 2}{(n+2)(n+1)} a_n = \frac{n-2}{n+2} a_n .$$

This relation enables us to determine $a_2, a_4, a_6, \ldots$ successively in terms of a_0. Similarly, we can compute $a_3, a_5, a_7, \ldots$ in terms of a_1. For the coefficients with even subscripts, we find

$$a_2 = -a_0 , \qquad a_4 = 0 \cdot a_2 = 0 , \qquad a_6 = a_8 = a_{10} = \cdots = 0 .$$

The odd coefficients are

$$a_3 = \frac{1-2}{1+2} a_1 = \frac{-1}{3} a_1 , \qquad a_5 = \frac{3-2}{3+2} a_3 = \frac{1}{5} \cdot \frac{(-1)}{3} a_1 ,$$

$$a_7 = \frac{5-2}{5+2} a_5 = \frac{3}{7} \cdot \frac{1}{5} \cdot \frac{(-1)}{3} a_1 = \frac{-1}{7 \cdot 5} a_1$$

and, in general,

$$a_{2n+1} = \frac{2n-3}{2n+1} a_{2n-1} = \frac{2n-3}{2n+1} \cdot \frac{2n-5}{2n-1} \cdot \frac{2n-7}{2n-3} \cdot \cdots \cdot \frac{3}{7} \cdot \frac{1}{5} \cdot \frac{(-1)}{3} a_1 .$$

When the common factors are canceled this simplifies to

$$a_{2n+1} = \frac{-1}{(2n+1)(2n-1)} a_1 .$$

Therefore the series for y can be written as follows:

$$y = a_0(1 - x^2) - a_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)(2n-1)} x^{2n+1} .$$

The ratio test may be used to verify the convergence of this series for $|x| < 1$. The work just carried out shows that the series actually satisfies the differential equation in (9.72), where a_0 and a_1 may be thought of as arbitrary constants. The reader should note that in this particular example the polynomial which multiplies a_0 is itself a solution of (9.72), and the series which multiplies a_1 is another solution.

The procedure just described is called the *method of undetermined coefficients*. Another way to find these coefficients is based on the result of Section 9.21 which states that

$$a_n = \frac{f^{(n)}(0)}{n!} \qquad \text{if} \quad y = f(x)\,.$$

Sometimes the higher derivatives of y at the origin can be computed directly from the differential equation. For example, putting $x = 0$ in (9.72) we immediately obtain

$$f''(0) = -2f(0) = -2a_0\,,$$

and hence

$$a_2 = \frac{f''(0)}{2!} = -a_0\,.$$

To find the higher derivatives we differentiate the differential equation to obtain

$$(1 - x^2)y''' - 2xy'' = -2y'\,. \tag{9.75}$$

Putting $x = 0$, we see that $f'''(0) = -2f'(0) = -2a_1$, and hence $a_3 = f'''(0)/3! = -a_1/3$. Differentiation of (9.75) leads to the equation

$$(1 - x^2)y^{(4)} - 4xy''' = 0\,.$$

When $x = 0$, this yields $f^{(4)}(0) = 0$ and hence $a_4 = 0$. Repeating the process once more, we find

$$(1 - x^2)y^{(5)} - 6xy^{(4)} - 4y''' = 0\,,$$

$$f^{(5)}(0) = 4f'''(0) = -8a_1\,, \qquad a_5 = \frac{f^{(5)}(0)}{5!} = -\frac{a_1}{15}\,.$$

It is clear that the process may be continued as long as desired.

9.25 Exercises

In each of Exercises 1, 2, and 3, the power series is used to define the function f. Determine the interval of convergence in each case and show that f satisfies the differential equation indicated, where $y = f(x)$.

1. $f(x) = \sum_{n=0}^{\infty} \frac{x^{4n}}{(4n)!}\,; \qquad \frac{d^4y}{dx^4} = y.$

2. $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{(n!)^2}\,; \qquad xy'' + y' - y = 0.$

3. $f(x) = 1 + \sum_{n=1}^{\infty} \frac{1 \cdot 4 \cdot 7 \cdot \cdots \cdot (3n-2)}{(3n)!}\, x^{3n}\,; \qquad y'' = x^a y + b\,.$

(Find a and b.)

4. The functions J_0 and J_1 defined by the series

$$J_0(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(n!)^2 2^{2n}}, \qquad J_1(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{n!(n+1)!2^{2n+1}}$$

are called *Bessel functions of the first kind* of orders zero and one, respectively. These functions arise in many problems in both pure and applied mathematics. Show that (a) both series converge for all real x; (b) $J_0'(x) = -J_1(x)$; (c) $j_0(x) = j_1'(x)$, where $j_0(x) = xJ_0(x)$ and $j_1(x) = xJ_1(x)$.

5. The differential equation

$$x^2y'' + xy' + (x^2 - n^2)y = 0$$

is called *Bessel's equation*. Show that J_0 and J_1 (as defined in Exercise 4) are solutions when $n = 0$ and 1, respectively.

In each of Exercises 6, 7, and 8, assume the given differential equation has a power-series solution and find the first four nonzero terms.

6. $y' = x^2 + y^2$, with $y = 1$ when $x = 0$.
7. $y' = 1 + xy^2$, with $y = 0$ when $x = 0$.
8. $y' = x + y^2$, with $y = 0$ when $x = 0$.

In Exercises 9, 10, and 11, assume the given differential equation has a power-series solution of the form $y = \Sigma a_n x^n$, and determine the nth coefficient a_n.

9. $y' = \alpha y$.
10. $y'' = xy$.
11. $y'' + xy' + y = 0$.

12. In this exercise we outline another method for showing that the binomial series $\Sigma\binom{\alpha}{n}x^n$ represents $(1 + x)^\alpha$ in the interval $|x| < 1$.

(a) Use the ratio test to show that the binomial series converges in the open interval $(-1, 1)$.

(b) For a fixed real α, define a function f_α by means of the equation

$$f_\alpha(x) = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \qquad \text{if } |x| < 1,$$

and prove that $f_\alpha'(x) = \alpha f_{\alpha-1}(x)$.

(c) Show that $(1 + x)f_\alpha'(x) = \alpha f_\alpha(x)$ if $|x| < 1$. Solve this differential equation to obtain $f_\alpha(x) = (1 + x)^\alpha$.

9.26 Applications of power series to the computation of function values

The partial sums of a power series, being *polynomials*, are relatively easy to compute. For this reason, power series are especially useful for calculating the values of the functions they represent. The idea, of course, is to use the partial sums as approximations to the sum of the series. To make this idea meaningful in practice, we must have a convenient way of estimating the error committed in replacing an infinite series by a finite sum. Although there is no *universal* method for estimating these errors, there are several useful procedures to choose from.

Taylor's formula (Theorem 7–6) expresses the error in terms of one of the higher derivatives of the function, and in many cases the size of this derivative may be estimated with ease. If the terms of a power series alternate in sign and decrease in absolute value, we have recourse to the estimate provided in Leibniz's rule for alternating series (Theorem 9–13). Another useful device is to find a simpler series (such as a geometric series) which dominates the error. Some of these techniques are illustrated in the following examples.

Example 1: Calculation of the Euler number e. If we put $x = 1$ in Taylor's formula (with remainder) for e^x, we find $e = s_n + r_n$, where

$$s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}, \qquad r_n = \frac{e^c}{(n+1)!}, \qquad \text{and} \quad 0 < c < 1 .$$

Since $1 < e^c < e < 3$† we have the estimate

$$\frac{1}{(n+1)!} < r_n < \frac{3}{(n+1)!} \tag{9.76}$$

which enables us to compute e to any desired degree of accuracy. For example, if we want the value of e correct to seven decimal places, we must choose n so that $3/(n+1)! < \frac{1}{2}10^{-8}$. We shall see presently that $n = 12$ suffices. A table of values of $1/n!$ may be computed rather quickly because $1/n!$ may be obtained from $1/(n-1)!$ by simply dividing by n. The following table for $3 \leq n \leq 12$ contains these numbers rounded off to nine decimals. The "round-off error" in each case is indicated by a plus or minus sign which tells whether the recorded value exceeds or is less than the correct value. (In any case, this error is less than one-half unit in the last decimal place.)

n	$\frac{1}{n!}$	n	$\frac{1}{n!}$
3	0.166 666 667 −	8	0.000 024 802 −
4	0.041 666 667 −	9	0.000 002 756 −
5	0.008 333 333 +	10	0.000 000 276 −
6	0.001 388 889 −	11	0.000 000 025 +
7	0.000 198 413 −	12	0.000 000 002 +

The terms corresponding to $n = 0, 1, 2$ have sum 5/2. Adding this to the sum of the entries in the table (for $n \leq 12$), we obtain a total of 2.718281830. If we take into account the round-off errors, the *actual* value of s_{12} may be less than this by as much as 7/2 of a unit in the last decimal place (due to the seven minus signs) or may exceed this by as much as $\frac{3}{2}$ unit in the last place (due to the three plus signs). Therefore, all we can assert by this calculation is the inequality $2.718281826 < s_{12} < 2.718281832$. Also, the remainder r_{12} itself satisfies (9.76) which becomes $0.000000000 < r_{12} < 0.000000001$. Since $e = s_{12} + r_{12}$ this calculation leads to the following inequality:

$$2.718281826 < e < 2.718281833 .$$

This tells us that the value of e, *correct to seven decimals*, is $e = 2.7182818$, or that the value of e, *rounded off to eight decimals*, is $e = 2.71828183$.

Example 2: Irrationality of e. When we put $x = -1$ in the series (9.70) for e^x, we obtain an alternating series for e^{-1} whose terms decrease monotonically in absolute value. If $s_n = \sum_{k=0}^{n} (-1)^k/k!$, then the inequalities (9.48) of Theorem 9–13 tell us that

$$0 < e^{-1} - s_{2k-1} < \frac{1}{(2k)!},$$

† To prove that $e < 3$ we may subtract the power series for $\log(1-x)$ from that for $\log(1+x)$ to obtain

$$\log\frac{1+x}{1-x} = 2\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \qquad \text{if} \quad |x| < 1 .$$

Therefore $\log[(1+x)/(1-x)] > 2x$ if $0 < x < 1$. When $x = \frac{1}{2}$ this states that $\log 3 > 1$ and hence $e < 3$.

from which we obtain

$$0 < (2k-1)!(e^{-1} - s_{2k-1}) < \frac{1}{2k} \leq \frac{1}{2} \tag{9.77}$$

if $k \geq 1$. The product $(2k-1)!s_{2k-1}$ is an integer for every choice of k. If e^{-1} were rational, we could choose k so large that $(2k-1)!e^{-1}$ would also be an integer. But then (9.77) would tell us that the difference of these two integers is a positive number not exceeding $\frac{1}{2}$, which is impossible. Therefore e^{-1} cannot be rational, and this proves that e is irrational.

Example 3: Calculation of π. A convenient formula for computing π may be deduced from Gregory's series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + (-1)^{n-1}\frac{x^{2n-1}}{2n-1} + \cdots, \tag{9.78}$$

which we have shown to be valid for $-1 < x < 1$. Actually, this equation also holds when $x = 1$ (although we have not proved this), and it yields the series

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots + \frac{(-1)^{n-1}}{2n-1} + \cdots. \tag{9.79}$$

Since this is an alternating series, the error committed by stopping at the nth term does not exceed $1/(2n+1)$ in absolute value. Therefore, to compute $\frac{1}{4}\pi$ to eight decimals from (9.79) would require $n > 10^8$ and this would entail the evaluation of 100 million terms. This is an example of a series that converges "too slowly" to be of any practical value for numerical work. Actually, by putting $x = 1$ in (9.78) we have lost the benefit of the powers $x^3, x^5, x^7, \ldots$ which tend to increase the rapidity of convergence for smaller values of x.

Fortunately, we can still make use of (9.78) with a smaller value of x to compute π, if we exploit some of our advance knowledge about π. It is not difficult to decide that π is nearly 3 so that $\frac{1}{4}\pi$ is approximately 0.8 or $\frac{4}{5}$, and this is nearly 4 arc tan $\frac{1}{5}$, since arc tan x and x are nearly equal for small x. Therefore let us put

$$\alpha = \arctan \tfrac{1}{5}, \qquad \text{so that} \qquad \tan \alpha = \tfrac{1}{5},$$

and let $\beta = 4\alpha - \frac{1}{4}\pi$. The number β should be small since 4α is nearly $\frac{1}{4}\pi$. We know $\tan \alpha = \frac{1}{5}$. Let us try to compute $\tan \beta$. Using the identity

$$\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

with $A = B = \alpha$ and then again with $A = B = 2\alpha$, we find

$$\tan 2\alpha = \frac{1/5 + 1/5}{1 - 1/25} = \frac{5}{12} \qquad \text{and} \qquad \tan 4\alpha = \frac{5/12 + 5/12}{1 - 25/144} = \frac{120}{119}.$$

Since $\beta = 4\alpha - \frac{1}{4}\pi$ we have

$$\tan \beta = \tan (4\alpha - \tfrac{1}{4}\pi) = \frac{\tan 4\alpha - \tan \frac{1}{4}\pi}{1 + \tan 4\alpha \tan \frac{1}{4}\pi} = \frac{120/119 - 1}{1 + 120/119} = \frac{1}{239}.$$

Therefore $\beta = \arctan (1/239)$ and the relation $\frac{1}{4}\pi = 4\alpha - \beta$ yields the following remarkable identity discovered in 1706 by John Machin (1680–1751):

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}. \tag{9.80}$$

If we use the arc tangent series with $x = 1/5$ and $x = 1/239$, this gives us

$$\pi = 16\left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - + \cdots\right) - 4\left(\frac{1}{239} - \frac{1}{3 \cdot 239^3} + \frac{1}{5 \cdot 239^5} - + \cdots\right).$$

Taking six terms of the first series and two terms of the second (retaining nine decimal places in the calculations) and paying attention to the remainders and round-off errors, we are finally led to the inequalities

$$3.141592629 < \pi < 3.141592668 \tag{9.81}$$

which tell us that the value of π, *correct to seven decimals*, is 3.1415926. If we want a value that is rounded off, we can only assert from this calculation that $\pi = 3.141593$ since the inequalities in (9.81) do not give us enough information for rounding off beyond the sixth decimal place. Machin used this method to compute π to 100 decimal places.

There are many other arc tangent relations like (9.80) that are helpful in computing π. An article by D. H. Lehmer in the *American Mathematical Monthly* (vol. 45, 1938, pp. 657–667) contains a long list of such formulas with a discussion of their relative merits in computational work.

9.27 Exercises

1. (a) Obtain the number $r = \sqrt{15} - 3$ as an approximation to the nonzero root of the equation $x^2 = \sin x$ by using the first two nonzero terms of the power-series expansion for $\sin x$.

(b) Show that the approximation in part (a) satisfies the inequality

$$|\sin r - r^2| < \frac{1}{200}$$

given that $\sqrt{15} - 3 < 0.9$. Is the difference $(\sin r - r^2)$ positive or negative? Give full details of your reasoning.

2. (a) Show that the first six terms of the binomial series for $(1 - x)^{-1/2}$ are:

$$1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \frac{35}{128}x^4 + \frac{63}{256}x^5 .$$

(b) Let a_n denote the nth term of this series when $x = 1/50$, and let r_n denote the remainder after n terms; that is, for $n \geq 0$ let

$$r_n = a_{n+1} + a_{n+2} + a_{n+3} + \cdots .$$

Show that $0 < r_n < a_n/49$. [*Hint.* Show that $a_{n+1} < a_n/50$, and dominate r_n by a suitable geometric series.]

(c) Verify the identity

$$\sqrt{2} = \frac{7}{5}\left(1 - \frac{1}{50}\right)^{-1/2}$$

and use it to compute the first ten correct decimals of $\sqrt{2}$. [*Hint.* Use parts (a) and (b), retain twelve decimals during the calculations, and take into account round-off errors.]

3. (a) Show that

$$\sqrt{3} = \frac{1732}{1000}\left(1 - \frac{176}{3{,}000{,}000}\right)^{-1/2} .$$

(b) Proceed as suggested in Exercise 2 and compute the first fifteen correct decimals of $\sqrt{3}$.

4. Use Gregory's series to complete the following four-place table of $f(x) = \arctan x$.

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.10
$f(x)$	0.0000	0.0100	0.0200	0.0300	0.0400			

5. (a) Use the identities

$$\log \frac{10}{9} = -\log\left(1 - \frac{1}{10}\right), \qquad \log \frac{25}{24} = -\log\left(1 - \frac{4}{100}\right), \qquad \log \frac{81}{80} = \log\left(1 + \frac{1}{80}\right)$$

and enough terms of the power series for $\log(1 + x)$ [or $\log(1 - x)$] to secure six decimals, and show that

$$\log \frac{10}{9} = 0.105360 + ; \qquad \log \frac{25}{24} = 0.040822 - ; \qquad \log \frac{81}{80} = 0.012423 -$$

(b) Use the results of part (a), along with the identity

$$2 = \left(\frac{10}{9}\right)^7 \left(\frac{25}{24}\right)^{-2} \left(\frac{81}{80}\right)^3$$

to compute log 2, rounded off to five decimals.

(c) Express 3 and 5 similarly as products of powers of 10/9, 25/24, and 81/80 and compute log 3 and log 5, rounded off to five decimals.

9.28 Improper integrals

The concept of an integral $\int_a^b f(x)\,dx$ was introduced in Chapter 1 under the restriction that the function f is *defined and bounded* on a *finite interval* $[a, b]$. The scope of integration theory may be extended by relaxing these restrictions.

To begin with, we may study the behavior of $\int_a^b f(x)\,dx$ as $b \to +\infty$. This leads to the notion of an *infinite integral* (also called an *improper integral of the first kind*) denoted by the symbol $\int_a^\infty f(x)\,dx$. Another extension is obtained if we keep the interval $[a, b]$ finite and allow f to become unbounded at one or more points. The new integrals so obtained (by a suitable limit process) are called *improper integrals of the second kind.*†

Many important functions in analysis appear as improper integrals of one kind or another and a detailed study of such functions is ordinarily undertaken in courses in advanced calculus. We shall be concerned here only with the most elementary aspects of the theory. In fact, we shall merely state some definitions and theorems and give some examples.

It will be evident presently that the definitions pertaining to improper integrals bear a strong resemblance to those for infinite series. Therefore it is not surprising that many of the elementary theorems on series have direct analogs for improper integrals.

If the proper integral $\int_a^b f(x)\,dx$ exists for every $b \geq a$, we may define a new function I as follows:

(9.82) $$I(b) = \int_a^b f(x)\,dx \qquad \text{for each } b \geq a\,.$$

The function I defined in this way is called an *infinite integral*, or an *improper integral of the first kind*, and it is denoted by the symbol $\int_a^\infty f(x)\,dx$. The integral is said to *converge* if the limit

(9.83) $$\lim_{b\to+\infty} I(b) = \lim_{b\to+\infty} \int_a^b f(x)\,dx$$

exists and is finite. Otherwise the integral $\int_a^\infty f(x)\,dx$ is said to *diverge*. If the limit in (9.83) exists and equals A, the number A is called the *value* of the integral and we write

† To distinguish the integrals of Chapter 1 from improper integrals, the former are often called "proper" integrals.

$$\int_a^\infty f(x)\,dx = A\,.$$

These definitions are similar to those given for infinite series. The function values $I(b)$ play the role of the "partial sums" and may be referred to as "partial integrals." Note that the symbol $\int_a^\infty f(x)\,dx$ is used both for the integral and for the value of the integral when the integral converges. (Compare with the remarks near the end of Section 9.5.)

Example 1. The improper integral $\int_1^\infty x^{-s}\,dx$ converges if $s > 1$ and diverges if $s \le 1$. To prove this we note that

$$I(b) = \int_1^b x^{-s}\,dx = \begin{cases} \dfrac{b^{1-s} - 1}{1 - s} & \text{if } \; s \ne 1\,, \\ \log b & \text{if } \; s = 1\,. \end{cases}$$

Therefore $I(b)$ tends to a finite limit if and only if $s > 1$, in which case the limit is

$$\int_1^\infty x^{-s}\,dx = \frac{1}{s-1}\,.$$

The behavior of this integral is analogous to that of the series for the zeta-function, $\zeta(s) = \sum_{n=1}^\infty n^{-s}$.

Example 2. The integral $\int_0^\infty \sin x\,dx$ diverges because

$$I(b) = \int_0^b \sin x\,dx = 1 - \cos b\,,$$

and this does not tend to a limit as $b \to +\infty$.

Infinite integrals of the form $\int_{-\infty}^b f(x)\,dx$ are similarly defined. Also, if $\int_{-\infty}^c f(x)\,dx$ and $\int_c^\infty f(x)\,dx$ are *both convergent* for some c, we say that the integral $\int_{-\infty}^\infty f(x)\,dx$ is convergent, and its value is defined to be the sum

(9.84)
$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^c f(x)\,dx + \int_c^\infty f(x)\,dx\,.$$

(It is easy to show that the choice of c is unimportant.) The integral $\int_{-\infty}^\infty f(x)\,dx$ is said to diverge if at least one of the integrals on the right of (9.84) is divergent.

Example 3. The integral $\int_{-\infty}^\infty e^{-a|x|}\,dx$ converges if $a > 0$, for if $b > 0$, we have

$$\int_0^b e^{-a|x|}\,dx = \int_0^b e^{-ax}\,dx = \frac{e^{-ab} - 1}{-a} \to \frac{1}{a} \qquad \text{as} \quad b \to \infty\,.$$

Hence $\int_0^\infty e^{-a|x|}\,dx$ converges and has the value $1/a$. Also, if $b > 0$ we have

$$\int_{-b}^0 e^{-a|x|}\,dx = \int_{-b}^0 e^{ax}\,dx = -\int_b^0 e^{-at}\,dt = \int_0^b e^{-at}\,dt.$$

Therefore $\int_{-\infty}^0 e^{-a|x|}\,dx$ also converges and has the value $1/a$. Hence we have $\int_{-\infty}^\infty e^{-a|x|}\,dx = 2/a$. Note, however, that the integral $\int_{-\infty}^\infty e^{-ax}\,dx$ *diverges* because $\int_{-\infty}^0 e^{-ax}\,dx$ diverges.

As in the case of series, we have various convergence tests for improper integrals. The simplest of these refers to a positive integrand.

9–21 THEOREM. Assume that the proper integral $\int_a^b f(x)\,dx$ exists for each $b \geq a$ and suppose that $f(x) \geq 0$ for all $x \geq 0$. Then $\int_a^\infty f(x)\,dx$ converges if and only if there is a constant $M > 0$ such that

$$\int_a^b f(x)\,dx \leq M \qquad \text{for every } b \geq a\,.$$

This theorem forms the basis for the following comparison tests:

9–22 THEOREM. Assume the proper integral $\int_a^b f(x)\,dx$ exists for each $b \geq a$ and suppose that

$$0 \leq f(x) \leq g(x) \qquad \text{for all } x \geq a\,,$$

where $\int_a^\infty g(x)\,dx$ converges. Then $\int_a^\infty f(x)\,dx$ also converges and

$$\int_a^\infty f(x)\,dx \leq \int_a^\infty g(x)\,dx\,.$$

Note. The integral $\int_a^\infty g(x)\,dx$ is said to *dominate* the integral $\int_a^\infty f(x)\,dx$.

9–23 THEOREM. *Limit comparison test.* Assume both proper integrals $\int_a^b f(x)\,dx$ and $\int_a^b g(x)\,dx$ exist for each $b \geq a$, where $f(x) \geq 0$ and $g(x) > 0$ for all $x \geq a$. If

(9.85)
$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = c\,, \qquad \text{where} \quad c \neq 0\,,$$

then both integrals $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)\,dx$ converge or both diverge.

Note. If the limit in (9.85) is 0, we can conclude only that convergence of $\int_a^\infty g(x)\,dx$ implies convergence of $\int_a^\infty f(x)\,dx$.

The proofs of Theorems 9–21 through 9–23 are similar to the corresponding results for series and are left as exercises.

Example 4. For each real s, the integral $\int_1^\infty e^{-x}x^s\,dx$ converges. This is seen by comparison with $\int_1^\infty x^{-2}\,dx$ since $e^{-x}x^s/x^{-2} \to 0$ as $x \to +\infty$.

Improper integrals of the second kind may be introduced as follows: Suppose f is defined on the half-open interval $(a, b]$, and assume that the integral $\int_x^b f(t)\,dt$ exists for each x satisfying $a < x \leq b$. Define a new function I as follows:

$$I(x) = \int_x^b f(t)\,dt \qquad \text{if} \quad a < x \leq b\,.$$

The function I so defined is called an *improper integral of the second kind* and is denoted by the symbol $\int_{a+}^b f(t)\,dt$. The integral is said to *converge* if the limit

(9.86)
$$\lim_{x\to a+} I(x) = \lim_{x\to a+} \int_x^b f(t)\,dt$$

exists and is finite. Otherwise the integral $\int_{a+}^{b} f(t)\,dt$ is said to *diverge*. If the limit in (9.86) exists and equals A, the number A is called the *value* of the integral and we write

$$\int_{a+}^{b} f(t)\,dt = A\,.$$

Example 5. Let $f(t) = t^{-s}$ if $t > 0$. If $b > 0$ and $x > 0$, we have

$$I(x) = \int_x^b t^{-s}\,dt = \begin{cases} \dfrac{b^{1-s} - x^{1-s}}{1-s} & \text{if } s \neq 1\,, \\ \log b - \log x & \text{if } s = 1\,. \end{cases}$$

When $x \to 0+$, $I(x)$ tends to a finite limit if and only if $s < 1$. Hence the integral $\int_{0+}^{b} t^{-s}\,dt$ converges if $s < 1$ and diverges if $s \geq 1$.

This example may be dealt with in another way. If we introduce the substitution $t = 1/u$, we obtain

$$\int_x^b t^{-s}\,dt = \int_{1/b}^{1/x} u^{s-2}\,du\,.$$

When $x \to 0+$, $1/x \to +\infty$ and hence $\int_{0+}^{b} t^{-s}\,dt = \int_{1/b}^{\infty} u^{s-2}\,du$, provided the last integral converges. By Example 1, this converges if and only if $s - 2 < -1$, which means $s < 1$.

The foregoing example illustrates a remarkable geometric fact. Consider the function f defined by the equation $f(x) = x^{-3/4}$ if $0 < x \leq 1$. The integral $\int_{0+}^{1} f(x)\,dx$ converges but the integral $\int_{0+}^{1} \pi f^2(x)\,dx$ diverges. Geometrically, this means that the ordinate set of f has a finite area but the solid obtained by rotating this ordinate set about the x-axis has an infinite volume.

Improper integrals of the form $\int_a^{b-} f(t)\,dt$ are defined in a similar fashion. If the two integrals $\int_{a+}^{c} f(t)\,dt$ and $\int_c^{b-} f(t)\,dt$ *both* converge, we write†

$$\int_{a+}^{b-} f(t)\,dt = \int_{a+}^{c} f(t)\,dt + \int_c^{b-} f(t)\,dt\,.$$

The definition can be extended (in an obvious way) to cover the case of any finite number of summands. For example, if f is undefined at two points $c < d$ interior to an interval $[a, b]$ we say the improper integral $\int_a^b f(t)\,dt$ converges and has the value $\int_a^{c-} f(t)\,dt + \int_{c+}^{d-} f(t)\,dt + \int_{d+}^{b} f(t)\,dt$, provided that each of these integrals converges. Furthermore, we can consider "mixed" combinations such as $\int_{a+}^{b} f(t)\,dt + \int_b^{\infty} f(t)\,dt$ which we write as $\int_{a+}^{\infty} f(t)\,dt$, or mixed combinations of the form $\int_a^{b-} f(t)\,dt + \int_{b+}^{c} f(t)\,dt + \int_c^{\infty} f(t)\,dt$ which we write simply as $\int_a^{\infty} f(t)\,dt$.

Example 6. If $s > 0$ the integral $\int_{0+}^{\infty} e^{-t}t^{s-1}\,dt$ converges. This must be interpreted as a sum, say

(9.87) $$\int_{0+}^{1} e^{-t}t^{s-1}\,dt + \int_1^{\infty} e^{-t}t^{s-1}\,dt\,.$$

The second integral converges for all real s, by Example 4. To test the first integral we put $t = 1/u$ and note that

$$\int_x^1 e^{-t}t^{s-1}\,dt = \int_1^{1/x} e^{-1/u}u^{-s-1}\,du\,.$$

† Some authors write $\int_a^b$ where we have written $\int_{a+}^{b-}$.

But $\int_1^\infty e^{-1/u}u^{-s-1}\,du$ converges for $s > 0$ by comparison with $\int_1^\infty u^{-s-1}\,du$. Therefore the integral $\int_{0+}^1 e^{-t}t^{s-1}\,dt$ converges for $s > 0$. When $s > 0$ the sum in (9.87) is denoted by $\Gamma(s)$. The function Γ so defined is called the *gamma function*, first introduced by Euler in 1729. It has the interesting property that $\Gamma(n + 1) = n!$ when n is any integer ≥ 0. (See Exercise 17 of Section 9.29 for an outline of the proof.)

The convergence tests given in Theorems 9–21 through 9–23 have straightforward analogs for improper integrals of the second kind. The reader should have no difficulty in formulating these tests for himself.

9.29 Exercises

In each of Exercises 1 through 10, test the improper integral for convergence.

1. $\displaystyle\int_0^\infty \frac{x}{\sqrt{x^4+1}}\,dx.$

2. $\displaystyle\int_{-\infty}^\infty e^{-x^2}\,dx.$

3. $\displaystyle\int_0^\infty \frac{1}{\sqrt{x^3+1}}\,dx.$

4. $\displaystyle\int_0^\infty \frac{1}{\sqrt{e^x}}\,dx.$

5. $\displaystyle\int_{0+}^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx.$

6. $\displaystyle\int_{0+}^1 \frac{\log x}{\sqrt{x}}\,dx.$

7. $\displaystyle\int_{0+}^{1-} \frac{\log x}{1-x}\,dx.$

8. $\displaystyle\int_{-\infty}^\infty \frac{x}{\cosh x}\,dx.$

9. $\displaystyle\int_{0+}^{1-} \frac{dx}{\sqrt{x}\log x}.$

10. $\displaystyle\int_2^\infty \frac{dx}{x(\log x)^s}.$

11. For a certain value of α the integral

$$\int_0^\infty \left(\frac{1}{\sqrt{1+2x^2}} - \frac{\alpha}{x+1}\right)dx$$

is convergent. Find this α and evaluate the integral.

12. Find the values of a and b such that

$$\int_1^\infty \left(\frac{2x^2+bx+a}{x(2x+a)} - 1\right)dx = 1\,.$$

13. For what values of the constants a and b will the following limit exist and be equal to 1?

$$\lim_{p\to+\infty}\int_{-p}^{p} \frac{x^3+ax^2+bx}{x^2+x+1}\,dx\,.$$

14. (a) Prove that

$$\lim_{h\to 0+}\left(\int_{-1}^{-h}\frac{dx}{x} + \int_h^1 \frac{dx}{x}\right) = 0 \quad \text{and that} \quad \lim_{h\to+\infty}\int_{-h}^{h} \sin x\,dx = 0\,.$$

(b) Do the following improper integrals converge or diverge?

$$\int_{-1}^1 \frac{dx}{x}\,; \qquad \int_{-\infty}^\infty \sin x\,dx\,.$$

15. (a) Use the power series for $\sin x$ to show that the integral $\int_{0+}^{1} (\sin x)/x\, dx$ converges.
(b) Prove that $\lim_{x\to 0+} x\int_x^1 (\cos t)/t^2\, dt = 1$.
(c) Does the integral $\int_{0+}^{1} (\cos t)/t^2\, dt$ converge or diverge?

16. (a) If f is monotonic decreasing for all $x \geq 1$ and if $f(x) \to 0$ as $x \to +\infty$, prove that the integral $\int_1^\infty f(x)\, dx$ and the series $\Sigma f(n)$ both converge or both diverge. [*Hint*. Recall the proof of the integral test.]
(b) Give an example of a nonmonotonic f for which the series $\Sigma f(n)$ converges and the integral $\int_1^\infty f(x)\, dx$ diverges.

17. Let $\Gamma(s) = \int_{0+}^{\infty} t^{s-1}e^{-t}\, dt$, if $s > 0$. (The gamma function.) Use integration by parts to show that $\Gamma(s + 1) = s\Gamma(s)$. Then use induction to prove that $\Gamma(n + 1) = n!$, if n is a positive integer.

Each of Exercises 18 through 23 contains a statement, not necessarily true, about a function f defined for all $x \geq 1$. In each of these exercises n denotes a positive integer, and I_n denotes the integral $\int_1^n f(x)\, dx$, which is always assumed to exist. For each statement either give a proof or provide a counterexample.

18. If f is monotonic decreasing and if $\lim_{n\to\infty} I_n$ exists, then the integral $\int_1^\infty f(x)\, dx$ converges.

19. If $\lim_{x\to\infty} f(x) = 0$ and $\lim_{n\to\infty} I_n = A$, then $\int_1^\infty f(x)\, dx$ converges and has the value A.

20. If the sequence $\{I_n\}$ converges, then the integral $\int_1^\infty f(x)\, dx$ converges.

21. If f is positive and if $\lim_{n\to\infty} I_n = A$, then $\int_1^\infty f(x)\, dx$ converges and has the value A.

22. Assume $f'(x)$ exists for each $x \geq 1$ and suppose there is a constant $M > 0$ such that $|f'(x)| \leq M$ for all $x \geq 1$. If $\lim_{n\to\infty} I_n = A$, then the integral $\int_1^\infty f(x)\, dx$ converges and has the value A.

23. If $\int_1^\infty f(x)\, dx$ converges, then $\lim_{x\to\infty} f(x) = 0$.

★9.30 Remarks on term-by-term integration and differentiation of series

Assume that a function f can be expressed as a convergent series of other functions, say

$$f(x) = \sum_{n=0}^{\infty} u_n(x)$$

for each x in an interval $[a, b]$. Power-series expansions are a special case in which the terms $u_n(x)$ have the form $u_n(x) = a_n x^n$. In the case of power series, we know that the integral of f may be obtained by integrating the series term by term. This is also true for some series expansions other than power series, but it is not true in general. For example, suppose $u_k(x)$ is the polynomial given by

$$u_k(x) = (k + 1)^2x(1 - x)^{k+1} - k^2x(1 - x)^k .$$

Then $\Sigma u_k(x)$ is a telescoping series with partial sums $s_n(x) = u_0(x) + \cdots + u_{n-1}(x) = n^2x(1 - x)^n$. For each fixed x in $[0, 1]$ we find that $s_n(x) \to 0$ as $n \to \infty$ so we have

$$\sum_{k=0}^{\infty} u_k(x) = 0 \qquad \text{and hence} \qquad \int_0^1 \sum_{k=0}^{\infty} u_k(x)\, dx = 0 .$$

On the other hand, we have

$$\sum_{k=0}^{n-1} \int_0^1 u_k(x)\, dx = \int_0^1 s_n(x)\, dx = n^2 \int_0^1 x(1 - x)^n\, dx = \frac{n^2}{(n + 1)(n + 2)} \to 1 \qquad \text{as} \quad n \to \infty .$$

Therefore $\sum_{k=0}^{\infty} \int_0^1 u_k(x)\,dx = 1$, and hence

$$\int_0^1 \sum_{k=0}^{\infty} u_k(x)\,dx \neq \sum_{k=0}^{\infty} \int_0^1 u_k(x)\,dx\,.$$

This example shows that term-by-term integration of a series of functions is not always permissible, in direct contrast to the analogous situation for *finite* sums where the operations of integration and summation can always be interchanged. George G. Stokes (1819–1903), Phillip L. v. Seidel (1821–1896), and Karl Weierstrass were first to realize that some extra condition is needed to justify interchanging these operations when dealing with infinite series. In 1848, Stokes and Seidel (independently and almost simultaneously) introduced a concept now known as *uniform convergence* and showed that uniformly convergent series could be integrated term by term. Weierstrass later showed that the concept is of great importance in advanced analysis. Since uniform convergence is ordinarily treated in detail in advanced calculus, we shall not develop the concept here. Instead, we shall discuss a more stringent condition which justifies term-by-term integration in a great number of cases that arise in practice.

9–24 THEOREM. Assume f is given by a convergent series of the form

$$f(x) = \sum_{k=0}^{\infty} u_k(x) \tag{9.88}$$

for each x in $[a, b]$. Assume also that the integrals $\int_a^b f(x)\,dx$ and $\int_a^b u_k(x)\,dx$ exist for each $k \geq 0$. If there is a sequence of nonnegative constants $\{M_n\}$ such that $|u_k(x)| \leq M_k$ for all x in $[a, b]$ and for all $k \geq 0$, and if the series ΣM_k is convergent, then term-by-term integration of (9.88) is permissible. That is to say, we have

$$\int_a^b f(x)\,dx = \int_a^b \sum_{k=0}^{\infty} u_k(x)\,dx = \sum_{k=0}^{\infty} \int_a^b u_k(x)\,dx\,. \tag{9.89}$$

Proof. The linearity property of the integral for finite sums implies

$$\int_a^b f(x)\,dx - \sum_{k=0}^{n} \int_a^b u_k(x)\,dx = \int_a^b \left(f(x) - \sum_{k=0}^{n} u_k(x)\right) dx = \int_a^b \sum_{k=n+1}^{\infty} u_k(x)\,dx\,.$$

Therefore we have

$$\left|\int_a^b f(x)\,dx - \sum_{k=0}^{n} \int_a^b u_k(x)\,dx\right| \leq \int_a^b \sum_{k=n+1}^{\infty} |u_k(x)|\,dx \leq \int_a^b \sum_{k=n+1}^{\infty} M_k\,dx$$

$$= (b - a) \sum_{k=n+1}^{\infty} M_k\,.$$

But $\sum_{k=n+1}^{\infty} M_k \to 0$ as $n \to \infty$ since ΣM_k converges, and this proves that

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=0}^{n} \int_a^b u_k(x)\,dx$$

which is the same as (9.89).

Example. Theorem 9–24 may be used to give an alternative proof that it is permissible to integrate a power series term by term. Suppose a power series $f(x) = \sum a_n x^n$ has a positive radius of convergence r, and let $[-b, b]$ be a closed subinterval of $(-r, r)$. Integrability of f on $[-b, b]$ may be established as in the proof of Theorem 9–18. Now, for each x in $[-b, b]$ we have

$$|a_n x^n| \leq |a_n| b^n .$$

Since $\sum |a_n| b^n$ converges, we may use $M_n = |a_n| b^n$ in Theorem 9–24 and thereby deduce that the power series may be integrated term by term over $[-b, b]$ and hence over an arbitrary closed subinterval of $(-r, r)$.

Term-by-term differentiation of an arbitrary series of functions is even less promising than term-by-term integration. For example, the series $\sum_{n=1}^{\infty} (\sin nx)/n^2$ converges for all real x because it is dominated by $\sum 1/n^2$. However, the series obtained by differentiating term by term is $\sum (\cos nx)/n$, and this *diverges* when $x = 0$. This example shows that term-by-term differentiation may destroy convergence. Therefore the problem of justifying the interchange of the operations of differentiation and summation is more serious than in the case of integration. We mention these facts so the reader may realize that familiar manipulations with finite sums do not always carry over to infinite series, even if the series involved are convergent.

9.31 Miscellaneous exercises

1. A sequence $\{x_n\}$ is defined by the following recursion formula:

$$x_1 = 1, \qquad x_{n+1} = \sqrt{1 + x_n} .$$

Prove that the sequence converges and find its limit.

2. A sequence $\{x_n\}$ is defined by the following recursion formula:

$$x_0 = 1, \qquad x_1 = 1, \qquad \frac{1}{x_{n+2}} = \frac{1}{x_{n+1}} + \frac{1}{x_n} .$$

Prove that the sequence converges and find its limit.

3. Let $\{a_n\}$ and $\{b_n\}$ be two sequences such that for each n we have

$$e^{a_n} = a_n + e^{b_n} .$$

(a) Show that $a_n > 0$ implies $b_n > 0$.

(b) If $a_n > 0$ for all n and if $\sum a_n$ converges, show that $\sum (b_n/a_n)$ converges.

In Exercises 4 through 8, test the given series for convergence.

4. $\sum_{n=1}^{\infty} (\sqrt{1 + n^2} - n)$.

5. $\sum_{n=1}^{\infty} n^s(\sqrt{n+1} - 2\sqrt{n} + \sqrt{n-1})$.

6. $\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}}$.

7. $\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}$.

8. $\sum_{n=1}^{\infty} a_n$, where $a_n = 1/n$ if n is odd, $a_n = 1/n^2$ if n is even.

9. Show that the infinite series

$$\sum_{n=0}^{\infty} (\sqrt{n^a + 1} - \sqrt{n^a})$$

converges for $a > 2$ and diverges for $a = 2$.

10. Let $n_1 < n_2 < n_3 < \cdots$ denote those positive integers that do not involve the digit 0 in their decimal representations. Thus $n_1 = 1, n_2 = 2, \ldots, n_9 = 9, n_{10} = 11, \ldots, n_{18} = 19, n_{19} = 21$, etc. Show that the series of reciprocals $\sum_{k=1}^{\infty} 1/n_k$ converges and has a sum less than 90. [*Hint.* Dominate the series by $9 \sum_{n=0}^{\infty} (9/10)^n$.]

11. Assume f is monotonic decreasing for all $x \geq 1$ and also assume that $\sum_{n=1}^{\infty} f(n)$ converges and has sum S. Let $s_n = \sum_{k=1}^{n} f(k)$, and show that

$$\int_{n+1}^{\infty} f(x)\,dx \leq S - s_n \leq f(n+1) + \int_{n+1}^{\infty} f(x)\,dx\,.$$

12. (a) If $a > 0$, use Exercise 11 to show that

$$\frac{1}{a} \leq \sum_{n=1}^{\infty} \frac{1}{n^{1+a}} \leq \frac{a+1}{a}\,.$$

(b) If $\zeta(s)$ denotes the Riemann zeta-function, show that $\zeta(s) \to \infty$ as $s \to 1+$, but that $(s-1)\zeta(s) \to 1$ as $s \to 1+$.

13. If a is an arbitrary real number, let $s_n(a) = 1^a + 2^a + \cdots + n^a$. Determine the following limit:

$$\lim_{n\to\infty} \frac{s_n(a+1)}{n s_n(a)}\,.$$

(Consider both positive and negative a, as well as $a = 0$.)

14. Compute the value of the coefficient a_{98} in the power-series expansion

$$\sin\left(2x + \frac{\pi}{4}\right) = \sum_{n=0}^{\infty} a_n x^n\,.$$

15. Find the intervals of convergence and of absolute convergence of the following power series: $\sum[\log(n+1) - \log n](x-2)^n$.

16. Let $f(x) = \sum_{n=1}^{\infty} x^2(1+x^2)^{-n}$. Show that the series converges for all real x, that $f(0) = 0$, and that $f(x) = 1$ if $x \neq 0$. This is an example of a convergent series of continuous functions whose sum is a discontinuous function.

17. If $f_n(x) = nxe^{-nx^2}$ for $n = 1, 2, \ldots$ and x real, show that

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx \neq \int_0^1 \left[\lim_{n\to\infty} f_n(x)\right] dx\,.$$

This example shows that the operations of integration and limit cannot always be interchanged.

18. Let $f_n(x) = (\sin nx)/n$, and for each fixed real x let $f(x) = \lim_{n\to\infty} f_n(x)$. Show that

$$\lim_{n\to\infty} f_n'(0) \neq f'(0)\,.$$

This example shows that the operations of differentiation and limit cannot always be interchanged.

19. Show that the series $\sum_{n=1}^{\infty} (\sin nx)/n^2$ converges for every real x, and denote its sum by $f(x)$. Given that f is integrable on $[0, \pi]$, use Theorem 9–24 to prove that

$$\int_0^{\pi} f(x)\,dx = 2 \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3}\,.$$

20. It is known that

$$\sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = \frac{x^2}{4} - \frac{\pi x}{2} + \frac{\pi^2}{6} \qquad \text{if} \quad 0 \leq x \leq 2\pi .$$

Use this formula and Theorem 9–24 to deduce the following formulas:

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$;

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^3} = \frac{\pi^3}{32}$.

ANSWERS TO EXERCISES

Chapter 1

1.3 Exercises (page 8)

1. (a) $\frac{2}{3}b^3$
   (b) b^3
   (c) $\frac{1}{12}b^3$
   (d) $\frac{2}{3}b^3 + b$
   (e) $\frac{1}{3}ab^3 + bc$
2. (c) $\frac{1}{4}ab^4 + bc$
3. (b) $s_n < \dfrac{b^{k+1}}{k+1} < S_n$
   (c) $\dfrac{ab^{k+1}}{k+1} + bc$

1.18 Exercises (page 27)

2. $1 - 4 + 9 - 16 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1}(1 + 2 + 3 + \cdots + n)$
3. $1 + \dfrac{1}{2} + \dfrac{1}{4} + \cdots + \dfrac{1}{2^n} = 2 - \dfrac{1}{2^n}$
4. $\left(1 - \dfrac{1}{2}\right)\left(1 - \dfrac{1}{3}\right)\cdots\left(1 - \dfrac{1}{n}\right) = \dfrac{1}{n}$
5. $\dfrac{n+1}{2n}$
6. (b) $A(1)$ is false
   (c) $1 + 2 + \cdots + n < \dfrac{(2n+1)^2}{8}$

1.20 Exercises (page 30)

1. (a) 10
   (b) 15
   (c) 170
   (d) 288
   (e) 36
   (f) $\frac{5}{6}$
8. (b) $n + 1$
9. Constant $= 2$

1.22 Exercises (page 33)

3. $(a_1, b_2), (a_2, b_5), (a_3, b_7), (a_4, b_{10}), (a_5, b_3), (a_6, b_8), (a_7, b_9), (a_8, b_4), (a_9, b_6), (a_{10}, b_1)$

★1.23 Miscellaneous exercises involving induction (page 33)

1. (a) 10
   (b) 1
   (c) 7
   (d) 21
   (e) 680
   (f) 1
2. (b) 17
   (c) 9
   (d) No
5. $\prod_{k=1}^{0} a_k = 1;\quad \prod_{k=1}^{n+1} a_k = a_{n+1} \cdot \prod_{k=1}^{n} a_k$
8. $\prod_{k=2}^{n}\left(1 - \frac{1}{k}\right) = \frac{1}{n};\quad \prod_{k=2}^{n}\left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$
9. 2^n
10. True if each $a_k \geq 0$
12. $n \geq 4$

1.32 Exercises (page 56)

1. (a) 2 (b) 4 (c) 4 (d) 6 (e) 6 (f) -6
2. $\frac{7}{2}$
4. One example: $s(x) = \frac{5}{2}$ if $0 \leq x < 2$, $s(2) = 0$, $s(x) = -1$ if $2 < x \leq 5$.
12. (a), (d), (e)

1.37 Exercises (page 71)

1. 3
2. 11
3. 21/8
4. 18
5. $\frac{1}{3}$
6. $-\frac{1}{3}$
7. -78
8. 14/3
9. $5 + (7/1296)$
10. 2592/35
11. $5^6/21$
12. $-2^{11}/11$
13. 32/3
14. 32/3
15. $\frac{4}{3}$
16. $\frac{4}{3}$

17. $5\sqrt{5}/6$
18. 2
19. (a) $0, 3/2$ (b) 0
20. (a) $5/6$ (b) $c/2$
22. $(1/A)\int_{Aa+B}^{Ab+B} f(x)\,dx$ if $A \neq 0$; $(b-a)f(B)$ if $A = 0$.
25. (a) $\displaystyle\sum_{k=0}^{2n}\binom{2n}{k}x^k$ (b) $\displaystyle\sum_{k=0}^{n} x^k$ (c) $\displaystyle\sum_{k=0}^{2^{n+1}-1} x^k$

1.39 Exercises (page 75)

6. $\tan(x+y) = \dfrac{\tan x + \tan y}{1 - \tan x \tan y}$; $\cot(x+y) = \dfrac{\cot x \cot y - 1}{\cot x + \cot y}$
9. $A = \frac{3}{2}$, $B = \frac{3}{2}\sqrt{3}$
10. $A = C\cos\alpha$, $B = C\sin\alpha$
11. $C = (A^2 + B^2)^{1/2}$. If $A^2 + B^2 \neq 0$, choose α so that $\cos\alpha = A/C$, $\sin\alpha = B/C$. If $A = B = 0$, choose any α.
12. $C = 2\sqrt{2}$, $\alpha = 5\pi/4$
13. $C = \sqrt{2}$, $\alpha = -\pi/4$

Note: In Exercises 15 through 20, n denotes an arbitrary integer.

15. (b) $(\pi/2) + n\pi$
16. (a) $(\pi/2) + 2n\pi$ (b) $2n\pi$
17. (a) $(3\pi/2) + 2n\pi$ (b) $(2n+1)\pi$
18. (a) $(\pi/6) + 2n\pi$; $(5\pi/6) + 2n\pi$ (b) $(\pi/3) + 2n\pi$; $(5\pi/3) + 2n\pi$
19. $(\pi/4) + n\pi$
20. $(\pi/2) + 2n\pi$; $\pi + 2n\pi$

1.43 Exercises (page 85)

1. (a) $x + \frac{1}{2}x^2 + \frac{1}{3}x^3$ (b) $\frac{8}{3}x^3 + 2x^2 + 2x + \frac{5}{6}$ (c) $-2x + 2x^2 - x^3$
   (d) $\frac{1}{5}x^{10} + \frac{2}{3}x^6 - \frac{1}{5}x^5 - \frac{2}{3}x^3 + x^2 - x$ (e) $0, \pm\sqrt{2}$
2. (d) $P(x) = \frac{1}{2}(x - [x])^2 - \frac{1}{2}(x - [x])$ (e) $\frac{1}{12}$
3. $2\pi^2$
4. $(1/24)\pi^3 + 1$
5. $4\sqrt{2}$
6. $1 + \sqrt{2}$
7. $(\pi/3) + 2\sqrt{3}$
8. π
9. $\frac{1}{15}(3x^5 + 5x^3 + 136)$
10. $\frac{1}{3}(x^6 - x^3 + \cos 3x - \cos 3x^2)$
11. $(\pi/6) + \sqrt{3} + \frac{1}{2}x + \sin x$ if $0 \leq x \leq 2\pi/3$; $(5\pi/6) + 2\sqrt{3} - \frac{1}{2}x - \sin x$ if $2\pi/3 \leq x \leq \pi$

1.45 Exercises (page 90)

1. $\frac{1}{3}\pi c^2 b^3$
2. $(\pi h/3)(a^2 + ab + b^2)$
3. $\pi b^5/5$
4. π^2
5. $\frac{4}{3}\pi r^3$
6. $\pi^2/2$
7. (a) $\pi/2$ (b) $(2\pi/3)(r^2 - c^2)^{3/2}$
8. $[(32/3) - 4\sqrt{3}]\pi r^3$
10. $\frac{4}{5}a^5$
11. $(h/6)(B_1 + 4M + B_2)$
12. (a) $8\pi/5$ (b) 2π (c) $10\pi/3$ (d) $16\pi/15$

Chapter 2

2.4 Exercises (page 104)

1. (b) $v_0/32$ sec (c) $-v_0$ ft/sec (d) 16 ft/sec; 160 ft/sec; $16T$ ft/sec
   (f) $f(t) = v_0 t - 10t^2$ is one example
2. (a) 1 (b) $2x$ (c) $3x^2$ (d) $4x^3 + 6x^2h + 4xh^2 + h^3$; $4x^3$
   (e) $5x^4 + 10x^3h + 10x^2h^2 + 5xh^3 + h^4$; $5x^4$
   (f) $-1/x^2$ (g) $-2/x^3$ (h) $-\dfrac{3}{x(x+h)^3} - \dfrac{h(3x+h)}{x^3(x+h)^3}$; $-\dfrac{3}{x^4}$
   (i) $\frac{1}{2}x^{-1/2}$ (j) $\frac{1}{2}x^{-3/2}$ (k) $\dfrac{3x^2 + h(3x+h)}{(x+h)^{3/2} + x^{3/2}}$; $\frac{3}{2}x^{1/2}$
   (l) $-\dfrac{3x^2 + h(3x+h)}{x^3(x+h)^{3/2} + (x+h)^3x^{3/2}}$; $-\frac{3}{2}x^{-5/2}$

2.9 Exercises (page 116)

1. $\frac{1}{4}$
2. -1
3. 4
4. 1
5. $2t$
6. -1
7. 1
8. 0
9. 0
10. 0
14. $f(x) \to 1$ as $x \to 0$. Define $f(0) = 1$ for continuity at 0
24. No. (Infinite discontinuity)
25. No. (Oscillating discontinuity)
26. $f(x) \to 0$ as $x \to 0$. Define $f(0) = 0$ for continuity at 0

2.12 Exercises (page 124)

1. $2x + 3$
2. $4x^3 + \cos x$
3. $4x^3 \sin x + x^4 \cos x$
4. $-1/(x+1)^2$
5. $-2x/(x^2+1)^2 + 5x^4 \cos x - x^5 \sin x$
6. $-1/(x-1)^2$
7. $\sin x/(2 + \cos x)^2$
8. $-\dfrac{2x^5 + 9x^4 + 8x^3 + 3x^2 + 2x - 3}{(x^4 + x^2 + 1)^2}$
9. $\dfrac{1 - 2(\sin x + \cos x)}{(2 - \cos x)^2}$
10. $\dfrac{\sin x + x\cos x}{1 + x^2} - \dfrac{2x^2 \sin x}{(1 + x^2)^2}$
11. 1; 0; -1; 21
12. (a) -2, 1
    (b) 0, -1
    (c) 3, -4

13. $mn[x^{m-1} + x^{n-1} + (m+n)x^{m+n-1}]$
14. $-(x^{-2} + 4x^{-3} + 9x^{-4})$
15. $\dfrac{2(1+x^2)}{(1-x^2)^2}$
16. $\dfrac{2(1-2x)}{(1-x+x^2)^2}$
17. $\dfrac{1-x+4x^2}{(1-x)^3(1+x)^4}$
18. $1 + \frac{1}{2}x^{-1/2} + \frac{1}{3}x^{-2/3} + \frac{1}{4}x^{-3/4}$
19. $\dfrac{1}{2}\left[\dfrac{2+\sqrt{x}}{(1+\sqrt{x})^2} + \dfrac{1-x}{\sqrt{x}(1+x)^2}\right]$
20. $-(x^{-2} + \frac{1}{2}x^{-3/2} + \frac{1}{3}x^{-4/3} + \frac{1}{4}x^{-5/4})$
22. (b) $-\csc^2 x$
    (d) $-\cot x \csc x$
    (f) $x \sec^2 x + \tan x$
    (g) $\dfrac{x\cos x - \sin x}{x^2}$
    (h) $-\dfrac{1+\cos x}{(x+\sin x)^2}$
    (i) $\dfrac{ad-bc}{(cx+d)^2}$
    (j) $-\dfrac{(2x^2+3)\sin x + 4x\cos x}{(2x^2+3)^2}$
    (k) $\dfrac{(2ax+b)(\sin x + \cos x) + (ax^2+bx+c)(\sin x - \cos x)}{1+\sin 2x}$
23. $3x^2$, where x is the length of an edge
25. $a = -2$, $b = 4$
26. (a) $x_1 + x_2 + a$
    (b) $\frac{1}{2}(x_1 + x_2)$
27. Tangent at $(3, -3)$; also intersect at $(0, 0)$
28. $a = d = 1$; $b = c = 0$
29. $a = c = e = 0$; $b = f = 2$; $d = -1$
30. (a) $\dfrac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}$
    (b) $\dfrac{n^2x^{n+3} - (2n^2+2n-1)x^{n+2} + (n+1)^2x^{n+1} - x^2 - x}{(x-1)^3}$

2.16 Exercises (page 133)

1. $\frac{5}{4}(b^4 - a^4)$
2. $\frac{4}{5}(b^5 - a^5) - 6(a^2 - b^2)$
3. $\frac{1}{5}(b^5 - a^5) + \frac{1}{4}(b^4 - a^4) - (b^2 - a^2) - 2(b - a)$
4. $\dfrac{1}{2}(b^2 - a^2) - \left(\dfrac{1}{b} - \dfrac{1}{a}\right) + \dfrac{3}{2}\left(\dfrac{1}{b^2} - \dfrac{1}{a^2}\right)$
5. $(b - a) + \frac{4}{3}(b^{3/2} - a^{3/2}) + \frac{1}{2}(b^2 - a^2)$
6. $\sqrt{2}(b^{3/2} - a^{3/2})$
7. $\frac{2}{5}(b^{5/2} - a^{5/2}) - 2(b^{3/2} - a^{3/2}) + 7(b^{1/2} - a^{1/2})$
8. $\frac{3}{2}(b^{4/3} - a^{4/3} - b^{2/3} + a^{2/3})$
9. $\frac{1}{3}(b^6 - a^6) - 3(\cos b - \cos a)$
10. $\frac{3}{7}(b^{7/3} - a^{7/3}) - 5(\sin b - \sin a)$

14. $f(t) = -\sin t;\quad c = \pi/3$
15. $f(t) = \sin t - 1;\quad c = 0$
16. $f(x) = 2x^{15};\quad c = -\frac{1}{9}$
17. (a) and (b) No such function
    (c) One solution is $f(x) = 1/(1 - x)$ for $x \le 0$, $f(x) = x^2 + x + 1$ for $x > 0$

2.18 Exercises (page 139)

1. $x^2 - 1$, all x
2. $(x - 1)^2$, all x
3. $|x|$, all x
4. 0, defined only at $x = 0$
5. x, $x \ge 0$
6. $-x$, $x \ge 0$
7. $\sin\sqrt{x}$, $x \ge 0$
8. $\sqrt{\sin x}$, $2k\pi \le x \le (2k + 1)\pi$, k an integer
9. x, all x
10. $\cos(\cos x)$, all x
11. $\sqrt{x + \sqrt{x}}$, $x > 0$
12. $\sqrt{x + \sqrt{x} + \sqrt{x + \sqrt{x}}}$, $x > 0$
13. (a) $\frac{1}{2}(x + |x|)$
    (b) and (c) x^2 if $x \ge 0$, 0 if $x < 0$
    (d) x^4 if $x \ge 0$, x if $x < 0$
14. (a) 1, all x
    (b) 1 if $|x| \le 1$, 2 if $|x| > 1$
    (c) 1 if $1 \le |x| \le \sqrt{3}$, 0 otherwise
    (d) $2 - (2 - x^2)^2$ if $|x| \le 2$, -2 if $|x| > 2$
15. $-2\cos^2 x\,(1 + 2\sin x)$
16. $x/\sqrt{1 + x^2}$
17. $(2x^3 - 4x)\sin x^2 - 2x\cos x^2 + 2\sin x^3 + 6x^3\cos x^3$
18. $-\sin 2x\cos(\cos 2x)$
19. $n\sin^{n-1}x\cos(n + 1)x$
20. $\cos x\cos(\sin x)\cos[\sin(\sin x)]$
21. $2/(\sin^2 x)$
22. $\dfrac{2\sin x\,(\cos x\sin x^2 - x\sin x\cos x^2)}{\sin^2 x^2}$
23. $-\dfrac{16\cos 2x}{\sin^3 2x}$
24. $\dfrac{1 + 2x^2}{\sqrt{1 + x^2}}$
25. $4(4 - x^2)^{-3/2}$
26. $\dfrac{2x^2}{1 - x^6}\left(\dfrac{1 + x^3}{1 - x^3}\right)^{1/3}$
27. $\dfrac{6 + 3x + 8x^2 + 4x^3 + 2x^4 + 3x^5}{(2 + x^2)^{1/2}\,(3 + x^3)^{2/3}}$
28. $-(1 + x^2)^{-3/2}$
29. $\dfrac{1 + 2\sqrt{x} + 4\sqrt{x}\,g(x)}{8\sqrt{x}\,g(x)\sqrt{x + g(x)}}$, where $g(x) = \sqrt{x + \sqrt{x}}$
30. (a) $2xf'(x^2)$
    (b) $[f'(\sin^2 x) - f'(\cos^2 x)]\sin 2x$

(c) $f'[f(x)]f'(x)$
(d) $f'(x)f'[f(x)]f'\{f[f(x)]\}$

31. (a) $(1 + x^2)^{-3}$
(b) $2x(1 + x^4)^{-3}$
(c) $2x(1 + x^4)^{-3} - 3x^2(1 + x^6)^{-3}$
32. (a) $75 \text{ cm}^3/\text{sec}$ (b) $300 \text{ cm}^3/\text{sec}$ (c) $3x^2 \text{ cm}^3/\text{sec}$
33. (a) and (b) $5/(4\pi)$ ft/min
34. (a) $20\sqrt{5}$ ft/sec (b) $50\sqrt{2}$ ft/sec
35. 36/5 mi/hr
36. (a) $x = \frac{1}{2}, y = \frac{1}{4}$ (b) $\frac{1}{2}\sqrt{3}$

2.20 Exercises (page 145)

1. (a) $\Delta f(1, 1) = 5, \quad df(1, 1) = 1$
(b) $\Delta f = 0.131, \quad df = 0.100$
(c) $\Delta f = 0.010301, \quad df = 0.010000$
2. (a) $\Delta f = -0.1000, \quad df = -0.0975$
(b) $\Delta f = 0.00998, \quad df = 0.01000$
3. (a) $\Delta f = 1/51 = 0.0196, \quad df = 1/50 = 0.0200$
(b) $\Delta f = 1/49 = 0.0204, \quad df = 1/50 = 0.0200$
4. 1.0066
5. 0.4848
6. -0.8746
7. 16.0041
10. (a) $dA(s, h) = 2sh$
(b) $dV(s, h) = 3s^2h$
(c) $dA(r, h) = 2\pi rh$
(d) $dA(r, h) = 8\pi rh$
(e) $dV(r, h) = 4\pi r^2h$
(f) $dV(r, h) = 2\pi rah$
(g) $dV(a, h) = \pi r^2h$

2.21 Exercises (page 147)

1. $g^{(k)}(0) = 0$ if $0 \le k \le n - 1; \quad g^{(n)}(0) = n!$
2. $6x^5 - 15x^4 + 10x^3 + 1$
8. $-\dfrac{1}{\sqrt{x}(1 + \sqrt{x})^2}; \quad \dfrac{1 + 3\sqrt{x}}{2(x + \sqrt{x})^3}; \quad -\dfrac{3}{4}\dfrac{1 + 4\sqrt{x} + 5x}{\sqrt{x}(x + \sqrt{x})^4}$
10. $a = 2c, b = -c^2$
11. $a = \dfrac{3}{2c}, b = -\dfrac{1}{2c^3}$
12. 3
13. 67/5
15. (b) $F'(x) = f[g(x)]g'(x)$ (c) $F'(x) = f[g(x)]g'(x) - f[h(x)]h'(x)$
16. $y = (16/9)x^2$
17. (b) $f'(0) = 0$
18. (a) $D^*(f + g) = (1 + g/f)D^*f + (1 + f/g)D^*g$ when $f(x)$ and $g(x)$ are not 0;
$D^*(f \cdot g) = g^2\, D^*f + f^2\, D^*g$;
$D^*(f/g) = (g^2\, D^*f - f^2\, D^*g)/g^4$ when $g(x) \neq 0$
(b) $D^*f(x) = 2f(x)\, Df(x)$
(c) $f(x) = c$ for all x

20. (a) $P_1(x) = x - \frac{1}{2}$; $P_2(x) = x^2 - x + \frac{1}{6}$; $P_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$;
$P_4(x) = x^4 - 2x^3 + x^2 - \frac{1}{30}$; $P_5(x) = x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{1}{6}x$

2.24 Exercises (page 155)

1. $\frac{1}{3}(2x + 1)^{3/2} + C$
2. $(2/45)(1 + 3x)^{5/2} - (2/27)(1 + 3x)^{3/2} + C$
3. $\frac{2}{7}(x + 1)^{7/2} - \frac{4}{5}(x + 1)^{5/2} + \frac{2}{3}(x + 1)^{3/2} + C$
4. $-2/27$
5. $-\dfrac{1}{4(x^2 + 2x + 2)^2} + C$
6. $\frac{1}{3}\cos^3 x - \cos x + C$
7. $\frac{3}{7}(z - 1)^{7/3} + \frac{3}{4}(z - 1)^{4/3} + C$
8. $-\frac{1}{2}\csc^2 x + C$
9. $\frac{8}{3} - \sqrt{3}$
10. $\dfrac{1}{3 + \cos x} + C$
11. $\dfrac{2}{\sqrt{\cos x}} + C$
12. $2(\cos 2 - \cos 3)$
13. $-\dfrac{\cos x^n}{n} + C$
14. $-\frac{1}{3}\sqrt{1 - x^6} + C$
15. $\frac{4}{9}(1 + t)^{9/4} - \frac{4}{5}(1 + t)^{5/4} + C$
16. $x(x^2 + 1)^{-1/2} + C$
17. $\frac{1}{40}(8x^3 + 27)^{5/3} + C$
18. $\frac{3}{2}(\sin x - \cos x)^{2/3} + C$
19. $2\sqrt{1 + \sqrt{1 + x^2}} + C$
20. $-\frac{5}{2}(x - 1)^{2/5} + C$

2.26 Exercises (page 158)

1. $\sin x - x\cos x$
2. $x^3\sin x + 3x^2\cos x - 6x\sin x - 6\cos x$
3. $2x\sin x + 2\cos x - x^2\cos x$
4. $-x^3\cos x + 3x^2\sin x + 6x\cos x - 6\sin x$
6. $\frac{2}{3}(3\sqrt{31} + \sqrt{3} - 11.35)$
12. $K_n = I_n$ of Exercise 9
13. $\tan x - x$; $\frac{1}{3}\tan^3 x - \tan x + x$
14. (b) $(5\pi/32)a^6$
22. $-3^{10}/20$
23. $37/8281$
24. $\frac{1}{30}(1 + x^5)^6$
25. $1/265650$
26. $\cos\frac{1}{2} - \cos 1$
27. $[12(x - 1)^{1/2} - 24]\sin(x - 1)^{1/4} - 4[(x - 1)^{3/4} - 6(x - 1)^{1/4}]\cos(x - 1)^{1/4}$
28. $\frac{1}{4}\sin^2 x^2$
29. $-\frac{2}{9}(1 + 3\cos^2 x)^{3/2}$
30. $8/15, 16/35, 128/315, 256/693$
31. $\frac{1}{13}x^{13} + \frac{1}{6}x^{12} + \frac{1}{11}x^{11}$
33. 3

34. $\frac{1}{2}\left(\frac{1}{2} + \frac{1}{\pi + 2} - A\right)$

2.28 Exercises (page 165)

1. $\frac{\partial f}{\partial x} = 4x^3 - 8xy^2;\quad \frac{\partial f}{\partial y} = 4y^3 - 8x^2y;\quad \frac{\partial^2 f}{\partial x^2} = 12x^2 - 8y^2;$
   $\frac{\partial^2 f}{\partial y^2} = 12y^2 - 8x^2;\quad \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = -16xy$
2. $f_x = \sin(x + y) + x\cos(x + y);\quad f_y = x\cos(x + y);\quad f_{yy} = -x\sin(x + y);$
   $f_{xx} = 2\cos(x + y) - x\sin(x + y);\quad f_{xy} = f_{yx} = \cos(x + y) - x\sin(x + y)$
3. $D_1f = y + y^{-1};\quad D_2f = x - xy^{-2};\quad D_{1,1}f = 0;\quad D_{2,2}f = 2xy^{-3};\quad D_{1,2}f = D_{2,1}f = 1 - y^{-2}$
4. $f_x = x(x^2 + y^2)^{-1/2};\quad f_y = y(x^2 + y^2)^{-1/2};\quad f_{xx} = y^2(x^2 + y^2)^{-3/2};$
   $f_{yy} = x^2(x^2 + y^2)^{-3/2};\quad f_{xy} = f_{yx} = -xy(x^2 + y^2)^{-3/2}$
5. $f_{yy} = 6x^2y\cos(x^2y^3) - 9x^4y^4\sin(x^2y^3);$
   $f_{xy} = f_{yx} = 6xy^2\cos(x^2y^3) - 6x^3y^5\sin(x^2y^3)$
6. $f_{xy} = f_{yx} = 6\cos(2x - 3y)\cos[\cos(2x - 3y)] + 6\sin^2(2x - 3y)\sin[\cos(2x - 3y)]$
7. $\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = -2(x + y)(x - y)^{-3};\quad \frac{\partial^2 f}{\partial x^2} = 4y(x - y)^{-3};\quad \frac{\partial^2 f}{\partial y^2} = 4x(x - y)^{-3}$
8. $f_{xx} = -3xy^2(x^2 + y^2)^{-5/2};\quad f_{yy} = -x(x^2 - 2y^2)(x^2 + y^2)^{-5/2};$
   $f_{xy} = f_{yx} = y(2x^2 - y^2)(x^2 + y^2)^{-5/2}$

Chapter 3

3.5 Exercises (page 180)

3. (a) 4
   (b) $(a + b)/(1 + ab)$
7. $(2x)/(1 + x^2)$
8. $x/(1 + x^2)$
9. $x/(x^2 - 4)$
10. $1/(x \log x)$
11. $(2/x) + 1/(x \log x)$
12. $x/(x^4 - 1)$
13. $1/[2(1 + \sqrt{x + 1})]$
14. $\log(x + \sqrt{x^2 + 1})$
15. $1/(a - bx^2)$
16. $2\sin(\log x)$
17. (e) $x - \frac{x^2}{2} + \cdots - \frac{x^{2n}}{2n} < \log(1 + x) < x - \frac{x^2}{2} + \cdots + \frac{x^{2n+1}}{2n + 1}$
18. (c) $x + \frac{x^2}{2} + \cdots + \frac{x^n}{n} < -\log(1 - x)$ if $0 < x < 1;$
    $-\log(1 - x) \le x + \frac{x^2}{2} + \cdots + \frac{x^n}{n} + \frac{2x^{n+1}}{n + 1}$ if $0 < x \le \frac{1}{2}$
20. (c) $1.382 < \log 4 < 1.396;\quad 1.774 < \log 6 < 1.813;\quad 2.166 < \log 9 < 2.230$

3.11 Exercises (page 191)

1. (a) 0
   (b) $\frac{e - 1}{e + 1}$

(c) $\dfrac{(e^2-1)^2}{4e^2}$

2. $b = e^a$, a arbitrary

3. (d) $e^x > 1 + x + \cdots + \dfrac{x^n}{n!}$;

$$1 - x + \frac{x^2}{2!} - \cdots - \frac{x^{2n-1}}{(2n-1)!} < e^{-x} < 1 - x + \frac{x^2}{2!} - \cdots + \frac{x^{2n}}{(2n)!}$$

6. $3e^{3x-1}$
7. $8xe^{4x^2}$
8. $-2xe^{-x^2}$
9. $\dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$
10. $-\dfrac{e^{1/x}}{x^2}$
11. $(\cos x)e^{\sin x}$
12. $-(\sin 2x)e^{\cos^2 x}$
13. 1
14. $e^x e^{e^x}$
15. $e^x e^{e^x} e^{e^{e^x}}$
20. $2^x \log 2$
21. $x^x(1 + \log x)$
22. $1 + (1 + 2x + 2x^2)e^{x^2}$
23. $4(e^x + e^{-x})^{-2}$
24. $2^{1+x^2} x \log 2$
25. $a^a x^{a^a-1} + ax^{a-1}a^{x^a} \log a + a^x a^{a^x} (\log a)^2$
26. $1/[x \log x \log(\log x)]$
27. $e^x(1 + e^{2x})^{-1/2}$
28. $x^x x^{x^x}\left[\dfrac{1}{x} + \log x + (\log x)^2\right]$
29. $(\log x)^x\left(\log\log x + \dfrac{1}{\log x}\right)$
30. $2x^{-1+\log x} \log x$
31. $-(\log_x e)^2/x$
32. $\dfrac{(\log x)^{x-1}}{x^{1+\log x}}[x - 2(\log x)^2 + x \log x \log(\log x)]$
33. $(\sin x)^{1+\cos x}[\cot^2 x - \log(\sin x)] - (\cos x)^{1+\sin x}[\tan^2 x - \log(\cos x)]$
34. $\dfrac{n(x + \sqrt{1+x^2})^n}{\sqrt{1+x^2}}$
35. $x^{-2+1/x}(1 - \log x)$
36. $\dfrac{54x - 36x^2 + 4x^3 + 2x^4}{3(1-x)^2(3-x)^{2/3}(3+x)^{5/3}}$
37. $\displaystyle\prod_{i=1}^{n}(x - a_i)^{b_i}\sum_{k=1}^{n}\frac{b_k}{x - a_k}$
41. $-e^{-x}(x + 1) + C$
42. $-\frac{1}{2}(x^2 + 1)e^{-x^2} + C$
43. $\frac{1}{2}x^2(-\frac{1}{2} + \log|x|) + C$
44. $\frac{1}{3}x^3(-\frac{1}{3} + \log|x|) + C$
45. $-\frac{1}{2}e^{-2x}(x^2 + x + \frac{1}{2}) + C$
46. $2(\sqrt{x} - 1)e^{\sqrt{x}} + C$

47. $\frac{2}{3}(-2 + \log|x|)\sqrt{1 + \log|x|} + C$
48. $-\log|\cos x| + C$
49. $\log|\sin x| - \frac{1}{2}\sin^2 x + C$
50. $(1/b)\log|a + bx| + C$
51. $\frac{1}{3}(x^3 - 1)e^{x^3} + C$
52. $\frac{2}{3}\sqrt{4 + x^3} + C$
53. $\log(x + \sqrt{x^2 + a^2}) + C$
55. (d) $\displaystyle\int_0^x e^{-t}\, t^n\, dt = n!\, e^{-x}\left(e^x - \sum_{k=0}^{n} \frac{x^k}{k!}\right)$

3.13 Exercises (page 195)

16. $5/3$
17. $3/4$
18. $\sinh x = 5/12$, $\cosh x = 13/12$
19. $37/12$
20. $24/25$

3.15 Exercises (page 199)

1. $g(y) = y - 1$; all y
2. $\frac{1}{2}(y - 5)$; all y
3. $1 - y$; all y
4. $y^{1/3}$; all y
5. $\frac{1}{2}\log y$; $y > 0$
6. $(\log y)^{1/3}$; $y > 0$
7. $\log(y + \sqrt{y^2 + 1})$; all y
8. $\frac{1}{2}\log\dfrac{1 + y}{1 - y}$; $|y| < 1$
9. $g(y) = y$ if $y < 1$; $\sqrt{y}$ if $1 \le y \le 16$; $\log_2 y$ if $y > 16$
10. The answers given are $f'(x)$ and $g'(y)$, respectively.
    (1) $1;\ 1$
    (2) $2;\ \frac{1}{2}$
    (3) $-1;\ -1$
    (4) $3x^2;\ 1/(3y^{2/3})$
    (5) $2e^{2x};\ 1/2y$
    (6) $3x^2 e^{x^3};\ 1/[3y(\log y)^{2/3}]$
    (7) $\cosh x;\ 1/\sqrt{1 + y^2}$
    (8) $\operatorname{sech}^2 x;\ 1/(1 - y^2)$
    (9) $f'(x) = 1$ if $x < 1$, $2x$ if $1 \le x \le 4$, $2^x \log 2$ if $x > 4$;
    $g'(y) = 1$ if $y < 1$, $1/(2\sqrt{y})$ if $1 \le y \le 16$, $1/(y \log 2)$ if $y > 16$
11. $g(y) = -e^y$; all y
12. (b) constant $= \frac{3}{2}$

3.17 Exercises (page 204)

15. $\dfrac{1}{\sqrt{4 - x^2}}$ if $|x| < 2$
16. $\dfrac{1}{\sqrt{1 + 2x - x^2}}$ if $|x - 1| < \sqrt{2}$

17. $\dfrac{1}{|x|\sqrt{x^2-1}}$ if $|x| > 1$

18. $\dfrac{\cos x}{|\cos x|}$ if $x \neq (k + \frac{1}{2})\pi$, k an integer

19. $\dfrac{\sqrt{x}}{2(1+x)}$ if $x \geq 0$

20. $\dfrac{1+x^4}{1+x^6}$

21. $-\dfrac{2x}{|x|(1+x^2)}$ if $x \neq 0$

22. $\dfrac{\sin 2x}{\sin^4 x + \cos^4 x}$ if $x \neq (k + \frac{1}{2})\pi$

23. $\dfrac{1}{2(1+x^2)}$

24. $\dfrac{\cos x + \sin x}{\sqrt{\sin 2x}}$ if $k\pi < x < (k + \frac{1}{2})\pi$

25. $\dfrac{x}{|x|\sqrt{1-x^2}}$ if $0 < |x| < 1$

26. $1/(1+x^2)$ if $x \neq 1$

27. $\dfrac{4x}{\sqrt{1-x^4}(\text{arc cos } x^2)^3}$ if $|x| < 1$

28. $\dfrac{1}{2x\sqrt{x-1}\text{ arc cos}(1/\sqrt{x})}$ if $x > 1$

30. $\dfrac{3x}{(1-x^2)^2} + \dfrac{(1+2x^2)\text{ arc sin } x}{(1-x^2)^{5/2}}$

31. (b) $f^{(2n)}(0) = 0$; $f^{(2n+1)}(0) = 1^2 \cdot 3^2 \cdot 5^2 \cdot \cdots \cdot (2n-1)^2$ for $n \geq 1$; $f'(0) = 1$

32. $\text{arc sin } \dfrac{x}{|a|} + C$

33. $\text{arc sin } \dfrac{x+1}{\sqrt{2}} + C$

34. $\dfrac{1}{a} \text{ arc tan } \dfrac{x}{a} + C$

35. $\dfrac{1}{\sqrt{ab}} \text{ arc tan } \left(\sqrt{\dfrac{b}{a}}\, x\right) + C$ if $ab > 0$;

$\dfrac{a}{2|a|\sqrt{-ab}} \log \left| \dfrac{\sqrt{|a|} + x\sqrt{|b|}}{\sqrt{|a|} - x\sqrt{|b|}} \right| + C$ if $ab < 0$

36. $\dfrac{2}{\sqrt{7}} \text{ arc tan } \dfrac{2x-1}{\sqrt{7}} + C$

37. $\frac{1}{2}[(1+x^2)\text{ arc tan } x - x] + C$

38. $\dfrac{x^3}{3} \text{ arc cos } x - \dfrac{2+x^2}{9}\sqrt{1-x^2} + C$

39. $\frac{1}{2}(1+x^2)(\text{arc tan } x)^2 - x \text{ arc tan } x + \frac{1}{2}\log(1+x^2) + C$

40. $(1+x)\text{ arc tan } \sqrt{x} - \sqrt{x} + C$

41. $(\text{arc tan } \sqrt{x})^2 + C$

42. $\frac{1}{2}(\text{arc sin } x + x\sqrt{1-x^2}) + C$

43. $\dfrac{(x-1)\, e^{\text{arc tan } x}}{2\sqrt{1+x^2}} + C$

44. $\dfrac{(x+1)\,e^{\arctan x}}{2\sqrt{1+x^2}} + C$

45. $\dfrac{1}{2}\left(\arctan x - \dfrac{x}{1+x^2}\right) + C$

46. $\arctan e^x + C$

47. $\frac{1}{2}\log(1+e^{-2x}) - \dfrac{\operatorname{arc\,cot} e^x}{e^x} + C$

48. $a \arcsin \dfrac{x}{a} - \sqrt{a^2 - x^2} + C$

49. $\dfrac{2(b-a)}{|b-a|} \arcsin \sqrt{\dfrac{x-a}{b-a}} + C$

50. $\frac{1}{4}|b-a|(b-a) \arcsin \sqrt{\dfrac{x-a}{b-a}} + \frac{1}{4}\sqrt{(x-a)(b-x)}\,[2x-(a+b)] + C$

3.19 Exercises (page 213)

1. $\log|x-2| + \log|x+5| + C$
2. $\frac{1}{2}\log\left|\dfrac{(x+2)^4}{(x+1)(x+3)^3}\right| + C$
3. $-\dfrac{1}{3(x-1)} + \frac{2}{9}\log\left|\dfrac{x-1}{x+2}\right| + C$
4. $\frac{1}{2}x^2 - x + \log\left|\dfrac{x^3(x+2)}{x-1}\right| + C$
5. $\log|x+1| - \dfrac{3}{(2x+1)^2} + \dfrac{3}{2x+1} + C$
6. $2\log|x-1| + \log(x^2+x+1) + C$
7. $x + \frac{1}{3}\arctan x - \frac{8}{3}\arctan(x/2) + C$
8. $2\log|x| - \log|x+1| + C$
9. $\log|x| - \frac{1}{2}\log(x^2+1) + \dfrac{1}{2(x^2+1)} + C$
10. $\dfrac{9x^2+50x+68}{4(x+2)(x+3)^2} + \frac{1}{8}\log\left|\dfrac{(x+1)(x+2)^{16}}{(x+3)^{17}}\right| + C$
11. $\dfrac{1}{x+1} + \log|x+1| + C$
12. $\frac{1}{2}\log|x^2-1| - \log|x| + C$
13. $x + \frac{4}{5}\log|x-2| - \frac{9}{5}\log|x+3| + C$
14. $\log|x-2| - \dfrac{4}{x-2} + C$
15. $\dfrac{1}{2-x} - \arctan(x-2) + C$
16. $4\log|x+1| - \frac{3}{2}\log|x| - \frac{5}{2}\log|x+2| + C$
17. $\frac{1}{4}\log\left|\dfrac{x+1}{x-1}\right| - \dfrac{x}{2(x^2-1)} + C$
18. $\frac{1}{3}\log\dfrac{(x-1)^2}{x^2+x+1} + C$
19. $\log|x| + \dfrac{1}{x^2+1} + C$
20. $\dfrac{1}{4x} + \dfrac{1}{4x^2} + \dfrac{1}{8}\log\left|\dfrac{x-2}{x}\right| + C$

21. $\log \dfrac{|x|}{\sqrt{1+x^2}} - x + \arctan x + C$
22. $\frac{1}{4}\log|(x-1)/(x+1)| - \frac{1}{2}\arctan x + C$
23. $\dfrac{1}{4\sqrt{2}}\log \dfrac{x^2 + x\sqrt{2}+1}{x^2 - x\sqrt{2}+1} + \dfrac{1}{2\sqrt{2}}\arctan \dfrac{x\sqrt{2}}{1-x^2} + C$
24. $(x^2+2x+2)^{-1} + \arctan(x+1) + C$
25. $-\,x/(x^5+x+1) + C$
26. $\dfrac{1}{\sqrt{5}}\arctan \dfrac{1+3\tan(x/2)}{\sqrt{5}} + C$
27. $\dfrac{2}{\sqrt{1-a^2}}\arctan\left(\sqrt{\dfrac{1-a}{1+a}}\tan\dfrac{x}{2}\right) + C$
28. $\dfrac{1}{\sqrt{a^2-1}}\log\left|\dfrac{a+\cos x+\sqrt{a^2-1}\sin x}{1+a\cos x}\right| + C$
29. $x - \frac{1}{2}\sqrt{2}\arctan(\sqrt{2}\tan x) + C$
30. $\dfrac{1}{ab}\arctan\left(\dfrac{a}{b}\tan x\right) + C$
31. $-\dfrac{\cos x}{a(a\sin x + b\cos x)} + C$
32. $(\pi/4) - \frac{1}{2}\log 2$

3.20 Miscellaneous review exercises (page 214)

1. $f(x) + f(1/x) = \frac{1}{2}(\log x)^2$
2. $f(x) = \log\sqrt{3/(2+\cos x)}$
4. 1
5. (a) $-\,7/12$
   (b) $V = \displaystyle\int_1^4 \frac{\pi(4x+2)}{x(x+1)(x+2)}\,dx$
6. (a) $x \geq 1$
   (c) $F(ax) - F(a)$; $F(x) - \dfrac{e^x}{x} + e$; $xe^{1/x} - e - F\left(\dfrac{1}{x}\right)$
7. (a) No such function
   (b) $-\,2^x\log 2$
   (c) $\frac{1}{2}x \pm 1$
9. (a) $g(3x) = 3e^{2x}g(x)$
   (b) $g(nx) = ne^{(n-1)x}g(x)$
   (c) 2
   (d) $C = 2$
10. $f(x) = b^{x/a}g(x)$, where g is periodic with period a
12. (a) $-Ae^{-a}$ (b) $\frac{1}{2}A$ (c) $A + 1 - \frac{1}{2}e$ (d) $e\log 2 - A$
13. (b) $c_0 + nc_1 + n(n-1)c_2 + n(n-1)(n-2)c_3$
    (c) If $p(x) = \displaystyle\sum_{k=0}^{m} c_kx^k$, then $f^{(n)}(0) = \displaystyle\sum_{k=0}^{m} k!\binom{n}{k}c_k$
16. (a) $\frac{2}{3}x^2(x + |x|)$
    (b) $x - \frac{1}{3}x^3$ if $|x| \leq 1$; $x - \frac{1}{2}x\,|x| + \frac{1}{6}\dfrac{|x|}{x}$ if $|x| > 1$
    (c) $1 - e^{-x}$ if $x \geq 0$; $e^x - 1$ if $x < 0$

(d) x if $|x| \le 1$; $\frac{1}{3}x^3 + \frac{2}{3}\frac{|x|}{x}$ if $|x| > 1$

17. $f(x) = \sqrt{(2x + 1)/\pi}$
19. (a) $f(t) = 2\sqrt{t} - 1$ if $t > 0$
(b) $f(t) = t - \frac{1}{2}t^2 + \frac{1}{2}$ if $0 \le t \le 1$
(c) $f(t) = t - \frac{1}{3}t^3 + \frac{1}{3}$ if $|t| \le 1$
(d) $f(t) = t$ if $t \le 0$; $f(t) = e^t - 1$ if $t > 0$

20. (b) $C_n = -2 \sum_{k=1}^{n} \frac{(k-1)!}{\log^k 2}$
(c) $b = \log 2$
(d) $e^2 \operatorname{Li}(e^{2x-2})$

Chapter 4

4.4 Exercises (page 222)

2. $x + y = -1$ is both an integral curve and an isocline
3. $y = Cx + C^2$; envelope: $y = \frac{1}{4}x^2$

4.7 Exercises (page 226)

1. $y^3 = \frac{3}{4}x^4 + C$
2. $\cos x = Ce^{1/\cos y}$
3. $y(C + \log|x + 1|) = 1$
4. $y - 2 = C(y - 1)e^x$
5. $y^2 + 2\sqrt{1 - x^2} = C$
6. $y = C(x - 1)e^x$
7. $\arctan y + \arcsin x = C$
8. $(1 + y^2)(1 + x^2) = Cx^2$
9. $y^4(x + 2) = C(x - 2)$
10. $1 + y^2 = Cx^2e^{x^2}$
11. $(y + \frac{1}{2})e^{-2y} = e^x(\cos x - \sin x) + C$
12. $x^2 - 1 = C(y^2 + 1)$
13. $f(x) = 2e^{x-1}$
14. $f(x) = \sqrt{5x^2 + 1}$
15. $f(x) = -\log(1 + x^2)$
16. $f(x) = \pm 1$; $f(x) = \sin(x + C)$; also, those continuous functions whose graphs may be obtained by piecing together portions of the curves $y = \sin(x + C)$ with portions of the lines $y = \pm 1$. One such example is $f(x) = -1$ for $x \le 0$, $f(x) = \sin(x - \frac{1}{2}\pi)$ for $0 \le x \le 3\pi$, $f(x) = 1$ for $x \ge 3\pi$
17. $f(x) = C$
18. $f(x) = Ae^{x/C}$
19. $f(x) = 0$
20. $f(x) = 0$

4.9 Exercises (page 228)

1. $y = -e^{2x} + Ce^{3x}$
2. $y \sin x = x + C$
3. $y = x \int (\sin x)/x \, dx + Cx$

4. $x = \frac{1}{3}e^{2t} + Ce^{-t}$
5. $y = C\cos x - 2\cos^2 x$
6. $y = Cx^2 + \frac{1}{3}x^5$
7. $y = \frac{1}{2}\left(1 + \dfrac{1}{x}\right)(C - e^{-x^2})$
8. $y = \sin x + \dfrac{C}{\sin x}$
9. $y = \left(\dfrac{x-2}{x-3}\right)^2\left(x + \dfrac{1}{x-2} + C\right)$
10. $y = x^2 - 2 + 2e^{-x^2/2}$
11. $y = e^{-\int P(x)\,dx}\,[(1-n)\int Q(x)e^{(1-n)\int P(x)\,dx}\,dx + C]^{1/(1-n)}$
12. $y = (Ce^{2x} - e^x)^2$
13. $y(Ce^{-x} + x^2 - x + 2) = 1$
14. $y = (x^3 + Cx)^2$
15. $y(x^2 - x^2\log x + Cx) = 1$
16. $\sin y = e^{-x^2/2}\left(\int x^2 e^{x^2/2}\,dx + C\right)$
17. $y^2 = \dfrac{e^x + Ce^{-x}}{2x}$
18. (a) v satisfies $v' - [P(x) + 2Q(x)u(x)]v = Q(x)$

    (b) $u = 1$ gives $y = \dfrac{Ce^{3x} + 2}{Ce^{3x} - 1}$ (includes $y = -2$ but not $y = 1$);

    $u = -2$ gives $y = \dfrac{e^{3x} + 2C}{e^{3x} - C}$ (includes $y = 1$ but not $y = -2$)
19. $f(x) = 1 + \log x$
20. Only the function given

4.11 Exercises (page 234)

1. $100(1 - 2^{-1/16})$ percent $= 95.8$ percent
2. $256(1 - e^{-t/8})$ if $0 \le t \le 10$; $16 + 166e^{20-2t}$ if $t \ge 10$
5. (b) $v \to \sqrt{mg/k}$
6. (c) 54.5 min

    (d) $T = \dfrac{1}{10k}[1 + (600 - t)k + (1400k - 1)e^{-kt}]$
7. 55°
8. 19.5 lb
9. 54.7 lb
10. 59.6 sec
12. $\dfrac{2\pi R^2\sqrt{h}}{9A_0}$ sec, where R is the radius of the base and h is the height of the cone (in feet)
13. For Equation (4.33), $x = x_0e^{k(t-t_0)}$; for Equation (4.35), $\alpha = Mk$
15. $x = M\left[1 + \exp\left(-M\int_{t_0}^{t} k(u)\,du\right)\right]^{-1}$
16. (a) 200 million (b) 217 million
17. (a) 0.026 per year (b) 0.011 per year; 260 million; 450 million
18. $dx/dt = kx(1 - at)$; $x = x_0\,e^{k(t-at^2/2)}$; curve (d)

4.13 Exercises (page 239)

1. $2\sqrt{2}$
2. $\pm 140\pi$

3. $A = C,\ m = k,\ \beta = \alpha - (\pi/2)$
4. $y = 3\cos 4\pi x$
5. $C = (y_0^2 + v_0^2)^{1/2}$
6. (b) No (c) If $k \neq 0$ the condition is $a_1 - a_2 \neq n\pi/k$
8. (c) $a = A - B,\ b = A + B$
10. (b) (i) $c = k = 1$, (ii) $c = \frac{3}{2},\ k = \sqrt{2}$, (iii) $c = 1,\ k = \sqrt{5}$

4.15 Miscellaneous exercises on differential equations (page 243)

2. $x^2 + y^2 = C$
3. $y = x \log|Cx|$
4. $x^2 + y^2 = Cx^4$
5. $(x^2 + y^2)^3 = Cy^2$
6. $x^2 + 2Cy = C^2,\ C > 0$
7. $y(Cx^2 - 1) = x$
8. $\arctan \dfrac{x}{y} + \log|y| = C$
9. $\dfrac{y}{x} - \dfrac{x}{y} + \log \dfrac{y^3}{x} = C$
10. $\tan \dfrac{y}{2x} = Ce^x$
11. $(x + y)^3 = Cx^4y^4$
12. $y = e^x$
13. $y^3 = -\frac{1}{3}x + Cx^{-1/2}$ for $x > 0$, or $y^3 = -\frac{1}{3}x$ for all x
14. $m = -1$; $y^2 \log|y| = \frac{1}{2}e^{-2x} + Cy^2$
15. (a) $a = 0,\ b = \frac{1}{4}$
    (b) $f(x) = 2x^{1/2}$
16. (b) $y = e^{4x} - e^{-x^3/3}$
18. (a) $1/(t + 1)$ grams in t years
    (b) 1 $\text{gm}^{-1}\text{yr}^{-1}$
19. $[1 - \frac{1}{2}(2 - \sqrt{2})t]^2$ grams in t years; $2 + \sqrt{2}$ years
20. (a) $365e^{t-1.825t^2}$ citizens in t years
    (b) $365e^t(1 - e^{-1.825t^2})$ fatalities in t years
21. 6.96 mi/sec = 25,056 mi/hr
22. Four times the initial amount

Chapter 5

5.7 Exercises (page 255)

9. $x = \frac{3}{5}c_1 - \frac{1}{5}c_2,\quad y = \frac{2}{5}c_2 - \frac{1}{5}c_1$
10. $x = \dfrac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1},\quad y = \dfrac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$
12. (c) $|\vec{A}|$ can be zero for $\vec{A} \neq \vec{0}$
18. A circle about C with radius r/k

5.11 Exercises (page 263)

3. (a) $\pm(\vec{i} - \vec{j})$
   (b) $\pm(\vec{i} + \vec{j})$

(c) $\pm (3\vec{i} + 2\vec{j})$
(d) $\pm (b\,\vec{i} - a\,\vec{j})$
4. (a) $\frac{6}{7}, \frac{3}{7}, -\frac{2}{7}$
(b) $\pm (\frac{6}{7}\vec{i} + \frac{3}{7}\vec{j} - \frac{2}{7}\vec{k})$
6. One solution is $\vec{C} = 8\,\vec{i} + \vec{j} + \vec{k}$
7. $C = a(\vec{i} - 5\vec{j} - 3\vec{k})$, a any scalar
8. (a) $x = \frac{1}{4}$, $y = \frac{7}{8}$
(b) $x = 3$, $y = -4$
9. (a) One solution is $\vec{B} = \vec{j} + 2\vec{k}$, $\vec{C} = 5\vec{i} - 2\vec{j} + \vec{k}$
(b) $x = \vec{V} \cdot \vec{B}/|\vec{B}|^2$, $y = \vec{V} \cdot \vec{C}/|\vec{C}|^2$
10. $\vec{C} = a(\vec{i} + 5\vec{j} - 4\vec{k})$, a any scalar
11. $\pm \dfrac{\sqrt{2}}{2}(\vec{j} + \vec{k})$
12. (a) $\vec{C} = -\frac{4}{9}(\vec{i} + 2\vec{j} - 2\vec{k})$, $\vec{D} = \frac{1}{9}(22\vec{i} - \vec{j} + 10\vec{k})$
(b) $\vec{C} = \dfrac{\vec{A} \cdot \vec{B}}{|\vec{B}|^2}\vec{B}$, $\vec{D} = \vec{A} - \dfrac{\vec{A} \cdot \vec{B}}{|\vec{B}|^2}\vec{B}$
13. $\sqrt{35/41}$, $\sqrt{6/41}$, 0
21. (c) All pairs except $\vec{B}$, $\vec{D}$
22. (c) $x = 5$, $y = -14/3$, $z = 5/3$
26. (b) $\pi/6$
27. 0

5.13 Exercises (page 269)

1. (a) $\pm(\frac{1}{3}\,\vec{i} + \frac{2}{3}\vec{j} - \frac{2}{3}\,\vec{k})$
(b) $-7, -\frac{7}{2}, \frac{7}{2}$
(c) $\frac{7}{3}$
(d) $(-\frac{7}{9}, -\frac{14}{9}, \frac{14}{9})$
2. $3x - y + 2z + 5 = 0$; $9/\sqrt{14}$
3. (b) $19/(3\sqrt{6})$
4. $\pi/3$, $2\pi/3$
5. (a) $\vec{i} + 2\vec{j} - 2\vec{k}$
(b) $x + 2y - 2z = 5$
(c) $5/3$
6. $10x - 3y - 7z + 17 = 0$
7. $2x - y + 2z - 8 = 0$
8. $x + 2y + 9z + 55 = 0$
9. $x + \sqrt{2}\,y + z = 2 + \sqrt{2}$
10. 6
11. Any s and t with $s = -t$
12. (a) $7\vec{i} - 2\vec{j} + 2\vec{k}$
(b) $7x - 2y + 2z = 0$
(e) $s = -2$, $t = 3$

5.15 Exercises (page 273)

1 $4x - 3y + 2 = 0$
2. $3x + 4y = 11$
3. $2x + 5y = 26$
4. $x - 4y = 4$

5. $bx + ay = ab$
6. $3x - y = 5$
7. $x + y + 2 = 0$
8. $\vec{P} = \vec{P}_0 + t(3\vec{i} - \vec{j} - 3\vec{k}) = (x_0 + 3t)\,\vec{i} + (y_0 - t)\,\vec{j} + (z_0 - 3t)\,\vec{k}$
9. $\vec{P} = \vec{P}_0 + t(3\vec{i} + 2\vec{j} - 5\vec{k}) = (x_0 + 3t)\,\vec{i} + (y_0 + 2t)\,\vec{j} + (z_0 - 5t)\,\vec{k}$
11. (b) $\vec{i} + 3\vec{j} - 2\vec{k}$
    (c) $t = 1$
    (d) $2x + 3y + 2z + 15 = 0$
    (e) $x + 3y - 2z + 19 = 0$
    (f) $x - 1 = (y - 2)/3 = (z + 1)/(-2)$
14. (b) $(5/9, 26/9, 19/9)$; $\frac{1}{3}\sqrt{65}$
15. (a) $2x + 7y + 2z + 10 = 0$
    (b) $(3/95)\sqrt{57}$
16. (a) $3x - 2y = C$
    (b) $y' = \frac{3}{2}$
17. (a) $2x + 3y = C$
    (b) $y' = -\frac{2}{3}$
18. (a) $y - 1 = C(x - 2)$ gives all lines except $x = 2$;
    $C(x - 2) + (1 - C^2)(y - 1) = 0$ gives all lines
    (b) $y'(x - 2) = y - 1$ for all except the line $x = 2$
19. (a) $y = Cx - \frac{1}{4}C^2$
    (b) $y = xy' - \frac{1}{4}(y')^2$
20. (b) $(y - xy')^2 = 1 + (y')^2$
21. $\vec{P} = (2 + 4t)\,\vec{i} + (1 - 3t)\,\vec{j} + (t - 3)\,\vec{k}$
22. $3x - z = 0$
23. $2x - 3z = 2$

5.18 Exercises (page 281)

1. (a) $-2\vec{i} + 3\vec{j} - \vec{k}$
   (b) $4\vec{i} - 5\vec{j} + 3\vec{k}$
   (c) $4\vec{i} - 4\vec{j} + 2\vec{k}$
   (d) $8\vec{i} + 10\vec{j} + 4\vec{k}$
   (e) $8\vec{i} + 3\vec{j} - 7\vec{k}$
   (f) $10\vec{i} + 11\vec{j} + 5\vec{k}$
   (g) $-2\vec{i} - 8\vec{j} - 12\vec{k}$
2. $\pm \dfrac{1}{\sqrt{26}}(4\vec{i} - 3\vec{j} - \vec{k})$
3. (a) $2\vec{i} - \vec{j} + 2\vec{k}$
   (b) $15/2$
4. $8\vec{i} + \vec{j} - 2\vec{k}$
7. (a) One solution is $\vec{B} = -\vec{i} - 3\vec{k}$
   (b) $\vec{B} = \vec{i} - \vec{j} - \vec{k}$
   (c) $\vec{B} = \dfrac{\vec{A} + \vec{C} \times \vec{A}}{|\vec{A}|^2}$
12. (b) 2
13. (b) $\sqrt{2005/41}$
14. (b) 1
15. (a) $|(\vec{C} - \vec{A}) \times (\vec{B} - \vec{A})|/|\vec{B} - \vec{A}|$
    (b) $22/7$

20. 2
23. (b) $23\vec{i} + 6\vec{j} - 7\vec{k}$

Chapter 6

6.3 Exercises (page 290)

1. (a) $y^2 - 4x = 0$
   (b) $0 \le y \le 4$
2. (a) $x^2 + y - 1 = 0$
   (b) $-1 \le x \le 1$
3. (a) $2x^2 - y = 0$
   (b) $-1 \le x \le 1$
4. (a) $3x - 2y = 0$
   (b) $2 \le x \le 10$
5. (a) $x^2 - y^2 - 1 = 0$
   (b) $1 \le x \le \frac{5}{3}, 0 \le y \le \frac{4}{3}$
6. (a) $x = \frac{1}{2}t^{1/2}, y = \frac{1}{2}t^{3/2}, t \ge 0$
   (b) $x = (1 + t^2)^{-1/2}, y = t(1 + t^2)^{-1/2}$, all t
   (c) $x = t/(1 + t^3), y = t^2/(1 + t^3), t \ne -1$
7. (a) $\vec{v}(t) = 2t\,\vec{i} + 2\vec{j}, \quad \vec{a}(t) = 2\vec{i}, \quad \cos\theta = \frac{1}{2}\sqrt{2}$
   (b) $\vec{v}(t) = -\sin t\,\vec{i} + \sin 2t\,\vec{j}, \quad \vec{a}(t) = -\cos t\,\vec{i} + 2\cos 2t\,\vec{j}, \quad \cos\theta = 0$
   (c) $\vec{v}(t) = \cos t\,\vec{i} + 2\sin 2t\,\vec{j}, \quad \vec{a}(t) = -\sin t\,\vec{i} + 4\cos 2t\,\vec{j}, \quad \cos\theta = 0$
   (d) $\vec{v}(t) = 2e^t\,\vec{i} + 3e^t\,\vec{j}, \quad \vec{a}(t) = 2e^t\,\vec{i} + 3e^t\,\vec{j}, \quad \cos\theta = 1$
   (e) $\vec{v}(t) = \sinh t\,\vec{i} + \cosh t\,\vec{j}, \quad \vec{a}(t) = \cosh t\,\vec{i} + \sinh t\,\vec{j}, \quad \cos\theta = 15/17$
8. $(7/10)\sqrt{2}$
12. $\vec{r}(t) = \frac{1}{6}t^3\vec{A} + \frac{1}{2}t^2\vec{B} + t\,\vec{v}_0 + \vec{r}_0$
13. $X(t) = \cos 2t + 2\sin 2t, \quad Y(t) = 2\cos 2t - \sin 2t$
14. $\vec{g}'(t) = \vec{f}(t) \times \vec{f}''(t)$

6.5 Exercises (page 293)

1. (a) $(x - 4)^2 + (y + 3)^2 = 36$
   (b) $(x - 1)^2 + (y - 1)^2 = 2$
   (c) $(x - 3)^2 + (y - 2)^2 = 4$
   (d) $(x - 3)^2 + (y - 2)^2 = 9$
   (e) $(x - 4)^2 + (y - 4)^2 = 16$
2. $A^2 + B^2 > 4C$; radius $= \frac{1}{2}(A^2 + B^2 - 4C)^{1/2}$; center at $(-A/2, -B/2)$
3. $(x + 2)^2 + (y - 1)^2 = 100$
4. $\left(x - \frac{7}{5}\right)^2 + \left(y + \frac{7}{10}\right)^2 = \frac{181}{4}$
7. $x^2 + y^2 = C$
8. $(x^2 - y^2 - 1)y' = 2xy$
9. $(y - xy')^2 = 1 + (y')^2$
10. $(4x - 3y)^2[1 + (y')^2] = (3 + 4y')^2; \quad y = \frac{1}{3}(4x \pm 5)$

6.8 Exercises (page 296)

3. $A = ab\omega^3, B = a^2\omega^3$
4. (a) $y^2 + z^2 = a^2$
   (b) $(x - 2)^2 + (y - 3)^2 = a^2$

5. $x - z + 4 = 0$
6. $z = x^3/8$
7. (b) $2x^2 + 9z^2 - 21 = 0$
8. $\frac{2}{3}r^3 \tan\theta$
9. $\frac{16}{3}r^3$
10. (a) $\frac{8}{3}\sqrt{3}\, r^3$
    (b) $\frac{8}{3}nr^3 \tan(\pi/2n)$; $\frac{4}{3}\pi r^3$

6.11 Exercises (page 303)

1. Center at $(0, 0)$; foci at $(\pm 8, 0)$; vertices at $(\pm 10, 0)$; $e = \frac{4}{5}$
2. Center at $(0, 0)$; foci at $(0, \pm 8)$; vertices at $(0, \pm 10)$; $e = \frac{4}{5}$
3. Center at $(2, -3)$; foci at $(2 \pm \sqrt{7}, -3)$; vertices at $(6, -3)$, $(-2, -3)$; $e = \sqrt{7}/4$
4. Center at $(0, 0)$; foci at $(\pm \frac{4}{3}, 0)$; vertices at $(\pm \frac{5}{3}, 0)$; $e = \frac{4}{5}$
5. Center at $(0, 0)$; foci at $(\pm \sqrt{3}/6, 0)$; vertices at $(\pm \sqrt{3}/3, 0)$; $e = \frac{1}{2}$
6. Center at $(-1, -2)$; foci at $(-1, 1)$, $(-1, -5)$; vertices at $(-1, 3)$, $(-1, -7)$; $e = \frac{3}{5}$
7. $7x^2 + 16y^2 = 7$
8. $\dfrac{(x+3)^2}{16} + \dfrac{(y-4)^2}{9} = 1$
9. $\dfrac{(x+3)^2}{9} + \dfrac{(y-4)^2}{16} = 1$
10. $\dfrac{(x+4)^2}{9} + (y-2)^2 = 1$
11. $\dfrac{(x-8)^2}{25} + \dfrac{(y+2)^2}{9} = 1$
12. $\dfrac{(x-2)^2}{16} + \dfrac{(y-1)^2}{4} = 1$
24. $(2, 1)$, $(-2, -1)$
25. $\frac{1}{2}\sqrt{2}$

6.13 Exercises (page 308)

2. Center at $(0, 0)$; foci at $(\pm 2\sqrt{41}, 0)$; vertices at $(\pm 10, 0)$; $e = \sqrt{41}/5$
3. Center at $(0, 0)$; foci at $(0, \pm 2\sqrt{41})$; vertices at $(0, \pm 10)$; $e = \sqrt{41}/5$
4. Center at $(-3, 3)$; foci at $(-3 \pm \sqrt{5}, 3)$; vertices at $(-1, 3)$, $(-5, 3)$; $e = \sqrt{5}/2$
5. Center at $(0, 0)$; foci at $(\pm 5, 0)$; vertices at $(\pm 4, 0)$; $e = 5/4$
6. Center at $(0, 0)$; foci at $(0, \pm 3)$; vertices at $(0, \pm 2)$; $e = 3/2$
7. Center at $(1, -2)$; foci at $(1 \pm \sqrt{13}, -2)$; vertices at $(3, -2)$, $(-1, -2)$; $e = \sqrt{13}/2$
8. $\dfrac{x^2}{4} - \dfrac{y^2}{12} = 1$
9. $y^2 - x^2 = 1$
10. $\dfrac{x^2}{4} - \dfrac{y^2}{16} = 1$
11. $(y-4)^2 - \dfrac{(x+1)^2}{3} = 1$
12. $\dfrac{8(y+3)^2}{27} - \dfrac{5(x-2)^2}{27} = 1$
13. $\pm\sqrt{23/3}$
14. $4x^2 - y^2 = 11$
25. $y^2 - x^2 = 9$

6.15 Exercises (page 312)

2. Vertex at (0, 0); directrix $x = 2$; axis $y = 0$
3. Vertex at (0, 0); directrix $x = -\frac{3}{4}$; axis $y = 0$
4. Vertex at $(\frac{1}{2}, 1)$; directrix $x = -\frac{5}{2}$; axis $y = 1$
5. Vertex at (0, 0); directrix $y = -\frac{3}{2}$; axis $x = 0$
6. Vertex at (0, 0); directrix $y = 2$; axis $x = 0$
7. Vertex at $(-2, -9/4)$; directrix $y = -13/4$; axis $x = -2$
8. $x^2 = -y$
9. $y^2 = 8x$
10. $(x + 4)^2 = -8(y - 3)$
11. $(y + 1)^2 = 5(x - \frac{7}{4})$
12. $(x - \frac{3}{2})^2 = 2(y + \frac{1}{8})$
13. $(y - 3)^2 = -8(x - 1)$
14. $x^2 - 4xy + 4y^2 + 40x + 20y - 100 = 0$
16. $(c/m^2, 2c/m)$
17. (a) $y - y_0 = m(x - x_0) + c/m$; tangent at $(x_0 + c/m^2, y_0 + 2c/m)$
    (b) $y - y_0 = m(x - x_0) - cm^2$; tangent at $(x_0 + 2cm, y_0 + cm^2)$
19. $(y_1 - y_0)(y - y_0) = 2c(x + x_1 - 2x_0)$; $\quad x_1 y = 2y_1 x - x_1 y_1$;
    $(x_1 - x_0)(y - y_0) = 2(y_1 - y_0)(x - x_0) - (x_1 - x_0)(y_1 - y_0)$
20. (4, 8)
22. $x + y + 4 = 0$
24. $\frac{2}{3}bh$
25. 16π
26. (a) $\frac{8}{3}$
    (b) 2π
    (c) $(48/5)\pi$
30. $y^2 = 3x$
31. $y^2 = 4x$
32. $y = Cx^2$, $C \neq 0$

6.19 Exercises (page 321)

1. Ellipse; center at $(2, -2)$, semiaxes $\frac{1}{3}\sqrt{3}$ and $\frac{1}{2}\sqrt{2}$
2. Parabola; vertex at $(-1, 2)$, focus at $(-1, 0)$
3. Hyperbola; center at (3, 0)
4. Hyperbola; center at $(-1, 5)$
5. Two intersecting lines
6. Hyperbola; center at $(-1, 2)$; rotate through $\pi/4$ radians
7. No locus
8. Two intersecting lines
9. Ellipse; center at (0, 0)
10. Hyperbola; center at $(-5/2, -5/2)$
11. Parabola; vertex at $(5/16, -15/16)$
12. Ellipse; center at (0, 0)
13. Ellipse; center at $(6, -4)$
14. Parabola; vertex at (2/25, 11/25)
15. Hyperbola; center at (0, 0)
16. -14
19. (c) $125X^2 + 50Y^2 = 1$

6.20 Miscellaneous exercises on conic sections (page 322)

1. $x^2 - 2xy + y^2 - 2x - 2y = 1$
2. $y^2 - 4x^2 - 4y + 4x = 0$
3. (a) Counterclockwise
   (b) $Y'(t) = 3X(t)$
   (c) $2\pi/\sqrt{3}$
4. $3T/4$
5. $(x - \frac{2}{5})^2 + (y - \frac{4}{5})^2 = \frac{4}{5}$
6. (a) $x = \frac{4}{3}a$
   (b) $27pq^2 = 4a^3$
7. $\tan \alpha = \dfrac{\tan \theta}{2 + \tan^2 \theta}$
9. (a) $2xy + 4x + 6y - 13 = 0$
   (b) Same as (a) in third quadrant; $2xy - 4x - 6y + 13 = 0$ in other quadrants
13. $3x - 2y = C$
14. $x^2 + y^2 + Cy = 0$
15. $x^2 + y^2 - Cx + 1 = 0$
16. $2x^2 + y^2 = C$
17. $2y^2 - x^2 = C$
18. $y^2 = x + C$
19. $xy = C$
20. $x^2 + y^2 - C(x + y) + 2 = 0$

6.22 Exercises (page 328)

3. $(x - 3)^2 + (y + 2)^2 + (z - 1)^2 = 25$; $(x + \frac{1}{3})^2 + (y - \frac{4}{3})^2 + (z - \frac{13}{3})^2 = 25$
4. $(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 3$
5. $z = \pm(\sqrt{35}/10)x$
6. $(Ax + By + Cz)^2 = D^2\left(\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2}\right)$
7. $-\dfrac{(x - 2)^2}{12} - \dfrac{(y + 1)^2}{12} + \dfrac{(z + 1)^2}{4} = 1$; hyperboloid of two sheets; center at $(2, -1, -1)$
8. $\frac{1}{4}x^2 + \frac{1}{9}y^2 + z^2 = 1$; ellipsoid
9. Hyperboloid of one sheet
10. Hyperbolic paraboloid
11. Hyperboloid of two sheets; $(1, 2, -1)$
12. Ellipsoid; $(1, -1, 2)$
13. Right circular cone; $(0, 0, 0)$
14. Hyperboloid of two sheets; $(0, 0, 0)$
15. Hyperboloid of one sheet; $(1, 2, 0)$
16. Hyperbolic paraboloid; $(0, 0, 0)$
20. (a) $\pi ab\left(1 - \dfrac{t^2}{c^2}\right)$
    (b) $\frac{4}{3}\pi abc$

6.25 Exercises (page 334)

1. (a) $\vec{v} = -9\vec{i} + 12\vec{j} + 15\vec{k}$; $\vec{a} = -12\vec{i} + 6\vec{j} + 12\vec{k}$;
   $\vec{T} = -\frac{3}{10}\sqrt{2}\,\vec{i} + \frac{2}{5}\sqrt{2}\,\vec{j} + \frac{1}{2}\sqrt{2}\,\vec{k}$; $\vec{N} = -\frac{4}{5}\vec{i} - \frac{3}{5}\vec{j}$

(b) $\kappa = 1/75$
(c) $\vec{a} = 12\sqrt{2}\,\vec{T} + 6\vec{N}$

2. (a) $\vec{v} = -\vec{j} + e^{\pi}\vec{k}$; $\vec{a} = \vec{i} + e^{\pi}\vec{k}$; $\vec{T} = -(1 + e^{2\pi})^{-1/2}\vec{j} + e^{\pi}(1 + e^{2\pi})^{-1/2}\vec{k}$;
$$\vec{N} = \frac{(1 + e^{2\pi})\,\vec{i} + e^{2\pi}\vec{j} + e^{\pi}\vec{k}}{(1 + e^{2\pi})^{1/2}(1 + 2e^{2\pi})^{1/2}}$$
(b) $\kappa = (1 + 2e^{2\pi})^{1/2}(1 + e^{2\pi})^{-3/2}$
(c) $\vec{a} = (1 + e^{2\pi})^{-1/2}[e^{2\pi}\vec{T} + (1 + 2e^{2\pi})^{1/2}\vec{N}]$

3. (a) $\vec{v} = 3\vec{i} + 4\vec{k}$; $\vec{a} = 6\vec{j}$; $\vec{T} = \frac{3}{5}\vec{i} + \frac{4}{5}\vec{k}$; $\vec{N} = \vec{j}$
(b) $\kappa = 6/25$
(c) $\vec{a} = 6\vec{N}$

4. (a) $\vec{v} = 2\vec{i}$; $\vec{a} = -\vec{j} - \vec{k}$; $\vec{T} = \vec{i}$; $\vec{N} = -\dfrac{1}{\sqrt{2}}(\vec{j} + \vec{k})$
(b) $\kappa = \frac{1}{4}\sqrt{2}$
(c) $\vec{a} = \sqrt{2}\,\vec{N}$

5. (a) $\vec{v} = 6\vec{i} + 6\vec{j} + 3\vec{k}$; $\vec{a} = 6\vec{i} + 12\vec{j}$; $\vec{T} = \frac{2}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{1}{3}\vec{k}$; $\vec{N} = \frac{1}{3}\vec{i} + \frac{2}{3}\vec{j} - \frac{2}{3}\vec{k}$
(b) $\kappa = 2/27$
(c) $\vec{a} = 12\vec{T} + 6\vec{N}$

7. (a) $x = z$
8. $2\sqrt{2}\,\pi$
9. 50
10. $\sqrt{2}\log(1 + \sqrt{2})$
11. $|\omega|\sqrt{a^2 + b^2}\,(t_1 - t_0)$
15. $\kappa = |\vec{a}|/|\vec{v}|^2$
17. $\sqrt{2}\,\vec{i} + \sqrt{2}\,\vec{j}$

6.29 Exercises (page 341)

3. (b) $r^2 + R^2 - 2rR\cos(\theta - \alpha) = a^2$
4. Radius $= \frac{1}{2}\sqrt{a^2 + b^2}$; center at $x = b/2$, $y = a/2$
5. $\vec{v}(t) = \vec{u}_r + t\,\vec{u}_\theta$; $\vec{a}(t) = -t\,\vec{u}_r + 2\vec{u}_\theta$; $\kappa(t) = (2 + t^2)(1 + t^2)^{-3/2}$
6. (a) $x^2 + y^2 = z^2$
(b) $\vec{v}(t) = \vec{u}_r + t\,\vec{u}_\theta + \vec{k}$; $\vec{a}(t) = -t\,\vec{u}_r + 2\vec{u}_\theta$; $\kappa(t) = \dfrac{\sqrt{t^4 + 5t^2 + 8}}{(t^2 + 2)^{3/2}}$
(c) $\arccos\sqrt{2/(2 + t^2)}$
7. (a) $x^2 + (y - \frac{1}{2})^2 = \frac{1}{4}$
(b) $(\pi/2) - t$
8. (b) $\frac{1}{2}\sqrt{2}\,\vec{k}$
9. $\pi^3/48$
10. $\frac{1}{4}$
11. $\frac{1}{2} + \frac{2}{3}\sqrt{2}$
12. $\frac{1}{4}(e^{2\pi} - 1)$
13. $\dfrac{3\pi}{16} + \dfrac{1}{8} + \dfrac{1}{2}\sqrt{2}$
14. $\dfrac{\pi}{9\sqrt{3}} - \dfrac{1}{12}$
16. 32
18. $r = r_0 e^{-\theta\cot\alpha}$; target at origin, missile starts at $r = r_0$, $\theta = 0$; α denotes the angle, $0 < \alpha < \pi$, determined by $\vec{v}$ and $-\vec{r}$; for $0 < \alpha < \pi/2$ the path is a spiral for which $r \to 0$ as θ increases indefinitely; for $\alpha = \pi/2$ it is a circle about the origin; for $\pi/2 < \alpha < \pi$ it is a spiral for which r increases indefinitely as θ increases indefinitely.

19. (a) $3\pi/16$

(b) $2 + \dfrac{1}{\sqrt{3}}\log(2 + \sqrt{3})$

21. $\log \sqrt{x^2 + y^2} + \arctan(y/x) = C$
22. Use as positive x-axis the line from ground crew to position sighted four miles away. Proceed three miles along this line (to allow for the possibility that the missile is returning to base) and then follow the spiral $r = 3e^{\theta/\sqrt{8}}$

6.34 Exercises (page 352)

1. $8a$
2. $\sqrt{2}\,(e^2 - 1)$
3. $2\pi^2 a$
4. $4(a^3 - b^3)/(ab)$
5. $2a\left(\cosh \dfrac{T}{2}\sqrt{\cosh T} - 1\right) - \sqrt{2}\,a \log\left(\dfrac{\sqrt{2}\cosh (T/2) + \sqrt{\cosh T}}{1 + \sqrt{2}}\right)$
7. $\int_c^d \sqrt{1 + [g'(y)]^2}\, dy$
9. $\dfrac{26\sqrt{13} - 16}{27}$
11. (a) $\int_0^1 \sqrt{1 + e^{2x}}\, dx$

(b) $\displaystyle\int_1^e \sqrt{2 + \frac{2}{t^2}}\; dt$

12. (c) $c \sinh \dfrac{2}{c}$
14. $f(x) = k \cosh\left(\dfrac{x}{k} + C\right)$, or $f(x) = k$
15. (a) $f(\theta) = k \sin(\theta + C)$, or $f(\theta) = k$

(b) $f(\theta) = Ce^{\theta/\sqrt{k^2-1}}$, where $k^2 > 1$

(c) $f(\theta) = (2/k)\sec(\theta + C)$, or $f(\theta) = 2/k$

Chapter 7

7.3 Exercises (page 358)

3. (b) $c = \frac{1}{2}$, $c = \sqrt{2}$
7. (b) f has at most $k + r$ zeros in $[a, b]$
8. (a) $\theta = \frac{1}{2}$, $\theta \to \frac{1}{2}$

(b) $\theta = \dfrac{x + \frac{1}{3}h}{x + \sqrt{x^2 + xh + \frac{1}{3}h^2}}$; $\quad \theta \to \frac{1}{2}$ if $x > 0$

10. (a) $\theta = (n + 1)^{-1/n}$

(b) $\theta = \dfrac{1}{x}\log \dfrac{e^x - 1}{x}$

11. Decreasing for $x < \frac{1}{2}$; increasing for $x > \frac{1}{2}$
12. Increasing everywhere
13. Increasing everywhere
14. Increasing for $x < 50$; decreasing for $x > 50$

15. Increasing for $0 < x < \dfrac{2}{\log 2}$; decreasing for $x < 0$ and for $x > \dfrac{2}{\log 2}$
16. Increasing for $0 < x < n$; decreasing for $x > n$

7.5 Exercises (page 361)

1. $c = \frac{1}{2}(a + b)$
2. $c = \dfrac{2(a^2 + ab + b^2)}{3(a + b)}$
3. $c = \frac{1}{2}(a + b)$
4. $c = \frac{1}{2}(a + b)$

7.8 Exercises (page 369)

5. $\arctan x = \displaystyle\sum_{k=0}^{n-1} (-1)^k \frac{x^{2k+1}}{2k + 1} + (-1)^n \int_0^x \frac{t^{2n}}{1 + t^2}\, dt$ for all x;

   $|\text{remainder}| \le \dfrac{1}{2n + 1}$ when $|x| \le 1$
6. (b) $\dfrac{55\sqrt{2}}{672} + R$, where $|R| \le \dfrac{\sqrt{2}}{7680} < 2 \cdot 10^{-4}$
7. $0.9461 + R$, where $|R| < 2 \cdot 10^{-4}$

7.10 Exercises (page 373)

1. $f(x) = O(0)$ for x in S means $f(x) = 0$ for every x in S
3. (b) Converse not true when $f(x) = x^2$
4. (b) False when $f(x) = \sin x$ and $g(x) = 1$
6. (e) $|x| \le 1$
7. $-\frac{2}{3}$
8. In first case, $g[f(x)] = 1 + x + \frac{1}{2}x^2 + O(x^3)$ for $x \le 1$; $g[f(x)] = O(x^9)$ for $x \ge 1$.
   In second case, $g[f(x)] = 1 + x + \frac{1}{2}x^2 + O(x^4)$ for $x \le 1$; $g[f(x)] = O(x^{20})$ for $x \ge 1$.
9. $g[f(x)] = 1 + x + \frac{1}{2}x^2 + o(x^3)$ in first case; $g[f(x)] = 1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^4 + o(x^4)$ in second case.
12. (a) $-\frac{1}{6}$
    (b) $\frac{2}{3}$
    (c) $\frac{1}{2}$
    (d) 1

Chapter 8

8.2 Exercises (page 381)

1. (a) ± 1
   (b) Increasing for $|x| > 1$; decreasing for $|x| < 1$
   (c) Relative max at $(-1, 4)$, relative min at $(1, 0)$
2. (a) 1, 3
   (b) Increasing for $x < 1$ and for $x > 3$; decreasing for $1 < x < 3$
   (c) Relative max at $(1, 9)$, relative min at $(3, 5)$
3. (a) 1
   (b) Increasing for $x > 1$; decreasing for $x < 1$
   (c) Relative min at $(1, 2)$

4. (a) ± 1
   (b) Increasing for $|x| < 1$; decreasing for $|x| > 1$
   (c) Relative max at $(1, \frac{1}{2})$, relative min at $(-1, -\frac{1}{2})$
5. Note that $f(x) = \sqrt{2} \sin [(7\pi/4) + x]$
6. (a) $-(3\pi/8) + 3k\pi$; $(3\pi/4) + 3k\pi$; $(9\pi/8) + 3k\pi$; k any integer.
   (c) Relative max at $x = (3\pi/4) + 6k\pi, (21\pi/8) + 6k\pi, (33\pi/8) + 6k\pi$
   Relative min at $x = -(3\pi/8) + 6k\pi, (9\pi/8) + 6k\pi, (15\pi/4) + 6k\pi$
7. (a) 1
   (b) Increasing for $x < 1$; decreasing for $x > 1$
   (c) Relative max at $(1, e^{-1})$
8. (a) ± 1
   (b) Increasing for $|x| > 1$; decreasing for $-1 < x < 0$ or $0 < x < 1$
   (c) Relative max at $(-1, -2)$, relative min at $(1, 2)$
9. Absolute min $\frac{1}{2}$; absolute max 32
10. Absolute min 1; absolute max 3
11. Absolute min $-(109/16)$; absolute max -4
12. Absolute min 2; absolute max 101/10
13. Absolute min $-\frac{1}{4}$; absolute max 72
14. Absolute min 0; absolute max 72
15. Absolute min 0; absolute max $\frac{1}{3} \log^2 3$
16. Absolute min $-(\sqrt{2}/2)\, e^{7\pi/4}$; absolute max $(\sqrt{2}/2)\, e^{3\pi/4}$
17. Local max $= 10$ when $x = k\pi$; local min $= 5$ when $x = (k + \frac{1}{2})\pi$, k any integer
18. Local max $= e^{-2}$ when $x = -1$; local min $= 0$ when $x = 0$; local max $= 1$ when $x = 1$
19. Local max $= 1$ when $x = 0$ if n is odd
20. Local max $= (\pi/4) - \frac{1}{2} \log 2$ when $x = 1$

8.5 Exercises (page 386)

1. $(\pm \frac{1}{2}\sqrt{5}, \frac{1}{2})$
2. A rectangle whose base is twice its altitude
3. Isosceles trapezoid, lower base the diameter, upper base equal to the radius
5. (a) $6\frac{2}{3}, 6\frac{2}{3}, \frac{5}{3}$
   (b) $8 + 2\sqrt{7}, 2 + 2\sqrt{7}, 5 - \sqrt{7}$
6. $\sqrt{5}$
7. Radius of sphere $= \frac{1}{4}\sqrt{2}(1 + \log 2)$
8. $V = 48\pi$ for $0 \le h < 2$; $V = 4\pi(4 + h)^3/(9h)$ for $h \ge 2$
9. $a = e^{-2}$
10. Maximum $\tan \alpha = \frac{1}{4}\sqrt{2}$
11. $\pi/4$
12. Crease $= (9\sqrt{3}/2)$ inches; angle $= \arctan (\sqrt{2}/2)$
13. (a) $20\sqrt{3}$ mi/hr; \$10.39
    (b) $40\sqrt{2}$ mi/hr; \$16.97
    (c) 60 mi/hr; \$22.00
    (d) 60 mi/hr; \$27.00
    (e) 60 mi/hr; \$32.00
14. $2\sqrt{4bc - a^2}$
15. (a) max $= 3\sqrt{3}\, r$; min $= 4r$
    (b) $\frac{1}{4} L$
17. Rectangle has base $4\pi/(3\pi + 8)$, altitude $P(4 + \pi)/(6\pi + 16)$

20. (b) Minimum
    (c) $\frac{1}{2}$

8.7 Exercises (page 390)

1. Concave downward for $x < 0$; concave upward for $x > 0$; point of inflection at $x = 0$
2. Concave downward for $x > 0$; concave upward for $x < 0$; point of inflection at $x = 0$
3. Concave downward for $|x| < \frac{1}{2}\sqrt{2}$; concave upward for $|x| > \frac{1}{2}\sqrt{2}$; points of inflection at $x = \pm \frac{1}{2}\sqrt{2}$
4. Concave downward for $|x| > 1$; concave upward for $|x| < 1$; points of inflection at $x = \pm 1$
5. Concave upward for all x
6. Concave downward for $e^{\pi[2k+(1/4)]} < x < e^{\pi[2k+(5/4)]}$;
   concave upward for $e^{\pi[2k-(3/4)]} < x < e^{\pi[2k+(1/4)]}$;
   points of inflection at $x = e^{\pi[k+(1/4)]}$, k any integer
7. Concave downward for all $x < 2$; no points of inflection
8. Concave downward for $|x| < 3$; concave upward for $|x| > 3$; no points of inflection

8.10 Exercises (page 396)

1. a/b
2. $14/3$
3. $(a/b)^2$
4. $-\frac{1}{6}$
5. $\frac{1}{6}$
6. 1
7. $\log a \log b$
8. $\frac{1}{3}$
9. $\frac{1}{2}$
10. $\frac{1}{6}$
11. -1
12. -1
13. $\frac{1}{6}$
14. 1
15. $\frac{1}{6} \log a$
16. $\frac{1}{6}$
17. $1/\sqrt{2a}$
18. -2
19. -2
20. 1
21. $-\frac{1}{3}$
22. $\dfrac{n(n+1)}{2}$
23. $\dfrac{a^2 - b^2}{3a^2b^2}$
24. $a = 2$; limit $= \frac{3}{2}$
25. $a = 4$; $b = 1$

8.15 Exercises (page 402)

1. 0
2. 1

3. $\frac{1}{3}$
4. $1/\sqrt{b}$
5. $1/24$
6. Meaningless because $\sin 2x < 0$ for $\pi/2 < x < \pi$
7. 0
8. $e/2$
9. $+\infty$
10. 1
11. 0
12. 0
13. $\frac{1}{2}$
14. $c = -1$; limit $= \frac{1}{2}\sqrt{3}$

8.16 Exercises (page 406)

1. $\frac{1}{2}$
2. $\frac{1}{2}$
3. $\frac{1}{2}$
4. 0
5. 0
6. 1
7. -1
8. 1
9. e
10. 1
11. e^{-1}
12. 1
13. e^3
14. e^e
15. e^{-1}
16. $e^{2/\pi}$
17. $-e/2$
18. $e^{-1/2}$
19. $e^{1/6}$
20. $-\frac{1}{2}$

8.18 Miscellaneous review exercises (page 408)

1. If $c \geq 0$, f increases when $x > 0$ and decreases when $x < 0$, minimum $= 1$ when $x = 0$. If $c < 0$, f increases when $-\sqrt{-\frac{c}{2}} < x < 0$ or when $x > \sqrt{-\frac{c}{2}}$, and decreases when $x < -\sqrt{-\frac{c}{2}}$ or when $0 < x < \sqrt{-\frac{c}{2}}$; relative max $= 1$ at $x = 0$, relative min $= 1 - \frac{1}{4}c^2$ at $x = \pm\sqrt{-\frac{c}{2}}$
2. Increasing when $\pi/6 < x < \pi/2$ and when $\pi/2 < x < 5\pi/6$; decreasing when $0 < x < \pi/6$ and when $5\pi/6 < x < \pi$; relative min $= \sqrt{3}$ at $x = \pi/6$; relative max $= -\sqrt{3}$ at $x = 5\pi/6$; $f(x) \to +\infty$ as $x \to \pi/2-$; $f(x) \to -\infty$ as $x \to \pi/2+$
3. (a) Local max when $x = \frac{1}{2}$
4. 22

6. $r = \dfrac{1 + c}{2} \to \dfrac{1}{2}$ as $c \to 0$
7. $3x^2 - y^2 = 3a^2$; $\dfrac{A(r)}{r^3} \to \dfrac{1}{36a}$
8. (a) $(0, \frac{1}{2})$
   (b) Write $Q = (0, b(x))$. If $f''(0) \neq 0$ then $b(x) \to f(0) + \dfrac{1}{f''(0)}$ as $x \to 0$.
   Otherwise, $|b(x)| \to +\infty$ as $x \to 0$.
9. (a) 6 as $x \to 0$; $4/\pi$ as $x \to \pi/2$
   (b) $a = -3$, $b = 9/2$
10. 3
12. (a) Relative minimum at 0
   (b) $a = 2/3$, $b = 20/9$
   (d) 2/3

Chapter 9

9.4 Exercises (page 418)

1. (a) Converges
   (b) 0
2. (a) Converges
   (b) -1
3. (a) Diverges
4. (a) Converges
   (b) $\frac{1}{3}$
5. (a) Converges
   (b) 0
6. (a) Diverges
7. (a) Converges
   (b) 0
8. (a) Diverges
9. (a) Converges
   (b) 1
10. (a) Diverges
11. (a) Converges
   (b) 0
12. (a) Converges
   (b) $\frac{1}{3}$
13. (a) Converges
   (b) 0
14. (a) Converges
   (b) 0
15. (a) Converges
   (b) 0
16. (a) Converges
   (b) 0
17. (a) Converges
   (b) e^2
18. (a) Diverges
19. $N > 1/\epsilon$

20. $N > 1/\epsilon$
21. $N > 1/\epsilon$
22. $N > 1/\epsilon$
23. $N > \sqrt{2/\epsilon}$
24. $N > \dfrac{\log \epsilon}{\log (9/10)}$

9.9 Exercises (page 426)

22. (a) 1
    (b) $2e - 3$
    (c) $e + 1$
23. (b) 5
24. (a) Identical
    (b) Not identical
    (c) Not identical
    (d) Identical

9.10 Exercises on decimal expansions (page 428)

1. 4/9
2. 51/99
3. 200/99
4. 41/333
5. 1/7

9.13 Exercises (page 433)

1. Divergent
2. Convergent
3. Convergent
4. Convergent
5. Convergent
6. Convergent
7. Convergent
8. Convergent
9. Divergent
10. Convergent
11. Divergent
12. Convergent
13. Divergent
14. Convergent
15. Convergent for $s > 1$; divergent for $s \leq 1$
16. Convergent
17. Convergent
18. Convergent

9.15 Exercises (page 437)

1. Convergent
2. Convergent
3. Convergent

4. Divergent
5. Divergent
6. Divergent
7. Divergent
8. Convergent
9. Convergent
10. Divergent
11. Convergent
12. Divergent
13. Convergent
14. Convergent if $0 < r < 1$, or when $x = k\pi$, k any integer

9.18 Exercises (page 442)

1. Conditionally convergent
2. Conditionally convergent
3. Divergent for $s \leq 0$; conditionally convergent for $0 < s \leq 1$; absolutely convergent for $s > 1$
4. Absolutely convergent
5. Absolutely convergent
6. Absolutely convergent
7. Divergent
8. Divergent
9. Divergent
10. Conditionally convergent
11. Absolutely convergent
12. Divergent
13. Absolutely convergent
14. Absolutely convergent
15. Divergent
16. Absolutely convergent
17. Absolutely convergent
18. Absolutely convergent
19. Conditionally convergent
20. Conditionally convergent
21. Divergent
22. Conditionally convergent
23. Divergent
24. Conditionally convergent
25. Divergent for $s \leq 0$; conditionally convergent for $0 < s \leq 1$; absolutely convergent for $s > 1$
26. Absolutely convergent
27. Absolutely convergent
28. Divergent
29. Absolutely convergent
30. Absolutely convergent
31. Absolutely convergent
32. Absolutely convergent
33. $x = 0$
34. Absolutely convergent for all x
35. Absolutely convergent for $x > -\frac{1}{3}$ or $x < -1$

36. Absolutely convergent for $x > -\frac{1}{3}$ or $x < -1$
37. Conditionally convergent for any x not a negative integer
38. Absolutely convergent if $|\sin x| \leq \frac{1}{2}\sqrt{2}$
39. Absolutely convergent if $|x| < 1$; conditionally convergent if $x = -1$
40. Absolutely convergent if $-2 < x < -1$; conditionally convergent if $x = -2$
41. Absolutely convergent if $|x| < e^{-1/85}$
42. Absolutely convergent if $x > 0$; conditionally convergent if $x = 0$
43. Absolutely convergent if $|x - k\pi| \leq \pi/6$, k any integer
44. Absolutely convergent if $0 < x < 2$; conditionally convergent if $x = 2$

9.20 Exercises (page 446)

1. $-5 \leq x < -1$
2. $-1 < x < 5$
3. $-2 \leq x \leq 0$
4. $-e < x < e$
5. $|x| \leq \frac{1}{2}$
6. $|x| < \frac{1}{2}$
7. All x if $a = k\pi$, k an integer; $|x| < 1$ if $a \neq k\pi$
8. $|x| < e^{-a}$
9. All x
10. $|x| < e^{-1}$
11. $|x| < r$, where $r = \min(1/a, 1/b)$; if $a \geq b$, converges at $x = -r$ and diverges at $x = r$; if $a < b$, converges at both end points
12. $|x| < \max(a, b)$
13. $|x| < 4$
14. $|x| < 1$
15. $-1 \leq x \leq 3$

9.25 Exercises (page 457)

1. All x
2. All x
3. All x; $a = 1, b = 0$
6. $y = 1 + x + x^2 + \frac{4}{3}x^3 + \cdots$
7. $y = x + \dfrac{1}{4}x^4 + \dfrac{1}{14}x^7 + \dfrac{23}{1120}x^{10} + \cdots$
8. $y = \dfrac{1}{2}x^2 + \dfrac{1}{20}x^5 + \dfrac{1}{160}x^8 + \dfrac{7}{8800}x^{11} + \cdots$
9. $y = \displaystyle\sum_{n=0}^{\infty} \frac{\alpha^n x^n}{n!}$
10. $$y = c_0\left(1 + \sum_{n=1}^{\infty} \frac{x^{3n}}{(2 \cdot 3)(5 \cdot 6) \cdot \cdots \cdot [(3n-1)\cdot(3n)]}\right) + c_1\left(x + \sum_{n=1}^{\infty} \frac{x^{3n+1}}{(3 \cdot 4)(6 \cdot 7) \cdot \cdots \cdot [(3n)\cdot(3n+1)]}\right)$$
11. $$y = c_0\left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2 \cdot 4 \cdot \cdots \cdot (2n)}\right) + c_1 \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n-1}}{1 \cdot 3 \cdot \cdots \cdot (2n-1)}$$

9.27 Exercises (page 461)

2. (c) $\sqrt{2} = 1.4142135623$
3. (b) $\sqrt{3} = 1.732050807568877$
4. 0.0500; 0.0599; 0.0997
5. (b) $\log 2 = 0.69315$
   (c) $\log 3 = 1.09861$; $\log 5 = 1.60944$

9.29 Exercises (page 466)

1. Divergent
2. Convergent
3. Convergent
4. Convergent
5. Convergent
6. Convergent
7. Convergent
8. Convergent
9. Divergent
10. Convergent if $s > 1$; divergent if $s \leq 1$
11. $\alpha = \frac{1}{2}\sqrt{2}$; $\frac{3}{\sqrt{2}} \log \sqrt{2}$
12. $a = b = 2e - 2$
13. $a = 1$; $b = 1 - \frac{\sqrt{3}}{\pi}$
14. (b) Both diverge
15. (c) Diverges

9.31 Miscellaneous exercises (page 469)

1. $\frac{1}{2}(1 + \sqrt{5})$
2. 0
4. Divergent
5. Convergent if $s < \frac{1}{2}$; divergent if $s \geq \frac{1}{2}$
6. Convergent
7. Divergent
8. Divergent
13. When $a \geq -1$, limit is $\frac{a+1}{a+2}$; when $a \leq -1$, limit is 0
14. $-\frac{\sqrt{2}\, 2^{97}}{98!}$
15. Absolutely convergent for $1 < x < 3$; conditionally convergent for $x = 1$

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